

THE FRACTIONAL DE GIORGI CONJECTURE IN DIMENSION FOUR FOR $1/2 < s < 1$

ABSTRACT. We prove that every bounded entire solution of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^4$, strictly increasing in one coordinate, is a planar layer when $1/2 < s < 1$. No assumptions are made on its directional limits. The proof first classifies stable entire solutions in $\mathbb{R}^3$. We obtain separation estimates for actual transition sheets, control their profiles and projected equations, and retain curvature under rescaling through an exact tangential trace operator. Surface stability and a capacity argument then exclude every exceptional zero. The resulting three-dimensional classification identifies the directional limits of a monotone four-dimensional solution. A comparison argument gives global minimality, and minimizer rigidity yields the claimed one-dimensional form.

Draft status. This candidate proof was obtained by our team on September 19, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. INTRODUCTION

The De Giorgi problem connects monotonicity of an entire phase transition with the geometry of its level sets. For the Allen–Cahn equation, the expected conclusion is that a monotone solution depends on a single Euclidean coordinate. In the fractional equation the interaction extends over the whole space, and the symmetry problem depends on both the dimension and the order of the operator. We prove the four-dimensional monotone classification throughout $1/2 < s < 1$ without imposing limits at infinity. The argument first establishes the corresponding classification of stable solutions in dimension three.

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1.1. Main results. We use the Fourier normalization $|\xi|^{2s}$ for $(-\Delta)^s$. Let q_s denote the increasing one-dimensional solution of

$$(-\partial_t^2)^s q_s = q_s - q_s^3, \quad q_s(0) = 0, \quad \lim_{t \rightarrow \pm\infty} q_s(t) = \pm 1.$$

Existence and uniqueness up to translation are given by Cabré–Sire [8, Theorems 2.4 and 2.7]. Our principal result is the following.

Theorem 1.1. *Let $1/2 < s < 1$. Every bounded entire classical solution of*

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^4, \quad \partial_4 u > 0 \quad \text{in } \mathbb{R}^4$$

has the form

$$u(x) = q_s(e \cdot x - a), \quad e \in S^3, \quad e_4 > 0, \quad a \in \mathbb{R}.$$

In particular its directional limits are -1 and 1 .

The directional limits in this theorem are conclusions, rather than hypotheses. They are obtained by classifying the stable solutions that arise as translation limits in the increasing direction. Here stability means nonnegativity of the second variation for compactly supported perturbations; its precise normalized form is recorded in Section 2. The two constant wells are -1 and 1 .

Theorem 1.2. *Every bounded stable entire classical solution of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^3$, $1/2 < s < 1$, is a constant well or has the form*

$$u(x) = q_s(e \cdot x - a), \quad e \in S^2, \quad a \in \mathbb{R}.$$

The proof of Theorem 1.2 occupies the three-dimensional part of the paper. It supplies the endpoint classification needed to make the monotone solution in Theorem 1.1 a global minimizer. Both proofs are given after their analytic and geometric ingredients have been established.

1.2. The classical De Giorgi problem. For the local equation

$$-\Delta v = v - v^3 \quad \text{in } \mathbb{R}^N, \quad \partial_N v > 0,$$

De Giorgi’s conjecture predicts one-dimensionality of bounded entire solutions for $N \leq 8$. Nasif Ghoussoub and Changfeng Gui proved the two-dimensional conjecture in [24]; Ambrosio and Cabré proved the three-dimensional result in [1, Theorem 1.2]. Neither result requires prescribed directional well limits. Ghoussoub and Gui’s second paper, published in the *Annals of Mathematics* [25], treated dimensions four and five under additional hypotheses. In particular, their Theorem 1.2 assumes the pointwise limits $v(x', t) \rightarrow \pm 1$ as $t \rightarrow \pm\infty$ and, after translation and rotation, the antisymmetry

$$v(y, z) = -v(y, -z), \quad (y, z) \in \mathbb{R}^{N-k} \times \mathbb{R}^k, \quad 1 \leq k \leq N.$$

Their Theorem 1.1 gives a separate criterion in terms of quantitative convergence of the normalized energy. These are conditional results in the higher dimensions.

Savin [32, Theorems 2.3 and 2.4(a)] classified global minimizers for $N \leq 7$ and proved the monotone conjecture for $N \leq 8$ under pointwise directional limits to the two wells. The pointwise condition here means that x' is fixed when $t \rightarrow \pm\infty$. In dimensions $N \geq 9$, del Pino, Kowalczyk, and Wei [14, Theorem 1] constructed bounded strictly monotone solutions with nonplanar level sets, showing that the conjectured dimension threshold cannot be increased.

Recent work has advanced the stable formulation, in which monotonicity is replaced by nonnegativity of the second variation. Florit–Simon and Serra [22, Theorem 1.3] classified stable solutions in $\mathbb{R}^4$ under the cubic energy-growth bound $E(B_R) = O(R^3)$. Two independent preprints of September 2026, by Liu–Luo–Wang–Wei–Wu [28, Theorems 1.1 and 1.2] and Chan–Fernández–Real–Figalli–Florit–Simon–Serra [11, Theorem 1.1 and Corollary 1.2], give the classification of arbitrary bounded stable solutions in $\mathbb{R}^3$ and consequently the unrestricted monotone result in $\mathbb{R}^4$. The energy bound in the four-dimensional stable theorem and the end-state hypotheses in the higher-dimensional monotone theorems are distinct restrictions.

1.3. Gibbons' conjecture and uniform end states. Gibbons' conjecture asks for symmetry under the stronger condition

$$\lim_{t \rightarrow \pm\infty} \sup_{x' \in \mathbb{R}^{N-1}} |v(x', t) \mp 1| = 0. \quad (1.1)$$

For the classical Allen–Cahn equation, bounded entire solutions satisfying (1.1) are one-dimensional in every dimension, with

$$v(x', t) = \tanh((t - a)/\sqrt{2})$$

for some $a \in \mathbb{R}$. Independent proofs were given by Berestycki–Hamel–Monneau [4, Theorem 1], Barlow–Bass–Gui [2, Theorem 1], and Farina [17, Theorem 2.1]. Strict monotonicity is a consequence in this setting and need not be assumed. Farina and Valdinoci [18, Theorem 2] proved the fractional analogue for all $0 < s < 1$ and every dimension in their $W^{3,\infty}$ solution class.

Uniformity in the transverse variables is essential to this formulation. A tilted planar layer has the pointwise directional limits used in De Giorgi-type theorems, but generally fails (1.1). Theorem 1.1 allows every orientation with $e_4 > 0$ and assumes neither pointwise nor uniform end states.

1.4. The fractional De Giorgi problem. The layer construction, regularity, and qualitative properties of fractional Allen–Cahn solutions were developed by Cabré and Sire [7, 8]. Their stable two-dimensional symmetry theorem holds for every $0 < s < 1$ [8, Theorem 2.12]. In dimension three, Cabré and Cinti [5, Theorem 1.5] proved one-dimensionality for bounded monotone solutions and global minimizers when $1/2 \leq s < 1$. Dipierro, Farina, and Valdinoci [15, Theorem 1.1] proved the monotone result for $0 < s < 1/2$. Together these results cover monotone three-dimensional solutions for the full fractional range without prescribed end states. At $s = 1/2$, Figalli and Serra [20, Theorem 1.5 and Corollary 1.6] classified bounded stable solutions in dimension three and monotone solutions in dimension four.

For $1/2 \leq s < 1$, Savin [33, 34, Theorems 1.2 and 1.3] classified global minimizers in dimensions at most seven and monotone solutions with pointwise directional well limits in dimensions at most eight. Below the half-order threshold, Dipierro–Serra–Valdinoci [16, Theorems 1.5 and 1.6] obtained the same respective dimension bounds when $s < 1/2$ is sufficiently close to $1/2$. In this regime Cabré–Cinti–Serra [6, Corollaries 1.10 and 1.11] proved the stable three-dimensional and unrestricted monotone four-dimensional classifications. The perimeter-order parameter in these latter works is $\sigma = 2s$, so $\sigma \uparrow 1$ corresponds to physical $s \uparrow 1/2$ from below.

The connection with nonlocal minimal surfaces has also led to recent geometric rigidity results. Chan–Dipierro–Serra–Valdinoci [10, Theorem 1.5 and Corollary 1.6] proved flatness of stable nonlocal minimal hypercones in $\mathbb{R}^4$ with nonempty smooth spherical trace, and of smooth stable entire hypersurfaces in $\mathbb{R}^4$, for perimeter order $\sigma < 1$ sufficiently close to 1. These results belong to the below-half regime. The present paper treats the interval above it: Theorem 1.2 gives the stable three-dimensional classification for every $1/2 < s < 1$, and Theorem 1.1 gives the four-dimensional monotone conclusion throughout that interval without an end-state assumption.

1.5. The proof mechanism. Savin's minimizer theorem [33, Theorem 1.2] is available in the range considered here, but monotonicity alone does not provide its minimizing hypothesis. The comparison argument of Cabré–Cinti–Serra [6, Proposition 7.9 and the following observation] supplies that hypothesis once the directional limits are known to be wells or planar layers. This is why the stable three-dimensional problem precedes the four-dimensional monotone argument.

The quantitative input is the uniform variation estimate of Gui–Jin–Zhao [27]. With $p = 2s \in (1, 2)$, it gives

$$\int_{B_R(x_0)} |\nabla u| \leq CR^P, \quad P = 3 - \frac{2}{p+1} > 2.$$

Applying the vector stability identity with a direct ambient cutoff therefore produces a cost of order R^{P-2} , which does not tend to zero. The argument must use the geometry of the transition region to construct a better test. Clearing estimates and point selection first give complete local graphs throughout a fixed central phase band. Compactness on these graphs supplies an initial

divergent separation between consecutive zeros, independently of the subsequent quantitative estimates.

We next apply stability to two actual neighboring sheets. The exterior row is kept signed, and every target in the nonlocal interaction is retained, including portions of sheets which leave and return to the chart. This gives an inverse-gap estimate and then logarithmic separation. The separation allows the reference profile to be fixed by the actual zero surface. Projection of the actual equation in Fermi coordinates controls the associated flux; applied to the corrected gap, it improves separation to the scale $R^{2/(p+1)}$. These steps are developed in Sections 4–7.

Two curvature arguments are then needed. A harmonic quadratic limit at the physical scale transfers curvature between central phases. An exact trace symbol, its finite-second-moment kernel, and a quantitative mean-curvature estimate give convergence of the Hessian at the curvature scale. The latter step retains a marked nonzero curvature in a complete intrinsic limit. Ambient stability becomes surface stability, and the Fischer-Colbrie–Schoen theorem [21] rules out that limit. We consequently obtain curvature control by the distance to the fixed exceptional set.

The remaining global argument removes this exceptional set. A fixed detector records a positive amount of vector defect at each exceptional zero. Sublinear total defect bounds give isolated finite clusters. On the regular exterior, a metric completion and the integrated Fiala inequality produce a logarithmic capacity test. Joining it to an ambient amplitude near the cluster preserves the negative defect contribution and makes every positive error small. The cluster and capacity strategy is related to the classical stable Allen–Cahn argument of Liu–Luo–Wang–Wei–Wu [28]; the nonlocal rows, normal cutoffs, and profile estimates required here are derived explicitly. The scales are chosen before the cluster, so the estimates do not depend on its diameter or packing count. This closes the stable three-dimensional classification in Sections 13 and 14.

Finally, a positive Jacobi field gives stability of the original monotone solution. Translation compactness and exact cylindrical reduction give stable directional limits in $\mathbb{R}^3$. Their classification makes the minimizer comparison applicable. After matching both the operator and energy normalizations, Savin’s minimizer theorem yields the four-dimensional layer.

1.6. Outline of the paper. Section 2 fixes the operator, energy, and layer normalizations, and Section 3 gives the compactness and stability tools. Sections 4–7 construct actual transition sheets, prove their separation, and derive the projected equation. Sections 8–10 develop the physical-scale cancellation, exact trace operator, and curvature retention. Sections 11 and 12 pass to complete surface limits and obtain distance-scale curvature control. Section 13 excludes exceptional zeros, and Section 14 proves Theorem 1.2. Finally, Section 15 classifies directional limits and establishes global minimality, and Section 16 proves Theorem 1.1. The proofs follow this dependency order throughout.

2. THE EQUATION AND ITS VARIATIONAL NORMALIZATION

Fix $s \in (1/2, 1)$ and put $p = 2s$. In every dimension we use the Fourier multiplier $|\xi|^{2s}$, or equivalently

$$(-\Delta)^s u(x) = c_{n,s} \text{PV} \int_{\mathbb{R}^n} \frac{u(x) - u(y)}{|x - y|^{n+p}} dy, \quad c_{n,s} = \frac{4^s \Gamma(n/2 + s)}{\pi^{n/2} |\Gamma(-s)|}. \quad (2.1)$$

We write

$$f(t) = t - t^3, \quad F(t) = \frac{1}{4}(1 - t^2)^2, \quad F' = -f, \quad (2.2)$$

so the equation under consideration is

$$(-\Delta)^s u = f(u) \quad \text{in } \mathbb{R}^n. \quad (2.3)$$

A bounded entire classical solution is called stable if

$$Q_u(\psi) = \frac{c_{n,s}}{2} \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{(\psi(x) - \psi(y))^2}{|x - y|^{n+p}} dx dy + \int_{\mathbb{R}^n} (3u^2 - 1)\psi^2 dx \geq 0 \quad (\psi \in C_c^\infty(\mathbb{R}^n)). \quad (2.4)$$

For a bounded open set Ω let $\mathcal{Q}_\Omega = \mathbb{R}^{2n} \setminus (\Omega^c \times \Omega^c)$ and set

$$E_\Omega(v) = \frac{c_{n,s}}{4} \iint_{\mathcal{Q}_\Omega} \frac{|v(x) - v(y)|^2}{|x - y|^{n+p}} dx dy + \int_\Omega F(v) dx. \quad (2.5)$$

An admissible competitor is a measurable function v with $v = u$ almost everywhere on Ω^c and $E_\Omega(v) < \infty$. Global minimality means that $E_\Omega(u) \leq E_\Omega(v)$ for every bounded open Ω and every such competitor. We retain the coefficient $c_{n,s}$ throughout; it will determine the rescalings in the final minimizer argument.

2.1. Uniform regularity and the energy variations. Bounded solutions of (2.3) have uniform local $C^{2,\beta}$ bounds at every translation center. To see this, fix $M \geq \|u\|_{L^\infty(\mathbb{R}^n)}$. In the notation of [30], the operator $L = -(-\Delta)^s$ has the constant angular density $a = c_{n,s}/2$ in their formula (1.3), and satisfies their ellipticity and smoothness hypotheses. Since $|f(u)| \leq M + M^3$, [30, Theorem 1.1(a)], applied on $B_2(x_0)$ after a fixed-radius rescaling, gives

$$\|u\|_{C^p(B_1(x_0))} \leq C(n, s, M).$$

Here $p \in (1, 2)$ is nonintegral, so this estimate includes a uniform gradient bound. Taking all centers gives $G := \|\nabla u\|_{L^\infty(\mathbb{R}^n)} \leq C(n, s, M)$, and hence

$$|u(x) - u(y)| \leq \min\{G|x - y|, 2M\} \quad (x, y \in \mathbb{R}^n).$$

For $|x - y| \leq 1$ use the first bound, and for $|x - y| \geq 1$ use the second. Thus the global C^α norm of u is controlled for every fixed $0 < \alpha < 1$. Since f' is bounded on $[-M, M]$, composition gives a uniformly bounded C^α norm for $f(u)$. We choose

$$\alpha_0 = \frac{3-p}{2} \in (2-p, 1).$$

Since $p + \alpha_0 = 2 + \beta$ with $\beta = (p-1)/2 \in (0, 1/2)$, this sum is not an integer. [30, Corollary 3.5] now gives

$$\sup_{x_0 \in \mathbb{R}^n} \|u\|_{C^{2,\beta}(B_{1/2}(x_0))} \leq C(n, s, M).$$

The forcing in that reference is $-f(u)$, because its operator has the opposite sign. The constant density belongs to C^{α_0} on the sphere, as required by the corollary. The constants depend only on n, s, M , and not on the center. These estimates use the original finite bound M ; after Lemma 2.1, we can take $M = 1$ for every bounded entire solution.

We can now justify the variational assertions. For a bounded open Ω , set

$$L_\Omega = \sup_{\text{dist}(x, \Omega) \leq 1} |\nabla u(x)| < \infty.$$

For $x \in \Omega$ and $|x - y| < 1$, the segment joining x and y lies in this neighborhood and $|u(x) - u(y)| \leq L_\Omega |x - y|$. For $|x - y| \geq 1$, the bound is $|u(x) - u(y)| \leq 2M$. Symmetry and polar integration give

$$\iint_{\mathcal{Q}_\Omega} \frac{|u(x) - u(y)|^2}{|x - y|^{n+p}} dx dy \leq 2|\Omega| |S^{n-1}| \left(\frac{L_\Omega^2}{2-p} + \frac{4M^2}{p} \right).$$

The potential is also integrable on Ω , so $E_\Omega(u) < \infty$; no boundary regularity of Ω is needed.

Let $\psi \in C_c^\infty(\Omega)$ be real. Then $u + \varepsilon\psi$ is an admissible competitor for every real ε . The mixed increment integral below is absolutely convergent by Cauchy–Schwarz, using the preceding finite energy and the finite fractional seminorm of ψ . For the latter, apply the preceding kernel estimate with ψ in place of u : both $\|\psi\|_\infty$ and $\|\nabla\psi\|_\infty$ are finite, and its increments vanish on $\Omega^c \times \Omega^c$, so the restricted integral is the full fractional seminorm. Since the increments of ψ vanish on

$\Omega^c \times \Omega^c$, expansion of the kinetic term and differentiation of the quartic potential give

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_\Omega(u + \varepsilon\psi) \right|_{\varepsilon=0} &= \frac{c_{n,s}}{2} \iint_{\mathbb{R}^n \times \mathbb{R}^n} \frac{(u(x) - u(y))(\psi(x) - \psi(y))}{|x - y|^{n+p}} dx dy - \int_\Omega f(u)\psi dx \\ &= \int_\Omega [(-\Delta)^s u - f(u)]\psi dx, \\ \left. \frac{d^2}{d\varepsilon^2} E_\Omega(u + \varepsilon\psi) \right|_{\varepsilon=0} &= Q_u(\psi). \end{aligned}$$

For the integration by parts in the second line, put

$$T_\delta u(x) = c_{n,s} \int_{|z|>\delta} \frac{u(x) - u(x+z)}{|z|^{n+p}} dz.$$

The truncated integrals are absolutely convergent, and kernel symmetry gives

$$\frac{c_{n,s}}{2} \iint_{|x-y|>\delta} \frac{(u(x) - u(y))(\psi(x) - \psi(y))}{|x - y|^{n+p}} dx dy = \int_\Omega T_\delta u \psi dx.$$

The left side converges by the absolute integrability of the mixed increments. On the support of ψ , symmetrization and the uniform local C^2 bounds give

$$|T_\delta u(x) - (-\Delta)^s u(x)| \leq C \int_0^\delta r^{1-p} dr = \frac{C}{2-p} \delta^{2-p} \quad (0 < \delta < 1).$$

The bounded far tail is integrable since $p > 0$. This proves the integration-by-parts identity and justifies its use above. The first variation therefore vanishes for a solution of (2.3). If u is a global minimizer, the scalar function $\varepsilon \mapsto E_\Omega(u + \varepsilon\psi)$ has a minimum at zero, and its second derivative is nonnegative. Choosing Ω to contain any prescribed compact test proves (2.4). This verifies both the sign and the coefficient of the stability form.

2.2. Translation compactness and the constant end cases. The preceding bounds apply equally to any sequence of entire solutions with the same global bound M . On each compact ball, the functions and their derivatives of order at most two are uniformly bounded; the second derivatives are equicontinuous by the uniform $C^{2,\beta}$ estimate. Arzelà–Ascoli and a diagonal extraction over the balls give convergence of all derivatives through order two. Passing to the limit in the fundamental theorem of calculus identifies these derivative limits with the derivatives of a function v . Consequently $u_j \rightarrow v$ in C_{loc}^2 and $|v| \leq M$. In particular this applies to arbitrary translates of one bounded solution.

Translations preserve the equation: changing variables in the truncated kernel integral and then taking its principal value gives

$$(-\Delta)^s[u(\cdot + a)](x) = ((-\Delta)^s u)(x + a) = f(u(x + a)) \quad (a \in \mathbb{R}^n).$$

To pass the equation to the limit, use the absolutely convergent symmetric formula

$$(-\Delta)^s u_j(x) = \frac{c_{n,s}}{2} \int_{\mathbb{R}^n} \frac{2u_j(x) - u_j(x+z) - u_j(x-z)}{|z|^{n+p}} dz.$$

For x in a fixed compact set and $0 < r < 1 < A$, the integral over $|z| < r$ is bounded in absolute value by Cr^{2-p} , uniformly in j , by the local second derivative bounds. The part over $|z| > A$ is bounded by CMA^{-p} . On the annulus $r \leq |z| \leq A$, the arguments $x, x+z, x-z$ remain in one fixed compact set, so local uniform convergence permits passage to the limit, uniformly for x in the chosen compact set. The same small-radius and far-tail bounds hold for v . Taking first $j \rightarrow \infty$, then $r \downarrow 0$ and $A \uparrow \infty$, proves local uniform convergence of the fractional Laplacians. Since $f(u_j) \rightarrow f(v)$ locally uniformly, $(-\Delta)^s v = f(v)$.

If all u_j are stable, fix a real compact smooth test ψ . Its kinetic integral is finite by the preceding variational argument and is independent of j , while $3u_j^2 - 1 \rightarrow 3v^2 - 1$ uniformly on $\text{supp } \psi$. Thus $Q_{u_j}(\psi) \rightarrow Q_v(\psi)$ and the limit is stable. Translates of a stable solution are stable by a change of variables in (2.4), so the same conclusion holds for translation limits.

At a global maximum or minimum of a C^2 function, its gradient vanishes. Hence its increment from the contact point is $O(|z|^2)$. Since $p < 2$, the unsymmetrized integral in (2.1) is absolutely convergent at such a point for a bounded function. We may therefore read its sign directly from the positive kernel in the following two proofs.

Lemma 2.1. *Every bounded entire classical solution of (2.3) takes values in $[-1, 1]$. A nonconstant solution takes values in $(-1, 1)$.*

Proof. If $U = \sup_{\mathbb{R}^n} u > 1$, choose x_j with $u(x_j) \rightarrow U$ and apply the preceding compactness to $u_j(x) = u(x + x_j)$, using the fixed bound $\|u\|_\infty$. The limit satisfies $v(0) = U$ and $v \leq U$ everywhere. At this attained maximum, the absolutely convergent integral in (2.1) is nonnegative, whereas the equation gives $(-\Delta)^s v(0) = f(U) < 0$. This is impossible. Linearity of the operator and oddness of f give $(-\Delta)^s(-u) = -f(u) = f(-u)$. Thus $-u$ is also a bounded entire solution, and applying the upper-bound argument to it gives $u \geq -1$. If $u(x_0) = 1$, the equation says that the absolutely integrable nonnegative function $(1 - u(y))|x_0 - y|^{-n-p}$ has zero integral. Thus $u = 1$ almost everywhere, and continuity gives $u \equiv 1$. The same reasoning at -1 proves the strict bounds for a nonconstant solution. $\square$

Lemma 2.2. *A stable entire solution of (2.3) which has one sign is a constant well: $u \equiv 1$ if $u \geq 0$, and $u \equiv -1$ if $u \leq 0$.*

Proof. Suppose $u \geq 0$. By Lemma 2.1, $m = \inf u \in [0, 1]$. Translate a sequence approaching m to obtain a stable limit v with $m \leq v \leq 1$ and $v(0) = m$. At this minimum $(-\Delta)^s v(0) \leq 0$, excluding $0 < m < 1$ since $f(m) > 0$. If $m = 0$, the equation gives

$$0 = (-\Delta)^s v(0) = -c_{n,s} \int_{\mathbb{R}^n} \frac{v(y)}{|y|^{n+p}} dy.$$

The integral is absolutely convergent: $v(0) = 0$ is a global minimum, so $\nabla v(0) = 0$ and the local C^2 bound gives $0 \leq v(y) \leq C|y|^2$ near zero. Its local radial integral is bounded by $C \int_0^1 r^{1-p} dr < \infty$, while boundedness of v controls the far tail because $p > 0$. Since $v \geq 0$ and the kernel is strictly positive, $v = 0$ almost everywhere and, by continuity, $v \equiv 0$. This limit is unstable. Indeed, take a nonzero real $\eta \in C_c^\infty(\mathbb{R}^n)$ and put $\eta_L(x) = \eta(x/L)$. Set

$$A_\eta = \frac{c_{n,s}}{2} \iint \frac{|\eta(X) - \eta(Y)|^2}{|X - Y|^{n+p}} dX dY < \infty, \quad D_\eta = \int \eta^2 dX > 0.$$

The preceding variational estimate gives $A_\eta < \infty$. Changing variables $x = LX$, $y = LY$ in the kinetic integral and $x = LX$ in the potential integral gives

$$Q_0(\eta_L) = L^{n-p} A_\eta - L^n D_\eta = L^n (L^{-p} A_\eta - D_\eta) < 0$$

when L is sufficiently large. Therefore $m = 1$, and Lemma 2.1 gives $u \equiv 1$. Since f is odd and f' is even, $-u$ has the same equation and stability form; this also proves the assertion for $u \leq 0$. $\square$

2.3. The reference layer. Recall the increasing layer q_s introduced in Section 1. We use the existence and uniqueness theorem [8, Theorem 2.4] for the one-dimensional equation

$$(-\partial_t^2)^s q_s = q_s - q_s^3, \quad \lim_{t \rightarrow -\infty} q_s(t) = -1, \quad \lim_{t \rightarrow +\infty} q_s(t) = 1.$$

Its hypotheses hold because f is smooth on a neighborhood of $[-1, 1]$, $f(\pm 1) = 0$, $f'(\pm 1) = -2 < 0$, and $F > F(-1) = F(1) = 0$ on $(-1, 1)$. To meet the stated global $C^{1,\gamma}(\mathbb{R})$ hypothesis, take, for example, $\gamma = 1/2 > \max\{0, 1 - 2s\}$ and replace f outside a neighborhood of $[-1, 1]$ by an odd smooth compactly supported extension. The layer stays in $(-1, 1)$ by strict monotonicity and its end limits, so this extension does not change its equation or any of the potential conditions on $[-1, 1]$. The cited theorem gives $q'_s > 0$ and uniqueness among increasing layers up to translation. Since the layer has a unique zero, we fix that translation by $q_s(0) = 0$.

The same theorem gives oddness about some point because f is odd; that point is a zero of the layer. Its unique zero is the origin, so q_s is odd. We shall use

$$\phi(t) = q'_s(t) > 0,$$

which is even by differentiation. [8, Theorem 2.7] gives positive constants a_s, A_s such that

$$a_s(1 + |t|)^{-1-p} \leq \phi(t) \leq A_s(1 + |t|)^{-1-p}, \quad a_s t^{-p} \leq 1 - q_s(t) \leq A_s t^{-p} \quad (t \geq 1).$$

The theorem gives the derivative bounds for $|t| \geq 1$; continuity and strict positivity of ϕ extend them to all t , with $1 + |t|$ in place of $|t|$ and adjusted constants. The positive tail can also be read directly from

$$1 - q_s(t) = \int_t^\infty \phi(r) dr \quad (t \geq 1),$$

since integrating $(1 + r)^{-1-p}$ gives $p^{-1}(1 + t)^{-p}$, which is comparable to t^{-p} there. Shrinking a_s and enlarging A_s if necessary makes the same pair of constants valid in both displayed estimates. The tail at $-\infty$ follows by oddness. The fundamental theorem of calculus and the end limits also give

$$\int_{\mathbb{R}} \phi(t) dt = \lim_{T \rightarrow \infty} (q_s(T) - q_s(-T)) = 2.$$

These results use the Fourier multiplier $|\xi|^{2s}$, as in (2.1). The exact dimensional reduction is proved once, in Lemma 3.2: integrating the normalized kernel in the transverse variables gives $c_{1,s}|h|^{-1-p}$, with no additional dimensional factor. Applying that lemma to the bounded C^2 layer and using orthogonal invariance gives

$$(-\Delta)^s [q_s(e \cdot x - b)] = f(q_s(e \cdot x - b)) \quad (e \in S^{n-1}, b \in \mathbb{R}).$$

Subsequent profiles are pinned to the actual zero set, so the chosen zero normalization fixes their translation mode.

Unless additional dependence is displayed, constants may depend on the fixed s , fixed phase bands and chart buffers. They are independent of the number of sheets, the manner in which sheets reconnect, and the diameter or packing count of a bad cluster. No estimate is asserted uniformly as $s \downarrow 1/2$.

3. COMPACTNESS AND STABILITY IN THREE DIMENSIONS

The auxiliary stable problem will be studied in $\mathbb{R}^3$. Throughout its analysis $p = 2s \in (1, 2)$, $x = (x', t) \in \mathbb{R}^2 \times \mathbb{R}$, and D_r denotes the radius- r disk in $\mathbb{R}^2$. The normalization in Lemma 2.1 is used in every dimension. In particular, $|u| \leq 1$ for a bounded entire solution, and stability is the nonnegativity of Q_u . Recall the derivative-tail estimate for the normalized layer from Section 2:

$$\phi(t) \asymp (1 + |t|)^{-1-p}, \quad \int_{\mathbb{R}} \phi(t) dt = 2. \quad (3.1)$$

For use with nonsmooth compact tests, let

$$\mathcal{B}_d(v, w) = \frac{c_{d,s}}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{(v(X) - v(Y))(w(X) - w(Y))}{|X - Y|^{d+p}} dX dY.$$

If v is bounded and locally Lipschitz near the support of the compact Lipschitz function w , this integral converges absolutely. Indeed, compactness of $K = \text{supp } w$ gives a radius $\rho_0 > 0$ on whose neighborhood the local Lipschitz constants of v have a common bound. Whenever $w(X) - w(Y) \neq 0$, at least one point belongs to K . For $|X - Y| < \rho_0$, both increments are therefore bounded by $C|X - Y|$, giving the integrable radial factor $\int_0^{\rho_0} \rho^{1-p} d\rho$. For larger distances, boundedness of v, w and compactness of K give $\int_{\rho_0}^\infty \rho^{-1-p} d\rho$.

Let ψ be a real Lipschitz function compactly supported in a bounded open set Ω . Its mollifications ψ_δ have support in one fixed compact subset of Ω , for small δ , and converge uniformly to ψ . Their Lipschitz constants are uniformly bounded. The increments of $\psi_\delta - \psi$ are bounded by $C \min\{|X - Y|, 1\}$, and vanish when both points are outside a fixed compact set. Dominated convergence, with the preceding radial bounds, proves

$$\|\psi_\delta - \psi\|_2 \longrightarrow 0, \quad \mathcal{B}_d(\psi_\delta - \psi, \psi_\delta - \psi) \longrightarrow 0.$$

Thus the convergence is in $H^s(\mathbb{R}^d)$. Choose one bounded open $\Omega_0 \Subset \Omega$ containing all these supports. Since the increments of $\psi_\delta - \psi$ vanish on $\Omega_0^c \times \Omega_0^c$, Cauchy–Schwarz gives

$$|\mathcal{B}_d(u, \psi_\delta - \psi)| \leq \left(\frac{c_{d,s}}{2} \iint_{\Omega_0} \frac{|u(X) - u(Y)|^2}{|X - Y|^{d+p}} dX dY \right)^{1/2} \mathcal{B}_d(\psi_\delta - \psi, \psi_\delta - \psi)^{1/2} \rightarrow 0.$$

The first factor is finite by Section 2. The potential terms converge by uniform convergence on the common compact support. Therefore the smooth-test equation and stability inequality pass to ψ . The finite polynomial energy expansion, with the mixed kinetic term controlled by the same inequality, extends the variation identities of that section to these tests:

$$\frac{d}{d\epsilon} E_\Omega(u + \epsilon\psi)|_{\epsilon=0} = \mathcal{B}_d(u, \psi) - \int_\Omega f(u)\psi = 0, \quad \frac{d^2}{d\epsilon^2} E_\Omega(u + \epsilon\psi)|_{\epsilon=0} = Q_u(\psi).$$

In particular $Q_u(\psi) \geq 0$ when u is stable.

Starting from the uniform $C^{2+(p-1)/2}$ bound established in Section 2, we obtain every fixed higher derivative bound. Here and below, for a noninteger $r > 0$, C^r means $C^{\lfloor r \rfloor, r - \lfloor r \rfloor}$. Center-uniform local C^r bounds give a global C^r bound: sufficiently nearby points lie in one controlled ball, and differences at larger distances are bounded by the derivative suprema. Suppose a noninteger exponent r is already controlled. For any $0 < \epsilon < \min\{r/2, p/2\}$ outside the finitely many exceptional choices for which $\alpha = r - \epsilon$ or $\alpha + p$ is an integer, the cubic product estimates give a uniform C^α bound for $f(u)$ on every fixed-radius ball. The operator $L = -(\Delta)^s$ has the smooth constant angular density $c_{3,s}/2$. Thus [30, Corollary 3.5], after a fixed-radius rescaling, gives

$$\|u\|_{C^{r+p-\epsilon}(B_1(P))} \leq C(\|u\|_{L^\infty(\mathbb{R}^3)} + \|f(u)\|_{C^\alpha(B_2(P))}),$$

with a constant independent of P and u . The forcing for L is $-f(u)$, which has the same norm. Taking all centers restores a global bound at the new noninteger exponent. Each gain exceeds $p/2$; after finitely many steps the exponent exceeds any prescribed integer j . Since $|u| \leq 1$, the resulting C^j bounds depend only on j, s in dimension three.

For a sequence of bounded entire solutions, these bounds apply uniformly. The bound of order $j + 1$ gives equicontinuity of all derivatives through order j on a compact ball. Arzelà–Ascoli, a diagonal extraction over balls and derivative orders, and identification of the derivative limits by the fundamental theorem of calculus give one subsequence converging in every fixed C_{loc}^j norm. Denote this limit by v ; normalization also gives $|v| \leq 1$. Apply the compactness result of Section 2 to the extracted subsequence. It gives a further subsequence converging in C_{loc}^2 to a bounded entire solution w , which is stable if the sequence is stable. This further subsequence also converges in C_{loc}^2 to v . Uniqueness of the limit gives $w = v$, so the smooth limit satisfies the equation and preserves stability.

A related estimate will apply to differences whose normalized exterior values need not be bounded. Let v be a measurable function satisfying

$$[(-\Delta)^s + a]v = G \quad \text{in } B_4(P), \quad a, G \in L^\infty(B_4(P)),$$

and define

$$M_P(v) = \|v\|_{L^\infty(B_4(P))} + \int_{\mathbb{R}^d} \frac{|v(Y)|}{(1 + |Y - P|)^{d+p}} dY < \infty.$$

The equation is understood distributionally in $B_4(P)$. Indeed, for $\eta \in C_c^\infty(B_4(P))$, the function $(-\Delta)^s \eta$ is bounded and satisfies $|(-\Delta)^s \eta(Y)| \leq C_\eta(1 + |Y - P|)^{-d-p}$ outside a fixed ball. Hence the weighted integrability makes $\int v(-\Delta)^s \eta$ absolutely convergent. This includes the classical differences used below. The local regularity conclusions will concern the continuous representative of v , rather than arbitrary values assigned on a null set. Choose $0 \leq \zeta \leq 1$ in $C_c^\infty(B_4(P))$, equal to one on $B_3(P)$. For $X \in B_2(P)$,

$$[(-\Delta)^s + a](\zeta v) = G + H_\zeta, \quad H_\zeta(X) = c_{d,s} \int_{\mathbb{R}^d} \frac{(1 - \zeta(Y))v(Y)}{|X - Y|^{d+p}} dY.$$

Indeed, $v - \zeta v$ vanishes on $B_3(P)$, so its fractional Laplacian on $B_2(P)$ is $-H_\zeta$. This identity also holds distributionally: the separation and weighted tail bound justify Fubini against every compact smooth test there. Thus the tail enters with a positive sign. Its kernel is separated from the source, and for every multi-index β ,

$$|D_X^\beta |X - Y|^{-d-p}| \leq C_\beta (1 + |Y - P|)^{-d-p} \quad \text{on } B_2(P) \times \{1 - \zeta \neq 0\}.$$

For $|Y - P| \geq 4$ this follows from $|X - Y| \geq |Y - P|/2$; the remaining annulus is separated by distance one. Dominated differentiation gives $\|H_\zeta\|_{C^j(B_2(P))} \leq C_j M_P(v)$. Writing $w = \zeta v$, the localized equation is

$$(-\Delta)^s w = G - aw + H_\zeta \quad \text{in } B_2(P).$$

The right side is bounded by $\|G\|_\infty + C(1 + \|a\|_\infty)M_P(v)$, and $\|w\|_{L^\infty(\mathbb{R}^d)} \leq M_P(v)$. Consequently [30, Theorem 1.1(a)], with the constant angular density $c_{d,s}/2$ and the forcing sign reversed, gives the interior C^p estimate for bounded weak solutions. Its application to the present distributional solution can be justified explicitly. Extend the bounded function $h = G - aw + H_\zeta$ from $B_2(P)$ by zero, and let $w_\delta = \rho_\delta * w$ for a nonnegative compactly supported smooth approximate identity. The translation-invariant operator commutes with convolution, so

$$(-\Delta)^s w_\delta = \rho_\delta * h \quad \text{in } B_{2-\delta}(P),$$

where the mollifier is supported in B_δ . Also $\|w_\delta\|_\infty \leq \|w\|_\infty$ and $\|\rho_\delta * h\|_\infty \leq \|h\|_\infty$. A fixed covering by interior balls therefore gives, uniformly for small δ ,

$$\|w_\delta\|_{C^p(B_{3/2}(P))} \leq C(M_P(v) + \|G\|_{L^\infty(B_4(P))}).$$

Arzelà–Ascoli gives convergence on compact subsets in $C^{1,\alpha}$ for every $0 < \alpha < p - 1$. Since mollification also converges in L^1_{loc} , the limit equals w , and hence v , almost everywhere there. The uniform $(p - 1)$ -Hölder bound for the first derivatives passes to the limit for each pair of points, giving the same C^p estimate for this representative. Two continuous representatives equal almost everywhere coincide everywhere. From now on all local regularity norms of v refer to this unique representative. In particular,

$$\|v\|_{C^{1,\alpha}(B_1(P))} \leq C_\alpha(M_P(v) + \|G\|_{L^\infty(B_4(P))}), \quad 0 < \alpha < p - 1. \quad (3.2)$$

The constant depends on d, s, α and a bound for $\|a\|_{L^\infty(B_4(P))}$.

For the higher estimate, fix an integer $j \geq 1$, assume $G = 0$ and $\|a\|_{C^j(B_4(P))} \leq A_j$. We can retain the same localization $w = \zeta v$ throughout: $\|w\|_{L^\infty(\mathbb{R}^d)} \leq M_P(v)$, and w equals the selected representative of v on $B_{3/2}(P)$, and its forcing there is $-av + H_\zeta$. The preceding tail calculation controls every fixed derivative of H_ζ by $C_j M_P(v)$ on $B_2(P)$. Start at the noninteger exponent $r_0 = p$. If $r_0 > j$, the desired bound already follows. Otherwise, whenever $r_\ell < j$, choose

$$0 < \varepsilon_\ell < \min\{r_\ell/2, p/2\}, \quad \alpha_\ell = r_\ell - \varepsilon_\ell, \quad r_{\ell+1} = \alpha_\ell + p,$$

with both α_ℓ and $r_{\ell+1}$ noninteger. Fix these choices as functions of s, j only, independently of v, a, P . Then $\alpha_\ell < j$ and $r_{\ell+1} > r_\ell + p/2$. The sequence therefore reaches $r_N > j$ after finitely many steps, without an integer endpoint. Choose fixed radii $3/2 = R_0 > R_1 > \dots > R_N = 1$ after fixing this finite sequence. If $\|v\|_{C^{r_\ell}(B_{R_\ell}(P))} \leq C_\ell M_P(v)$, the product estimate gives

$$\| -av + H_\zeta \|_{C^{\alpha_\ell}(B_{R_\ell}(P))} \leq C(d, s, j, A_j, \ell) M_P(v).$$

Here only coefficient derivatives of order at most j are used. On these balls w is locally C^p , hence locally Lipschitz. For a real $\eta \in C_c^\infty(B_{R_\ell}(P))$, the bilinear estimate proved above makes $\mathcal{B}_d(w, \eta)$ absolutely convergent. Let T_δ denote the operator in (2.1) with the integration restricted to $|X - Y| > \delta$. Truncated kernel symmetry gives

$$\frac{c_{d,s}}{2} \iint_{|X-Y|>\delta} \frac{(w(X) - w(Y))(\eta(X) - \eta(Y))}{|X - Y|^{d+p}} dX dY = \int_{\mathbb{R}^d} w T_\delta \eta.$$

The left side tends to $\mathcal{B}_d(w, \eta)$ by absolute integrability. On the right, the symmetric Taylor estimate gives $\|T_\delta \eta - (-\Delta)^s \eta\|_\infty \leq C_\eta \delta^{2-p}$. Since w is bounded with compact support, it belongs

to L^1 , so this pairing also converges. The distributional equation therefore gives exactly the required local weak identity

$$\mathcal{B}_d(w, \eta) = \int_{\mathbb{R}^d} w(-\Delta)^s \eta = \int_{B_{R_\ell}(P)} (-av + H_\zeta) \eta.$$

Apply [30, Corollary 3.5] to the bounded function w on interior balls whose doubled balls lie in $B_{R_\ell}(P)$. A fixed covering of $B_{R_{\ell+1}}(P)$ gives the $C^{r_{\ell+1}}$ bound there. No global regularity of the exterior values of v is needed: the corollary uses the global L^∞ norm of w , and the exterior contribution has already been put into H_ζ . Finite induction proves

$$\|v\|_{C^j(B_1(P))} \leq C_j M_P(v). \quad (3.3)$$

Here C_j depends only on d, s, j, A_j and the chosen positive radius gaps; the case $j = 0$ follows from the local essential supremum and continuity. A merely bounded nonzero G does not justify this higher estimate.

The uniform gradient bound just obtained lets the variation estimate of Gui–Jin–Zhao enter with no solution-dependent constant.

Proposition 3.1 (Uniform variation). *There is $C = C(s)$ such that every bounded stable entire solution in $\mathbb{R}^3$ satisfies*

$$\int_{B_R(P)} |\nabla u| \, dX \leq C R^{P_3}, \quad P_3 = 3 - \frac{2}{p+1} = 2 + \frac{p-1}{p+1}, \quad R \geq 2.$$

Proof. Normalization and the uniform interior bound give $|u| \leq 1$ and $L = \|\nabla u\|_\infty \leq L_*(s)$. We use [27, Theorem 5.1] in the Fourier normalization $|\xi|^p$: for a classical bounded stable solution of the cubic equation, with $|u| \leq 1$ and stability as in (2.4), it gives

$$\int_{B_R} |\nabla u| \leq C_{\text{GJZ}}(3, s) \left(R^2 + L^{(p-1)/(p+1)} R^{3-2/(p+1)} \right), \quad R \geq 2.$$

Put $\gamma = (p-1)/(p+1) > 0$. Since $P_3 = 2 + \gamma > 2$, $R^2 \leq R^{P_3}$ for $R \geq 2$. The desired constant is therefore $C = C_{\text{GJZ}}(3, s)(1 + L_*(s)^\gamma)$. Applying the estimate to $u(P + \cdot)$ proves the assertion at every center, since translation preserves the equation, stability, and gradient norm. $\square$

For the next lemma, $m \geq 1$ and $k \geq 0$ are integers, and C_b^2 denotes functions with bounded continuous derivatives through order two.

Lemma 3.2 (Equation and stability for cylindrical functions). *If $v \in C_b^2(\mathbb{R}^m)$ and $U(x, z) = v(x)$ on $\mathbb{R}^m \times \mathbb{R}^k$, then*

$$(-\Delta_{\mathbb{R}^{m+k}})^s U = (-\Delta_{\mathbb{R}^m})^s v.$$

A solution v is stable in $\mathbb{R}^m$ if and only if its cylindrical extension is stable in $\mathbb{R}^{m+k}$. Both assertions are invariant under orthogonal changes of variables.

Proof. If $k = 0$ the assertions are identities; suppose $k \geq 1$. Use the symmetric increment representation from Section 2, whose coefficient is $c_{d,s}/2$:

$$(-\Delta_{\mathbb{R}^d})^s w(X) = \frac{c_{d,s}}{2} \int_{\mathbb{R}^d} \frac{2w(X) - w(X+H) - w(X-H)}{|H|^{d+p}} \, dH.$$

For $w \in C_b^2$, its numerator is bounded by $C \min\{|H|^2, 1\}$, so the integral is absolutely convergent. For $U(x, z) = v(x)$, the numerator depends only on the increment $h \in \mathbb{R}^m$. For $h \neq 0$, polar coordinates and the beta integral give

$$\int_{\mathbb{R}^k} (|h|^2 + |z|^2)^{-(m+k+p)/2} \, dz = \pi^{k/2} \frac{\Gamma((m+p)/2)}{\Gamma((m+k+p)/2)} |h|^{-m-p}.$$

Indeed the radial substitution $r = |h|\rho$, followed by $y = \rho^2$, produces $\frac{1}{2}|S^{k-1}|B(k/2, (m+p)/2)$. Multiplication by $c_{m+k,s}$ gives exactly $c_{m,s}$. Fubini is justified by $C \min\{|h|^2, 1\}|h|^{-m-p}$ after integrating out z . This proves the equation identity.

For a real smooth compactly supported test ψ use the unitary Fourier convention $\widehat{\psi}(\xi) = (2\pi)^{-d/2} \int e^{-iX \cdot \xi} \psi(X) \, dX$. The integration-by-parts identity of Section 2, the prescribed multiplier, and Plancherel give

$$\mathcal{B}_d(\psi, \psi) = \int \psi(-\Delta)^s \psi = \int |\xi|^p |\widehat{\psi}(\xi)|^2 \, d\xi.$$

For $\psi \in C_c^\infty(\mathbb{R}^{m+k})$, the inequality $(|\xi|^2 + |\eta|^2)^s \geq |\xi|^{2s}$ and partial Plancherel give

$$\int_{\mathbb{R}^k} \mathcal{B}_m(\psi(\cdot, z), \psi(\cdot, z)) \, dz = \iint |\xi|^p |\widehat{\psi}(\xi, \eta)|^2 \, d\xi \, d\eta.$$

The potential term is exactly the integral of the slice potential terms, because $f'(U(x, z)) = f'(v(x))$. It follows that

$$Q_U^{m+k}(\psi) \geq \int_{\mathbb{R}^k} Q_v^m(\psi(\cdot, z)) \, dz.$$

Every slice is a smooth compactly supported test in $\mathbb{R}^m$; hence stability of v makes the last integral nonnegative and proves stability of U . Conversely, suppose U is stable. Fix real $\varphi \in C_c^\infty(\mathbb{R}^m)$ and nonzero real $\chi \in C_c^\infty(\mathbb{R}^k)$, and use $\psi_L(x, z) = \varphi(x) L^{-k/2} \chi(z/L)$. Its kinetic energy is

$$\iint \left(|\xi|^2 + \frac{|\theta|^2}{L^2} \right)^s |\widehat{\varphi}(\xi)|^2 |\widehat{\chi}(\theta)|^2 \, d\xi \, d\theta.$$

The potential term equals $-\|\chi\|_2^2 \int f'(v) \varphi^2$ exactly. The elementary inequality $0 \leq (a+b)^s - a^s \leq b^s$ for $a, b \geq 0$ gives the quantitative identity

$$Q_U^{m+k}(\psi_L) = \|\chi\|_2^2 Q_v^m(\varphi) + r_L, \quad 0 \leq r_L \leq L^{-p} \|\varphi\|_2^2 \mathcal{B}_k(\chi, \chi).$$

All these Fourier integrals are finite because the tests are smooth and compactly supported. The displayed inequality follows, for example, by integrating $s(a+t)^{s-1} \leq st^{s-1}$ over $0 < t < b$. Since $r_L \rightarrow 0$, the inequality $Q_U^{m+k}(\psi_L) \geq 0$ gives $\|\chi\|_2^2 Q_v^m(\varphi) \geq 0$ in the limit, proving stability of v . Only the tests have been Fourier transformed; no L^2 assumption is imposed on either solution. An orthogonal change of variables preserves distances, Lebesgue measure, and compact smooth tests. It therefore preserves the operator and the stability form, proving both rotated assertions. $\square$

4. FROM THE PHASE BAND TO COMPLETE LOCAL GRAPHS

Throughout this section, u and the sequences u_j are bounded stable entire solutions of the normalized cubic equation in $\mathbb{R}^3$. The comparison functions introduced below are auxiliary functions. Fix $a \in (1/\sqrt{3}, 1)$. This interval is wide enough that the linearized potential is positive outside $|u| \leq a$. We first obtain isolation estimates without assuming any graph geometry.

For comparison arguments, let $D \subset \mathbb{R}^3$ be open, possibly unbounded, and let $w \in C_b(\mathbb{R}^3) \cap C_{\text{loc}}^2(D)$. Suppose $w \leq 0$ on D^c and $[(-\Delta)^s + c(X)]w \leq 0$ pointwise in D , with $c \geq c_* > 0$. Then $w \leq 0$. To see this, set $b(X) = \sqrt{1 + |X|^2}$. Its bounded Hessian controls the symmetric increment near the diagonal, and its global Lipschitz bound controls the increment at infinity, so

$$\|(-\Delta)^s b\|_\infty \leq C \left(\int_0^1 \rho^{1-p} \, d\rho + \int_1^\infty \rho^{-p} \, d\rho \right) < \infty.$$

If $\sup w = m > 0$, choose X_* with $w(X_*) > 3m/4$ and then choose $\epsilon > 0$ small enough that $w(X_*) - \epsilon b(X_*) > m/2$. Since w is bounded and $b \rightarrow \infty$, $w - \epsilon b$ attains a global maximum, whose value is greater than $m/2$. The condition $w \leq 0$ on D^c places its maximizer X_ϵ in D . At this point the gradient of the penalized function vanishes, giving an integrable quadratic increment near the diagonal; its linear growth at infinity is integrable because $p > 1$. Its fractional Laplacian is therefore nonnegative. Linearity gives $(-\Delta)^s w(X_\epsilon) \geq -C\epsilon$, while the assumed inequality and $w(X_\epsilon) > m/2$ give $c_* m/2 \leq C\epsilon$. Choosing ϵ also smaller than $c_* m/(2C)$ is a contradiction. In particular, applying this comparison to w and $-w$ shows that a bounded entire classical solution of $[(-\Delta)^s + 2]w = 0$ is zero.

By the sign classification in Lemma 2.2, translation compactness implies that stable solutions approach the appropriate well uniformly as their distance from $Z = \{u = 0\}$ tends to infinity. If

$Z = \emptyset$, continuity and the sign classification make u a constant well; we set $\text{dist}(X, \emptyset) = \infty$, and the clearing estimate is immediate. Otherwise, define $m(D)$ as the supremum, with zero included, of $1 - u(X)$ over all bounded stable solutions and points with $u(X) > 0$ and $\text{dist}(X, Z) \geq D$. Sign reversal preserves the equation and stability, so this class also includes $-u$. Range normalization gives $m(D) \in [0, 1]$, and it is nonincreasing. If it did not tend to zero, there would be $\varepsilon_0 > 0$, solutions u_j and points X_j with $\text{dist}(X_j, \{u_j = 0\}) \rightarrow \infty$, $u_j(X_j) > 0$, and $1 - u_j(X_j) \geq \varepsilon_0$. Every fixed ball centered at X_j eventually contains no zero and, by connectedness and continuity, has the positive sign. Smooth stable compactness applied to $u_j(X_j + \cdot)$ gives a nonnegative stable entire limit U with $U(0) \leq 1 - \varepsilon_0$. Lemma 2.2 gives $U \equiv 1$, a contradiction. Thus $m(D) \downarrow 0$ uniformly over the class. For $v = 1 - u$, put $h = v^2(3 - v)$. Then $0 \leq v \leq 2$, $0 \leq h \leq 4$, and

$$[(-\Delta)^s + 2]v = h.$$

We justify the resolvent representation and kernel bounds used below. Write $\nu(dr) = \frac{s}{\Gamma(1-s)} r^{-1-s} dr$ on $(0, \infty)$. On $(1, \infty)$ and on each interval $(2^{-k-1}, 2^{-k}]$, $k \geq 0$, take independent compound Poisson processes with intensity ν restricted to that interval: the jump rate is its finite ν -mass, and the jump distribution is the normalized restriction of ν . These are the compound Poisson processes of [31, Example 3.2(d), Theorem 3.4 and Corollary 3.5]. The expected sum of the jumps at most one before time t is $t \int_0^1 r \nu(dr) = ts/((1-s)\Gamma(1-s)) < \infty$, and the jumps greater than one have finite number on bounded time intervals. The sum S_t of all these positive jumps is therefore finite almost surely at every fixed t . Intersecting the corresponding probability-one events at integer times gives finiteness simultaneously for all $t \geq 0$, by monotonicity. On $[0, T]$, the tails of the small-jump sums are bounded by their value at T , so the partial sums converge uniformly to a nondecreasing càdlàg process. Its increments are independent and stationary, since they use disjoint time increments of the independent Poisson processes. Conditioning on the number of jumps in each finite truncation and then passing to the positive sum gives

$$\mathbb{E} e^{-\lambda S_t} = \exp \left[-t \int_0^\infty (1 - e^{-\lambda r}) \nu(dr) \right] = e^{-t\lambda^s}, \quad \lambda > 0,$$

because integration by parts gives

$$\frac{s}{\Gamma(1-s)} \int_0^\infty (1 - e^{-\lambda r}) r^{-1-s} dr = \frac{\lambda}{\Gamma(1-s)} \int_0^\infty e^{-\lambda r} r^{-s} dr = \lambda^s.$$

The boundary terms vanish since $s \in (0, 1)$. For $t > 0$, letting $\lambda \rightarrow \infty$ shows $\mathbb{P}(S_t = 0) = 0$. Define

$$p_t^{(d)}(Y) = \mathbb{E} \left[(4\pi S_t)^{-d/2} e^{-|Y|^2/(4S_t)} \right], \quad G_2^{(d)}(Y) = \int_0^\infty e^{-2t} p_t^{(d)}(Y) dt.$$

The Gaussian mixture is finite and continuous, since for every $b > 0$ the identity

$$\mathbb{E} S_t^{-b} = \frac{1}{\Gamma(b)} \int_0^\infty \lambda^{b-1} e^{-t\lambda^s} d\lambda = \frac{\Gamma(b/s)}{s\Gamma(b)} t^{-b/s} < \infty$$

follows by Tonelli and the substitution $z = t\lambda^s$. The heat kernel is even, positive and has mass one. Independent stationary increments of S_t and Gaussian convolution give the convolution semigroup property. In the unitary Fourier convention, its convolution multiplier is $\mathbb{E} e^{-S_t|\xi|^2} = e^{-t|\xi|^{2s}} = e^{-t|\xi|^p}$. Thus $G_2^{(d)}$ is even and positive almost everywhere, has mass 1/2, and its convolution multiplier is $\int_0^\infty e^{-t(2+|\xi|^p)} dt = (2 + |\xi|^p)^{-1}$.

For the kernel estimate, the Laplace identity just proved gives

$$\mathbb{P}(S_t \geq r) \leq \frac{\mathbb{E}(1 - e^{-S_t/r})}{1 - e^{-1}} = \frac{1 - e^{-tr^{-s}}}{1 - e^{-1}} \leq Ctr^{-s}.$$

For $\rho = |Y| > 0$, split the Gaussian expectation into $I_k^+ = [2^k \rho^2, 2^{k+1} \rho^2)$ and $I_k^- = [2^{-k-1} \rho^2, 2^{-k} \rho^2)$, $k \geq 0$. On I_k^+ the Gaussian is bounded by $C2^{-kd/2} \rho^{-d}$, and the probability of

this interval is at most $Ct2^{-ks}\rho^{-p}$. On I_k^- the Gaussian is bounded by $C2^{(k+1)d/2}\rho^{-d}e^{-2^k/4}$, and the probability is at most $Ct2^{(k+1)s}\rho^{-p}$. Summing gives

$$p_t^{(d)}(Y) \leq Ct\rho^{-d-p} \left(\sum_{k \geq 0} 2^{-k(d/2+s)} + \sum_{k \geq 0} 2^{(k+1)(d/2+s)} e^{-2^k/4} \right) \leq Ct\rho^{-d-p}.$$

The preceding negative-moment identity with $b = d/2$ gives, at every Y including $Y = 0$, $p_t^{(d)}(Y) \leq (4\pi)^{-d/2} \mathbb{E} S_t^{-d/2} = C_{d,s} t^{-d/p}$. Comparing the cases $\rho \leq t^{1/p}$ and $\rho \geq t^{1/p}$ yields $p_t^{(d)}(Y) \leq Ct/(t^{1/p} + |Y|)^{d+p}$. For $d > p$ and $0 < \rho \leq 1$, the contributions to the resolvent integral from $t \leq \rho^p$ and $t \geq \rho^p$ are bounded respectively by $C\rho^{-d-p} \int_0^{\rho^p} t \, dt$ and $C \int_{\rho^p}^\infty t^{-d/p} \, dt$. For $\rho \geq 1$, use the bound $Ct\rho^{-d-p}$ for all t and $\int_0^\infty te^{-2t} \, dt < \infty$. Hence

$$G_2^{(d)}(Y) \leq \begin{cases} C|Y|^{p-d}, & 0 < |Y| \leq 1, \\ C|Y|^{-d-p}, & |Y| \geq 1. \end{cases}$$

Differentiating the Gaussian at a positive radius gives the exact dimension-shift identity $-\partial_\rho G_2^{(d)}(\rho) = 2\pi\rho G_2^{(d+2)}(\rho)$. Indeed, on a compact radius interval $[a, b] \subset (0, \infty)$ its absolute radial derivative is bounded by $2\pi b(4\pi S_t)^{-(d+2)/2} e^{-a^2/(4S_t)}$. The preceding resolvent bound in dimension $d + 2$ makes this majorant integrable after expectation and integration in t . For $G_2 = G_2^{(3)}$, polar integration therefore gives

$$\int_{|Y| > D} G_2(Y) \, dY \leq CD^{-p}, \quad \nabla G_2 \in L^1, \quad \int_{|Y| > D} |\nabla G_2(Y)| \, dY \leq CD^{-p-1}, \quad D \geq 1.$$

The gradient bounds are $C|Y|^{p-4}$ near zero and $C|Y|^{-4-p}$ at infinity; their local integrability uses $p > 1$. These classical derivatives away from zero are also weak derivatives. For $\eta \in C_c^\infty(\mathbb{R}^3)$, integrate by parts on $\mathbb{R}^3 \setminus B_\delta$, with an outer boundary outside the support of η . The inner boundary term is at most $C\|\eta\|_\infty \delta^2 \delta^{p-3} = C\|\eta\|_\infty \delta^{p-1} \rightarrow 0$. Both volume integrals converge because G_2 and its displayed derivatives are integrable. Thus $G_2 \in W^{1,1}(\mathbb{R}^3)$, with no distributional residue at zero.

For any bounded measurable h , the convolution $r = G_2 * h$ solves $[(-\Delta)^s + 2]r = h$ distributionally. Indeed, for $\eta \in C_c^\infty(\mathbb{R}^3)$, put $\psi = [(-\Delta)^s + 2]\eta$. The symmetric integral formula makes ψ bounded, and outside a ball containing the support of η its fractional-Laplacian term is $O(|X|^{-3-p})$. Hence $\psi \in L^1$. The prescribed operator multiplier gives $\hat{\psi} = (2 + |\xi|^p)\hat{\eta}$, while the convolution multiplier of G_2 is $(2 + |\xi|^p)^{-1}$. Both $G_2 * \psi$ and η belong to L^1 , so Fourier uniqueness gives $G_2 * \psi = \eta$. Evenness of G_2 and Fubini then give

$$\int_{\mathbb{R}^3} r [(-\Delta)^s + 2]\eta = \int_{\mathbb{R}^3} h (G_2 * \psi) = \int_{\mathbb{R}^3} h \eta.$$

The absolute Fubini bound is $\|h\|_\infty \|G_2\|_1 \|\psi\|_1 < \infty$. Only the integrable kernel and test functions have been Fourier transformed. If h is smooth with all derivatives bounded, then dominated differentiation gives $D^\alpha r = G_2 * D^\alpha h$ for every multi-index α . Thus r is smooth with bounded derivatives. Its symmetric fractional-Laplacian integral is continuous, by the uniform quadratic increment near zero and boundedness at infinity. Fubini in the symmetric formula identifies this continuous integral with the distributional fractional Laplacian: the paired integrals are bounded using $\min\{|Y|^2, 1\}|Y|^{-3-p}$, the bounded derivatives of r , and the L^1 norms of the compact test and its second derivatives. Its distributional equation is consequently pointwise. Our cubic $h = v^2(3-v)$ has these derivative bounds by the preceding solution bounds. Bounded comparison applied to $v - r$ and $r - v$ therefore gives

$$v = G_2 * h = G_2 * (3v^2 - v^3).$$

If $u(X) > 0$ and $\text{dist}(X, Z) \geq D$, the ball $B_{D/2}(X)$ has positive sign and its points have zero-set distance at least $D/2$. On this ball $h \leq 3m(D/2)^2$; outside it use $h \leq 4$. Positivity and mass $1/2$ of G_2 give $m(D) \leq \frac{3}{2}m(D/2)^2 + CD^{-p}$ for $D \geq 2$. Choose $D_0 \geq 2$ so that $\lambda = \frac{3}{2}m(D_0/2) < 2^{-p}$.

Monotonicity gives $m(D) \leq \lambda m(D/2) + CD^{-p}$ for $D \geq D_0$. For $D \geq D_0$, choose $k \geq 0$ with $D/2^k \in [D_0, 2D_0)$. Iterating gives

$$m(D) \leq \lambda^k m(D/2^k) + CD^{-p} \sum_{j=0}^{k-1} (2^p \lambda)^j \leq CD^{-p}.$$

The sum is bounded because $2^p \lambda < 1$, and $\lambda^k \leq 2^{-pk} \leq (2D_0/D)^p$ bounds the terminal term using $m \leq 1$; for $k = 0$ the sum is empty and the same bound holds. Since $G_2 \in W^{1,1}$ and h is bounded, $\nabla v = (\nabla G_2) * h$ distributionally, and hence classically. For example, the distributional identity follows by Fubini and the weak derivative formula for G_2 ; the convolution on the right is continuous because h is uniformly continuous. The same ball split gives

$$|\nabla v(X)| \leq 3\|\nabla G_2\|_1 m(D/2)^2 + 4 \int_{|Y| > D/2} |\nabla G_2(Y)| dY \leq C(D^{-2p} + D^{-p-1})$$

for $D \geq 2D_0$, using the bound for $m(D/2)$. Since $2p > p + 1$, the second power controls the first for $D \geq 1$. Sign reversal covers negative points; points of Z and all remaining bounded distances are covered by range normalization and the uniform gradient bound. The constants D_0 and C depend only on s in dimension three, since m was defined over the entire normalized class. This proves the following estimate, with constants independent of the solution.

Proposition 4.1 (Clearing estimates). *Uniformly over bounded stable solutions in $\mathbb{R}^3$,*

$$1 - |u(X)| \leq C(1 + \text{dist}(X, Z))^{-p}, \quad |\nabla u(X)| \leq C(1 + \text{dist}(X, Z))^{-p-1}. \quad (4.1)$$

We first record a differentiation fact for later use. In every fixed dimension n , a bounded entire solution has bounded derivatives of every order. Indeed, the translation-uniform $C^{2+(p-1)/2}$ bound of Section 2 starts a finite Schauder bootstrap. If a noninteger exponent r is controlled globally, choose $0 < \delta < \min\{r/2, p/2\}$ so that $\alpha = r - \delta$ and $\alpha + p$ are nonintegers. The cubic product estimate controls $f(u)$ in C^α , and [30, Corollary 3.5], applied with the smooth constant angular density $c_{n,s}/2$ and forcing $-f(u)$, gives translation-uniform $C^{r+p-\delta}$ bounds. Taking all centers restores a global bound at that exponent. Each gain exceeds $p/2$, so finitely many steps give every prescribed derivative bound. This argument uses only boundedness and the equation, and applies in particular to the one-dimensional reference layer. By Lemma 2.1, all bounds depend only on the prescribed derivative order, n and s .

For a constant unit vector e , put $w = \partial_e u$. Differentiating the symmetric integral formula is legitimate: the derivative of its second increment is bounded by $C|h|^2$ for $|h| < 1$, using the bounded third derivatives of u , and by $4\|w\|_\infty$ for $|h| \geq 1$. The corresponding radial integrals are finite because $0 < p < 2$. Thus

$$(-\Delta)^s w = f'(u)w, \quad \mathcal{B}_n(w, \eta) = \int_{\mathbb{R}^n} f'(u)w\eta \quad (\eta \text{ compactly supported and Lipschitz}). \quad (4.2)$$

To justify the weak identity, truncate at $|h| = \rho$ and denote the truncated operator by T_ρ . Fubini and symmetry give $\int \eta T_\rho w = \mathcal{B}_{n,\rho}(w, \eta)$, where $\mathcal{B}_{n,\rho}$ is restricted to $|X - Y| > \rho$. The bounded Hessian of w bounds the omitted symmetric increment by $C\rho^{2-p}$, so $T_\rho w \rightarrow (-\Delta)^s w$ uniformly on the compact support of η . The mixed bilinear integral converges absolutely by the compact-test estimates of Section 3. Passing to the limit proves the identity without a global L^2 assumption on w .

Lemma 4.2 (One-dimensional stable classification). *The stable bounded profiles of the normalized cubic equation on $\mathbb{R}$ consist of the two wells and the shifted increasing or decreasing layers q_s and $-q_s$.*

Proof. Lemma 2.1 gives $|v| \leq 1$, with strict bounds for a nonconstant solution. The constant solutions are $-1, 0, 1$. Lemma 2.2 excludes zero from the stable class, while $Q_{\pm 1}(\psi) = \mathcal{B}_1(\psi, \psi) + 2\|\psi\|_2^2 \geq 0$.

For a nonconstant solution, its Poisson extension is

$$\mathcal{V}(t, z) = \int_{\mathbb{R}} H_s(\xi) v(t - z\xi) d\xi, \quad H_s(\xi) = C_s(1 + \xi^2)^{-(1+p)/2}, \quad \int_{\mathbb{R}} H_s = 1.$$

It is nonconstant and strictly between the wells, since the kernel is positive and its boundary trace is v . Moreover $\int |\xi| H_s(\xi) d\xi < \infty$ because $p > 1$. The one-dimensional derivative bound just established permits differentiation of this formula and gives $|\mathcal{V}_t| + |\mathcal{V}_z| \leq C_s \|v'\|_\infty$. Thus the extension is continuous up to the boundary and has finite local weighted energy, since $\int_0^R z^{1-p} dz < \infty$.

For this bounded trace, [7, Remark 3.8] identifies the Poisson extension equation and its conormal derivative. Its exact normalization, given in [7, Remark 3.11], is

$$(-\partial_t^2)^s v = -\kappa_s \lim_{z \downarrow 0} z^{1-p} \mathcal{V}_z, \quad \kappa_s = 2^{p-1} \frac{\Gamma(s)}{\Gamma(1-s)} > 0.$$

No L^2 assumption is made on v . For a real compact C^1 extension test Ψ , its trace ψ is compactly supported and Lipschitz, hence belongs to H^s . Let $P_s \psi$ be its energy-minimizing extension. The weak Dirichlet-to-Neumann identity for this minimizer, together with [7, (3.2) and Theorem 3.1], gives

$$\kappa_s \int_{\mathbb{R}_+^2} z^{1-p} |\nabla P_s \psi|^2 = \langle (-\partial_t^2)^s \psi, \psi \rangle = \mathcal{B}_1(\psi, \psi).$$

Minimization and the compact Lipschitz stability extension of Section 3 now give

$$\kappa_s \int_{\mathbb{R}_+^2} z^{1-p} |\nabla \Psi|^2 \geq \mathcal{B}_1(\psi, \psi) \geq \int_{\mathbb{R}} (1 - 3v^2) \psi^2.$$

The conormal convention of [8, (1.3)] uses coefficient $2 - p$; hence its nonlinearity and potential here are

$$f_{\text{ext}}(r) = \frac{2-p}{\kappa_s} (r - r^3), \quad G_{\text{ext}}(r) = \frac{2-p}{\kappa_s} \frac{(1-r^2)^2}{4}.$$

Dividing the preceding inequality by κ_s gives exactly the extension stability inequality, whose boundary coefficient is $f'_{\text{ext}}/(2-p)$. The only critical points of G_{ext} in $[-1, 1]$ are $-1, 0, 1$. Among distinct critical points, only the two wells have equal potential, and the potential is strictly larger between them. These facts verify the implication (2.13)–(2.15) required by [8, Theorem 2.9(c)]. A smooth modification of f_{ext} outside a neighborhood of $[-1, 1]$ meets its global $C^{1,\gamma}$ hypothesis for any $0 < \gamma < 1$; since $p > 1$, the required inequality $\gamma > \max\{0, 1-p\}$ holds. The modification does not change the equation or the potential conditions on $[-1, 1]$. The theorem makes $v(t)$ or $v(-t)$ an increasing layer. The uniqueness and oddness of q_s recalled in Section 2 identify the stated translates of q_s and $-q_s$.

Conversely, put $\phi = q'_s > 0$. Equation (4.2) gives $(-\partial_t^2)^s \phi = f'(q_s) \phi$, including its compact weak formulation. For real $\psi \in C_c^\infty(\mathbb{R})$ set $\zeta = \psi/\phi$. The positive minimum of ϕ on the compact support of ψ makes ζ and $\phi \zeta^2$ compact smooth tests. Test the derivative equation with $\phi \zeta^2$. Subtracting it from $Q_{q_s}(\phi \zeta)$ cancels the potential term. The pointwise algebra

$$(aA - bB)^2 - (a-b)(aA^2 - bB^2) = ab(A - B)^2$$

with $a = \phi(t)$, $b = \phi(r)$, $A = \zeta(t)$ and $B = \zeta(r)$ then gives

$$Q_{q_s}(\psi) = \frac{c_{1,s}}{2} \iint_{\mathbb{R}^2} \frac{\phi(t)\phi(r)(\zeta(t) - \zeta(r))^2}{|t-r|^{1+p}} dt dr \geq 0.$$

The mixed integrals converge absolutely by the compact-test estimates; the displayed nonnegative integral also converges, since ϕ is bounded and ζ is compactly supported and Lipschitz. Translations preserve the equation and stability form, and sign reversal does so because f is odd and f' is even. This proves both directions of the classification. $\square$

At a regular point of a level surface, its unit normal is $\nu = \nabla u / |\nabla u|$. If v, w are tangent vectors, then $\nabla u \cdot w = 0$, so differentiating the normal gives

$$A(v, w) = \pm \langle D_v \nu, w \rangle = \pm \frac{D^2 u[v, w]}{|\nabla u|}.$$

The sign depends only on the convention for A . In an orthonormal tangent basis this yields $|A| \leq C |D^2 u| / |\nabla u|$.

Let $C_{\text{cl}} > 0$ be a uniform constant in Proposition 4.1. Choose $D_a > 0$ so that $C_{\text{cl}}(1 + D_a)^{-p} < 1 - a$. If $|u(X)| \leq a$, then $1 - |u(X)| \geq 1 - a$, and clearing forces $\text{dist}(X, \{u = 0\}) < D_a$. Thus every band point lies within the fixed distance D_a of a zero. Choose $L_* > D_a + 2$ and define

$$B_\epsilon(u) = \left\{ z \in Z : \inf_{e \in S^2} \|u(z + \cdot) - q_s(e \cdot \cdot)\|_{C^2(B_{L_*})} \geq \epsilon \right\}, \quad \Omega_\epsilon = \{X : \text{dist}(X, B_\epsilon) > D_a + 1\}.$$

The translation-uniform $C^{2,\beta}$ bound of Section 2 applies in dimension three to u and in dimension one to the bounded normalized layer q_s . The lifted layer has gradient $q'_s e$ and Hessian $q''_s e \otimes e$, so every comparison norm on the fixed ball is finite. An empty bad set gives $\Omega_\epsilon = \mathbb{R}^3$. There are $\epsilon_0, \gamma_a, C_a > 0$, depending only on s, a, L_* , such that for every bounded stable solution and $0 < \epsilon \leq \epsilon_0$,

$$|\nabla u| \geq \gamma_a > 0, \quad M(X) := |A_{\{u=u(X)\}}(X)| \leq C_a \epsilon \quad (X \in \Omega_\epsilon, |u(X)| \leq a). \quad (4.3)$$

The norm in the definition is jointly continuous in z and e : on the closure of the fixed ball all derivatives through order two depend continuously on these parameters, uniformly on each compact parameter set. The uniform solution bounds and the smooth layer provide the required bounds there. Compactness of S^2 makes the infimum a minimum and its value continuous in z . Since Z is closed, B_ϵ is closed and Ω_ϵ is open.

We use the elementary nearest-point fact for a nonempty closed set $F \subset \mathbb{R}^3$. A distance-minimizing sequence from a fixed point X eventually lies in the ball of radius $\text{dist}(X, F) + 1$ about X . A convergent subsequence exists by finite-dimensional compactness, and its limit belongs to F and attains the distance. For a phase-band point X , clearing makes Z nonempty, and Z is closed by continuity of u . The fact just proved gives a nearest zero z . If $X \in \Omega_\epsilon$, then $|X - z| \leq D_a$ and $\text{dist}(z, B_\epsilon) > 1$, so z is good. Choose its layer approximation with error less than ϵ . It covers X because $L_* > D_a + 2$. Taking $\epsilon_0 \leq (1 - a)/2$, the corresponding layer value lies in $[-a', a']$, where $a' = (1 + a)/2 < 1$. Let $m_a > 0$ be the minimum of q'_s on $q_s^{-1}([-a', a'])$. With the fixed norm-comparison constant C_0 in $|\nabla u - q'_s e| \leq C_0 \epsilon$, reduce ϵ_0 so that $C_0 \epsilon_0 \leq m_a/2$. Then we may take $\gamma_a = m_a/2$. For an actual unit tangent v , $\nabla u \cdot v = 0$ and $\nabla u = q'_s e + O(\epsilon)$ imply $|e \cdot v| \leq C\epsilon$. The tangential restriction of $q''_s e \otimes e$ is $O(\epsilon^2)$, while the Hessian error is $O(\epsilon)$. Division by $|\nabla u|$ proves the curvature estimate with the uniform constants in (4.3).

We will need a scale chosen from the whole phase band. Let Ω be open, let u be a bounded stable entire solution, suppose $|\nabla u| \geq \gamma > 0$ on $\Omega \cap \{|u| \leq a\}$, and take X_0 in this set. For $L > 0$, if

$$M(X_0) > 0, \quad M(X_0) \text{dist}(X_0, \Omega^c) > 4L,$$

there exists $P \in \Omega \cap \{|u| \leq a\}$, with $b = M(P) \geq M(X_0)$, such that

$$B_{L/b}(P) \subset \Omega, \quad M \leq 2b \quad \text{on } B_{L/b}(P) \cap \{|u| \leq a\}. \quad (4.4)$$

Write $m_0 = M(X_0) > 0$. At stage k , the current point X_k is in the band and $m_k = M(X_k) > 0$. Before selecting a next point, we verify that its comparison ball lies in Ω . Inductively, $m_k \geq 2^k m_0$, and the preceding moves give

$$|X_k - X_0| \leq \sum_{\ell < k} \frac{L}{m_\ell} < \frac{2L}{m_0}, \quad \frac{L}{m_k} \leq \frac{L}{m_0}.$$

This includes $k = 0$, when the sum is zero. Therefore every tested ball is contained in $B_{3L/m_0}(X_0) \subset \Omega$, using the initial margin greater than $4L/m_0$. When $\Omega = \mathbb{R}^3$ this containment and the distance condition are automatic. If the bound $M \leq 2m_k$ holds on this ball's phase band, stop with $P = X_k$. Otherwise choose X_{k+1} in that comparison ball's band with $m_{k+1} > 2m_k$. Containment, already proved, puts X_{k+1} in Ω , and $|X_{k+1} - X_k| < L/m_k$ verifies the induction for the next stage. The regular-level identity just proved, the uniform Hessian bound of Section 2 and the lower bound $|\nabla u| \geq \gamma$ give $M \leq C\|D^2 u\|_\infty/\gamma$ throughout this band. Infinite iteration would make $m_k \geq 2^k m_0$ exceed this finite bound. Thus the process stops and proves (4.4).

This selection describes precisely the sequences relevant to a curvature contradiction. Consider the uniform assertion that there exist $0 < \epsilon_* \leq \epsilon_0$ and $C < \infty$ such that every bounded stable entire solution satisfies $M(X) \text{dist}(X, B_{\epsilon_*}(u)) \leq C$ for $X \in \Omega_{\epsilon_*}(u) \cap \{|u| \leq a\}$ whenever its bad

set is nonempty. For an empty bad set the assertion means $M = 0$ on that band. Its negation says that for every $\epsilon \in (0, \epsilon_0]$ and every finite C there are a bounded stable solution and a good-band point violating this bound, or, when the bad set is empty, a good-band point with positive curvature. Thus we may choose $\epsilon_j \downarrow 0$ and $L_j \rightarrow \infty$. For nonempty B_ϵ and $X \in \Omega_\epsilon$,

$$\text{dist}(X, \Omega_\epsilon^c) = \text{dist}(X, B_\epsilon) - (D_a + 1).$$

The one-Lipschitz distance function proves the lower bound. Since B_ϵ is closed and nonempty, the nearest-point fact just proved gives a nearest bad point z . Put $d = |X - z|$. At distance t along the segment from X to z the distance to B_ϵ equals $d - t$: the Lipschitz lower bound and the distance to z coincide. At $t = d - (D_a + 1)$ the segment therefore reaches Ω_ϵ^c , proving the upper bound. By (4.3), subtracting $D_a + 1$ changes the curvature-distance product by at most $C_a \epsilon (D_a + 1)$. At threshold ϵ_j choose a violating point X_j with

$$M(X_j) \text{dist}(X_j, B_{\epsilon_j}(u_j)) > 4L_j + C_a \epsilon_j (D_a + 1).$$

It satisfies the hypothesis of (4.4) with $\Omega = \Omega_{\epsilon_j}(u_j)$. If the bad set is empty, failure supplies $M(X_j) > 0$; apply selection in an artificial ball about X_j of radius greater than $4L_j/M(X_j)$, which lies wholly in the good region. In either case selection gives a point P_j with $b_j = M(P_j)$ and $0 < b_j \leq C_a \epsilon_j \rightarrow 0$. Consequently

$$R_j = b_j^{-1} \rightarrow \infty, \quad |A| \leq 2/R_j \quad \text{on } B_{L_j R_j}(P_j) \cap \{|u_j| \leq a\}, \quad (4.5)$$

where the comparison ball is contained in $\Omega_{\epsilon_j}(u_j)$ and the band has the same lower gradient bound γ_a . The vanishing thresholds arise from the exact uniform negation; they are not an improvement asserted for a fixed ϵ . We next turn the whole-band curvature bound into a complete list of local graphs. The continuation argument is essential: a pointwise implicit-function theorem alone does not supply a radius proportional to K^{-1} .

Theorem 4.3 (Complete phase charts). *Prescribe the phase interval, the lower gradient bound γ , and the slope to be retained, with the uniform solution derivative bounds fixed. One can choose $K_* > 0$, chart fractions, and an ambient buffer constant such that the following assertion holds. Suppose $0 < K \leq K_*$, P belongs to the phase band, and throughout $B_{C_{\text{buf}}/K}(P) \cap \{|u| \leq a\}$,*

$$|\nabla u| \geq \gamma, \quad |A_{\{u=u(X)\}}| \leq K.$$

After choosing one vertical direction, a cylinder of horizontal and vertical sizes proportional to K^{-1} admits finitely many compatible families $u(x', F_{i,\tau}(x')) = \tau$, $-a \leq \tau \leq a$. They contain every phase-band point in a fixed smaller cylinder, have small slopes, and on buffered subdisks satisfy

$$|D^2 F_{i,\tau}| \leq CK, \quad |F_{i,\tau} - F_{i,\rho}| \leq C|\tau - \rho|/\gamma, \quad |\partial_3 u(x', F_{i,\tau})| \geq \gamma/2. \quad (4.6)$$

The derivative sign is fixed in each family. The list includes local returning branches of globally connected surfaces, and all constants are independent of the number of branches.

Proof. Translate P to zero, put $\tau_* = u(P)$, and choose its increasing normal as vertical. Start with the local graph of this phase through P . While its normal angle is small, a horizontal lift has length at most twice the base length. The shape-operator identity bounds normal variation by K times lifted length. Thus a radial lift to z stays of slope at most $CK|z|$ on a sufficiently small disk of radius proportional to K^{-1} . Integration gives height at most $CK|z|^2$. These estimates prevent both loss of the angle bound and escape from the ambient ball.

For a direct single-valued construction, use the radial ODE

$$\dot{h}(t) = -\frac{\nabla' u(tz, h(t)) \cdot z}{\partial_3 u(tz, h(t))}, \quad h(0) = 0,$$

The preceding continuation estimates preserve $|\partial_3 u| \geq \gamma/2$ and keep the lift in a compact regular part of the phase surface, so the ODE extends to time one. Set $F_*(z) = h(1; z)$. ODE uniqueness, after reparametrizing a radial segment, gives $h(t; z) = F_*(tz)$; smooth parameter dependence gives a smooth function on the disk. In particular $u(z, F_*(z)) = \tau_*$. Differentiating this identity

in every base direction gives $DF_* = -\nabla' u / \partial_3 u$, so the radial normal estimate controls the full gradient:

$$|DF_*(z)| \leq CK|z|, \quad |F_*(z)| \leq CK|z|^2, \quad |D^2F_*| \leq CK.$$

For the last estimate the small-slope graph metric and its inverse are uniformly bounded; the second fundamental form in graph coordinates is $D^2F_*/\sqrt{1+|DF_*|^2}$.

Retain a disk D_{ρ_0} over which the graph is defined, and a vertical buffer $(-\rho_0, \rho_0)$ containing it strictly. Its graph Γ_* is relatively open and closed in

$$S_* := \{u = \tau_*\} \cap (D_{\rho_0} \times (-\rho_0, \rho_0)).$$

Openness is the implicit-function theorem and local uniqueness. For closedness, a sequence of graph points converging to a point of S_* has an interior limiting base coordinate, and continuity of F_* identifies its limiting height. The preimage of Γ_* under a same-phase path confined to S_* is therefore open and closed in the path interval. Such a path cannot enter from another local branch, even when the two branches are globally connected.

Choose a sufficiently small $\theta \leq 10^{-2}$ for the prescribed slope, then small fixed $\eta, \alpha > 0$, in that order. Set

$$\rho_0 = \eta\theta/K, \quad r = \alpha\rho_0, \quad H = \theta r, \quad t_* = \rho_0/4.$$

Choose η so the reference slope $CK\rho_0$ is much smaller than θ , and then α so

$$8r + t_* < \rho_0/2, \quad 3H + t_* < \rho_0, \quad 3H + CK(8r)^2 \ll \theta t_*, \quad CKt_*^2 \ll \theta t_*.$$

The constants in the last two inequalities will be smaller than the fixed tangent-component constant used below. These choices are compatible: $K\rho_0/\theta = \eta$, and $H/(\theta t_*) = 4\alpha$. Fix C_{buf} to contain all these disks and paths. If a band point $Q \in D_{8r} \times (-3H, 3H)$ has normal line making angle at least $\theta/20$ with vertical, some unit tangent T has vertical component of magnitude at least $c\theta$. The surface geodesic with this tangent is obtained directly from

$$\dot{X} = T, \quad \dot{T} = -\frac{D^2u(X)[T, T]}{|\nabla u(X)|^2} \nabla u(X).$$

Indeed

$$\frac{d}{dt}(T \cdot \nabla u(X)) = \dot{T} \cdot \nabla u + D^2u[T, T] = 0.$$

Since this constraint is initially zero, it stays zero. Consequently

$$\frac{d}{dt}u(X) = T \cdot \nabla u = 0, \quad \frac{d}{dt}|T|^2 = -2\frac{D^2u[T, T]}{|\nabla u|^2} T \cdot \nabla u = 0.$$

Thus the path stays on its initial phase and has unit speed. Its acceleration has magnitude $|D^2u[T, T]|/|\nabla u| \leq K$, by the curvature bound for this unit tangent. Unit speed and the buffer inequalities confine the state (X, T) to a compact regular part of the surface through time t_* . The smooth ODE therefore exists for that whole interval.

Write X_{hor} for its horizontal coordinates and set $\delta(t) = X_3(t) - F_*(X_{\text{hor}}(t))$. Choose the initial tangent sign toward the reference graph. Since its slope is much smaller than $c\theta$, the estimates give

$$|\delta(0)| \leq 3H + CK(8r)^2, \quad \delta(0)\delta'(0) < 0, \quad |\delta'(0)| \geq c\theta/2, \quad |\delta''| \leq CK.$$

More explicitly,

$$\delta'' = \ddot{X}_3 - D^2F_*(X_{\text{hor}})[\dot{X}_{\text{hor}}, \dot{X}_{\text{hor}}] - DF_*(X_{\text{hor}}) \cdot \ddot{X}_{\text{hor}},$$

so $|\delta''| \leq C_1K$, using $|\dot{X}| = 1$, $|\ddot{X}| \leq K$ and the reference graph bounds. If $\delta(0) = 0$, the point already lies on the reference graph, whose small slope excludes the large angle. Otherwise put $\varsigma = \text{sgn } \delta(0)$ and choose the scale constants so that

$$|\delta(0)| \leq c\theta t_*/8, \quad C_1Kt_*^2 \leq c\theta t_*/8.$$

The chosen tangent sign gives $\varsigma\delta'(0) \leq -c\theta/2$. Taylor's formula with its integral remainder now gives

$$\varsigma\delta(t_*) \leq |\delta(0)| - \frac{c\theta t_*}{2} + \frac{C_1Kt_*^2}{2} \leq -\frac{5c\theta t_*}{16} < 0.$$

Since $\varsigma\delta(0) > 0$, continuity forces a crossing strictly before t_* . Horizontal and vertical confinement place that crossing in the reference buffer. Different phases cannot meet at one point; for the same phase the relatively open-and-closed property forbids entry from a distinct branch. Thus every phase in the alignment cylinder has normal line within $\theta/20$ of vertical. In particular its graph slope is at most $\theta/10$, and $|\partial_3 u| \geq \gamma/2$.

Enumerate every root of $u(0, t) = \tau_*$ in $(-H, H)$. At a root $|\partial_3 u| \geq \gamma/2$. Between any two roots Rolle's theorem gives $\partial_3 u = 0$, so the bound $\|D^2 u\|_\infty \leq M_s$ gives separation at least $\gamma/(2M_s)$. There are therefore finitely many roots. Between consecutive roots the continuous function $u(0, t) - \tau_*$ has one strict sign. At a simple root its sign on the right is the derivative sign, and its sign on the left is the opposite derivative sign. The derivative signs at consecutive roots therefore alternate. Each root extends over D_{4r} by radial lifting and local uniqueness. Alignment preserves the denominator, and horizontal travel changes height by at most $4r\theta/10 = 2H/5$, preventing escape from the alignment cylinder. If two such graphs met, local implicit-function uniqueness would make their coincidence set both open and closed in the connected disk. They would then agree at the axis, contradicting the distinct roots. Thus they are disjoint and ordered.

Continue each graph by

$$\partial_\tau F_{i,\tau} = 1/\partial_3 u(x', F_{i,\tau}).$$

Alignment gives a total phase displacement at most $4a/\gamma$. Since $H = \alpha\eta\theta^2/K$, it is sufficient to decrease K_* so that

$$K_* \leq \frac{\alpha\eta\theta^2\gamma}{32a}.$$

Then $4a/\gamma \leq H/8$ whenever $K \leq K_*$. Including the initial root height and the horizontal travel across D_{4r} , the height stays less than $H + 2H/5 + H/8 < 3H$. Thus no loss of alignment or escape can stop the phase ODE, and it continues throughout $[-a, a]$. Derivative sign cannot change. If two families met at the same phase, uniqueness of the vertical phase ODE, run back to τ_* , would contradict their disjointness there. Their ordering is therefore preserved for every phase. The graph metric and curvature bound give the Hessian estimate, and integration of the phase ODE gives the offset estimate in (4.6).

Conversely, from a band point in $D_r \times (-H/4, H/4)$, return vertically to phase τ_* at a cost at most $H/8$, then radially follow that phase to the axis at a cost at most $H/10$. Both chosen paths remain inside the alignment cylinder. Their axis height has magnitude at most $H/4 + H/8 + H/10 < H$, so this intersection is one of the enumerated roots. Reverse that same radial path: uniqueness of its ODE identifies the returned point with the root's graph at phase τ_* . Reverse the same vertical phase path: uniqueness of its ODE identifies the original point with that family's graph at the original phase. This proves completeness without requiring independence of arbitrary continuation paths or any assumption on global component connectivity. No estimate has depended on the number of roots. $\square$

The whole-band bounds in (4.5) satisfy these hypotheses with $K = 2/R_j$. For large j , every band center $X \in B_{L_j R_j/2}(P_j)$ has

$$B_{C_{\text{buf}} R_j/2}(X) \subset B_{L_j R_j}(P_j).$$

The same chart fractions and constants therefore apply at every such center. Moving a central-phase point to its zero phase costs bounded physical distance; preservation of its rescaled curvature will be proved only after a more precise profile limit.

In the following interior estimates and planar-limit assertion, buffered charts include both a larger common graph domain and a region where the list is complete. Allowed centers lie in a strictly smaller retained cylinder, at distance at least cR from both boundaries, with $c > 0$ fixed. In particular the actual zero chosen as origin lies in this controlled interior region. All uniform claims are on further fixed interior subcylinders with these margins. This convention ensures that translations and small rotations still leave complete chart neighborhoods of size proportional to R .

A useful consequence of positivity and the graph Hessian bound is the following interior estimate for any positive graph difference h at points whose common disk-boundary distance is at

least cR :

$$|Dh| \leq C(\sqrt{h/R} + h/R). \quad (4.7)$$

Indeed $|D^2h| \leq C/R$. If $Dh \neq 0$, following $-Dh/|Dh|$ a distance ℓ inside the buffer gives $0 < h(X - \ell Dh/|Dh|) \leq h(X) - \ell|Dh(X)| + C\ell^2/R$. Thus $|Dh| \leq h/\ell + C\ell/R$. For $h \leq c_1 R$, choose c_1 sufficiently small in terms of c and take $\ell = \sqrt{Rh}$. For $h > c_1 R$, the fixed slope bound is at most $C(\sqrt{h/R} + h/R)$. The case $Dh = 0$ is immediate. If the two zero graphs coincide, their difference and its gradient vanish instead. For a signed phase offset $w = F_{i,\tau} - F_{i,0}$, use $|w| \leq C_0$ and $|D^2w| \leq C/R$ instead. The same Taylor formula gives $|Dw| \leq 2C_0/\ell + C\ell/R$. With available disk buffer at least r , take $\ell = \min\{r, \sqrt{R}\}$, or a fixed fraction of it, to obtain

$$|D(F_{i,\tau} - F_{i,0})| \leq C(R^{-1/2} + r^{-1}). \quad (4.8)$$

Theorem 4.4 (Planar limits of complete charts). *Suppose complete buffered phase charts have fixed small slopes, uniform nondegeneracy and bounded phase offsets, sizes proportional to $R_j \rightarrow \infty$, and $|D^2 F_{j,i,\tau}| \leq C/R_j$. Choose coordinates whose origin is an actual zero and whose third axis is its increasing normal. Then $u_j \rightarrow q_s(t)$ in every local C^k norm. All neighboring zero gaps diverge uniformly on fixed interior subcylinders. These conclusions require no prior isolation assumption.*

Proof. Translations and orthogonal coordinate changes preserve this solution class. Indeed, if O is orthogonal and $v(X) = u(OX + b)$, changing variables in the normalized kernel gives $(-\Delta)^s v(X) = ((-\Delta)^s u)(OX + b)$, and changing variables in the stability form gives $Q_v(\psi) = Q_u(\psi \circ T^{-1})$, where $T(X) = OX + b$. Compact supports are preserved. The derivative bounds also persist, up to constants depending only on their order.

First justify the coordinate choice. The increasing normal at an interior zero makes a fixed small angle with either the old vertical or its negative; reverse the old vertical if necessary. Under the resulting small rotation, a small-slope graph remains a graph on a smaller disk of comparable size. More explicitly, its projection onto the new horizontal plane has the form $z \mapsto Az + bF(z)$, where the smallest singular value of A is at least $\cos \vartheta$, $|b| \leq \sin \vartheta$, and ϑ is the rotation angle. If the old slope is at most σ , then

$$|(Az + bF(z)) - (Ay + bF(y))| \geq (\cos \vartheta - \sigma \sin \vartheta)|z - y|.$$

Here $\vartheta \leq \arctan \sigma$. The fixed small slope therefore gives a lower bound $m_0 > 0$ for the displayed coefficient. The projection is injective and bi-Lipschitz, uniformly in the family and phase. We next check its image and the physical buffers uniformly. Write $R = R_j$ temporarily. After the translation, the old common base contains $D_{cR}(0)$ and its completeness region contains $B_{cR}(0)$, after decreasing the fixed interior fraction $c > 0$. Let $B = 2a/\gamma_0$ bound the old height displacement between any two phases. Choose a fixed $\delta > 0$ below, and take R large enough that $B \leq \delta R$. Retain only families for which some phase graph meets

$$\mathcal{C}_\delta = D_{\delta R} \times (-\delta R, \delta R)$$

in the new coordinates. The centered family is among them. For any such family, a meeting point has old coordinates (z_y, h_y) with $|z_y|, |h_y| \leq \sqrt{2}\delta R$. At its phase τ_y , $F_{i,\tau_y}(z_y) = h_y$. Hence, for every $\tau \in [-a, a]$,

$$|F_{i,\tau}(0)| \leq (1 + \sigma)\sqrt{2}\delta R + B \leq C_1\delta R, \quad C_1 = (1 + \sigma)\sqrt{2} + 1.$$

The projection $\Pi_{i,\tau}(z) = Az + bF_{i,\tau}(z)$ therefore satisfies $|\Pi_{i,\tau}(0)| \leq C_2\delta R$, with a fixed C_2 (for example $C_2 = C_1$). On $D_{cR/2}$ its boundary images are at distance at least $m_0 cR/2$ from $\Pi_{i,\tau}(0)$. Its local inverse extends along every segment of length less than this radius starting at $\Pi_{i,\tau}(0)$: otherwise the inverse path has an interior accumulation point, where the implicit-function theorem extends it, or approaches the boundary, contrary to the lower distance bound. Thus the image contains that open ball. Choose δ so that

$$(4 + C_2)\delta < m_0 c/4.$$

The common new disk $D_{4\delta R}$ then lies in every such image, with distance at least $m_0 cR/4$ from its boundary.

For $w \in D_{4\delta R}$ the inverse old base coordinate obeys

$$|z| \leq m_0^{-1}|w - \Pi_{i,\tau}(0)| \leq C_3 \delta R, \quad C_3 = (4 + C_2)/m_0.$$

The old height there is at most $(C_1 + \sigma C_3)\delta R$. Consequently the old physical point has norm at most $C_4 \delta R$, where $C_4 = C_3 + C_1 + \sigma C_3$ is fixed. Also require $C_4 \delta < c/4$. Every graph point just constructed, for every retained family and every phase, then lies in $B_{cR/4}$, with a fixed positive margin inside both old buffers. These choices of δ depend only on the fixed chart constants.

The unoriented normal of each old phase graph makes angle at most $\arctan \sigma$ with the old vertical. Its absolute scalar product with the new vertical is therefore at least $c_\sigma = \cos(2 \arctan \sigma) > 0$. Since $|\nabla u_j| \geq |\partial_{t_{\text{old}}} u_j| \geq \gamma_0$ there,

$$|\partial_{t_{\text{new}}} u_j| \geq \gamma_1 := c_\sigma \gamma_0$$

throughout these families over $D_{4\delta R}$ and the full phase interval. The uniform inverse-function theorem with its phase parameter defines jointly smooth new graph heights $G_{i,\tau}$. Their level identity gives

$$\partial_\tau G_{i,\tau} = 1/\partial_{t_{\text{new}}} u_j(w, G_{i,\tau}(w)), \quad |G_{i,\tau} - G_{i,\rho}| \leq |\tau - \rho|/\gamma_1.$$

The derivative sign is constant on each connected base-phase domain. This bounded phase continuation stays inside the old buffers by the preceding estimates, including at both endpoints of the phase interval. The new slopes are at most $\tan(2 \arctan \sigma)$. The old graph metric $g = I + DF \otimes DF$ and second fundamental form $\Pi_{\alpha\beta} = F_{\alpha\beta}/\sqrt{1 + |DF|^2}$ give $|A| \leq C/R$. Curvature is invariant under rotation; the bounded new slope and the same graph formula then give $|D^2 G| \leq C/R$.

Graphs at the same phase stay disjoint, since rotation preserves intersections and their old inverse coordinates belong to the common old base. Their order is constant on the connected new base-phase domain, so they may be reindexed once. The level identities and unique inverses preserve compatibility. Every band point in $\mathcal{C}_\delta$ belongs to an old family by old completeness; that family is retained by definition, and its unique inverse projection identifies the point with its new phase graph. Thus the new families are complete on $\mathcal{C}_\delta$ and have the common disk $D_{4\delta R}$ for every phase. A further fixed restriction gives positive interior margins. All these constants are independent of the number of families. Relabel their new heights as F . The centered zero graph $F_{j,0}$ now satisfies $F_{j,0}(0) = 0$ and $DF_{j,0}(0) = 0$. Consequently, on each fixed disk D_N ,

$$\sup_{D_N} |F_{j,0}| \leq CN^2/R_j, \quad \sup_{D_N} |DF_{j,0}| \leq CN/R_j.$$

Let X_j be arbitrary band points in a fixed ball B_N . Their associated zero graphs have heights at X'_j bounded by $N + C_0$, where C_0 is the common phase-offset bound. Their gaps from the reference zero graph are therefore bounded at these coordinates. Distinct zero graphs are ordered and disjoint. Apply (4.7) to the positive difference between each such graph and the reference graph; its gradient at X'_j tends to zero. For the reference graph this follows directly from the preceding estimate. Equation (4.8), with available buffer comparable to R_j , then makes the phase-graph slope through X_j tend to zero. Differentiating its level identity gives

$$\partial_\alpha u_j(X_j) = -\partial_3 u_j(X_j) \partial_\alpha F_{j,i,u_j}(X'_j) \longrightarrow 0, \quad \alpha = 1, 2,$$

using the uniform ambient first-derivative bound from Section 3. This applies also to branches with opposite derivative signs. Since the points were arbitrary, the vanishing is uniform over band points in each fixed ball.

Pass to any smooth stable subsequential limit u_∞ , using Section 3. Its tangential derivatives vanish where $|u_\infty| < a$, since each such point belongs to the band of u_j for large j . At a point with $|u_\infty| = a$, either it is approached from the interior band or a neighborhood avoids that band. In the first case use continuity of the derivative. In the second, continuity excludes the opposite outer phase locally, so the point is a local minimum at value a or a local maximum at value $-a$;

all its first derivatives vanish. The Jacobi differentiation justified above applies to this bounded smooth limit and makes each bounded tangential derivative w a pointwise solution of

$$[(-\Delta)^s + 3u_\infty^2 - 1]w = 0.$$

On the open set $\{|u_\infty| > a\}$ the potential is at least $3a^2 - 1 > 0$, and $w = 0$ on its complement. The bounded comparison proved at the start of this section, applied to both signs, gives $w = 0$ everywhere. The cylindrical reduction Lemma 3.2 transfers equation and stability to the one-dimensional trace. Lemma 4.2, the reference-layer normalization of Section 2, the centered zero, and the increasing-normal derivative at the origin, bounded below by nondegeneracy, identify it with q_s .

This proves convergence of the whole sequence in every local C^k norm. Indeed, if some fixed N, k and positive error persisted along a subsequence, smooth compactness would give a further subsequence with a smooth limit. The preceding argument identifies that limit with q_s , contradicting the positive $C^k(B_N)$ error.

For gap divergence, take two neighboring vertical zero crossings whose first zero lies in the retained cylinder. A bounded gap keeps the second inside its completeness buffer for large j . At each crossing $|\partial_3 u_j| \geq c\gamma$. Rolle's theorem and $\|D^2 u_j\|_\infty \leq M_s$ give a fixed lower spacing $\delta_0 = c\gamma/M_s > 0$. If their gap stayed bounded, recenter at the first zero and rotate to its increasing normal. The second zero has coordinates Y_j in a fixed ball, with $|(Y_j)_3| \geq \delta_0 \cos \vartheta \geq c\delta_0$. Extract $Y_j \rightarrow Y_*$. The centered profiles converge uniformly on that ball to $q_s(t)$, and continuity gives $q_s((Y_*)_3) = \lim_j u_j(Y_j) = 0$. But $(Y_*)_3 \neq 0$, contradicting the strict monotonicity and unique centered zero of q_s .

Finally, a failure of uniformity would give scales $R_j \rightarrow \infty$ and interior centers violating a fixed profile-error bound or a fixed lower bound on neighboring gaps. Recenter at those zeros. The stipulated positive margins preserve complete chart neighborhoods of size proportional to R_j ; the projection argument preserves all constants up to fixed factors. The entire limit argument just given therefore applies to this sequence and contradicts the violation. Equivalently, for each fixed physical radius N and derivative order k , the supremum of the tangent-centered $C^k(B_N)$ profile error over this class of charts with scale at least R tends to zero as $R \rightarrow \infty$. This common qualitative modulus will be used below. $\square$

5. A STABILITY INEQUALITY FOR TWO ACTUAL LAYERS

Throughout this section, u is a bounded stable entire solution of the normalized cubic equation in $\mathbb{R}^3$, and $p = 2s \in (1, 2)$. We use the uniform smooth derivative bounds and compact Lipschitz testing established above. All amplitudes below are real-valued. We now localize stability without replacing the actual solution by a sum of profiles. Put $g = |\nabla u|$ and

$$\mathcal{D}(X, Y) = g(X)g(Y) - \nabla u(X) \cdot \nabla u(Y) \geq 0, \quad D_u(X) = c_{3,s} \int \mathcal{D}(X, Y) |X - Y|^{-3-p} dY.$$

Indeed,

$$2\mathcal{D}(X, Y) = |\nabla u(X) - \nabla u(Y)|^2 - (g(X) - g(Y))^2, \quad 0 \leq \mathcal{D}(X, Y) \leq \frac{1}{2} |\nabla u(X) - \nabla u(Y)|^2.$$

The bounded Hessian gives $\mathcal{D}(X, Y) \leq C|X - Y|^2$ near the diagonal; bounded gradients control its tail. The radial integrals $\int_0^1 \rho^{1-p} d\rho$ and $\int_1^\infty \rho^{-1-p} d\rho$ are finite, so D_u is finite and uniformly bounded. Moreover, $|g(X) - g(Y)| \leq |\nabla u(X) - \nabla u(Y)|$ shows that g is bounded and globally Lipschitz, including at its zeros.

Lemma 5.1 (Vector identity). *For every compact Lipschitz amplitude η ,*

$$Q_u(g\eta) = \frac{c_{3,s}}{2} \iint g(X)g(Y)(\eta(X) - \eta(Y))^2 |X - Y|^{-3-p} dX dY - \int D_u \eta^2. \quad (5.1)$$

Proof. Write $u_j = \partial_j u$. The compact weak Jacobi identity already proved in (4.2), with test $u_j \eta^2$, gives

$$\mathcal{B}_3(u_j, u_j \eta^2) + \int (3u^2 - 1)u_j^2 \eta^2 = 0.$$

These tests are compactly supported and Lipschitz. The compact-test estimates of Section 3 give absolute convergence of each bilinear pairing. Also $g\eta$ is compactly supported and Lipschitz, hence belongs to $H^s(\mathbb{R}^3)$. The same section's fixed-support mollifications converge to $g\eta$ in H^s and in L^2 ; boundedness of $3u^2 - 1$ therefore passes the quadratic form and stability inequality to $g\eta$, including when its support meets $\{g = 0\}$.

Sum the three Jacobi identities and subtract them from $Q_u(g\eta)$. Their potential term is exactly $\int (3u^2 - 1)g^2\eta^2$. For $a, b \in \mathbb{R}^3$ and $q, t \in \mathbb{R}$,

$$(|a|q - |b|t)^2 - (a - b) \cdot (aq^2 - bt^2) = |a||b|(q - t)^2 - (|a||b| - a \cdot b)(q^2 + t^2).$$

Apply this with $a = \nabla u(X)$, $b = \nabla u(Y)$, $q = \eta(X)$ and $t = \eta(Y)$. The first term is integrable because g is bounded and $\eta \in H^s$. For the second term, symmetry and nonnegativity of $\mathcal{D}$ give

$$\frac{c_{3,s}}{2} \iint \mathcal{D}(X, Y)(\eta(X)^2 + \eta(Y)^2)|X - Y|^{-3-p} dX dY = \int D_u \eta^2 < \infty.$$

Thus the integrated algebra proves the claimed identity. No division by g is used. $\square$

Fix $a_0 \in (0, 1)$, $L_0, K_0 \geq 1$, and $0 \leq \theta_0 \leq 10^{-2}$. Consider two actual phase families on D_{4r} ,

$$u(x', F_{i,\tau}(x')) = \tau, \quad |\tau| \leq a_0, \quad i = 1, 2, \quad F_i = F_{i,0}, \quad F_1 < F_2,$$

with

$$|DF_{i,\tau}| \leq \theta_0, \quad |D^2 F_{i,\tau}| \leq K_0/R, \quad |F_{i,\tau} - F_i| \leq L_0. \quad (5.2)$$

Here a regular phase family means that the graphs depend C^1 on τ , are C^2 in x' , and satisfy $\sigma_i \partial_3 u(x', F_{i,\tau}(x')) > 0$ for one fixed $\sigma_i \in \{-1, 1\}$. In particular the phase map is strictly monotone and $\partial_\tau F_{i,\tau} = 1/\partial_3 u$. The argument needs no quantitative lower bound for $|\partial_3 u|$. Require that the enlarged tubes $\{|x'| < 4r, |x_3 - F_i(x')| < 6T\}$ be disjoint, and impose the quantitative tube bounds

$$\begin{aligned} g(x', F_i(x') + t) &\leq C_w(1 + |t|)^{-p-1}, \\ 1 - \sigma_i u(x', F_i(x') + 3T) &\leq C_w T^{-p}, \\ 1 + \sigma_i u(x', F_i(x') - 3T) &\leq C_w T^{-p}. \end{aligned} \quad (5.3)$$

The first bound holds for $x' \in D_{4r}$, $|t| < 6T$, and the endpoint bounds hold for $x' \in D_{4r}$. Their deficits are nonnegative by Lemma 2.1. We assume

$$T \geq T_0 \geq 10(1 + L_0), \quad T \leq c_0 \min\{r, R\}, \quad (5.4)$$

where $0 < c_0 \leq 1/12$ will be chosen from these fixed data. The bounds in (5.3) are standing hypotheses. For example, (4.1) supplies them, with a constant depending also on $c_* > 0$, if $\text{dist}((x', F_i(x') + t), Z) \geq c_*|t|$ throughout the enlarged tubes and $F_i(x')$ is the only zero on the vertical segment $[F_i(x') - 3T, F_i(x') + 3T]$. On the upper and lower parts of that segment the regular-family sign gives respectively $\sigma_i u > 0$ and $\sigma_i u < 0$, which identifies the endpoint deficits with $1 - |u|$.

Fix once and for all a smooth function b equal to one on $[-1, 1]$, supported in $(-2, 2)$, and between zero and one. Extend the real function $\xi \in C_c^1(D_r)$ by zero and set

$$\Omega_i = \{|x'| < 2r, |x_3 - F_i(x')| < 3T\}, \quad \eta_i(x', x_3) = \xi(x')b((x_3 - F_i(x'))/T), \quad \psi_i = g\eta_i.$$

The formula for η_i is used on $D_{4r} \times \mathbb{R}$; set it equal to zero outside that cylinder. Since ξ vanishes near ∂D_r , this gives a globally compactly supported C^1 function. Its derivatives on that cylinder are

$$\nabla' \eta_i = b \nabla \xi - T^{-1} \xi b' D F_i, \quad \partial_3 \eta_i = T^{-1} \xi b',$$

where b and b' are evaluated at $(x_3 - F_i(x'))/T$. Thus η_i and ψ_i are compactly supported Lipschitz functions. For $X \in \text{supp } \eta_i$ we have $|X'| < r$ and $|X_3 - F_i(X')| \leq 2T$. If $Y \notin \Omega_i$ and $|Y'| \geq 2r$, then $|X - Y| \geq r$. Otherwise $|Y_3 - F_i(Y')| \geq 3T$, and

$$T \leq |Y_3 - X_3| + \theta_0 |Y' - X'| \leq \sqrt{1 + \theta_0^2} |X - Y|.$$

Since $r \geq 12T$, these bounds show $\text{dist}(\text{supp } \eta_i, \Omega_i^c) \geq T/\sqrt{1 + \theta_0^2}$.

For $X \in \Omega_i$ define

$$D_i(X) = c_{3,s} \int_{\Omega_i} \mathcal{D}(X, Y) |X - Y|^{-3-p} dY, \quad V_i(X) = \int_{\Omega_i^c} \nabla u(Y) |X - Y|^{-3-p} dY,$$

and put

$$P_i = \frac{c_{3,s}}{2} \iint_{\Omega_i^2} g(X)g(Y)(\eta_i(X) - \eta_i(Y))^2 |X - Y|^{-3-p} dX dY.$$

We have $0 \leq D_i \leq D_u$ and $P_i < \infty$. The integral defining $V_i(X)$ is absolutely convergent at each interior point X . On $\text{supp } \eta_i$ the preceding buffer and boundedness of g give

$$\int_{\Omega_i^c} g(Y) |X - Y|^{-3-p} dY \leq CT^{-p}, \quad |V_i(X)| \leq CT^{-p}.$$

Consequently all the exterior terms below are absolutely integrable.

Splitting (5.1) at the tube boundary gives

$$Q_u(\psi_i) = P_i - \int_{\Omega_i} D_i \eta_i^2 + c_{3,s} \int_{\Omega_i} \eta_i^2 \nabla u \cdot V_i. \quad (5.5)$$

For completeness, write $A_i(X) = \int_{\Omega_i^c} g(Y) |X - Y|^{-3-p} dY$ for $X \in \Omega_i$. The two cross terms in the positive double integral are equal, so their sum is $c_{3,s} \int_{\Omega_i} \eta_i^2 g A_i$. Also

$$D_u(X) = D_i(X) + c_{3,s}(g(X)A_i(X) - \nabla u(X) \cdot V_i(X)).$$

Substitution cancels the two scalar terms exactly and proves (5.5). The remaining row retains the full signed vector $\nabla u(Y)$.

Put

$$E(R, T, r) = T^{-1-2p} + (RT)^{-(p+1)/2} + r^{-p-1}.$$

The next calculation controls that signed row independently of any remote transitions.

Lemma 5.2 (Exterior vector row). *Let $X = (a, F_i(a) + t)$, $a \in D_r$, $|t| \leq 2T$, and $n_i(a) = (-DF_i(a), 1)/\sqrt{1 + |DF_i(a)|^2}$. For a unit vector e with $e_3 \geq 0.9$, $|e'| \leq 1/10$, and $|e - n_i(a)| \leq C_e(T/R + R^{-1/2} + r^{-1})$,*

$$|V_i(X)| \leq CT^{-p-1}, \quad e \cdot \sigma_i V_i(X) \leq CE(R, T, r).$$

Constants may depend on C_e , and c_0 may be decreased accordingly.

Proof. The preceding smoothness bounds give a globally bounded gradient, so the separated exterior row is absolutely convergent. The boundary of Ω_i is piecewise C^1 , with its edges of surface measure zero. Integrate ∇u componentwise by parts over $\Omega_i^c \cap B_M(X)$, with M large enough to enclose $\overline{\Omega_i}$. Using $|u| \leq 1$ bounds the row by

$$\int_{\partial\Omega_i} |X - Y|^{-3-p} dS_Y + \int_{\Omega_i^c} |\nabla_Y |X - Y|^{-3-p}| dY.$$

The artificial boundary at radius M contributes $O(M^{-1-p})$, using $|u| \leq 1$, and vanishes as $M \rightarrow \infty$. The differentiated kernel is integrable on the separated exterior. For exterior points with $Y' \in D_{2r}$, $T \leq |Y_3 - X_3| + \theta_0 |Y' - a| \leq \sqrt{1 + \theta_0^2} |X - Y|$; for the remaining points the horizontal distance is at least r . Thus the source is at distance cT from the complement, and the volume term is bounded by $C \int_{cT}^\infty \rho^{-2-p} d\rho = CT^{-p-1}$. On each top or bottom graph, $|X - Y|^2 \geq c(|a - Y'|^2 + T^2)$, and

$$\int_{\mathbb{R}^2} (|z|^2 + T^2)^{-(3+p)/2} dz = \pi \frac{\Gamma((p+1)/2)}{\Gamma((p+3)/2)} T^{-p-1}.$$

Parameterizing the lateral boundary by its circle angle and vertical offset gives its area $24\pi rT$. It is at distance at least r , costing $CTr^{-p-2} \leq CT^{-p-1}$.

For the signed assertion decompose $Y = X + w + \lambda e$, with $w \perp e$ and $\rho = |w|$. Along every line the horizontal disk condition is an interval, and within it

$$\frac{d}{d\lambda}(Y_3 - F_i(Y')) = e_3 - DF_i(Y') \cdot e' \geq c > 0.$$

Its intersection with Ω_i is therefore empty or one interval. For example their sections are described by the measurable indicator $\mathbf{1}_{\Omega_i}(X + w + \lambda e)$, so the line integrals are measurable in w . Absolute convergence of $\nabla u(Y)|X - Y|^{-3-p}$ on the exterior permits Fubini in the orthogonal coordinates (w, λ) , whose Jacobian is one. Choose $c_1 > 0$ sufficiently small depending on C_e, K_0 and the fixed data, and then decrease c_0 . Set $L = c_1 \min\{\sqrt{RT}, r\}$. For $\rho \leq L$ and $|\lambda| \leq 8T$, one has $|Y'| \leq r + c_1 r + 0.8T < 2r$, so all horizontal points stay in D_{2r} . Taylor's formula gives

$$Y_3 - F_i(Y') = t + (w_3 - DF_i(a) \cdot w') + \lambda(e_3 - DF_i(a) \cdot e') + O((\rho^2 + T^2)/R).$$

Because $w \perp e$, its middle term is bounded by $C|e - n_i(a)|\rho$. The total perturbation of $t + \lambda q$, $q = e_3 - DF_i(a) \cdot e' \in [0.899, 1.01]$, is at most $T/4$. The ratios controlling this choice are

$$\frac{\rho^2}{RT} \leq c_1^2, \quad \frac{T}{R} \leq c_0, \quad \frac{(T/R)\rho}{T} \leq c_1 \sqrt{T/R}, \quad \frac{R^{-1/2}\rho}{T} \leq c_1 T^{-1/2}, \quad \frac{\rho}{rT} \leq c_1/T.$$

Thus the line stays inside for $|\lambda| \leq T/4$, and exits through the top or bottom before $|\lambda| = 8T$, with endpoints

$$-8T \leq b_-(w) \leq -T/4, \quad T/4 \leq b_+(w) \leq 8T.$$

The horizontal buffer excludes lateral exits on these lines. Strict monotonicity of the height gives unique endpoints, and b_+ is on $Y_3 = F_i(Y') + 3T$, whereas b_- is on $Y_3 = F_i(Y') - 3T$. Their measurability also follows from the implicit function theorem, since the height derivative is at least 0.899 throughout this part of each line.

Set $U(\lambda) = \sigma_i u(X + w + \lambda e)$ and $k_\rho(\lambda) = (\rho^2 + \lambda^2)^{-(3+p)/2}$. On the upper ray integration by parts gives

$$\int_{b_+}^{\infty} U' k_\rho = (1 - U(b_+))k_\rho(b_+) + \int_{b_+}^{\infty} (1 - U)k'_\rho \leq CT^{-p}(\rho^2 + cT^2)^{-(3+p)/2}.$$

Here the endpoint deficit is evaluated at the actual top graph $Y_3 = F_i(Y') + 3T$, so it is bounded by $C_w T^{-p}$. Also $k'_\rho \leq 0$ and $|U| \leq 1$. Boundedness of U makes the term at infinity vanish; no limit of U along the ray is needed. The lower ray similarly gives

$$\int_{-\infty}^{b_-} U' k_\rho = (1 + U(b_-))k_\rho(b_-) - \int_{-\infty}^{b_-} (1 + U)k'_\rho \leq CT^{-p}(\rho^2 + cT^2)^{-(3+p)/2}.$$

Its endpoint is on the actual bottom graph, with the corresponding deficit from (5.3). Integrating over $\rho \leq L$ and enlarging to the entire two-dimensional plane gives $CT^{-p}T^{-p-1} = CT^{-1-2p}$. For $\rho > L$, the exterior consists of at most two rays or a whole line. On each interval I , integration by parts and $|U| \leq 1$ give

$$\left| \int_I U' k_\rho \right| \leq \sum_{b \in \partial I \cap \mathbb{R}} k_\rho(b) + \int_I |k'_\rho|.$$

There are at most two finite endpoints, $k_\rho \leq \rho^{-3-p}$, and $\int_{\mathbb{R}} |k'_\rho| d\lambda = 2\rho^{-3-p}$. Boundedness of U removes every infinite endpoint term. Summing the intervals therefore bounds their signed integrals by $C\rho^{-3-p}$. The bad lines cost $C \int_L^\infty \rho^{-2-p} d\rho \leq C[(RT)^{-(p+1)/2} + r^{-p-1}]$. No absolute value has been put on U' inside a ray integral; subsequent transitions are accounted for by boundedness and the signs of the integrated kernel. $\square$

For each source above let $\ell_X = 1 + |t|$. Inside Ω_i take C_X to consist of $\ell_X \leq |Y' - a| \leq 2\ell_X$ and the actual phase interval $I_i(Y')$, whose endpoints are $F_{i,-a_0}(Y')$ and $F_{i,a_0}(Y')$, placed in increasing order. Since $T \geq 1$ and $c_0 \leq 1/12$, $2\ell_X \leq 6T \leq r/2$. Thus its horizontal annulus lies in D_{2r} , and its vertical offsets are at most $L_0 < T$. In particular $C_X \subset \Omega_i$. The exact phase change of variables gives

$$\int_{I_i(Y')} g(Y', z) dz = \int_{-a_0}^{a_0} \sqrt{1 + |DF_{i,\tau}(Y')|^2} d\tau \in [2a_0, 2a_0\sqrt{1 + \theta_0^2}]. \quad (5.6)$$

Indeed the phase parameterization is one-to-one, and differentiating the level identity gives $\nabla' u = -\partial_3 u DF_{i,\tau}$ along it. Consequently $g/|\partial_3 u| = \sqrt{1 + |DF_{i,\tau}|^2}$. Use the absolute Jacobian $|\partial_\tau F_{i,\tau}| = 1/|\partial_3 u|$; sorting the endpoints makes the formula valid for either σ_i . For each fixed Y' , the

derivative $\partial_3 u$ is continuous and nonzero along the compact phase interval, so this ordinary one-dimensional change of variables is legitimate. No lower bound uniform in Y' is used: the factors $|\partial_3 u|$ cancel exactly. Define

$$m_X = \int_{C_X} g(Y) |X - Y|^{-3-p} dY, \quad A_X = \int_{C_X} \sigma_i \nabla u(Y) |X - Y|^{-3-p} dY, \quad e_X = A_X / |A_X|.$$

The phase mass, an annulus of area comparable to ℓ_X^2 , and $\ell_X \leq |X - Y| \leq C(L_0, \theta_0) \ell_X$ imply $m_X \asymp \ell_X^{p-1}$. The upper distance estimate uses $|Y_3 - X_3| \leq L_0 + 2\theta_0 \ell_X + |t|$. Indeed the annulus has area $3\pi \ell_X^2$; multiplying the two bounds in (5.6) by the kernel bounds $(C\ell_X)^{-3-p}$ and ℓ_X^{-3-p} gives the two-sided estimate for m_X , in particular $m_X > 0$. Every vector in the average is in the upward cone $|v'| \leq \theta_0 v_3$, hence $|A_X| \geq c m_X$. More precisely, along a phase graph,

$$\sigma_i \nabla u = g n_{i,\tau}, \quad n_{i,\tau} = \frac{(-DF_{i,\tau}, 1)}{\sqrt{1 + |DF_{i,\tau}|^2}}, \quad (A_X)_3 \geq \frac{m_X}{\sqrt{1 + \theta_0^2}}.$$

Here $|A_X| \leq m_X$ by the triangle inequality, so A_X is nonzero and e_X is well-defined for every source, even when $g(X) = 0$. These quantities are measurable: the moving annulus and sorted phase endpoints define a measurable indicator, their distance from X is at least $\ell_X \geq 1$, and parameterized integration followed by normalization preserves measurability. Since $T \geq T_0 \geq 1$ and $T \leq c_0 R$, one has $R \geq 1$. Apply (4.8) to the signed function $F_{i,\tau} - F_i$ on D_{4r} : its absolute value is at most L_0 , its Hessian is at most $2K_0/R$, and every point in D_{2r} has disk-boundary distance at least $2r$. Together with the variation of DF_i over distance $2\ell_X$, this gives

$$|n_{i,\tau}(Y') - n_i(a)| \leq C(\ell_X/R + R^{-1/2} + r^{-1}).$$

The line segment between a and Y' lies in the convex common disk, so the Hessian bound gives $|DF_i(Y') - DF_i(a)| \leq 2K_0 \ell_X/R$. The normal map $v \mapsto (-v, 1)/\sqrt{1 + |v|^2}$ is Lipschitz on the fixed slope ball; this proves the displayed estimate uniformly in τ . Average it to obtain $|A_X/m_X - n_i(a)| \leq \delta_X$, where $\delta_X = C(\ell_X/R + R^{-1/2} + r^{-1})$. Since $|n_i(a)| = 1$, $|1 - |A_X|/m_X| \leq \delta_X$, and thus $|e_X - n_i(a)| \leq 2\delta_X$. The result is

$$|e_X - n_i(a)| \leq C(\ell_X/R + R^{-1/2} + r^{-1}).$$

The upward cone gives $(e_X)_3 \geq (1 + \theta_0^2)^{-1/2}$ and $|e'_X| \leq \theta_0$. In particular $(e_X)_3 \geq 0.9$ and $|e'_X| \leq 1/10$, since $\theta_0 \leq 1/100$. Since $\ell_X \leq 3T$, this is an admissible direction for Lemma 5.2, with a constant C_e fixed by the chart data. At a source with $g > 0$, write $\nu_X = \sigma_i \nabla u(X)/g(X)$. Since $|A_X| \leq m_X$,

$$D_i(X) \geq c g(X) \ell_X^{-p-1} |\nu_X - e_X|^2. \quad (5.7)$$

To see this, retain only C_X in the nonnegative defect integral. For the unit vectors ν_X and e_X ,

$$m_X - \nu_X \cdot A_X = (m_X - |A_X|) + |A_X|(1 - \nu_X \cdot e_X) \geq \frac{1}{2} |A_X| |\nu_X - e_X|^2.$$

Combining $D_i/c_{3,s} \geq g(m_X - \nu_X \cdot A_X)$, $|A_X| \geq m_X/\sqrt{1 + \theta_0^2}$ and the lower bound for m_X proves (5.7).

At $g(X) = |\nabla u(X)| = 0$, one has $\nabla u(X) = 0$. Thus $g(X)g(Y) - \nabla u(X) \cdot \nabla u(Y) = 0$ for every Y , and the defining integral gives $D_i(X) = 0$. The positive core mass still defines e_X at these sources; $V_i(X)$ need not vanish.

At sources with $g(X) > 0$, decomposing $\nabla u = \sigma_i g(e_X + \nu_X - e_X)$, the row estimates and Young's inequality show, for any fixed $\varepsilon > 0$,

$$c_{3,s} \eta_i^2 \nabla u \cdot V_i \leq C E g \eta_i^2 + \varepsilon D_i \eta_i^2 + C_\varepsilon g \eta_i^2 \ell_X^{p+1} T^{-2p-2}.$$

For the error use Young's inequality on $\sqrt{g} \ell_X^{-(p+1)/2} |\nu_X - e_X|$ and $\sqrt{g} \ell_X^{(p+1)/2} T^{-p-1}$, absorbing the square of the first through (5.7). Explicitly, $Cab \leq \varepsilon c_D a^2 + C_\varepsilon b^2$ with $c_D > 0$ the fixed coefficient in that bound; all constants depend only on p , the fixed chart and tube data, and, for C_ε , on ε . The signed row bound controls the e_X term, and its absolute row bound controls the

difference $\nu_X - e_X$. At zero-gradient sources all source terms in the pointwise estimate vanish by the preceding argument, so it holds there as well. The vertical integrals satisfy

$$\int_{-2T}^{2T} g(a, F_i(a) + t) dt \leq C, \quad \int_{-2T}^{2T} g(a, F_i(a) + t)(1 + |t|)^{p+1} dt \leq 4C_w T.$$

For the first, (5.3) gives the explicit upper bound $C_w \int_{\mathbb{R}} (1 + |t|)^{-p-1} dt = 2C_w/p$. For the second the integrand is at most C_w on an interval of length $4T$. The second turns the last Young remainder into CT^{-1-2p} . Graph-offset coordinates have Jacobian one, and $|b| \leq 1$. Consequently

$$c_{3,s} \int_{\Omega_i} \eta_i^2 \nabla u \cdot V_i \leq \varepsilon \int_{\Omega_i} D_i \eta_i^2 + C_\varepsilon E \int_{D_r} \xi^2. \quad (5.8)$$

The direction was chosen independently at each source and is never differentiated.

To estimate P_i , put $w(t) = (1 + |t|)^{-p-1}$. Normal integration, uniformly over an arbitrary vertical shift d , gives

$$\iint_{\mathbb{R}^2} \frac{w(t)w(v)}{(\rho^2 + (d + t - v)^2)^{(3+p)/2}} dt dv \leq C \min\{\rho^{-2-p}, \rho^{-3-p}\}, \quad \rho > 0.$$

Thus the horizontal kernel majorant is

$$k_0(\rho) = C \min\{\rho^{-2-p}, \rho^{-3-p}\}, \quad \int_{\mathbb{R}^2} |z|^2 k_0(|z|) dz \leq C \left(\int_0^1 \rho^{1-p} d\rho + \int_1^\infty \rho^{-p} d\rho \right) < \infty.$$

Indeed $\int_{\mathbb{R}} w = 2/p$. For the first bound use $w(v) \leq 1$, translate $z = d + t - v$, and rescale $z = \rho\zeta$ to obtain

$$\int_{\mathbb{R}} (\rho^2 + z^2)^{-(3+p)/2} dz = \rho^{-2-p} \int_{\mathbb{R}} (1 + \zeta^2)^{-(3+p)/2} d\zeta.$$

Integrating the remaining weight gives the first estimate. For the second use $(\rho^2 + (d + t - v)^2)^{-(3+p)/2} \leq \rho^{-3-p}$ and both finite masses. Each argument is independent of d , so the majorant applies to arbitrary graph-height differences, including pairs on different graphs. For one graph, put $\rho = |a - b'|$ and $d = F_i(a) - F_i(b')$. The slope bound gives $|d| \leq \theta_0 \rho$, and $|t - v|^2 \leq 2|d + t - v|^2 + 2\theta_0^2 \rho^2$. Consequently

$$|a - b'|^2 + |F_i(a) - F_i(b') + t - v|^2 \geq \frac{1}{2}(|a - b'|^2 + |t - v|^2).$$

Graph-offset coordinates $(a, t) \mapsto (a, F_i(a) + t)$ have Jacobian one; this estimate also shows their uniform bi-Lipschitz control. The normal cutoff has cost

$$\iint w(t)w(v) \frac{|b(t/T) - b(v/T)|^2}{|t - v|^{1+p}} dt dv \leq CT^{-1-2p}. \quad (5.9)$$

First consider $|t| \leq T/2$. A nonzero difference forces $|v| > T$, and then $|t - v| \geq |v|/2$. Using $|b| \leq 1$ and the finite mass of $w(t)$ bounds this part, and its symmetric counterpart, by

$$C \int_T^\infty v^{-2p-2} dv = CT^{-1-2p}.$$

On the remaining region $|t|, |v| > T/2$, a nonzero difference requires $\min\{|t|, |v|\} < 2T$. Set $t = T\tau$, $v = Tv$ and use $w(T\tau) \leq T^{-p-1}|\tau|^{-p-1}$. This contribution is at most CT^{-1-3p} times

$$\iint_{\substack{|\tau|, |v| > 1/2 \\ \min\{|\tau|, |v|\} < 2}} \frac{|\tau|^{-p-1}|v|^{-p-1}|b(\tau) - b(v)|^2}{|\tau - v|^{1+p}} d\tau dv.$$

This integral is finite. The weights have no singularity at zero; near the diagonal the Lipschitz difference leaves the locally integrable power $|\tau - v|^{1-p}$ since $p < 2$. Far from the diagonal at least one variable is in the fixed annulus $1/2 < |\tau| < 2$ or $1/2 < |v| < 2$, and the other tail is $O(|v|^{-2p-2})$ or its symmetric counterpart. Finally $T^{-1-3p} \leq T^{-1-2p}$ for $T \geq 1$, proving (5.9).

Split the amplitude difference as

$$b(t/T)\xi(a) - b(v/T)\xi(b') = b(t/T)(\xi(a) - \xi(b')) + \xi(b')(b(t/T) - b(v/T)).$$

The translation estimate $\int |\xi(a + z) - \xi(a)|^2 da \leq |z|^2 \int |D\xi|^2$ follows from $\xi(a + z) - \xi(a) = \int_0^1 D\xi(a + \lambda z) \cdot z d\lambda$ by Cauchy-Schwarz and translation invariance of Lebesgue measure. Use

$(A + B)^2 \leq 2A^2 + 2B^2$ in the amplitude split. The shifted normal majorant and its second moment control the first square by $C \int |D\xi|^2$. Here ξ is extended by zero to $\mathbb{R}^2$, and enlarging the horizontal integrals to $\mathbb{R}^2$ is legitimate after replacing the kernel by k_0 . For the second square the same-graph distance bound gives, for $t \neq v$,

$$\int_{\mathbb{R}^2} (|z|^2 + |t - v|^2)^{-(3+p)/2} dz = \frac{2\pi}{p+1} |t - v|^{-1-p}.$$

Tonelli's theorem and (5.9) then give $CT^{-1-2p} \int \xi^2$; the set $t = v$ has measure zero. Thus $P_i \leq C \int |D\xi|^2 + CT^{-1-2p} \int \xi^2$. Use (5.8) with $\varepsilon = 1/2$ in (5.5). Since $D_i \geq 0$, dropping the remaining term $-\frac{1}{2} \int D_i \eta_i^2$ and using $T^{-1-2p} \leq E$ proves

$$Q_u(\psi_i) \leq C \int |D\xi|^2 + CE \int \xi^2. \quad (5.10)$$

The supports of ψ_1 and ψ_2 are compact and separated inside their disjoint enlarged tubes, so all mixed integrals below are absolutely convergent. The mixed potential vanishes by support disjointness. Expanding the bilinear kinetic form, the two off-diagonal products have equal integrals by symmetry. Thus

$$\mathcal{B}_3(\psi_1, \psi_2) = -c_{3,s} \iint_{\Omega_1 \times \Omega_2} \psi_1(X) \psi_2(Y) |X - Y|^{-3-p} dX dY,$$

and

$$Q_u(\psi_1 + \psi_2) = Q_u(\psi_1) + Q_u(\psi_2) - 2c_{3,s} I(\xi), \quad I(\xi) = \iint \xi(a) \xi(b') S(a, b') da db'.$$

Renaming horizontal variables in the absolutely convergent integral shows $\iint \xi(a) \xi(b') H(a, b') = \iint \xi(a) \xi(b') H(b', a)$. Their average is $I(\xi)$, for real ξ . Here $S(a, b') = \frac{1}{2}(H(a, b') + H(b', a))$ is the symmetric part of the nonnegative kernel

$$H(a, b') = \iint_{(-2T, 2T)^2} g(a, F_1(a) + t) b(t/T) g(b', F_2(b') + v) b(v/T) \\ \times (|a - b'|^2 + |F_1(a) + t - F_2(b') - v|^2)^{-(3+p)/2} dt dv,$$

extended by zero outside $D_{2r} \times D_{2r}$. The nonnegative measurable integrand makes H , and hence S , measurable by Tonelli's theorem. The same normal integration gives $S(a, b') \leq k_0(|a - b'|)$, after enlarging the constant in k_0 to absorb C_w^2 . We also need integrable rows, which follow from the actual tube separation. It gives $h(a) := F_2(a) - F_1(a) \geq 12T$. For $\rho = |a - b'|$ and $|t|, |v| \leq 2T$,

$$F_2(b') + v - F_1(a) - t \geq 8T - \theta_0 \rho.$$

If $\rho \leq 4T$, the vertical difference is at least $(8 - 4\theta_0)T$, while the horizontal term still contributes ρ^2 . If $\rho > 4T$, then $\rho^2 \geq (16/17)(\rho^2 + T^2)$. Hence the mixed squared distance is at least $c(\rho^2 + T^2)$ in all cases. The bounded vertical masses from (5.3) imply

$$H(a, b'), S(a, b') \leq C(T^2 + |a - b'|^2)^{-(3+p)/2}, \quad \sup_a \int S(a, b') db' \leq CT^{-p-1}.$$

The row bound holds also outside D_{2r} , where the kernel is zero. For real ξ extended by zero, the inequality $2|\xi(a)\xi(b')| \leq \xi(a)^2 + \xi(b')^2$ and symmetry give

$$\iint S(a, b') |\xi(a)\xi(b')| da db' \leq CT^{-p-1} \int \xi^2.$$

The same row estimate bounds the integrals with $\xi(a)^2$, $\xi(b')^2$, and $(\xi(a) - \xi(b'))^2$. Thus every subsequent polarization term is absolutely integrable, including for signed ξ . The lower bound for this row is

$$\int S(a, b') db' \geq ch(a)^{-p-1} - Cr^{-p-1}, \quad a \in D_r. \quad (5.11)$$

For $a \in D_r$ and $h(a) \leq r$, the disk $|b' - a| < h(a)/4$ is contained in D_{2r} . Restrict to that disk and restrict each normal variable to the actual phase offset interval $I_i(z') - F_i(z') \subset [-L_0, L_0]$. The

cutoff is one there and (5.6) gives mass at least $2a_0$ in each fiber. Disjoint enlarged tubes imply $h(a) \geq 12T \geq 12L_0$. For $|b' - a| < h(a)/4$ and the chosen offsets,

$$|F_2(b') + v - F_1(a) - t| \leq h(a) + \theta_0 h(a)/4 + 2L_0 \leq Ch(a).$$

Together with the horizontal bound, this makes every retained pair have distance at most $Ch(a)$. Thus the normal integral defining $H(a, b')$ is at least $ch(a)^{-3-p}$ on this horizontal disk of area $ch(a)^2$. Since $S \geq H/2$, symmetrization loses at most a factor two. If $h(a) > r$, choose the error coefficient at least as large as the positive coefficient, making the right side of (5.11) nonpositive.

Symmetry, the identity $2\xi(a)\xi(b') = \xi(a)^2 + \xi(b')^2 - (\xi(a) - \xi(b'))^2$, give the exact formula

$$I(\xi) = \int \xi(a)^2 \left(\int S(a, b') \, db' \right) da - \frac{1}{2} \iint S(a, b') (\xi(a) - \xi(b'))^2 da \, db'.$$

The finite-row estimate justifies separating these terms. The second-moment estimate and (5.11) therefore give

$$I(\xi) \geq c \int h^{-p-1} \xi^2 - C \int |D\xi|^2 - Cr^{-p-1} \int \xi^2.$$

This applies to real ξ without a positivity assumption on its pointwise products.

Theorem 5.3 (Actual-gap inequality). *Under (5.2)–(5.4), one has positive constants λ, C , fixed by p, a_0, L_0, K_0, C_w and the stipulated slope control, for which*

$$\lambda \int_{D_r} h^{-p-1} \xi^2 \leq C \int_{D_r} |D\xi|^2 + CE(R, T, r) \int_{D_r} \xi^2, \quad \xi \in C_c^1(D_r; \mathbb{R}). \quad (5.12)$$

No equation for h is assumed. Neither the size of the remaining layer family nor its global branch connections enters these constants.

Proof. The amplitudes ψ_i and their sum are compactly supported and Lipschitz, so stability applies for every real ξ under consideration. The mixed identity gives

$$2c_{3,s}I(\xi) \leq Q_u(\psi_1) + Q_u(\psi_2).$$

Insert the one-mode bounds (5.10) on the right and the preceding signed lower bound for I on the left. The result is

$$2c_{3,s}c \int_{D_r} h^{-p-1} \xi^2 \leq C \int_{D_r} |D\xi|^2 + C(E(R, T, r) + r^{-p-1}) \int_{D_r} \xi^2.$$

Since r^{-p-1} is already a term of E , this is (5.12) with $\lambda = 2c_{3,s}c > 0$, after enlarging C . All constants came from the stipulated two-tube data, the fixed cutoff, and p . No pointwise sign condition on ξ , equation for h , or bound on the number or connectivity of other branches was used. $\square$

6. ISOLATION AND A PROFILE FIXED BY THE ZERO SURFACE

The horizontal disks are two-dimensional, so a logarithmic cutoff has small Dirichlet energy. We first produce a common tube width on a larger buffer, before applying this cutoff to any selected gap. Throughout this section, $1 < p = 2s < 2$ is fixed and u belongs to the bounded stable solution class of the preceding sections.

Use the complete ordered phase charts of Section 4, with common graph domain $D_{\lambda R}$ and completeness cylinder $\mathcal{C}_{\text{comp}} = D_{\mu R} \times (-\nu R, \nu R)$, where $\lambda > \mu > 0$ and $\nu > 0$ are fixed. Denote the ordered zero graphs by $F_1 < \dots < F_N$, with N finite but allowed to vary. Retain a prescribed slope bound $\theta_0 \leq 10^{-2}$ and write $|D^2 F_{i,r}| \leq C_A/R$ for the fixed chart Hessian bound. Choose a fixed subphase interval $[-a_0, a_0]$ inside the available phase interval, and fixed $L_0, K_0 \geq 1$ bounding the inherited phase offsets and C_A , as in Section 5. For definite nested buffers put

$$\mathcal{C}_j = D_{2^{-j}\mu R} \times (-2^{-j}\nu R, 2^{-j}\nu R), \quad j = 1, 2, 3, 4.$$

Each smaller cylinder has distance at least $c_{\text{buf}}R$ from the boundary of its predecessor and from the boundary of the common graph domain, with a fixed $c_{\text{buf}} > 0$ after decreasing it. The argument applies to neighboring actual zeros over a base point a with at least one endpoint in the indicated retained cylinder. If their gap is $o(R)$, both endpoints and their midpoint retain

the next larger fixed buffer. A gap comparable to R already satisfies the bounds asserted below for large R .

Proposition 6.1 (Logarithmic separation). *On nested fixed interior subcylinders of the complete stable charts, for all sufficiently large selected scales R every neighboring actual zero gap satisfies*

$$h \geq c(R \log R)^{1/p}. \quad (6.1)$$

The fractions, constants, and threshold are independent of the selected branch and neighboring pair.

Proof. We first claim that there are $c_1 > 0$ and a uniform threshold such that every neighboring actual gap based at a zero in $\mathcal{C}_2$ satisfies $h \geq c_1 R^{1/p}$. Otherwise choose scales $R \rightarrow \infty$ and such gaps $h_0 = h(a)$ with $h_0/R^{1/p} \rightarrow 0$; then $h_0 = o(R)$. Both zeros are eventually inside $\mathcal{C}_1$, so completeness identifies them as consecutive members of the ordered list. By Theorem 4.4, all neighboring gaps whose zero centers retain this fixed buffer diverge uniformly as $R \rightarrow \infty$.

At the same base point, if an immediately exterior listed gap is less than half the selected gap, move to it. If the old gap has length d_k and the new one has length $d_{k+1} < d_k/2$, their midpoints differ by $(d_k + d_{k+1})/2 < 3d_k/4$. Thus the total midpoint travel is less than $3h_0/2$, and every endpoint visited is within $2h_0$ of the original midpoint. This remains inside $\mathcal{C}_1$ for large R . The finite ordered list makes the procedure terminate: each move strictly decreases the gap value, so no gap can be revisited. At the terminal gap $d \leq h_0$, each existing immediately exterior listed gap is at least $d/2$. An actual zero not in this list is outside $\mathcal{C}_{\text{comp}}$ and stays at distance at least $c_{\text{buf}}R$ from these retained points; it cannot trigger a further halving move. The terminal pair keeps a fixed interior margin, so Theorem 4.4 gives $d \rightarrow \infty$.

Center the disks at a and set $r_d = c_d \sqrt{Rd}$, with a fixed small $c_d > 0$. The selected gap is comparable to d on $D_{4r_d}(a)$, and its existing exterior gaps are at least a fixed multiple of d . Indeed, for a positive listed gap H with $H(a) = D \geq d/2$ and $D \leq R$, the interior gradient estimate (4.7) and Taylor's formula give

$$|H(x') - D| \leq Cc_d(\sqrt{dD} + D\sqrt{d/R}) + Cc_d^2d \leq C(c_d + c_d^2)D, \quad |x' - a| \leq 4r_d.$$

Here $d \leq R$ eventually and $d \leq 2D$. For $D > R$, the fixed slope bound instead gives $|H(x') - D| \leq Cc_d\sqrt{Rd} \leq Cc_dD$. Choose c_d so both bounds are at most $D/2$. This applies to the terminal gap $D = d$ and to each immediately exterior listed gap. By ordering, every other listed zero is also vertically separated from either selected graph by at least κd , for a fixed $\kappa > 0$. All required horizontal buffers remain available since $r_d/R = c_d\sqrt{d/R} \rightarrow 0$. The selected graph points on these disks move vertically by at most $4\theta_0 r_d$ from their values at a . Together with the descent travel $2h_0$ and a tube offset $O(d)$, this is $o(R)$, so the completeness buffer also remains.

For any of the listed graphs G and any two base points in its common convex domain, its slope bound gives

$$|z - G(x')| \leq |z - G(y')| + \theta_0|x' - y'| \leq \sqrt{1 + \theta_0^2}|(x', z) - (y', G(y'))|. \quad (6.2)$$

At $X = (x', F_i(x') + t)$, the own graph has vertical distance $|t|$. Every other listed graph has vertical distance at least $\kappa d - |t|$. Choose a fixed $\tau_d > 0$ with $6\tau_d \leq \kappa/2$. For $|t| \leq 6\tau_d d$, this last distance is at least $|t|$. Zeros inside $\mathcal{C}_{\text{comp}}$ belong to the listed graphs, so (6.2) applies to all of them, even when their base points are outside the small disk. Zeros outside $\mathcal{C}_{\text{comp}}$ are at distance at least $c_{\text{buf}}R$ from the tubes, which is larger than $|t|$ for large R . Consequently

$$\text{dist}((x', F_i(x') + t), Z) \geq c_*|t| \quad (x' \in D_{4r_d}(a), |t| \leq 6\tau_d d),$$

with a fixed $c_* > 0$. Decrease τ_d further so the enlarged tubes of the terminal pair are disjoint. Clearing (4.1) supplies their gradient weights and endpoint deficits with one fixed C_w . The signs of the endpoint values follow from the orientation and the absence of another zero along the intervening vertical segment. The regular phase families and their fixed offsets and derivative bounds are restrictions of the original families. Thus Theorem 5.3 applies with $T = \tau_d d$ and $r = r_d$: $T \geq T_0$ eventually, $T/r_d \rightarrow 0$, and $T/R \rightarrow 0$.

Take a fixed rescaled smooth cutoff supported in $D_{r_d}(a)$ and equal to one on $D_{r_d/2}(a)$. Its Dirichlet energy is bounded independently of r_d in dimension two, and its squared mass is at most Cr_d^2 . Since the selected gap is at most $2d$ on this disk, the left side of (5.12) is at least $cd^{-p-1}r_d^2 = cRd^{-p}$. The error is bounded by

$$E(R, \tau_d d, r_d) r_d^2 \leq C[Rd^{-2p} + (Rd)^{(1-p)/2} + r_d^{1-p}] \leq C[Rd^{-2p} + (Rd)^{-(p-1)/2}].$$

Relative to Rd^{-p} , the first term has coefficient Cd^{-p} , which tends to zero, so it is absorbed into the left side. The other error is bounded for large R . The gap inequality therefore gives $Rd^{-p} \leq C$, and hence $d \geq cR^{1/p}$. Since $d \leq h_0$, this contradicts the chosen sequence. The claim follows with constants and a threshold independent of the branch, pair, and number of listed graphs.

We next obtain a width common to every retained graph. At a fixed base point a , retain the indices whose zeros $(a, F_i(a))$ lie in $\mathcal{C}_3$. Choose $r_* = c_b R$ with fixed $c_b > 0$ small enough that graphs based at zeros in $\mathcal{C}_3$, over $D_{4r_*}(a)$, stay inside $\mathcal{C}_2$ with a fixed vertical margin. This is possible uniformly because their vertical variation is at most $4\theta_0 c_b R$, and the horizontal buffer loss is at most $4c_b R$. Set $T = \tau_* R^{1/p}$, with fixed $\tau_* > 0$ small compared to c_1 . For large R , the added offsets $6T = o(R)$ retain the same margin. At each base point on the larger disk, the first separation bound gives every neighboring actual gap from such a graph at least $c_1 R^{1/p}$. For an immediately exterior listed graph whose zero lies outside the completeness cylinder, the vertical gap is instead at least a fixed multiple of R , so it satisfies the same lower bound for large R . Ordering gives this lower separation from all other listed graphs. For $|t| \leq 6T$, choose $6\tau_* \leq c_1/2$, so every other listed graph is vertically at least $|t|$ away from $(x', F_i(x') + t)$. The single comparison (6.2) and the completeness buffer now give the same distance estimate $\text{dist}((x', F_i(x') + t), Z) \geq c_* |t|$ uniformly. Clearing again supplies one fixed set of tube weights and endpoint deficits. Decrease τ_* so the enlarged tubes of all retained graphs are disjoint. The inherited phase data are unchanged, and $T/r_* \rightarrow 0$, $T/R \rightarrow 0$ and $T \geq T_0$ eventually. Every selected pair of these graphs therefore satisfies all hypotheses of Theorem 5.3 on $D_{r_*}(a)$.

With $\alpha = (p-1)^2/(2p) > 0$, direct substitution gives

$$E(R, T, c_b R) \leq C(R^{-2-1/p} + R^{-(p+1)^2/(2p)} + R^{-p-1}) \leq CR^{-2-\alpha}. \quad (6.3)$$

Indeed $(p+1)^2/(2p) = 2 + \alpha$, $1/p > \alpha$ and $p+1 > 2 + \alpha$ for $1 < p < 2$.

Fix $\vartheta \in (1/p, 1)$ once in terms of p and take a neighboring actual gap $h_0 = h(a)$ based at a zero in $\mathcal{C}_4$. If $h_0 \geq R^\vartheta$, then $R^\vartheta/(R \log R)^{1/p} \rightarrow \infty$, which gives the claimed bound for large R . Otherwise $h_0 < R^\vartheta = o(R)$, so both endpoints lie in $\mathcal{C}_3$ and belong to consecutive listed graphs. The common-width construction just proved applies to this pair on $D_{r_*}(a)$.

The positive-gap gradient estimate and the Hessian bound give $h(x') \leq Ch_0$ on $D_\ell(a)$, where $\ell = c_\ell \sqrt{R h_0}$ and $c_\ell > 0$ is fixed and small. For example Taylor's formula bounds $|h - h_0|$ there by $Cc_\ell(h_0 + h_0 \sqrt{h_0/R}) + Cc_\ell^2 h_0$. Since $h_0 < R^\vartheta$, we have $\ell < r_*/2$ for large R . Writing $\rho = |x' - a|$, set

$$\xi(\rho) = \begin{cases} 1, & 0 \leq \rho \leq \ell, \\ \frac{\log((r_*/2)/\rho)}{\log((r_*/2)/\ell)}, & \ell < \rho < r_*/2, \\ 0, & \rho \geq r_*/2. \end{cases}$$

It is Lipschitz and compactly supported strictly inside $D_{r_*}(a)$. Polar integration gives

$$\int |D\xi|^2 = \frac{2\pi}{\log((r_*/2)/\ell)}, \quad \int \xi^2 \leq CR^2.$$

Mollify its zero extension to obtain smooth compactly supported functions in $D_{r_*}(a)$ converging in H^1 . Take the mollification radii below $r_*/4$, so all supports lie in the fixed compact disk $K = \overline{D_{3r_*/4}(a)} \Subset D_{r_*}(a)$. For this fixed R and pair, strict ordering and continuity of the two graph functions give $\min_K h > 0$. Thus h^{-p-1} is bounded on K , and the L^2 convergence gives convergence of $\int h^{-p-1} \xi^2$ and of the mass term; strong H^1 convergence gives convergence of the

Dirichlet term. Consequently (5.12) applies to this logarithmic cutoff. Its left side is at least $ch_0^{-p-1}\ell^2 = cRh_0^{-p}$. Using (6.3), we obtain

$$Rh_0^{-p} \leq \frac{C}{\log((r_*/2)/\ell)} + CR^{-\alpha}.$$

Moreover

$$\log \frac{r_*/2}{\ell} = \log \frac{c_b}{2c_\ell} + \frac{1}{2} \log \frac{R}{h_0} \geq \frac{1-\vartheta}{2} \log R + \log \frac{c_b}{2c_\ell} \geq \frac{1-\vartheta}{4} \log R$$

for a uniform sufficiently large threshold. Since $\alpha > 0$, $R^{-\alpha} \leq C/\log R$ beyond another fixed threshold. It follows that $Rh_0^{-p} \leq C/\log R$, or $h_0 \geq c(R \log R)^{1/p}$. All four nested cylinders and all constants were fixed before selecting any branch or pair. The number of descent steps may vary, but its geometric travel bound is uniform and introduces no further buffer reductions. This proves the proposition. $\square$

We next compare a solution with the layer on its own zero surface. Take an actual zero graph $t = f(x')$ on D_{8L} , with $f(0) = Df(0) = 0$, $|D^2f| \leq C/R$, and $L \leq c_0R$. Assume it contains every zero in B_{4L} and is increasingly oriented, meaning $\partial_3 u(x', f(x')) > 0$ along this actual regular zero graph. All zero centers in B_{2L} retain the fixed complete-chart buffers specified above. The implicit function theorem and the preceding regularity estimates make this graph smooth; the quantitative arguments below use only its Hessian bound. Choose $c_0 > 0$ sufficiently small in terms of the fixed chart data. Fix a smooth reference cutoff χ equal to one on D_4 and zero outside D_6 , with $0 \leq \chi \leq 1$. Set $\bar{f}(x') = \chi(x'/L)f(x')$ on D_{8L} and extend it by zero. Since the cutoff vanishes near the boundary of that disk, this is a smooth entire function. It produces an entire graph $\Gamma = \text{graph } \bar{f}$, with $0 \in \Gamma$, and

$$\|D\bar{f}\|_\infty \leq CL/R \ll 1, \quad \|D^2\bar{f}\|_\infty \leq C/R.$$

These follow from $|f| \leq C|x'|^2/R$, $|Df| \leq C|x'|/R$ and cutoff derivative sizes L^{-1}, L^{-2} . Let n denote the upward unit normal of the patched graph:

$$n(a, \bar{f}(a)) = \frac{(-D\bar{f}(a), 1)}{\sqrt{1 + |D\bar{f}(a)|^2}}.$$

Write $H_0 = \|D^2\bar{f}\|_\infty \leq C/R$. For graph points $A = (a, \bar{f}(a))$ and $B = (b', \bar{f}(b'))$, Taylor's formula gives

$$|(B - A) \cdot n(A)| + |(B - A) \cdot n(B)| \leq \|D^2\bar{f}\|_\infty |a - b'|^2.$$

Each of the two summands is at most $H_0|a - b'|^2/2$. If $A + tn(A) = B + vn(B)$ with $|t|, |v| < \rho$, scalar product with $A - B$ implies $|A - B|^2 \leq \rho H_0|a - b'|^2 \leq \rho H_0|A - B|^2$. Choose $\rho = cR$ with $\rho H_0 < 1/4$, using only the fixed Hessian and cutoff constants. Then $A = B$ and $t = v$, so the normal map is injective. The entire graph is closed and proper: its horizontal coordinates escape to infinity when its parameter does. For any source, a minimizing sequence of graph points is bounded, and a convergent subsequence gives a nearest graph point. At a minimizer the displacement is orthogonal to the tangent plane, so it is $tn(A)$, with $|t|$ equal to the distance. Points at distance less than ρ consequently have a unique nearest point. Conversely, the nearest-point representation of $A + tn(A)$, $|t| < \rho$, has distance at most $|t|$; injectivity shows that its nearest point is exactly A and its distance $|t|$.

Let $S_\Gamma = -Dn$ on the tangent plane, so $\|S_\Gamma\| \leq H_0$. This bound follows from the graph metric $G = I + D\bar{f} \otimes D\bar{f} \geq I$ and the second fundamental form $D^2\bar{f}/\sqrt{1 + |D\bar{f}|^2}$: the Euclidean norm of the shape operator is the norm of $G^{-1/2}D^2\bar{f}G^{-1/2}/\sqrt{1 + |D\bar{f}|^2}$. The tangential differential of the normal map is $I - tS_\Gamma$; its inverse has norm at most $(1 - |t|H_0)^{-1} \leq 4/3$, while its normal derivative is $n(A)$. The inverse-function theorem and global injectivity give smooth normal coordinates on the full tube. The coordinate t has the sign of being above or below the graph: for small positive t it is above, and the path $A + tn(A)$ cannot cross the graph for $0 < |t| < \rho$, by injectivity with the

zero normal coordinate. Thus this coordinate is precisely the signed distance δ_Γ . Differentiating gives

$$\nabla \delta_\Gamma(A + tn(A)) = n(A), \quad D^2 \delta_\Gamma|_{T_A \Gamma} = -S_\Gamma(I - tS_\Gamma)^{-1}, \quad D^2 \delta_\Gamma n(A) = 0.$$

Here the parallel tangent plane is identified with $T_A \Gamma$. In particular $|D^2 \delta_\Gamma| \leq (4/3)H_0 \leq C/R$.

Globally set $\delta_\Gamma(X) = \text{sign}(X_3 - \bar{f}(X')) \text{dist}(X, \Gamma)$, with value zero on the graph. This extends the preceding tube coordinate and is 1-Lipschitz. For two points on the same side this is the usual distance inequality. If the points lie on opposite sides, their segment meets the graph, and the sum of their distances to the graph is at most the segment length. No curvature derivative is used in these conclusions.

For $X \in B_L$, use $0 \in \Gamma$ to bound a nearest graph point A by $|A| \leq |X| + \text{dist}(X, \Gamma) \leq 2|X| < 2L$. It lies on the unchanged graph and is therefore an actual zero. Thus $\text{dist}(X, Z) \leq \text{dist}(X, \Gamma)$. Every actual zero in B_{4L} belongs to the unchanged graph, so none is closer than A , whereas every actual zero outside B_{4L} is more than $3L$ from X , greater than $\text{dist}(X, \Gamma) \leq |X| < L$. Consequently

$$\text{dist}(X, Z) = |\delta_\Gamma(X)|, \quad \text{sign } u(X) = \text{sign } \delta_\Gamma(X). \quad (6.4)$$

For the sign assertion, $|f(x')| \leq CL^2/R \leq L$ on D_L after decreasing c_0 . The vertical segment between $(x', f(x'))$ and $X = (x', z) \in B_L$ stays in $B_{\sqrt{2}L} \subset B_{4L}$ and has no other zero. The increasing orientation fixes its sign, proving the second equality as well.

Put $U = q_s(\delta_\Gamma)$. In coordinates tangent at the nearest point A to a source $X \in B_L$, that source is $(0, t)$, where $|t| \leq L$. Choose an orthogonal rotation sending $n(A)$ to the positive vertical direction; its angle ϑ from the old vertical is at most $\arctan \sigma$, where $\sigma = \|D\bar{f}\|_\infty$ is fixed small. The rotated horizontal projection of a graph parameter a is

$$\Psi(a) = B(a - a_A) + b_0(\bar{f}(a) - \bar{f}(a_A)), \quad s_{\min}(B) \geq \cos \vartheta, \quad |b_0| \leq \sin \vartheta,$$

where $A = (a_A, \bar{f}(a_A))$ and $\Psi(a_A) = 0$. Hence

$$|\Psi(a) - \Psi(\tilde{a})| \geq m|a - \tilde{a}|, \quad m = \cos \vartheta - \sigma \sin \vartheta \geq 1/2.$$

The same bound for its differential makes Ψ a local diffeomorphism. It is injective and proper, and its image is open by the inverse-function theorem. If $\Psi(a_j)$ converges, the lower Lipschitz bound makes a_j Cauchy, so continuity shows that its image is also closed. Connectedness of $\mathbb{R}^2$ makes it onto. Its smooth inverse therefore produces an entire graph $z = F(y')$, with $F(0) = DF(0) = 0$.

For completeness, write the rotated vertical coordinate as $\Phi(a) = c_v \cdot (a - a_A) + d_v(\bar{f}(a) - \bar{f}(a_A))$, where $|c_v| \leq \sin \vartheta$ and $|d_v| \leq 1$. Since $F \circ \Psi = \Phi$,

$$|DF| \leq m^{-1}(\sin \vartheta + \sigma) \leq C\sigma, \quad D^2 F[D\Psi v, D\Psi w] = (d_v - DF \cdot b_0)D^2 \bar{f}[v, w].$$

The bounded inverse of $D\Psi$ gives $|D^2 F| \leq C/R$. The new vertical component of the rotated upward normal at any point is its old scalar product with $n(A)$, which is at least $(1 - \sigma^2)/(1 + \sigma^2) > 0$. Each side of an entire graph is connected, as seen from the coordinates $(x', x_3 - \bar{f}(x'))$. The rotation therefore carries the old positive side to the side above F , first locally by the normal orientation and then globally by connectedness. Thus the same signed distance is positive above this rotated graph.

Choose a fixed small $b > 0$ and the box $\mathcal{B} = \{|y'| < bR, |z| < bR\}$ in these coordinates. Since $|F(y')| \leq C|y'|^2/R$, its points have distance at most $bR + Cb^2R$ from the graph. First choose b so that this is less than the reach ρ and its product with the rotated Hessian bound is less than $1/2$. These choices depend only on the fixed data, since all rotation bounds hold uniformly when the original slope is below a fixed small threshold. Then decrease c_0 to be below $b/2$ and the reach fraction. The source $(0, t)$, $|t| \leq L \leq c_0 R$, has distance at least $bR/2$ from $\mathcal{B}^c$. Within $\mathcal{B}$ the footpoint equations imply

$$|\delta_\Gamma(y', z) - z| \leq C|y'|^2/R. \quad (6.5)$$

Indeed, if ζ is the horizontal coordinate of the nearest point and $\delta = \delta_\Gamma(y', z)$, then

$$y' = \zeta - \frac{\delta DF(\zeta)}{\sqrt{1 + |DF(\zeta)|^2}}, \quad z = F(\zeta) + \frac{\delta}{\sqrt{1 + |DF(\zeta)|^2}}.$$

The first equation and $|DF(\zeta)| \leq C|\zeta|/R$ give $|\zeta| \leq 2|y'|$. The second then gives $|\delta - z| \leq C|\zeta|^2/R + C|\delta||\zeta|^2/R^2$, which proves (6.5).

The reference layer is smooth with bounded derivatives, by the preceding regularity results in dimension one. Hence $U = q_s(\delta_\Gamma)$ is globally bounded and Lipschitz, and is smooth within the reach tube. At a source in B_L , its symmetric second increment is $O(|H|^2)$ locally, while it is bounded for large H . The radial integrals $\int_0^1 \rho^{1-p} d\rho$ and $\int_1^\infty \rho^{-1-p} d\rho$ are finite. Thus its fractional Laplacian is well-defined by the symmetric principal value there. Let $P(y', z) = q_s(z)$. By the normalized cylindrical reduction, $(-\Delta)^s P - P + P^3 = 0$, and $U(X) = P(X) = q_s(t)$ at the source. The operator and the signed distance are unchanged by the chosen translation and rotation. Since $U(X) = P(X)$, their nonlinear terms agree at the source. The residual there is exactly $(-\Delta)^s(U - P)(X)$; after the following absolute-convergence bounds it can equally be written as $c_{3,s} \int_{\mathbb{R}^3} (P(Y) - U(Y))|X - Y|^{-3-p} dY$. For $r = |y'|$, monotonicity of q_s and $\int_{\mathbb{R}} q'_s = 2$ give

$$\int_{-bR}^{bR} |U(y', z) - P(y', z)| dz \leq \int_{\mathbb{R}} [q_s(z + Cr^2/R) - q_s(z - Cr^2/R)] dz \leq Cr^2/R.$$

More precisely, with $d_r = Cr^2/R$, nonnegative Fubini gives $\int_{\mathbb{R}} [q_s(z + d_r) - q_s(z - d_r)] dz = 2d_r \int_{\mathbb{R}} q'_s = 4d_r$. This does not subtract two divergent integrals. Take $R \geq R_0 = \max\{1, b^{-1}\}$. For $1 \leq r < bR$, bound the kernel by r^{-3-p} and use this one-dimensional integral. Polar integration in y' gives

$$\int_{\mathcal{B} \cap \{r \geq 1\}} \frac{|U - P|}{|(y', z) - (0, t)|^{3+p}} dy' dz \leq \frac{C}{R} \int_1^{bR} r^{-p} dr \leq C/R.$$

For $0 < r < 1$, boundedness of q'_s instead gives $|U - P| \leq Cr^2/R$ pointwise. Since

$$\int_{\mathbb{R}} (r^2 + (z - t)^2)^{-(3+p)/2} dz = Cr^{-2-p},$$

the corresponding contribution is at most $CR^{-1} \int_0^1 r^{1-p} dr \leq C/R$. These two integrals are finite precisely in the needed directions $p > 1$ and $p < 2$, respectively. On $\mathcal{B}^c$, the source separation and boundedness of U, P give a cost $CR^{-p} \leq C/R$. The difference integral is absolutely convergent, including at the source, and all constants are uniform in X . We have proved

$$|(-\Delta)^s U - U + U^3| \leq C/R \quad \text{on } B_L. \quad (6.6)$$

The same residual identity holds weakly on B_L . Indeed U is bounded and globally Lipschitz, so its mixed bilinear integral against a real compactly supported Lipschitz test is absolutely convergent by the preceding compact-test estimate. Symmetry for the kernel truncated at $|X - Y| = \varepsilon$ identifies that bilinear pairing with the truncated operator pairing. On the compact support of the test, the bounded local Hessian of U makes the omitted symmetric increment $O(\varepsilon^{2-p})$ uniformly. Passing to the limit gives the pointwise residual as the distributional forcing as well.

Writing $\mathcal{R} = (-\Delta)^s U - U + U^3$, the resulting identity is

$$\mathcal{B}_3(U, \psi) + \int_{B_L} (U^3 - U)\psi = \int_{B_L} \mathcal{R}\psi$$

for every such test ψ , with the same bound $|\mathcal{R}| \leq C/R$.

We also need the following precisely normalized external input.

Lemma 6.2 (Bounded flat Jacobi fields). *For $L_q = (-\Delta)^s + 3q_s(t)^2 - 1$ on $\mathbb{R}^3$, every bounded solution of $L_q v = 0$ is $c\phi(t)$.*

Proof. Recall that $\phi = q'_s$. We use Chan–Liu–Wei, [13, Lemma 4.4, pp. 20–21, equations (4.2) and (4.5)], and check its bounded-extension class and conormal normalization. The preceding weighted localization and higher regularity estimates, applied uniformly to translates of the trace

equation, give bounded first derivatives of v and a smooth representative. The same regularity applies to the bounded entire layer $q_s(t)$. For either trace h , its mass-one Poisson extension is

$$\mathcal{P}h(X, z) = \int_{\mathbb{R}^3} H_s(\xi) h(X - z\xi) \, d\xi, \quad H_s(\xi) = C_s(1 + |\xi|^2)^{-(3+p)/2}, \quad \int H_s = 1.$$

The kernel is positive, so these extensions are bounded and have the continuous traces h . Since $\int |\xi| H_s(\xi) \, d\xi < \infty$ for $p > 1$, differentiating this formula gives

$$|\nabla_X \mathcal{P}h| \leq \|\nabla h\|_\infty, \quad |\partial_z \mathcal{P}h| \leq \|\nabla h\|_\infty \int |\xi| H_s(\xi) \, d\xi.$$

Thus the extensions belong locally to $H^1(z^{1-p})$ up to the boundary, since $\int_0^B z^{1-p} \, dz < \infty$ for $p < 2$. Write $\mathcal{V} = \mathcal{P}v$ and $Q = \mathcal{P}(q_s)$. The normalized extension identity of [7, Remarks 3.8 and 3.11] gives

$$\operatorname{div}(z^{1-p} \nabla \mathcal{V}) = 0, \quad -\kappa_s \lim_{z \downarrow 0} z^{1-p} \mathcal{V}_z + (3q_s^2 - 1)v = 0, \quad \kappa_s = 2^{p-1} \frac{\Gamma(s)}{\Gamma(1-s)}.$$

The corresponding boundary equation for Q has nonlinear term $q_s^3 - q_s$. To match the unit-conormal convention of the cited lemma, put $a = \kappa_s^{1/p}$ and dilate all extension coordinates:

$$\tilde{Q}(X, Z) = Q(aX, aZ), \quad \tilde{\mathcal{V}}(X, Z) = \mathcal{V}(aX, aZ).$$

The weighted harmonic equation is preserved, while

$$-\lim_{Z \downarrow 0} Z^{1-p} \partial_Z \tilde{\mathcal{V}} = -a^p \lim_{z \downarrow 0} z^{1-p} \mathcal{V}_z.$$

Consequently $a^p = \kappa_s$ gives

$$\begin{aligned} -\lim_{Z \downarrow 0} Z^{1-p} \tilde{Q}_Z + \tilde{Q}(X, 0)^3 - \tilde{Q}(X, 0) &= 0, \\ -\lim_{Z \downarrow 0} Z^{1-p} \tilde{\mathcal{V}}_Z + (3\tilde{Q}(X, 0)^2 - 1)\tilde{\mathcal{V}}(X, 0) &= 0. \end{aligned}$$

Here $\tilde{Q}$ is the increasing one-dimensional layer in this unit-conormal convention: its derivative in X_3 is positive by positivity of the Poisson kernel and $\phi > 0$, and its trace tends to the two wells. These are the layer and Jacobi equations in the cited result, with its exponent $1 - 2s = 1 - p$ and $n = 3$. Lemma 4.4 applies to bounded extension solutions in arbitrary tangential dimension; its hypothesis imposes neither a global L^2 condition nor orthogonality. It gives $\tilde{\mathcal{V}} = c \partial_{X_3} \tilde{Q}$. Returning to the original coordinates absorbs the factor a into c , so $\mathcal{V} = cQ_t$. Taking traces proves the assertion. $\square$

Theorem 6.3 (Profile pinned at an actual zero graph). *Let $R_j, L_j \rightarrow \infty$ and $L_j \leq c_0 R_j$. Suppose the preceding single-graph isolation and fixed buffers hold for u_j, f_j , and that the uniform qualitative convergence of Theorem 4.4 holds at every buffered zero center. Then, for $0 < \alpha < p - 1$,*

$$\|u_j - q_s(\delta_{\Gamma_j})\|_{L^\infty(B_{L_j/2})} \leq C(R_j^{-1} + L_j^{-p}), \quad (6.7)$$

$$\|u_j - q_s(\delta_{\Gamma_j})\|_{C^{1,\alpha}(B_{L_j/4})} \leq C_\alpha(R_j^{-1} + L_j^{-p}). \quad (6.8)$$

Constants and thresholds are uniform in the selected zero center and may depend on the fixed chart data and common qualitative modulus. No C^2 estimate of this error is asserted.

Proof. Recall that c_0 was already chosen sufficiently small in the preceding distance-profile construction. Its geometry and residual estimate therefore apply. Let $W_j = u_j - U_j$, $U_j = q_s(\delta_{\Gamma_j})$. Lemma 2.1 gives $|u_j| \leq 1$, and the reference layer has $|q_s| \leq 1$. Hence $|U_j| \leq 1$ and $|W_j| \leq 2$ globally. In the comparison ball the exact secant equation is

$$[(-\Delta)^s + V_j]W_j = -\mathcal{R}_j, \quad V_j = u_j^2 + u_j U_j + U_j^2 - 1, \quad |\mathcal{R}_j| \leq C/R_j.$$

Assume the supremum estimate fails by an unbounded factor. Maximize $(L_j - |X|)^p |W_j(X)|$ over B_{L_j} and write its maximum as $M_j = m_j \ell_j^p$, with maximizer X_j , $\ell_j = L_j - |X_j|$, and $m_j = |W_j(X_j)|$. This maximum is attained at an interior point: the weighted function extends continuously as

zero to the boundary and is positive somewhere. Moreover $M_j \geq (L_j/2)^p \|W_j\|_{L^\infty(B_{L_j/2})}$, so the failure gives

$$M_j/(L_j^p/R_j + 1) \rightarrow \infty.$$

In particular $M_j \rightarrow \infty$, and $m_j \leq 2$ implies $\ell_j \rightarrow \infty$. Maximality bounds $|W_j|/m_j \leq 2^p$ on $B_{\ell_j/2}(X_j)$. Outside this ball,

$$\int_{\mathbb{R}^3 \setminus B_{\ell_j/2}(X_j)} \frac{|W_j(Y)|/m_j}{|Y - X_j|^{3+p}} dY \leq C/(m_j \ell_j^p) = C/M_j \rightarrow 0.$$

Between fixed radius A and $\ell_j/2$ the tail is at most CA^{-p} . The normalized forcing tends to zero, since

$$\frac{1}{R_j m_j} = \frac{\ell_j^p}{R_j M_j} \leq \frac{L_j^p}{R_j M_j} \rightarrow 0.$$

Define $v_j(X) = W_j(X_j + X)/m_j$. It satisfies

$$[(-\Delta)^s + V_j(X_j + X)]v_j(X) = -\mathcal{R}_j(X_j + X)/m_j \quad \text{in } B_{\ell_j}.$$

For every fixed translated center P , the ball $B_4(P)$ lies in $B_{\ell_j/2}$ for large j . Its local supremum and the weighted norm $M_P(v_j)$ in (3.2) are uniformly bounded. Indeed, the part $|Y| < \ell_j/2$ is controlled by 2^p and the integrable weight; for $|Y| \geq \ell_j/2 > 2|P| + 1$, the weight is at most $C|Y|^{-3-p}$ and the remaining integral is at most C/M_j . The same comparison shows, uniformly for P in a fixed compact set and $A > 2|P| + 1$, that

$$\limsup_j \int_{|Y-P|>A} \frac{|v_j(Y)|}{|Y-P|^{3+p}} dY \leq CA^{-p}.$$

The constants are uniform on compact sets of centers. Also $|V_j| \leq 4$. The same localization estimate gives uniform local $C^{1,\alpha}$ bounds, for $0 < \alpha < p-1$. A diagonal subsequence converges locally uniformly, with first derivatives on smaller compact sets if needed, to a function v with $|v(0)| = 1$ and $\|v\|_\infty \leq 2^p$.

The equation passes in distributions without a normalized C^2 bound. For a compact smooth test ψ , $|(-\Delta)^s \psi(Y)| \leq C_\psi(1 + |Y|)^{-3-p}$ outside a fixed ball. The preceding tails imply $\limsup_j \int_{|Y|>A} |v_j(Y)| |Y|^{-3-p} dY \leq CA^{-p}$. Local convergence followed by $A \rightarrow \infty$ passes the operator. Whenever the translated potentials converge locally uniformly, their products with v_j pass to the limit as well, and the normalized residual tends to zero. The limiting equations below have smooth bounded coefficients, so their bounded weak solutions are smooth by the preceding interior regularity estimates.

If X_j escapes every bounded distance from the graph, (6.4) holds on each fixed translated ball for large j , because $\ell_j \rightarrow \infty$. The 1-Lipschitz distance tends uniformly to infinity there. Clearing and sign agreement imply that u_j and U_j approach the same well on each such ball, so $V_j \rightarrow 2$. The limit satisfies $[(-\Delta)^s + 2]v = 0$ on $\mathbb{R}^3$. Applying the bounded positive-potential comparison to v and $-v$ forces $v = 0$, a contradiction.

Otherwise, after a subsequence, choose a nearest graph point A_j with $|X_j - A_j|$ bounded. For large j , $|A_j| \leq |X_j| + O(1) = L_j - \ell_j + O(1) < L_j$. In particular A_j is an actual zero with the required fixed buffers, and every fixed ball centered at A_j lies in the comparison ball for large j . Recenter at A_j and rotate its increasing normal vertically. Use the same divisor m_j for the normalized differences. Bounded recentering and orthogonal rotations preserve the local estimates and tail bounds just proved. In these coordinates the maximizing point is $t_j e_3$, with $|t_j|$ bounded; pass to $t_j \rightarrow t_*$. A further subsequence of the normalized differences has a bounded limit, still denoted v , for which $|v(t_* e_3)| = 1$.

On every fixed horizontal disk, write the recentered zero graph as $t = g_j(x')$, where $g_j(0) = Dg_j(0) = 0$ and $|D^2 g_j| \leq C/R_j$. Thus $g_j \rightarrow 0$ locally in C^1 . The tangent-coordinate distance estimate (6.5), whose fixed box contains every fixed compact set for large j , gives $\delta_{\Gamma_j}(x', t) \rightarrow t$ uniformly there. Hence $U_j \rightarrow q_s(t)$ locally, and the common qualitative hypothesis at A_j gives the same limit for u_j . Hence the limiting equation is $[(-\Delta)^s + 3q_s(t)^2 - 1]v = 0$. On this actual

zero graph, both u_j and U_j vanish, so the normalized difference is exactly zero at $(x', g_j(x'))$. Its uniform local C^1 bound and $g_j \rightarrow 0$ imply $v(x', 0) = 0$ for every x' . Lemma 6.2 gives $v = c\phi$; evaluating at $t = 0$, where $\phi(0) > 0$, gives $c = 0$. This contradicts the transported maximizing value. The supremum estimate follows, uniformly over the class with the stipulated common qualitative modulus.

For the first-derivative estimate, put $\varepsilon_j = R_j^{-1} + L_j^{-p}$ and take any $P \in B_{L_j/4}$. For large j , $B_4(P) \subset B_{L_j/2}$, where the local supremum of W_j is at most $C\varepsilon_j$ and its forcing is at most C/R_j . The contribution to $M_P(W_j)$ from $B_{L_j/2}$ is at most $C\varepsilon_j$ because its weight has finite integral. On the complement, $|W_j| \leq 2$ and $|Y - P| \geq L_j/4$, so the contribution is at most CL_j^{-p} . Equation (3.2) now gives $\|W_j\|_{C^{1,\alpha}(B_1(P))} \leq C_\alpha \varepsilon_j$, uniformly in P . For two points of $B_{L_j/4}$ separated by less than one, use such a unit-ball estimate centered at the first point. For separation at least one, the two gradient supremum bounds control the Hölder quotient. This proves the full $C^{1,\alpha}(B_{L_j/4})$ estimate. $\square$

For a zero P in a fixed inner subcylinder, let d^{actual} be a common lower bound for the neighboring vertical gaps throughout a larger retained complete-chart region, chosen to satisfy (6.1). Set $d^{\text{actual}} = \infty$ when the region has only the selected zero graph, and put $L = c \min\{d^{\text{actual}}, R\}$. Translate P to the origin and rotate its increasing normal vertically, first reversing the old vertical direction if necessary.

Here is a quantitative regraphing of the complete phase families. Choose a fixed $h > 0$ so that, after this translation, the old disk D_{4hR} stays in the common graph domain and B_{4hR} stays in the larger retained complete-chart region. These inclusions follow from the fixed margins at P . Write $\rho = hR$ and let $\theta \leq 10^{-2}$ bound all old phase slopes. The angle β of the rotation is at most $\arctan \theta$. In block form the rotation sends an old graph point $(\zeta, F(\zeta))$ to

$$y = \Phi_F(\zeta) = A\zeta + bF(\zeta), \quad z = e \cdot \zeta + dF(\zeta),$$

where $s_{\min}(A) \geq \cos \beta$, $|b| = |e| = \sin \beta$ and $d = \cos \beta$. Put $\kappa = \cos \beta - \theta \sin \beta > 0$. On the convex disk $D_{2\rho}$, every old phase graph satisfies

$$|\Phi_F(\zeta) - \Phi_F(\zeta')| \geq \kappa |\zeta - \zeta'|, \quad \|(D\Phi_F)^{-1}\| \leq \kappa^{-1}.$$

Thus its horizontal projection is globally injective and has a smooth local inverse. Moreover, if $D_\rho(\zeta_*) \subset D_{2\rho}$, its image contains $D_{\kappa\rho}(\Phi_F(\zeta_*))$. Indeed a radial segment of length less than $\kappa\rho$ from $\Phi_F(\zeta_*)$ lifts by the local inverse. A lift cannot reach $\partial D_\rho(\zeta_*)$, since the displayed lower bound places its image at distance at least $\kappa\rho$ from the initial image point. If continuation stopped earlier, compactness would give an interior limiting point, where the inverse-function theorem extends the lift. Injectivity makes this lift unique. This proves the claimed image inclusion.

Choose a fixed $\delta > 0$ much smaller than κh , and retain exactly the finitely many old families having some phase point in the new cylinder $D_{\delta R} \times (-\delta R, \delta R)$. For each retained index, the old base coordinate ζ_* of such a point has $|\zeta_*| \leq 2\delta R$, while its old height has magnitude at most $2\delta R$. At the same ζ_* every other phase in that family has height within $2C_0$ of this height. For large R , uniformly in the index and phase, $D_\rho(\zeta_*) \subset D_{2\rho}$ and $|\Phi_{F_{i,\tau}}(\zeta_*)| < \kappa\rho/4$. The preceding image inclusion therefore supplies all their regraphed functions $G_{i,\tau}$ on the common disk $D_{\kappa\rho/2}$. In particular it supplies the selected family through the origin. The inverse-function theorem with the phase parameter gives joint smoothness and retains the level identity $u(y, G_{i,\tau}(y)) = \tau$.

For completeness, the transformed differential estimates follow from $G(\Phi_F(\zeta)) = e \cdot \zeta + dF(\zeta)$:

$$DG = (e^T + dDF)(A + bDF)^{-1}, \quad |DG| \leq \frac{\sin \beta + \theta \cos \beta}{\kappa}.$$

Twice differentiating the same identity gives, for any new horizontal vectors v, w ,

$$D^2G[v, w] = (d - DG \cdot b)D^2F[(D\Phi_F)^{-1}v, (D\Phi_F)^{-1}w].$$

Hence $|D^2G| \leq C/R$, uniformly for all retained phases and indices. Their slopes stay uniformly small. Also

$$\partial_{\text{new}} 3u = \partial_{\text{old}} 3u (d - e \cdot DF), \quad d - e \cdot DF \geq \kappa,$$

so every orientation sign is preserved and the vertical derivative has magnitude at least $\kappa\gamma_0$. Differentiating the new level identity in phase bounds its phase offsets by $2a/(\kappa\gamma_0)$, a fixed constant. Its heights on the common disk are bounded by a fixed multiple of R : start from the above intersection point, use the fixed old phase offset, and integrate the uniform new slope bound. No two members of the same phase can intersect, since an intersection would also be an intersection of the old graphs. Their old ordering is preserved: a reversed new vertical separation $h_* > 0$ would have old vertical component $h_* \cos \beta > \theta h_* \sin \beta$, inconsistent with the old graph order and slope bound. Thus restriction keeps the ordered labels and compatible phase continuations.

The new cylinder is contained in $B_{2\delta R}$, hence in the old retained complete region. Every point of it with $|u| \leq a$ belongs to an old family; that family was retained by its very intersection with the new cylinder, and the inverse projection puts the point on its new graph. This proves completeness there. Restrict to a still smaller fixed fraction of this cylinder before selecting allowed centers. Both its horizontal and vertical boundaries then stay at fixed positive multiples of R from the origin, as does the boundary of the common graph disk of radius $\kappa h R/2$. Thus, if c is sufficiently small and $L \leq cR$, every zero center in B_{2L} retains positive margins proportional to R to the completeness and graph-domain boundaries, while D_{8L} and B_{4L} lie in the respective retained regions. The same argument centered at any such zero gives its buffered charts, with the same uniform projection bound and fixed interior fractions. These are precisely the translated and rotated complete buffers, with constants depending only on the original data. No separation assumption has been used to construct them.

The gap bound must be compared across this rotation. Let β be its small angle and θ an upper bound for the old slopes. Two graph points on the same new vertical line, with new vertical separation h_{new} , have old vertical separation $h_{\text{new}} \cos \beta$ and old horizontal separation at most $h_{\text{new}} \sin \beta$. Since $\cos \beta - \theta \sin \beta > 0$, the graph ordering is preserved. Comparing the old gap at a common old horizontal point and using the old slope bound gives

$$d^{\text{actual}} \leq (\cos \beta + \theta \sin \beta) h_{\text{new}}.$$

The relevant old points remain in the larger retained region by the fixed buffers. Thus the new vertical gaps are at least $c_{\text{rot}} d^{\text{actual}}$, with a uniform $c_{\text{rot}} > 0$. This comparison applies also to nonadjacent graphs, by ordering.

Choose c to satisfy the buffer inequalities just proved. The selected new graph f has $f(0) = Df(0) = 0$, since the origin is its actual zero and the new vertical is its increasing normal. The transformed Hessian estimate gives $|D^2 f| \leq C/R$ on D_{8L} , and the orientation calculation gives $\partial_3 u > 0$ there. Decrease c also so that $c \leq c_0$. If a different graph met B_{4L} at (x', z) , then $|f(x')| \leq C|x'|^2/R$ and $L \leq cR$ would give

$$c_{\text{rot}} d^{\text{actual}} \leq |z - f(x')| \leq (4 + 16Cc)L \leq c(4 + 16Cc) d^{\text{actual}},$$

which is impossible when $c(4 + 16Cc) < c_{\text{rot}}$. If $d^{\text{actual}} = \infty$, completeness directly says there is no other zero graph in this ball; no inequality with an infinite gap is needed. Unlisted zeros are outside the buffered complete region and hence cannot enter B_{4L} . These observations verify every isolation and buffer hypothesis of Theorem 6.3.

Let Σ denote the actual zero surface near P , with signed distance positive on the increasing side. On B_L , its distance agrees with that of the patched graph by (6.4). The pinned theorem and (6.1) now yield on $B_{L/4}$

$$\|u - q_s(\delta_\Sigma)\|_{C^{1,\alpha}} \leq C_\alpha(R^{-1} + L^{-p}) \leq C_\alpha/R. \quad (6.9)$$

When $d^{\text{actual}} \leq R$, its negative p th power is at most $C/(R \log R)$; otherwise $L^{-p} = O(R^{-p}) = o(R^{-1})$. In particular, along each fixed normal interval $|t| \leq N$,

$$|\nabla u(P + tn(P)) - \phi(t)n(P)| \leq C_N/R. \quad (6.10)$$

The comparison region has radius proportional to the capped isolation scale. Indeed $L \rightarrow \infty$, so each such interval lies in $B_{L/4}$ for large R and in the normal-coordinate tube. There $\delta_\Sigma(P + tn(P)) = t$ and $\nabla \delta_\Sigma(P + tn(P)) = n(P)$; differentiating $q_s(\delta_\Sigma)$ proves (6.10).

7. PROJECTION OF THE ACTUAL EQUATION AND POWER SEPARATION

The estimates established so far apply to complete charts with curvature C/R , nondegenerate wide phase band, and logarithmic separation. Fix a small chart-buffer constant c_{cap} and use the effective isolation length

$$d = \min\{d^{\text{actual}}, c_{\text{cap}}R\}, \quad d \geq c(R \log R)^{1/p}.$$

The same estimates apply to a fixed smaller fraction of this length, with only the constants changed. Here d^{actual} is a common gap lower bound throughout a larger retained region, as in Section 6; it is infinite if only the selected zero graph is present there. Orient each selected zero branch increasingly, replacing u by $-u$ if necessary. Keep a further fixed horizontal and completeness buffer around every patch used below, with every comparison ball and normal fiber inside those buffers. The quantitative isolated-profile result gives a fixed $\beta > 0$, uniform in every retained zero center P , such that

$$\|u - q_s(\delta_\Sigma)\|_{C^{1,\alpha}(B_{\beta d}(P))} \leq C_\alpha/R \quad (0 < \alpha < p-1).$$

In each such ball the actual increasing signed distance agrees with the patched distance from Section 6. That patched graph has reach at least $c_{\text{reach}}R$ and the quadratic tangent-plane distance estimate (6.5). Thus all constants below depend only on the fixed chart data and common qualitative modulus, p , and the retained fractions; the scale is sufficiently large uniformly for this fixed class. No constant depends on the number of layers.

For a surface Σ let $A = -D_\Sigma \nu$ and $H = \text{tr } A$. At the tangent origin of an upward graph, $A = D^2 f$ and $H = \Delta f$; H is the sum of the principal curvatures. With a parameterization $X(y)$, metric g_Σ , and injective normal coordinates $X_\Sigma(y, t) = X(y) + t\nu(y)$, put

$$(g_\Sigma)_t(v, w) = g_\Sigma((I - tA)v, (I - tA)w), \quad J = \det(I - tA), \quad H_t = \text{tr}(A(I - tA)^{-1}).$$

Then $\partial_t \log J = -H_t$ and $\Delta_x = \partial_{tt} - H_t \partial_t + \Delta_{(g_\Sigma)_t}$. Indeed $D_y X_\Sigma = (I - tA)D_y X$, so the normal direction is orthogonal to the tangential directions and the volume element is $J \, d\mu_{g_\Sigma} \, dt$. We always take $|t| \|A\| < 1/2$, so $J > 0$ and the displayed metric and its inverse are uniformly comparable to the metric at $t = 0$. The fractional operator is not obtained by taking fractional powers of these terms individually. In the integral formulation its tube part is

$$c_{3,s} \text{PV} \int \frac{v(y, t) - v(\bar{y}, \tau)}{|X_\Sigma(y, t) - X_\Sigma(\bar{y}, \tau)|^{3+p}} J(\bar{y}, \tau) \, d\mu_{g_\Sigma}(\bar{y}) \, d\tau,$$

with the exterior contribution retained and physical Euclidean balls used for the principal value.

We use the Poisson realization of the extension formulation of [9, Sections 2.4 and 3.1–3.2]. In any dimension n , its mass-one kernel and the conormal constant in our normalization are

$$P_z^{(n)}(h) = \frac{\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(p/2)} \frac{z^p}{(|h|^2 + z^2)^{(n+p)/2}}, \quad \kappa_s = 2^{p-1} \frac{\Gamma(s)}{\Gamma(1-s)}.$$

Let $\mathcal{U}(x, z) = P_z^{(3)} * u(x)$ be the extension of the actual solution, and let $Q(t, z) = P_z^{(1)} * q_s(t)$. These integrals exist for bounded traces. Differentiating the kernel for $z > 0$ gives $\text{div}(z^{1-p} \nabla \mathcal{U}) = 0$ and $D_p Q = 0$, where $D_p = \partial_{tt} + \partial_{zz} + (1-p)z^{-1} \partial_z$. Integrating the three-dimensional kernel in the two tangential variables gives exactly $P_z^{(1)}$, so Q is also the extension of every frozen tangent planar solution.

For completeness, the conormal identity holds here for bounded smooth traces, without a global finite-energy hypothesis. For such a trace h , use the zero mass of $\partial_z P_z^{(n)}$ and its evenness to write the differentiated convolution with symmetric second increments. After multiplication by z^{1-p} the derivative kernel is bounded by $C|Y|^{-n-p}$, and the increments are bounded by $C \min\{|Y|^2, 1\}$, locally uniformly in the source. This is integrable because $0 < p < 2$. Dominated convergence gives

$$-\lim_{z \downarrow 0} z^{1-p} (P_z^{(n)} * h)_z = \frac{p\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(p/2)c_{n,s}} (-\Delta)^s h = \kappa_s^{-1} (-\Delta)^s h,$$

locally uniformly. Here the operator constant $c_{n,s}$ is the Fourier-normalized constant from Section 2. The equality of constants follows from $p = 2s$ and $\Gamma(1-s) = s|\Gamma(-s)|$. The earlier uniform smoothness of u and of the planar layer supplies the bounded derivatives used here. Moreover, in the scaled Poisson formula the first moment is finite for $p > 1$; differentiating the trace gives bounded spatial derivatives and $|\partial_z(P_z^{(n)} * h)| \leq C\|\nabla h\|_\infty$. Consequently these extensions are locally in the weighted H^1 space up to $z = 0$, since z^{1-p} is integrable there.

Write $V(y, t, z) = \mathcal{U}(X_\Sigma(y, t), z)$ and $W = V - Q$, with trace $w = u(X_\Sigma(y, t)) - q_s(t)$. Then

$$\begin{aligned} \mathcal{L}_\Sigma W &= H_t Q_t, \\ -\kappa_s \lim_{z \downarrow 0} z^{1-p} W_z &= (1 - 3q_s^2)w - 3q_s w^2 - w^3, \end{aligned} \quad (7.1)$$

where $\mathcal{L}_\Sigma = D_p - H_t \partial_t + \Delta_{(g_\Sigma)_t}$. Indeed $\mathcal{L}_\Sigma V = 0$ and $\mathcal{L}_\Sigma Q = -H_t Q_t$, and the trace equation is the exact subtraction of $u - u^3$ and $q_s - q_s^3$. For a smooth test with compact (y, t) support and vanishing near the upper extension face,

$$\begin{aligned} &\kappa_s \int z^{1-p} J \left(\langle \nabla_{(g_\Sigma)_t} W, \nabla_{(g_\Sigma)_t} \psi \rangle + W_t \psi_t + W_z \psi_z \right) d\mu_{g_\Sigma} dt dz \\ &\quad + \int J(3q_s^2 - 1)w\psi(y, t, 0) d\mu_{g_\Sigma} dt \\ &= -\kappa_s \int z^{1-p} J H_t Q_t \psi d\mu_{g_\Sigma} dt dz - \int J(3q_s w^2 + w^3) \psi(y, t, 0) d\mu_{g_\Sigma} dt. \end{aligned} \quad (7.2)$$

Integrate first over $z > \epsilon$. The bottom outward normal is $-\partial_z$, so the conormal limit has exactly the displayed sign. The locally uniform conormal limit and bounded first derivatives, with the integrable weight, justify passage to zero. This also permits $\psi = \eta(y)Q_t\chi_T$: apply the same calculation for $z > \epsilon$, where the test is smooth, and take the limit using the bounded first derivatives of Q_t and its smooth trace ϕ . These bounds follow from the same scaled Poisson formula applied to ϕ , whose derivative is bounded. All remote layers remain encoded in the actual extension; none is discarded by this local change of coordinates.

We first estimate this difference on actual normal fibers. Let $\varepsilon \leq C/R$ be its fixed common upper bound, and choose $T_0 \geq 1$ and $T_0 \leq T \leq c_T d$, with fixed sufficient buffers. The number T is constant on the lateral patch. Choose c_T so that $|t| \leq 2T$ lies well inside each pinned comparison ball of radius $c'd$ and inside the normal-coordinate reach. For $|t| \leq 2T$, $0 < z \leq 2T$,

$$|W| \leq C[\varepsilon + z/R + z^p d^{-p}], \quad (7.3)$$

$$|\nabla_{(g_\Sigma)_t} W| \leq \begin{cases} C/R, & 0 < z < 1, \\ C[\varepsilon/z + R^{-1} + z^{p-1} d^{-p}], & 1 \leq z \leq 2T. \end{cases} \quad (7.4)$$

To prove this, fix an actual zero P and freeze its tangent plane. In its tangent coordinates the comparison solution is $q_s(\tau)$; its extension is $Q(\tau, z)$. At the source $P + t\nu(P)$ this planar profile equals the signed-distance profile. Inside a box of size $c'd$ contained in the pinned comparison ball, split $u - q_s(\tau)$ into $u - q_s(\delta_\Sigma)$ and $q_s(\delta_\Sigma) - q_s(\tau)$. The first has supremum at most ε . The footpoint calculation (6.5), applied to the patched graph whose distance agrees there with the actual distance, gives

$$\int |q_s(\delta_\Sigma(y', \tau)) - q_s(\tau)| d\tau \leq C|y'|^2/R.$$

Here the integral is over the vertical interval of the comparison box. If $a = C|y'|^2/R$, monotonicity bounds its integrand by $q_s(\tau + a) - q_s(\tau - a)$; the integral of this upper bound over the whole line is $2a \int \phi = 4a$, by nonnegative Fubini. This proves the stated vertical estimate without requiring the distance estimate outside the box. The value and derivative kernels of the three-dimensional Poisson extension are bounded respectively by

$$Cz^p(|Y|^2 + z^2)^{-(3+p)/2}, \quad Cz^{p-1}(|Y|^2 + z^2)^{-(3+p)/2}.$$

The derivative bound holds for every first spatial or extension derivative, by differentiating the kernel and using $|Y|/(|Y|^2 + z^2) \leq C/z$. For the geometric term, bound the kernel independently

of the normal variable and use its preceding vertical L^1 estimate. Since

$$\int_0^\infty \rho^3 (\rho^2 + z^2)^{-(3+p)/2} d\rho = C_p z^{1-p},$$

the value cost is $CR^{-1}z^p z^{1-p} = Cz/R$ and the derivative cost is C/R . The integral is finite at infinity because $p > 1$. The pinned error costs $C\varepsilon$ and $C\varepsilon/z$. Boundedness outside the comparison box costs $Cz^p d^{-p}$ and $Cz^{p-1}d^{-p}$; this uses only $\int_{cd}^\infty \rho^{-1-p} d\rho = Cd^{-p}$ and includes all other layers. The source is separated from the complement of the box by a fixed multiple of d , by the choice of c_T .

For $z < 1$, localize the difference of the actual solution and the fixed tangent planar solution near the normal source. Denote this trace difference by b_P . Write $X = P + t\nu(P)$. For large R , $B_4(X)$ lies inside the comparison box, with a fixed margin, because $d \rightarrow \infty$ and $2T \ll \beta d$. Its supremum on $B_4(X)$ is $O(\varepsilon + R^{-1}) = O(R^{-1})$: on such a ball the tangential displacement is bounded, so the distance error is $O(R^{-1})$. Its weighted tail has the same bound: the geometric part is at most

$$\frac{C}{R} \int_0^{cd} \rho^3 (1 + \rho)^{-3-p} d\rho \leq C/R,$$

the pinned part is $C\varepsilon$, and the exterior part is $Cd^{-p} \leq C/R$. More explicitly, the weighted norm is centered at X , so $1 + |Y - X| \geq 1 + |y'|$ for the geometric part; the source has distance at least cd from the box complement, which gives its Cd^{-p} contribution. The logarithmic lower bound for d implies $d^{-p} \leq C/(R \log R) \leq C/R$. This is a difference of two actual solutions, so its secant equation has zero forcing. Its coefficient $u^2 + uq_s + q_s^2 - 1$ has uniform C^2 bounds by the preceding entire-solution regularity, with this fixed planar solution in place of q_s . Equation (3.3) with $j = 2$ therefore gives $\|b_P\|_{C^2(B_1(X))} \leq C/R$; the hypotheses are zero forcing on $B_4(X)$, the preceding weighted norm $O(R^{-1})$, and the uniform coefficient C^2 bound. The Poisson formula now gives a spatial gradient bound C/R for its extension: localize b_P by a fixed cutoff, integrate the spatial derivative onto that localized trace, and bound the separated remainder by its weighted tail. For the extension derivative use the even kernel and its zero differentiated mass to write symmetric second increments. Its local contribution is bounded by

$$\frac{C}{R} z^{p-1} \int_0^1 \frac{r^4}{(r^2 + z^2)^{(3+p)/2}} dr \leq C/R \quad (0 < z < 1).$$

Splitting this integral at $r = z$ proves the bound because $p < 2$; the exterior contribution is at most $Cz^{p-1}/R \leq C/R$. At the frozen base point $D_y X_\Sigma = (I - tA)D_y X$ is orthogonal to $\nu(P)$; the tangential derivative of the fixed planar extension vanishes against it. This proves the Fermi tangential estimate without any derivative of A . It uses a C^2 estimate for a difference of two exact solutions, rather than a C^2 estimate for the pinned signed-distance error.

The layer extension also satisfies

$$|Q_t(t, z)| \leq \frac{C}{1 + \sqrt{t^2 + z^2}}, \quad |\nabla_{t,z} Q_t(t, z)| \leq \frac{C}{(1 + \sqrt{t^2 + z^2})^2}, \quad \int_{\mathbb{R}} Q_t(t, z) dt = 2. \quad (7.5)$$

Indeed $Q_t = P_z^{(1)} * \phi$, where $P_z^{(1)}(h) = c_p z^p (z^2 + h^2)^{-(1+p)/2}$. It is positive, and nonnegative Fubini gives $\int_{\mathbb{R}} Q_t(t, z) dt = \int_{\mathbb{R}} \phi = 2$. For either first derivative of this kernel,

$$\int_{\mathbb{R}} \partial P_z^{(1)}(h) dh = 0, \quad \int_{\mathbb{R}} |h| |\partial P_z^{(1)}(h)| dh \leq C.$$

These follow by scaling $h = z\eta$; the extension-derivative moment at infinity is finite because $p > 1$. Thus boundedness of ϕ, ϕ' and cancellation give uniform bounds for Q_t and its first derivatives up to $z = 0$ on bounded sets. If $z \geq |t|/2$ and $z \geq 1$, use the kernel suprema Cz^{-1} and Cz^{-2} and $\|\phi\|_1 = 2$. For $|t| > 2z$, $|t| \geq 2$, split at $|t - \tau| = |t|/2$. The near part uses $\phi(\tau) \leq C|t|^{-1-p}$; the far part uses the kernel bound $Cz^p |t|^{-1-p}$ and $\|\phi\|_1 = 2$. Both are at most $C/|t|$, proving the first decay estimate. For either first kernel derivative, subtract $\phi(t)$ using its zero integral. On the near part, the mean value theorem and the first absolute moment give a bound $C|t|^{-2-p}$, using $|\phi'| \leq C|t|^{-2-p}$ throughout that interval. On the far part, the contributions containing $\phi(\tau)$

are bounded respectively by $Cz^p|t|^{-2-p}$ and $Cz^{p-1}|t|^{-1-p}$. The subtracted constant contributes at most $C|\phi(t)|z^p|t|^{-1-p}$ and $C|\phi(t)|z^{p-1}|t|^{-p}$, respectively. Since $z < |t|/2$ and $p > 1$, all these terms are at most $C|t|^{-2}$. This proves the second decay estimate. For the derivative tail, use [12, Theorem 1.2 and Section 2.1, assumptions (W1)–(W3)]. That theorem applies for every $s \in (0, 1)$ and integer $k \geq 2$ when the potential is C^{k+1} ; it gives $|q_s^{(k)}(t)| \leq C_k(1 + |t|)^{-p-k}$. More explicitly, its potential $(1 - v^2)^2/4$ is even, nonnegative and smooth, has precisely the wells ± 1 , and has second derivative 2 at both wells. It satisfies that theorem's structural assumptions, and its normalized increasing profile is q_s by uniqueness. The operator constant in their Section 2.3 is $s2^{2s}\pi^{-n/2}\Gamma((n+2s)/2)/\Gamma(1-s)$, equal to our $c_{n,s}$. Only the one-dimensional derivative theorem is used here; the exponent restriction in their separate Willmore approximation theorem is not a hypothesis of this result.

Fix once and for all a smooth even function $\chi \in C_c^\infty(\mathbb{R})$, equal to one on $[-1, 1]$, supported in $(-2, 2)$, and satisfying $0 \leq \chi \leq 1$. Choose $\chi_T(t, z) = \chi(t/T)\chi(z/T)$, equal to one for $|t|, z \leq T$, supported in $|t|, z < 2T$, even in t , and constant in z near zero, with $0 \leq \chi \leq 1$. Put

$$a_T = \kappa_s \int_{\mathbb{R} \times (0, \infty)} z^{1-p} Q_t^2 \chi_T \, dt \, dz.$$

The layer mass and its supremum bound give

$$\int_{\mathbb{R}} Q_t(t, z)^2 \, dt \leq \|Q_t(\cdot, z)\|_\infty \int_{\mathbb{R}} Q_t(t, z) \, dt \leq C/(1+z).$$

Consequently $\int_0^\infty z^{1-p}(1+z)^{-1} \, dz < \infty$: near zero use $p < 2$, and at infinity use $p > 1$. Dominated convergence proves that a_T tends to its finite uncut integral as $T \rightarrow \infty$. That integral is positive because $Q_t > 0$. A fixed compact rectangle where $\chi_T = 1$ for $T \geq T_0$ gives $0 < c \leq a_T \leq C$ uniformly. Uniqueness applied to $-q_s(-t)$ makes q_s odd, so Q_t is even in t . In two tangential dimensions, with $K_\Sigma = \det A$,

$$J = 1 - tH + t^2 K_\Sigma, \quad JH_t = H - 2tK_\Sigma.$$

Hence the mean-curvature coefficient cancels exactly:

$$\kappa_s \int z^{1-p} JH_t Q_t^2 \chi_T \, dt \, dz = a_T H. \quad (7.6)$$

Insert $\psi = \eta(y)Q_t\chi_T$ in (7.2) and integrate by parts in (t, z) , with η smooth and compactly supported in the lateral patch. The bounds just proved make this a locally weighted H^1 test with a smooth trace. The reference-layer result in Section 2 gives the exact bounded solution q_s , with $|q_s| \leq 1$, $\phi = q'_s > 0$ and $\int \phi = 2$. Apply the shared Jacobi equation from Section 4 to this one-dimensional layer: ϕ is bounded and smooth with all derivatives bounded, and

$$(-\partial_t^2)^s \phi = (1 - 3q_s^2)\phi.$$

Thus $Q_t = P_z^{(1)} * \phi$ and $D_p Q_t = 0$. The same Poisson conormal calculation applies to this bounded smooth trace. Explicitly, evenness and zero differentiated mass yield

$$-\kappa_s \lim_{z \downarrow 0} z^{1-p} (Q_t)_z = \frac{c_{1,s}}{2} \int_{\mathbb{R}} \frac{2\phi(t) - \phi(t+h) - \phi(t-h)}{|h|^{1+p}} \, dh = (-\partial_t^2)^s \phi = (1 - 3q_s^2)\phi.$$

Here $c_{1,s}$ is the normalized operator constant from Section 2. The increments are bounded by $C \min\{|h|^2, 1\}$, and the weighted differentiated kernel by $C|h|^{-1-p}$, so dominated convergence proves the limit locally uniformly because $p < 2$. First integrate over $z > \epsilon$ and then take this limit; it cancels the linear trace potential exactly. The tangential term defines the intrinsic tangent vector

$$\mathcal{F} = \kappa_s \int z^{1-p} JQ_t \chi_T (I - tA)^{-2} \nabla_{g_\Sigma} W \, dt \, dz.$$

Its norm is measured in g_Σ . In graph coordinates its components $\mathcal{F}^a$ are contravariant, and $\operatorname{div}_\Sigma \mathcal{F} = \omega^{-1} \partial_a (\omega \mathcal{F}^a)$, where $\omega = \sqrt{\det g_\Sigma}$. Write $D_p = \partial_{tt} + \partial_{zz} + (1-p)z^{-1}\partial_z$, and retain the

scalar terms

$$\begin{aligned} B_{\text{geom}} &= \kappa_s \int z^{1-p} J H_t W \chi_T Q_{tt} \, dt \, dz, \\ C_\chi &= \kappa_s \int z^{1-p} W \{ J H_t Q_t \partial_t \chi_T - J (Q_t D_p \chi_T + 2 \nabla Q_t \cdot \nabla \chi_T) \} \, dt \, dz, \\ N &= \int J (3 q_s w^2 + w^3) \phi(t) \chi_T(t, 0) \, dt. \end{aligned}$$

Together with (7.6), this gives $a_T H = \text{div}_\Sigma \mathcal{F} + \mathcal{P}$, with $\mathcal{P} = -B_{\text{geom}} - C_\chi - N$. For completeness, $D_p Q_t = 0$ and $J_t = -J H_t$ give the normal integration identity

$$\begin{aligned} \kappa_s \int z^{1-p} J \{ W_t (Q_t \chi_T)_t + W_z (Q_t \chi_T)_z \} \, dt \, dz \\ = B_{\text{geom}} + C_\chi + \int J (1 - 3 q_s^2) w \phi \chi_T(t, 0) \, dt. \end{aligned}$$

Multiplication by η and integration on Σ cancels this last integral against the linear trace term in (7.2). The tangential term is $\int_\Sigma \mathcal{F} \cdot \nabla_\Sigma \eta$ and hence represents $-\text{div}_\Sigma \mathcal{F}$ when tested against η . This proves the stated signs in the distributional projection. All integrations have compact normal/extension cutoffs and lateral buffers, so no other boundary term is suppressed.

Metric factors are uniformly bounded for small T/R . Equation (7.4) and the layer mass show

$$|\mathcal{F}| \leq C \left[\varepsilon + R^{-1} \int_0^{2T} z^{1-p} \, dz + d^{-p} \int_1^{2T} 1 \, dz \right] \leq C [T^{2-p}/R + T d^{-p}].$$

The ε/z term is integrable after weighting since $p > 1$. For $0 < z < 1$, use the separate bound C/R in (7.4); for $z \geq 1$, its ε/z contribution is $C\varepsilon \int_1^{2T} z^{-p} \, dz \leq C\varepsilon$. For the scalar terms, polar integration in the (t, z) half-plane has weighted area $O(\rho^{2-p} \, d\rho)$, and $|Q_{tt}| \leq C\rho^{-2}$ for $\rho \geq 1$. Thus

$$\begin{aligned} |B_{\text{geom}}| &\leq \frac{C}{R} \left[\varepsilon + R^{-1} + d^{-p} + \int_1^{CT} \rho^{-p} (\varepsilon + \rho/R + \rho^p d^{-p}) \, d\rho \right] \\ &\leq \frac{C}{R} [\varepsilon + R^{-1} + d^{-p} + R^{-1} T^{2-p} + T d^{-p}]. \end{aligned}$$

The first three terms account for $\rho < 1$, where the layer derivatives are bounded and $|W| \leq C(\varepsilon + R^{-1} + d^{-p})$. On the cutoff annulus the second integrand in C_χ is $O(|W|T^{-3})$ before weighting, its weighted area is $O(T^{3-p})$, and the first has an extra factor T/R . Therefore

$$|C_\chi| \leq C[\varepsilon T^{-p} + R^{-1} T^{1-p} + d^{-p}],$$

For the trace estimate, $\delta_\Sigma(X_\Sigma(y, t)) = t$ on the prepared fibers, so the supplied pinned bound gives $|w| \leq \varepsilon$. After increasing the uniform scale threshold, $\varepsilon \leq 1$. Using $|q_s| \leq 1$, $J \leq C$, $0 \leq \chi_T \leq 1$ and $\phi > 0$ with mass 2 gives directly

$$|N| \leq C\varepsilon^2 \int_{\mathbb{R}} \phi(t) \, dt \leq C\varepsilon^2.$$

Use $\varepsilon \leq C/R$, $1 \leq T \leq cR$, and $d \leq c_{\text{cap}} R$. In particular $R^{-2} \leq CR^{-p} \leq Cd^{-p}$, and $R^{-2} T^{2-p} \leq CR^{-1} T^{1-p}$. Also $R^{-1} d^{-p} + R^{-1} T d^{-p} \leq Cd^{-p}$ and $\varepsilon T^{-p} \leq CR^{-1} T^{1-p}$. These inequalities absorb all scalar remainders, including the small-radius terms just recorded. We have proved the following controlled projection of the actual equation.

Proposition 7.1 (Projected equation). *On these prepared isolated patches,*

$$a_T H = \text{div}_\Sigma \mathcal{F} + \mathcal{P}, \quad |\mathcal{F}| \leq C[T^{2-p}/R + T d^{-p}], \quad |\mathcal{P}| \leq C[d^{-p} + R^{-1} T^{1-p}]. \quad (7.7)$$

The equation holds in distributions and constants are independent of other layers. It does not give a pointwise bound on H or on the divergence of the flux.

For an upward graph write

$$X(x') = (x', f(x')), \quad \omega = \sqrt{1 + |Df|^2}, \quad \nu = \frac{(-Df, 1)}{\omega}, \quad N(v) = \frac{v}{\sqrt{1 + |v|^2}}.$$

With the convention $A = -D_\Sigma \nu$, the second fundamental form is f_{ab}/ω , and contraction with $g^{ab} = \delta_{ab} - f_a f_b / (1 + |Df|^2)$ gives $H = \operatorname{div} N(Df)$. In particular this is the upward geometric orientation for both branches. Put $\mathcal{F} = \mathcal{F}^a \partial_a X$ and write the surface projection against a test η in these coordinates:

$$a_T \int H \eta \omega \, dx' = - \int \mathcal{F}^a \partial_a \eta \omega \, dx' + \int \mathcal{P} \eta \omega \, dx'.$$

For any $\varphi \in C_c^\infty(D_r)$, take $\eta = \varphi/\omega$. This proves, without differentiating the flux,

$$a_T \int H \varphi \, dx' = - \int \mathcal{F}^a \partial_a \varphi \, dx' + \int (\mathcal{P} + \mathcal{F}^a \partial_a \log \omega) \varphi \, dx'.$$

Here repeated base indices are summed. The inherited slope and Hessian bounds give

$$D \log \omega = \frac{D^2 f Df}{1 + |Df|^2}, \quad |D \log \omega| \leq C/R.$$

The metric $g_{ab} = \delta_{ab} + f_a f_b$ is uniformly comparable with the Euclidean metric, so the intrinsic bound on $\mathcal{F}$ also bounds its contravariant components. Thus the density term satisfies

$$C|\mathcal{F}|/R \leq C[T^{2-p}/R^2 + Td^{-p}/R] \leq C[R^{-1}T^{1-p} + d^{-p}],$$

since T/R is bounded by a fixed small constant. Choose the same constant T on the comparison disk, and set

$$G^a = \mathcal{F}^a/a_T, \quad P = (\mathcal{P} + \mathcal{F}^a \partial_a \log \omega)/a_T.$$

The numbers T and a_T are constant in the base variable, and $0 < c_a \leq a_T \leq C_a$. We obtain

$$\operatorname{div} N(Df) = \operatorname{div} G + P, \quad |G| \leq C[T^{2-p}/R + Td^{-p}], \quad |P| \leq C[d^{-p} + R^{-1}T^{1-p}]. \quad (7.8)$$

For a decreasing branch apply the preceding projection to $-u$. Linearity and oddness give $(-\Delta)^s(-u) = (-u) - (-u)^3$, while the stability potential $3(-u)^2 - 1 = 3u^2 - 1$ is unchanged. The zero graph and its upward geometric orientation remain the same, so the same base equation applies. The oddness of the reference layer gives $-q_s(\delta) = q_s(-\delta)$: reversing the increasing signed distance together with u preserves the prepared profile error and its stated norm bounds. The underlying zero set, reach and buffers are unchanged.

For two ordered graphs on D_r , put $h = f_2 - f_1$ and

$$A_h = \int_0^1 DN(Df_1 + \theta(Df_2 - Df_1)) \, d\theta.$$

The derivative

$$DN(v) = (1 + |v|^2)^{-1/2} I - (1 + |v|^2)^{-3/2} v \otimes v$$

has eigenvalues $(1 + |v|^2)^{-3/2}$ along v and $(1 + |v|^2)^{-1/2}$ perpendicular to v ; the formula also covers $v = 0$. The convex segment of gradients has norm at most θ_0 . Consequently A_h is symmetric and

$$\lambda_0 I \leq A_h \leq I, \quad \lambda_0 = (1 + \theta_0^2)^{-3/2} > 0.$$

The fundamental theorem of calculus gives $N(Df_2) - N(Df_1) = A_h Dh$, and subtraction of the weak base equations therefore yields

$$\operatorname{div}(A_h Dh) = \operatorname{div}(G_2 - G_1) + P_2 - P_1.$$

We record the bounded-divergence estimate with its weak interpretation. For $r > 0$ and $0 < \lambda \leq \Lambda < \infty$, let A be a measurable symmetric matrix on $D_r \subset \mathbb{R}^2$ with $\lambda I \leq A \leq \Lambda I$ almost everywhere. If $F \in L^\infty(D_r; \mathbb{R}^2)$, there is a unique $v \in H_0^1(D_r)$ such that

$$\int_{D_r} A Dv \cdot D\psi \, dx' = \int_{D_r} F \cdot D\psi \, dx' \quad (\psi \in H_0^1(D_r)), \quad \|v\|_\infty \leq C(\lambda, \Lambda) r \|F\|_\infty.$$

Here and below the norm of a weak solution is the essential supremum. For existence and uniqueness, use the gradient norm on $H_0^1(D_r)$; Poincaré's inequality makes it a Hilbert norm.

The bilinear form is bounded by Λ and coercive with constant λ , while the right-hand side is bounded by $\|F\|_\infty |D_r|^{1/2} \|D\psi\|_2$. These are the hypotheses of Lax–Milgram.

For the supremum estimate first work on D_1 . If $k \geq 0$, the Lipschitz truncation $w = (v - k)_+$ belongs to $H_0^1(D_1)$, with $Dw = \mathbf{1}_{\{v > k\}} Dv$. Write $E_k = \{v > k\}$ and $K_F = \|F\|_\infty$. Testing gives

$$\lambda \|Dw\|_2^2 \leq K_F |E_k|^{1/2} \|Dw\|_2, \quad \|Dw\|_2^2 \leq \lambda^{-2} K_F^2 |E_k|.$$

The planar Sobolev inequality on the fixed unit disk is $\|w\|_4 \leq S_4 \|Dw\|_2$. For $\ell > k$ it follows that

$$|E_\ell| \leq \frac{S_4^4 \lambda^{-4} K_F^4}{(\ell - k)^4} |E_k|^2.$$

When $K_F > 0$, fix $\beta > 4$ and set $k_j = M(1 - 2^{-j})$. Choose $M = CK_F$ so large that

$$S_4^4 \lambda^{-4} (K_F/M)^4 |D_1| 2^{4+\beta} \leq 1.$$

Starting with $|E_{k_0}| \leq |D_1|$, the recurrence

$$|E_{k_{j+1}}| \leq S_4^4 \lambda^{-4} (K_F/M)^4 2^{4(j+1)} |E_{k_j}|^2$$

gives $|E_{k_j}| \leq |D_1| 2^{-\beta j}$ by induction: the remaining factor is $2^{(4-\beta)j} \leq 1$. Thus $|\{v \geq M\}| = 0$, so $v \leq M$ almost everywhere. Apply the same argument to $-v$, with datum $-F$. If $K_F = 0$, the energy test with v and Poincaré's inequality give $v = 0$. Under $x' = rx$, the rescaled solution $v(rx)$ satisfies the unit-disk weak equation with matrix $A(rx)$ and vector datum $rF(rx)$. This proves the stated factor r .

Apply this estimate with $A = A_h$ and $F = G_2 - G_1$. Let $e \in H_0^1(D_r)$ be the resulting unique weak solution and set $\hat{h} = h - e$. The smooth graphs with bounded gradients have $h \in H^1(D_r)$, so $\hat{h} \in H^1(D_r)$, and

$$\|e\|_\infty \leq Cr[T^{2-p}/R + Td^{-p}], \quad \hat{h} = h - e, \quad \operatorname{div}(A_h D\hat{h}) = P_2 - P_1. \quad (7.9)$$

The last equation holds weakly against every compactly supported smooth base test. Its scalar forcing satisfies $\|P_2 - P_1\|_\infty \leq C[d^{-p} + R^{-1}T^{1-p}]$. Neither coefficient nor flux derivatives are used.

Fix $\tau > 0$ and take $T = \tau R^{1/p}$, $r \leq c_r R$, and $d \geq c_d (R \log R)^{1/p}$, with $c_r, c_d > 0$ fixed and $R \geq 3$. These choices meet the width bounds for all sufficiently large R , since $T \rightarrow \infty$, $T/d \leq C(\log R)^{-1/p}$ and $T/R \rightarrow 0$. From the preceding estimate,

$$\frac{\|e\|_\infty}{d} \leq C \left[\frac{T^{2-p}}{d} + \frac{RT}{d^{p+1}} \right].$$

Substitution gives

$$\frac{\|e\|_\infty}{d} \leq C \left[R^{(1-p)/p} (\log R)^{-1/p} + (\log R)^{-(p+1)/p} \right] \rightarrow 0. \quad (7.10)$$

The constants here may depend on the fixed τ, c_r, c_d and the prepared data. If $h \geq c_h d$ for a fixed $c_h > 0$, then $\hat{h} \geq c_h d/2$ almost everywhere for all sufficiently large R . Any continuous interior representative provided by the elliptic regularity used below satisfies this same lower bound everywhere. When additionally $d \leq C_D R^{2/(p+1)}$ with C_D fixed, put $\delta_p = (p-1)/(p(p+1)) > 0$. Then

$$R^{-1}T^{1-p} = \tau^{1-p} R^{-(2p-1)/p}, \quad \frac{R^{-1}T^{1-p}}{d^{-p}} \leq CR^{-\delta_p} \rightarrow 0.$$

Equivalently, $R^{-1}T^{1-p} = o(R^{-2p/(p+1)}) \leq Cd^{-p}$, uniformly in this range. The corrected gap consequently has scalar forcing bounded by Cd^{-p} .

Theorem 7.2 (Power separation). *In any fixed buffered interior subcylinder of the prepared complete charts, every neighboring actual zero gap satisfies*

$$h \geq cR^{2/(p+1)}. \quad (7.11)$$

The constant is uniform over branches and the scale remains the selected curvature scale R .

Proof. Suppose an interior gap h_0 satisfies $h_0/R^{2/(p+1)} \rightarrow 0$ along a sequence $R \rightarrow \infty$. All centers initially have fixed margins of order R in a larger prepared cylinder. Put $b = (p+1)/2$ and $L = (R/h_0^b)^{1/2}$. Then $L \rightarrow \infty$, $Lh_0^b = o(R)$, and $h_0 = o(R)$. Translate the initial lower endpoint to zero and restrict the inherited complete families to a common disk and complete cylinder of fixed sizes proportional to R around it. The original interior margins give uniform positive fractions; restriction and translation preserve the phase identities, order and analytic bounds. Choose the nested cylinders $\mathcal{C}_j$ of Section 6 in this neighborhood. Both initial endpoints and their midpoint lie in $\mathcal{C}_4$ with fixed remaining margins proportional to R for large R , because $h_0 = o(R)$. Locate each gap state at the physical midpoint of its two consecutive vertical zeros at its base point. Starting from h_0 , move to any state of less than half the current gap d_k within distance $C_*Ld_k^b$. Here C_* is fixed independently of the charts and the gap. The total movement is bounded by

$$\sum_k C_*Ld_k^b \leq \frac{C_*Lh_0^b}{1-2^{-b}} = o(R).$$

Every search ball also has radius at most $C_*Lh_0^b = o(R)$. Every candidate gap is at most h_0 , so its endpoints are within $h_0/2 = o(R)$ of its midpoint. Thus all candidate endpoints retain fixed margins in this same $\mathcal{C}_4$: the total movement, each search radius and the endpoint displacement are all $o(R)$. Proposition 6.1 gives their common positive lower bound $c(R \log R)^{1/p}$ at this fixed R , independently of the branch or pair. Infinite halving would contradict this lower bound, so the descent terminates. At the last state, of gap $d \leq h_0$, every gap whose midpoint lies in the search ball is at least $d/2$.

Translate the terminal base point to zero. Set $r = cLd^b = o(R)$. Logarithmic separation at this retained center gives $d \geq c(R \log R)^{1/p}$; in particular $d \rightarrow \infty$ and $d/r = (cLd^{b-1})^{-1} \rightarrow 0$. First consider the selected graphs on D_{8r} . Their heights differ from the terminal midpoint by at most $d/2 + 8\theta_0 r$, so their graph points and gap midpoints are within Cr of that midpoint. Fix C_* larger than these conversion constants and then choose c sufficiently small that this region lies inside the terminal search ball. The selected gap is therefore at least $d/2$ on all of D_{8r} . If an immediately exterior neighboring gap there were smaller than $d/2$, its inner endpoint would lie on a selected graph and its midpoint would be within another $d/4$ of that endpoint. It would also lie in the search ball, a contradiction. Ordering then controls the distance to every other listed graph.

We verify the local profile comparison separately from the inherited complete-chart buffers of size proportional to R . Put $\ell = c_{\text{iso}}d$ with a sufficiently small fixed $c_{\text{iso}} > 0$. For a selected zero $Q = (a, f_i(a))$ over D_{4r} , every fixed $O(d)$ enlargement stays horizontally in D_{8r} , since $d/r \rightarrow 0$. If another listed zero $Y = (y', F_k(y'))$ belonged to $B_{4\ell}(Q)$, then

$$|F_k(y') - f_i(y')| \leq |Y_3 - Q_3| + |f_i(y') - f_i(a)| \leq 4(1 + \theta_0)\ell < d/2,$$

contrary to the separation just established. Unlisted zeros are outside the old complete region, at distance at least a fixed multiple of R . Thus every zero of $B_{4\ell}(Q)$ lies on the selected sheet.

Here is the quantitative regraphing needed for pinning on this local scale. Translate Q to zero and reverse the old vertical direction if needed, relabeling the phase families. Its increasing normal has angle $\beta \leq \arctan \theta_0$ with the resulting vertical. The rotated horizontal projection and height of an old graph are

$$\Phi_F(\zeta) = B\zeta + bF(\zeta), \quad g(\Phi_F(\zeta)) = e \cdot \zeta + \delta F(\zeta),$$

where $s_{\min}(B) \geq \cos \beta$, $|b| = |e| = \sin \beta$, and $\delta = \cos \beta$. With $\kappa = \cos \beta - \theta_0 \sin \beta > 0$, on any convex retained disk,

$$|\Phi_F(\zeta) - \Phi_F(\zeta')| \geq \kappa|\zeta - \zeta'|, \quad \|(D\Phi_F)^{-1}\| \leq \kappa^{-1}.$$

The map is injective and locally invertible. Its image contains the disk of radius κH about the image of the center of any retained old disk of radius H : lift each radial segment by the local inverse; the displayed lower bound prevents reaching the old boundary before this radius, while compactness and the inverse-function theorem extend any prematurely stopping lift. The selected

old graph has a disk of radius hR with fixed $h > 0$, so its new image contains $D_{8\ell}$ for large R . Differentiating the height identity gives

$$|Dg| \leq (\sin \beta + \theta_0 \cos \beta)/\kappa, \quad D^2g[v, w] = (\delta - Dg \cdot b)D^2F[(D\Phi_F)^{-1}v, (D\Phi_F)^{-1}w].$$

In particular $|D^2g| \leq C/R$, $g(0) = Dg(0) = 0$, and $\partial_{\text{new } 3}u = \partial_{\text{old } 3}u(\delta - e \cdot DF)$ retains the regular increasing orientation because $\delta - e \cdot DF \geq \kappa$.

The quantitative complete-family argument in Section 6 applies to the original buffers of size proportional to R , rather than to the local gap region. Retain the old families meeting a sufficiently small new cylinder of fixed size proportional to R . The same projection bound gives their common new graph domain; phases in one family differ by at most the fixed offset $2C_0$. Their signs and order are preserved, vertical derivatives remain at least $\kappa\gamma_0$, and phase offsets remain bounded by $2a/(\kappa\gamma_0)$. Inverse rotation places this cylinder inside the old complete region, hence establishes completeness. After taking one fixed further interior, every zero in $B_{2\ell}(Q)$ retains complete-chart and graph-domain margins proportional to R , since $\ell/R \rightarrow 0$.

Theorem 4.4 supplies a common qualitative modulus at all these zero centers: an offending sequence would retain the same fixed complete buffers and contradict that theorem. The distance construction and Theorem 6.3, with $L = \ell \rightarrow \infty$ and the fixed smallness $\ell \leq c_0R$ valid eventually, therefore give

$$\|u - q_s(\delta_{\Sigma_i})\|_{C^{1,\alpha}(B_{\ell/4}(Q))} \leq C_\alpha(R^{-1} + \ell^{-p}) \leq C_\alpha/R, \quad 0 < \alpha < p - 1.$$

The last inequality follows from $d^p \geq cR \log R$. The patched distance agrees with the actual distance on $B_\ell(Q)$, has reach at least cR , and has the quadratic tangent-plane error established in that distance construction. Choose a fixed smaller effective isolation length $c'd$ so all comparison enlargements required in Section 7 fit these regions. Its curvature-scale cap is inactive since $d/R \rightarrow 0$. This supplies the prepared local comparison data for both branches; the sign reversal in the base projection handles a decreasing branch. Applying projection and correction with this fixed multiple of d changes only constants, independently of the number of layers. This local comparison fixes only one selected sheet and its separation from all others; it also applies when only one graph is retained.

Use the common width $T = \tau_* R^{1/p}$ constructed in the proof of Proposition 6.1 throughout this disk. Logarithmic separation gives $T/d \rightarrow 0$, and $r/d = cLd^{b-1} \rightarrow \infty$. Thus $1 \ll T \ll d \ll r \ll R$, with all needed normal and horizontal buffers. Equations (7.9)–(7.10) give

$$\hat{h} > 0, \quad \hat{h}(0) \leq Cd, \quad |\operatorname{div}(A_h D\hat{h})| \leq Cd^{-p}, \quad \|h - \hat{h}\|_\infty = o(d).$$

For the actual-gap inequality, reuse the common-width construction in the proof of Proposition 6.1. The terminal endpoints retain $\mathcal{C}_4 \subset \mathcal{C}_3$, so it supplies a disk D_{r_*} , $r_* = c_b R$, the same T , and every tube, weight, endpoint and phase hypothesis of Theorem 5.3 for this pair. Eventually $r < r_*$. Restrict its tests to $C_c^1(D_r)$; their zero extensions have the same integrals and $E(R, T, r_*) \leq E(R, T, r)$. Thus (5.12) holds on D_r with the latter error. This avoids repeating the isolation and clearing argument.

The corrected graph has a continuous interior representative by [23, Theorem 8.22, p. 200]: the coefficient is measurable, bounded and uniformly elliptic, all lower-order coefficients and vector forcing vanish, and the scalar forcing is bounded. In dimension two choose $q = 4 > 2$, so it belongs to the required $L^{q/2}$ on each finite interior disk. Since h is continuous, the essential error bound is pointwise for this representative.

Take $\rho = c_1 d^b$. On $D_{8\rho}$, rescale $\hat{h}(\rho x')/d$. Its forcing has bound $C\rho^2 d^{-p-1} = Cc_1^2$. The required Harnack bound is

$$\sup_{D_\rho} \hat{h} \leq C[\hat{h}(0) + \rho^2 d^{-p}] \leq Cd.$$

To justify the forcing term directly, put $f = P_2 - P_1$ on $D_{8\rho}$ and define, almost everywhere,

$$F(x_1, x_2) = \left(\int_0^{x_1} f(s, x_2) \, ds, 0 \right).$$

Each horizontal integration segment is inside the disk, so $\operatorname{div} F = f$ distributionally and $\|F\|_\infty \leq 8\rho\|f\|_\infty$. Solve $\operatorname{div}(A_h D\zeta) = f$ with $\zeta \in H_0^1(D_{8\rho})$. The bounded-divergence estimate already proved gives $\|\zeta\|_\infty \leq C\rho^2\|f\|_\infty$. Consequently $\hat{h} - \zeta + \|\zeta\|_\infty$ is nonnegative and A_h -harmonic. Homogeneous Harnack [23, Theorem 8.20, p. 199], with $\overline{D_{4\rho}} \subset D_{8\rho}$ and ellipticity constants independent of the charts, bounds its supremum on D_ρ by a fixed multiple of its value at zero. Both $\hat{h}$ and ζ have continuous interior representatives by the same bounded-forcing regularity. Undoing the correction proves the displayed estimate. The disk buffer is available because $r/\rho = cL/c_1 \rightarrow \infty$. The correction estimate then gives $h \leq Cd$ on D_ρ .

For $r > 4\rho$, choose a radial cutoff equal to one on D_ρ , equal to $\log(r/(2|x'|))/\log(r/(2\rho))$ on $\rho < |x'| < r/2$, and zero beyond $r/2$. Its energy is $2\pi/\log(r/(2\rho)) \rightarrow 0$ and its squared mass is at most Cr^2 . It is admissible by H^1 approximation: $h \geq d/2$ makes the weighted L^2 term continuous under this approximation. The left side of (5.12) is bounded below by $c\rho^2 d^{-p-1} = c' > 0$, whereas its error is at most

$$Cr^2[T^{-1-2p} + (RT)^{-(p+1)/2} + r^{-p-1}] \leq C[R^{-1/p} + R^{-(p-1)^2/(2p)} + r^{1-p}] \rightarrow 0.$$

Here $r = o(R)$ and $r \rightarrow \infty$. This contradiction proves the uniform lower bound: failure of a common constant and threshold over any fixed retained chart class would supply the violating sequence with which the proof began. $\square$

Lemma 7.3 (Uniform comparison after recentering). *For a sequence satisfying (4.5), fix $M < \infty$. Every zero $Q \in B_{MR_j}(P_j)$, after translating it and rotating its increasing normal vertically, admits a selected zero graph and comparison radius*

$$d_j = c_* R_j^{2/(p+1)}, \quad L_j^* = \eta d_j,$$

with constants independent of j, Q, M . The graph is defined on $D_{8L_j^*}$, contains every zero in $B_{4L_j^*}$, and its normal footpoint is single valued throughout a fixed smaller box. For $0 < \alpha < p-1$,

$$\|u_j - q_s(\delta_{\Sigma_Q})\|_{C^{1,\alpha}(B_{cL_j^*}(Q))} \leq C_\alpha/R_j, \quad |D^2 f_Q| \leq C/R_j, \quad |\partial_t u_j(Q)| \geq \gamma/2. \quad (7.12)$$

Thresholds are uniform over Q for each fixed M . Any fixed finite number of nested comparison buffers is allowed after decreasing η once.

Proof. Use the normalized stable solution class from Lemma 2.1 and the smoothness and uniform derivative bounds of Section 3. The curvature quantity defined in (4.3) is $M = |A|$, so (4.5) gives the level-surface curvature bound $2/R_j$ and nondegeneracy $|\nabla u_j| \geq \gamma$ on the selected phase band. The selected ball has radius $L_j R_j$, with $L_j \rightarrow \infty$. For fixed M , a ball of radius $C_{\text{buf}} R_j/2$ around every such Q is contained in it for large j , since $L_j > M + C_{\text{buf}}/2$ eventually. The complete-chart theorem therefore supplies a fixed class of complete phase families around every Q , including returning branches, with common small slopes, Hessian bounds C/R_j , regular signs and buffers. Only the threshold for ambient inclusion may depend on M .

Retain finitely many nested interiors of fixed size proportional to R_j . The qualitative normal comparison of Theorem 4.4 is uniform over the centers: any offending sequence would retain the same fixed complete data and contradict that theorem. Apply Theorem 7.2 on a larger interior, reserving a smaller one around Q . It gives a common neighboring gap bound $c_{\text{pow}} R_j^{2/(p+1)}$ there, with constants independent of M after ambient inclusion.

Center the old coordinates at Q and put $\beta_p = 2/(p+1)$,

$$d = c_{\text{pow}} R_j^{\beta_p}, \quad r = \sqrt{R_j d}.$$

Then $r/R_j = d/r = \sqrt{d/R_j} \rightarrow 0$. Since $\beta_p > 1/p$, $d/(R_j \log R_j)^{1/p} \rightarrow \infty$, so d meets the logarithmic lower bound in Proposition 6.1. The selected old graph F_Q on D_{8r} remains in the reserved smaller interior with fixed margins proportional to R_j : its displacement is $O(r) = o(R_j)$. At any such base point, every other listed zero in the larger interior is vertically separated from it by at least d , by power separation and ordering. A listed zero outside that larger interior has instead vertical separation at least a fixed multiple of $R_j \gg d$, by the reserved interior margin.

Thus F_Q is separated by at least $d/2$ from every other listed graph throughout D_{8r} ; when it is the only listed graph this condition is vacuous.

Apply the single-sheet local comparison verified in the preceding proof, with these r, d . It gives increasing-normal coordinates, the selected actual graph on $D_{8\ell}$, all-zero isolation on $B_{4\ell}(Q)$, complete buffers of size proportional to R_j at all zeros in $B_{2\ell}(Q)$, Hessian bound C/R_j , and the profile bound C_α/R_j on $B_{\ell/4}(Q)$, where $\ell = c_{\text{iso}}d$. Choose fixed $c_*, \eta > 0$ so that $L_j^* = \eta c_* R_j^{\beta_p} \leq \ell/16$, reducing them to meet the required fixed buffer fractions. Restriction gives the graph on $D_{8L_j^*}$ and includes every zero in $B_{4L_j^*}$. The actual signed distance and profile estimate apply on a fixed smaller ball $B_{cL_j^*}$. The scale error also satisfies $C_\alpha(R_j^{-1} + (L_j^*)^{-p}) \leq C_\alpha/R_j$, because $\beta_p p = 2p/(p+1) > 1$.

Use the restricted graph in the distance construction. Its value and slope at zero vanish, its orientation is increasing and regular, its Hessian is bounded by C/R_j , and it has the required isolation and complete buffers. Choose the fixed c_0 in terms of the graph and cutoff data; $L_j^*/R_j \rightarrow 0$ ensures its smallness requirement holds eventually. The distance geometry gives reach at least cR_j , agreement of patched and actual signed distances on $B_{L_j^*}$, and a unique actual normal footpoint there. A box $D_{\zeta L_j^*} \times (-\zeta L_j^*, \zeta L_j^*)$ with fixed small $\zeta > 0$ lies in this ball and the reach. At the center the third axis is the increasing normal, so $\partial_t u_j(Q) = |\nabla u_j(Q)| \geq \gamma$, implying the asserted $\gamma/2$ bound. Any prescribed finite number of further comparison enlargements is retained by decreasing η once. All constants and fractions are chosen from fixed data before Q and M . $\square$

8. THE PHYSICAL-SCALE CURVATURE CANCELLATION

A quadratic tangential displacement has a positive normal moment. Its finiteness across the full range $1 < p < 2$ is what permits the solvability pairing below.

Lemma 8.1 (Normal moment). *For $L_q = (-\Delta)^s + 3q_s(t)^2 - 1$, define*

$$\Xi_p(t) = \frac{c_{3,s}}{2} \int_{\mathbb{R}^2 \times \mathbb{R}} \frac{y_1^2 \phi(\tau)}{(|y|^2 + (t - \tau)^2)^{(3+p)/2}} dy d\tau = \frac{c_{1,s}}{2(p-1)} \int_{\mathbb{R}} |t - \tau|^{1-p} \phi(\tau) d\tau.$$

It is bounded and positive, and $0 < \sigma_p := \int_{\mathbb{R}} \phi(t) \Xi_p(t) dt < \infty$. For every quadratic polynomial P on $\mathbb{R}^2$,

$$L_q(P(x')\phi(t)) = -(\Delta P)\Xi_p(t). \quad (8.1)$$

Proof. The reference layer from Section 2 has $\phi = q'_s > 0$, $\int_{\mathbb{R}} \phi = 2$ and $\phi(t) \leq C(1 + |t|)^{-1-p}$. The shared Jacobi equation (4.2), applied to the bounded entire solution $q_s(t)$ in dimension three, gives bounded smooth ϕ , with all derivatives bounded, and

$$L_q \phi = 0.$$

Here and below the fractional Laplacian of a function with polynomial tangential growth uses spherical principal-value truncations in all three physical variables.

For $a \neq 0$, polar coordinates and the beta integral give

$$\int_{\mathbb{R}^2} \frac{y_1^2}{(|y|^2 + a^2)^{(3+p)/2}} dy = \frac{\pi \Gamma((p-1)/2)}{2 \Gamma((p+3)/2)} |a|^{1-p}.$$

Indeed, the angular integral is π and the substitution $r = |a|\sqrt{v}$ in the radial integral leaves $\frac{1}{2}|a|^{1-p}B(2, (p-1)/2)$. The beta integral is finite because $p > 1$. The normalized coefficients satisfy

$$c_{3,s}\pi \frac{\Gamma((p+1)/2)}{\Gamma((p+3)/2)} = c_{1,s}, \quad \Gamma((p+1)/2) = \frac{p-1}{2}\Gamma((p-1)/2).$$

Tonelli's theorem now gives the asserted one-dimensional formula for Ξ_p . The horizontal integral at $a = 0$ is infinite, but this single normal slice has measure zero, and its remaining singularity $|t - \tau|^{1-p}$ is locally integrable because $p < 2$. Splitting at $|t - \tau| = 1$ gives the uniform bound

$$\int_{\mathbb{R}} |t - \tau|^{1-p} \phi(\tau) d\tau \leq \frac{2\|\phi\|_\infty}{2-p} + \|\phi\|_1.$$

Since the integrand is positive almost everywhere, $\Xi_p(t) > 0$ for every t . Thus $0 < \sigma_p \leq 2\|\Xi_p\|_\infty < \infty$.

We make the operator and distributional identity explicit. If $v \in C_{\text{loc}}^2(\mathbb{R}^3)$ and

$$\int_{\mathbb{R}^3} |v(X)|(1+|X|)^{-3-p} dX < \infty,$$

its spherical principal value exists: symmetric second increments are $O(|H|^2)$ near each fixed X , and the weighted bound controls the tail. On compact sets the omitted ball of radius ϵ costs $O(\epsilon^{2-p})$, uniformly. For $\eta \in C_c^\infty(\mathbb{R}^3)$, put

$$T_\epsilon v(X) = c_{3,s} \int_{|X-Y|>\epsilon} \frac{v(X) - v(Y)}{|X-Y|^{3+p}} dY.$$

The truncated pairings are absolutely integrable, so symmetry and Fubini give $\int \eta T_\epsilon v = \int v T_\epsilon \eta$. For $0 < \epsilon < 1$, the symmetric second-increment formula on a fixed ball containing $\text{supp } \eta$ and the kernel tail outside a larger ball give

$$|T_\epsilon \eta(X)| \leq C_\eta (1+|X|)^{-3-p}.$$

Dominated convergence on the right and local uniform convergence on the left therefore show

$$\int \eta(-\Delta)^s v = \int v(-\Delta)^s \eta.$$

This proves consistency with the distributional operator; adding the bounded potential in L_q preserves the identity.

For a polynomial P of degree at most two, $v(x', t) = P(x')\phi(t)$ belongs to this weighted class. In fact $|P(x')| \leq C_P(1+|x'|^2)$, so integrating ϕ first bounds its weighted integral by

$$C_P \|\phi\|_1 \int_0^\infty (1+\rho^2)\rho(1+\rho)^{-3-p} d\rho < \infty.$$

The last integrand is $O(\rho)$ near zero and $O(\rho^{-p})$ at infinity. Subtract $P(x')L_q\phi = 0$ using the same spherical truncation $|(y, \tau - t)| > \epsilon$ in both operators. The remaining numerator is exactly

$$[P(x') - P(x' + y)]\phi(\tau) = -(DP(x') \cdot y + \frac{1}{2}y^T D^2 P y)\phi(\tau).$$

For each positive truncation the linear term is absolutely integrable and vanishes by reflection in y ; no absolute integrability of that term at the source is needed. The quadratic term is absolutely integrable without truncation. Near the source it is bounded by $C|(y, \tau - t)|^2$ before multiplication by the kernel, which gives $C \int_0^1 r^{1-p} dr < \infty$. At infinity the same weighted estimate, centered at the fixed evaluation point, applies to $|y|^2\phi(\tau)$. Reflection eliminates mixed terms, and interchanging the two horizontal coordinates makes the diagonal moments equal. Dominated convergence gives

$$L_q(P(x')\phi(t)) = -\frac{c_{3,s}}{2}(\Delta P) \int_{\mathbb{R}^2 \times \mathbb{R}} \frac{y_1^2 \phi(\tau)}{(|y|^2 + (t - \tau)^2)^{(3+p)/2}} dy d\tau = -(\Delta P)\Xi_p(t).$$

This holds pointwise and, by the preceding pairing, as an identity of distributions. $\square$

In fact $\Xi_p(t) \asymp |t|^{1-p}$ at infinity, with constants depending only on s . For $|t| \geq 2$, restrict $|\tau| \leq 1$ for the lower bound: $|t - \tau| \leq \frac{3}{2}|t|$ there, and $\int_{-1}^1 \phi > 0$. For the upper bound split the normal integral into three regions. On $|\tau| \leq |t|/2$, use $|t - \tau| \geq |t|/2$ and $\int \phi = 2$. On $|t|/2 < |\tau| \leq 2|t|$, use $\phi(\tau) \leq C|t|^{-1-p}$ and

$$\int_{|a| \leq 3|t|} |a|^{1-p} da \leq C|t|^{2-p}.$$

On $|\tau| > 2|t|$, use $|t - \tau| \geq |\tau|/2$ and integrate $C|\tau|^{-2p}$. The last two contributions are bounded by $C|t|^{1-2p}$, which is no larger than $C|t|^{1-p}$. Together with boundedness on compact sets, these two-sided estimates show that $\Xi_p \in L^2(\mathbb{R})$ precisely when $p > 3/2$; at $p = 3/2$ the squared tail diverges logarithmically. The positive, finite ϕ pairing in Lemma 8.1 therefore remains available across the full range $1 < p < 2$ even when $\Xi_p \notin L^2$.

In the next statement u_j are the bounded entire solutions of the standing normalized equation, and P_j denotes the original marked center before recentering at a zero. The pinned supremum bound means (6.7), with one constant for the fixed chart and buffer data; reducing the fixed comparison-ball fraction preserves it.

Theorem 8.2 (Harmonic quadratic limit). *Let $R_j \rightarrow \infty$. Center an actual zero and rotate its increasing normal vertically. Suppose its selected graph $t = f_j(x')$ is defined on D_{8L_j} , with $f_j(0) = Df_j(0) = 0$, $|D^2 f_j| \leq C/R_j$, contains every zero in B_{4L_j} , and has the fixed normal/graph buffers used in Theorem 6.3. Assume the pinned supremum bound on B_{cL_j} , where $L_j \rightarrow \infty$, $L_j \leq cR_j$, and*

$$R_j L_j^{-p} \rightarrow 0. \quad (8.2)$$

After extracting a subsequence the following convergences hold in all local C^k topologies, for finite k :

$$R_j(u_j(x', t) - q_s(t)) \rightarrow -P_*(x')\phi(t), \quad R_j f_j \rightarrow P_* \quad \text{in } C_{\text{loc}}^k, \quad (8.3)$$

The limit P_* is quadratic, satisfies $\Delta P_* = 0$, and obeys $P_*(0) = DP_*(0) = 0$. If the hypotheses hold uniformly after recentering at every zero in $B_{MR_j}(P_j)$ for each fixed M , then

$$\sup_{\substack{Q \in B_{MR_j}(P_j) \\ u_j(Q)=0}} R_j |H_{\Sigma_Q}(Q)| \rightarrow 0. \quad (8.4)$$

The polynomial is not asserted to vanish.

Proof. Write δ_j for the upward signed distance in the cutoff-graph construction of Section 6, and put $U_j = q_s(\delta_j)$, reserving scalar isolation lengths for d . The assumed pinned supremum bound uses this cutoff profile, as in (6.7), and gives

$$\|R_j(u_j - U_j)\|_{L^\infty(B_{cL_j})} \leq C(1 + R_j L_j^{-p}) \leq C$$

for all sufficiently large j . Denote the fixed upper bound for L_j/R_j by c_0 . We may decrease the comparison-ball fraction c once, preserving the assumed estimate, to remain inside the nearest-point tube of the cutoff graph. The cutoff-distance geometric argument preceding Theorem 6.3 provides a unique actual graph footpoint on this ball, with $|\delta_j(X)| \leq |X|$. The Hessian bound gives the cutoff graph reach at least $c_1 R_j$. Choose $c \leq 1/2$ also so that $cc_0 < c_1/2$ and $Ccc_0 < 1/2$. The nearest cutoff point then has norm at most $2cL_j \leq L_j$ and lies in the unchanged graph on D_{4L_j} . Every zero in B_{4L_j} is on that graph, whereas every zero outside B_{4L_j} is at distance at least $(4-c)L_j > cL_j \geq |\delta_j(X)|$. Thus the distance is the actual zero-set distance on B_{cL_j} ; its upward normal coordinate agrees with δ_j there. If $(b, f_j(b))$ is the footpoint of $X = (y', t)$, its normal equations are

$$y' = b - \frac{\delta_j Df_j(b)}{\sqrt{1 + |Df_j(b)|^2}}, \quad t = f_j(b) + \frac{\delta_j}{\sqrt{1 + |Df_j(b)|^2}}.$$

The footpoint lies on the original graph inside its buffered domain. Since $|Df_j(b)| \leq C|b|/R_j$, the first equation gives $|b| \leq 2|y'|$. The second equation, together with $|f_j(b)| \leq C|b|^2/R_j$, yields

$$|\delta_j(y', t) - t| \leq |f_j(b)| + C|\delta_j||Df_j(b)|^2 \leq C|y'|^2/R_j.$$

Range normalization gives $|u_j| \leq 1$, and the layer has $|q_s| \leq 1$. The layer derivative bound gives, for every $X = (x', t) \in B_{cL_j}$,

$$|q_s(\delta_j(X)) - q_s(t)| \leq \|\phi\|_\infty |\delta_j(X) - t| \leq C|x'|^2/R_j.$$

Using the exact decomposition $v_j = R_j(u_j - U_j) + R_j(q_s(\delta_j) - q_s(t))$ and the pinned bound, we obtain

$$|v_j(x', t)| \leq C(1 + |x'|^2) \quad (X \in B_{cL_j})$$

uniformly for all sufficiently large j . Every fixed ball eventually lies in this growing comparison ball.

More precisely, for all sufficiently large j and every $K \geq 2$,

$$\int_{|Y|>K} |v_j(Y)| |Y|^{-3-p} dY \leq C(K^{-p} + K^{1-p} + R_j L_j^{-p}).$$

Consequently

$$\limsup_j \int_{|Y|>K} |v_j(Y)| |Y|^{-3-p} dY \leq C(K^{-p} + K^{1-p}), \quad K \geq 2. \quad (8.5)$$

Inside B_{cL_j} , the pinned part costs CK^{-p} . For the remaining displacement put $a = C|y'|^2/R_j$. Monotonicity gives

$$|q_s(\delta_j(y', t)) - q_s(t)| \leq q_s(t+a) - q_s(t-a) = \int_{t-a}^{t+a} \phi(\tau) d\tau.$$

The right side is nonnegative. Tonelli and $\int \phi = 2$ imply

$$\int_{\{t:(y', t) \in B_{cL_j}\}} R_j |q_s(\delta_j(y', t)) - q_s(t)| dt \leq 4R_j a \leq C|y'|^2.$$

On $|y'| > K/2$, bound the kernel by $|y'|^{-3-p}$ and use horizontal polar integration, obtaining $C \int_{K/2}^\infty \rho^{-p} d\rho = CK^{1-p}$. On $|y'| \leq K/2$ with $|Y| > K$, the kernel is at most K^{-3-p} , while the integral of $|y'|^2$ over this horizontal disk is CK^4 ; this also costs CK^{1-p} . Outside B_{cL_j} , $|v_j| \leq 2R_j$, and the integral is at most $CR_j L_j^{-p}$. This proves both tail bounds, using $p > 1$ for horizontal integrability.

The cylindrical layer is an actual normalized solution on $\mathbb{R}^3$. Subtracting its equation from that of u_j gives

$$[(-\Delta)^s + c_j]v_j = 0, \quad c_j = u_j^2 + u_j q_s + q_s^2 - 1, \quad L_q v_j = -3q_s v_j^2/R_j - v_j^3/R_j^2.$$

These equations hold classically by the cubic secant identity. Each v_j is smooth and globally bounded, hence belongs to the locally C^2 , weighted L^1 class whose principal-value and distributional operator consistency was established in the proof of Lemma 8.1. The equations therefore hold distributionally as well. The coefficients c_j have uniform global derivatives of every fixed order, by the entire-solution estimates applied to u_j and the cylindrical layer. The local bound and the tail estimate with $K = 2$ bound

$$\int_{\mathbb{R}^3} |v_j(Y)| (1 + |Y|)^{-3-p} dY$$

uniformly for all sufficiently large j . For $|P| \leq M$, the weight centered at P is bounded by $(1 + M)^{3+p}$ times this weight, and $B_4(P) \subset B_{M+4}$. Thus the weighted local norms in (3.3) are uniform on each compact set of centers. That estimate, with $G = 0$ and $a = c_j$, bounds every fixed derivative order of v_j on every fixed ball. No higher graph derivative has been used in obtaining these estimates. Arzelà–Ascoli, using one higher derivative for each desired order, and a diagonal extraction over integer-radius balls and derivative orders give a single subsequence with $v_j \rightarrow v$ in every finite local C^k topology. In particular v is smooth and $|v(x', t)| \leq C(1 + |x'|^2)$.

Since $u_j = q_s + v_j/R_j$, the coefficients converge locally to $3q_s^2 - 1$. For $\psi \in C_c^\infty(\mathbb{R}^3)$ its fractional Laplacian is locally bounded and satisfies $|(-\Delta)^s \psi(Y)| \leq C_\psi (1 + |Y|)^{-3-p}$ outside a fixed ball. Fatou applied to the nonnegative tail integrands gives the same bound (8.5) for v . On a fixed ball local convergence passes $\int v_j (-\Delta)^s \psi$ and the potential term; on the complement the displayed tail bounds make the limsup of the error tend to zero as the ball grows. Thus $L_q v = 0$ distributionally. Its classical principal value exists and is continuous: local symmetric increments are bounded by $C|h|^2$ on each compact core, and the exterior integral is controlled by the weighted norm and a separated kernel. These two integrals converge because $p < 2$ and the tail is integrable. The smooth limit v lies in the same locally C^2 , weighted L^1 operator class. The pairing identity proved for that class in Lemma 8.1 identifies this continuous principal value with its distributional operator. Hence the continuous function $(-\Delta)^s v + (3q_s^2 - 1)v$ has zero distribution and vanishes pointwise.

Set $F_j = R_j f_j$. The exact zero equation is $G_j(x', F_j(x')) = 0$, where

$$G_j(x', a) = R_j q_s(a/R_j) + v_j(x', a/R_j).$$

On each fixed horizontal disk the initial Hessian bound and $f_j(0) = Df_j(0) = 0$ uniformly bound F_j . On bounded (x', a) sets, the chain rule and the smooth convergence of v_j give

$$G_j \longrightarrow \phi(0)a + v(x', 0) \quad \text{in every finite } C^k, \quad \partial_a G_j = \phi(a/R_j) + R_j^{-1} \partial_t v_j(x', a/R_j) \longrightarrow \phi(0) > 0.$$

Work on a slightly larger fixed horizontal disk and choose a bounded a interval containing all the supplied roots. For large j , $\partial_a G_j \geq \phi(0)/2$ there, so each supplied root is unique in that interval. The implicit-function theorem identifies its smooth representative. Uniform convergence first gives convergence of the roots; then $DF_j = -(\partial_a G_j)^{-1} D_{x'} G_j$ and the higher implicit chain-rule identities give convergence of each successive derivative. We obtain

$$R_j f_j \longrightarrow P_* = -v(x', 0)/\phi(0) \quad \text{in every } C_{\text{loc}}^k, \quad |D^2 P_*| \leq C \quad \text{on } \mathbb{R}^2.$$

The last bound passes pointwise on every fixed disk from $|D^2 F_j| \leq C$, with the same constant. The normalization gives $P_*(0) = DP_*(0) = 0$. At this stage P_* is a smooth function with a bounded Hessian; its polynomial character is established below.

Return to the normal equations on a fixed physical core. The C^3 bound for F_j gives $\|f_j\|_{C^3} = O(R_j^{-1})$ on a slightly larger fixed horizontal disk. Write the normal map as

$$\mathcal{N}_j(b, a) = \left(b - \frac{a Df_j(b)}{\sqrt{1 + |Df_j(b)|^2}}, f_j(b) + \frac{a}{\sqrt{1 + |Df_j(b)|^2}} \right).$$

On bounded (b, a) sets it is the identity plus $O_{C^2}(R_j^{-1})$. Its inverse is the previously identified footprint/distance map. The inverse derivative formula gives a uniformly bounded first inverse derivative, and $D^2 \mathcal{N}_j^{-1} = -D\mathcal{N}_j^{-1} D^2 \mathcal{N}_j [D\mathcal{N}_j^{-1}, D\mathcal{N}_j^{-1}]$ gives the second inverse derivative. Hence $b - x' = O_{C^2}(R_j^{-1})$ and $\delta_j - t = O_{C^2}(R_j^{-1})$. Now the chain and product rules, using $\|f_j\|_{C^3} = O(R_j^{-1})$, yield

$$f_j(b) - f_j(x') = O_{C^2}(R_j^{-2}), \quad (1 + |Df_j(b)|^2)^{-1/2} - 1 = O_{C^2}(R_j^{-2}).$$

For example, the second derivative of the first difference consists of $D^2 f_j(b) - D^2 f_j(x')$, terms with $Db - Dx'$, and $Df_j(b)D^2 b$, each $O(R_j^{-2})$. The second normal equation therefore implies

$$\delta_j = t - f_j(x') + O_{C^2}(R_j^{-2}), \quad R_j(\delta_j - t) \longrightarrow -P_* \quad \text{in } C_{\text{loc}}^2.$$

Taylor's formula for the smooth layer, with these C^2 bounds, gives $R_j(U_j - q_s(t)) \rightarrow -P_*\phi(t)$ in C_{loc}^2 . It follows that $R_j(u_j - U_j) \rightarrow W = v + P_*\phi$ locally in C^2 . Every fixed core eventually lies in the pinned ball B_{CL_j} , where the same constant bounds $R_j(u_j - U_j)$. Thus W is bounded on all of $\mathbb{R}^3$, and it is smooth by its formula in terms of v , P_* and ϕ . Its trace is zero by the definition $P_* = -v(x', 0)/\phi(0)$. We obtain

$$v(x', t) = -P_*(x')\phi(t) + W(x', t), \quad \|W\|_\infty \leq C, \quad W(x', 0) = 0. \quad (8.6)$$

For each $h \in \mathbb{R}^2$, denote the tangential second difference by $\Delta_h^2 v(x', t) = v(x' + h, t) + v(x' - h, t) - 2v(x', t)$. It is bounded: $|\Delta_h^2 P_*| \leq \|D^2 P_*\|_\infty |h|^2$ and $|\Delta_h^2 W| \leq 4\|W\|_\infty$ in (8.6). Tangential translations commute with L_q , so $L_q \Delta_h^2 v = 0$. Lemma 6.2 gives $\Delta_h^2 v = c_h \phi$. Evaluating at $t = 0$ yields

$$\Delta_h^2 P_*(x') = -c_h \quad \text{for every } x'.$$

Compare this identity at two arbitrary base points, set $h = \epsilon e$, divide their difference by ϵ^2 , and let $\epsilon \rightarrow 0$. Taylor's formula gives the same value of $e^T D^2 P_*(x') e$ at both points for every direction e ; polarization therefore gives a constant Hessian. Thus P_* is a polynomial of degree at most two, with its constant and linear parts zero. Substituting back gives $\Delta_h^2 W = 0$ for every h . Dividing by ϵ^2 again shows that each tangential Hessian of the smooth function W is zero. For each t it is affine in x' , and global boundedness eliminates its linear part. Hence $W = W(t)$.

Equation (8.1) gives $L_q W = -(\Delta P_*) \Xi_p$ on $\mathbb{R}^3$. We use the normalized cylindrical reduction distributionally, so no global derivative bound for W is needed here. Take $\eta \in C_c^\infty(\mathbb{R}^2)$ with $\int \eta = 1$ and $\psi \in C_c^\infty(\mathbb{R})$. For the compact smooth product $F = \eta\psi$, the L^1 norms of its symmetric second increments are bounded by $\min\{|H|^2 \|D^2 F\|_1, 4\|F\|_1\}$. Their integral against $|H|^{-3-p}$ is finite, since $0 < p < 2$. Thus Fubini is legitimate in the normalized symmetric-increment formula. Integration in x' leaves the second increment of ψ , because every horizontal translate of η has mass one, and polar integration gives

$$c_{3,s} \int_{\mathbb{R}^2} (|y|^2 + h^2)^{-(3+p)/2} dy = c_{1,s} |h|^{-1-p} \quad (h \neq 0).$$

Consequently

$$\int_{\mathbb{R}^2} (-\Delta)^s (\eta\psi)(x', t) dx' = (-\partial_t^2)^s \psi(t).$$

The same increment estimate proves that $(-\Delta)^s F$ belongs to $L^1(\mathbb{R}^3)$. Thus boundedness of $W(t)$ permits Fubini in this product-test identity. Testing the equation for W in this way gives

$$L_0 W = -(\Delta P_*) \Xi_p, \quad L_0 = (-\partial_t^2)^s + 3q_s^2 - 1.$$

A cutoff pairing with ϕ is legitimate without any L^2 assumption. The layer mass gives $\phi \in L^1$. For derivative orders $k = 2, 3$, [12, Theorem 1.2] applies with $\mathcal{W}(r) = (1-r^2)^2/4$: this potential is even, nonnegative, smooth, has exactly the wells ± 1 , and satisfies $\mathcal{W}''(\pm 1) = 2 > 0$. Their constant (2.2) equals our $c_{1,s}$, so their profile is our normalized layer. It gives $|\phi'(t)| \leq C(1+|t|)^{-2-p}$ and $|\phi''(t)| \leq C(1+|t|)^{-3-p}$, hence both derivatives are integrable. Choose $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[-1, 1]$, and $\chi = 0$ outside $[-2, 2]$, and set $\chi_N(t) = \chi(t/N)$ and $\psi_N = \phi\chi_N$. The terms with a derivative of χ_N have L^1 bounds $CN^{-1}\|\phi\|_1$, $CN^{-1}\|\phi'\|_1$, or $CN^{-2}\|\phi\|_1$. The other terms converge by dominated convergence, so $\psi_N \rightarrow \phi$ in $W^{2,1}$. Differentiation of the layer equation gives $L_0\phi = 0$.

For $\psi \in W^{2,1}(\mathbb{R})$, symmetric increments satisfy

$$\|2\psi - \psi(\cdot + h) - \psi(\cdot - h)\|_1 \leq \min\{|h|^2 \|\psi''\|_1, 4\|\psi\|_1\}.$$

Integrating against $|h|^{-1-p}$, separately on $|h| < 1$ and $|h| \geq 1$, proves

$$\|(-\partial_t^2)^s \psi\|_1 \leq C_s (\|\psi''\|_1 + \|\psi\|_1).$$

Both integrals converge, since $p < 2$ and $p > 0$. The bounded potential in L_0 therefore gives $L_0\psi_N \rightarrow L_0\phi = 0$ in L^1 . Testing the distributional equation for bounded W against the compact smooth function ψ_N yields the precise pairing

$$\int_{\mathbb{R}} W L_0 \psi_N dt = -(\Delta P_*) \int_{\mathbb{R}} \Xi_p \phi \chi_N dt.$$

The left side tends to zero by the L^1 convergence just proved. The right side tends to $-(\Delta P_*) \sigma_p$ by dominated convergence, since $\Xi_p \phi \in L^1$. Positivity of σ_p forces $\Delta P_* = 0$. Now $W(t)$, viewed on $\mathbb{R}^3$, is a bounded Jacobi field. Lemma 6.2 gives $W = c\phi$, and $W(0) = 0$, $\phi(0) > 0$ imply $c = 0$. This proves (8.3).

At the tangent origin $Df_j(0) = 0$, so the mean-curvature formula is exactly $R_j H_j(0) = \Delta(R_j f_j)(0) \rightarrow 0$. If (8.4) failed for some fixed M , there would be zeros $Q_j \in B_{MR_j}(P_j)$ along a subsequence with $R_j |H_{\Sigma_{Q_j}}(Q_j)| \geq \epsilon_0 > 0$. The stated uniform hypotheses supply the same constants, buffers, and vanishing normalized exterior tails after recentering at Q_j . Applying the preceding compactness argument to that sequence gives a further subsequence with rescaled mean curvature tending to zero, a contradiction. This proves the asserted uniformity. $\square$

The comparison in Lemma 7.3 meets these hypotheses with $L_j = cR_j^{2/(p+1)}$, since $R_j L_j^{-p} = c^{-p} R_j^{(1-p)/(p+1)} \rightarrow 0$. For condition (8.2), the logarithmic lower bound also suffices. Let d_j be the actual isolation length supplied by the logarithmic gap estimate, so $d_j^p \geq c_{\log} R_j \log R_j$ when $d_j \leq R_j$, and take $L_j = c \min\{d_j, R_j\}$. Then in this case $R_j L_j^{-p} \leq C/\log R_j$; when $d_j > R_j$, the expression is $c^{-p} R_j^{1-p} \leq C/\log R_j$ for large j . Both cases tend to zero.

For the following corollary, a compact central phase interval means $I \Subset (-1, 1)$. Curvature tensors on its different phase graphs are compared in the common horizontal coordinates, by pulling back under $x' \mapsto (x', f_{j,\tau}(x'))$, with the increasing normal.

Corollary 8.3 (Transfer between central phases). *Under Theorem 8.2, let $t_\tau = q_s^{-1}(\tau)$ on a fixed compact central phase interval. Then on the selected branch,*

$$R_j(f_{j,\tau}(x') - t_\tau) \longrightarrow P_*(x') \quad \text{in } C_{\text{loc}}^k, \quad R_j(A_{j,\tau} - A_{j,0}) \longrightarrow 0,$$

uniformly in τ . Also

$$R_j(u_j - q_s(\delta_{\Sigma_j})) \longrightarrow 0 \quad \text{locally in } C^2. \quad (8.7)$$

These conclusions give qualitative convergence as R_j grows.

Proof. Enlarge I , if necessary, to a compact interval $J \Subset (-1, 1)$ containing I and 0. The values $t_\tau = q_s^{-1}(\tau)$ for $\tau \in J$ form a compact set, and $\min_{\tau \in J} \phi(t_\tau) > 0$. For a scaled phase height $a = R_j(f_{j,\tau} - t_\tau)$, the exact phase equation is

$$G_{j,\tau}(x', a) := R_j[q_s(t_\tau + a/R_j) - q_s(t_\tau)] + v_j(x', t_\tau + a/R_j) = 0.$$

By (8.3), on bounded (x', a) sets this function converges, uniformly for $\tau \in J$ and with every fixed number of derivatives, to $\phi(t_\tau)(a - P_*(x'))$. Its a derivative is bounded below by a fixed positive constant for large j . On a fixed disk choose $A > \sup |P_*| + 1$ using a slightly larger disk. For large j , the values at $a = \pm A$ have opposite signs uniformly in $x', \tau \in J$, and the positive a derivative gives exactly one root between them. The implicit-function theorem gives its smooth dependence on x' and τ . At $\tau = 0$, uniqueness identifies this root with the selected zero graph; its continuation through the connected interval J is therefore the selected phase branch. The differentiated implicit identities give

$$f_{j,\tau} = t_\tau + R_j^{-1}P_* + o(R_j^{-1}) \quad \text{in every local } C^k \text{ norm, uniformly in } \tau \in J.$$

The constants follow from one compact phase interval and a slightly larger fixed spatial core; no threshold depends on the individual phase.

In the stated graph coordinates, the metric, second fundamental form, and shape operator are

$$(g_{j,\tau})_{ab} = \delta_{ab} + \partial_a f_{j,\tau} \partial_b f_{j,\tau}, \quad (a_{j,\tau})_{ab} = \frac{\partial_{ab} f_{j,\tau}}{\sqrt{1 + |Df_{j,\tau}|^2}}, \quad A_{j,\tau} = g_{j,\tau}^{-1} a_{j,\tau}.$$

For every fixed local C^k norm, the graph convergence with $k+2$ derivatives gives $g_{j,\tau} = I + O(R_j^{-2})$ and $A_{j,\tau} = D^2 f_{j,\tau} + O(R_j^{-3})$, uniformly in τ . Consequently $R_j A_{j,\tau} \rightarrow D^2 P_*$ in these coordinates, and subtracting the limit for $\tau = 0$ proves the curvature assertion. These coordinates and the induced metrics are uniformly equivalent.

On fixed physical cores, the preceding geometry identifies the cutoff distance with the actual zero-surface distance. The local C^1 convergence $u_j \rightarrow q_s$ gives $\partial_3 u_j > 0$ near the selected zero graph for large j , so the upward sign is its increasing-normal sign. Fix a compact physical core K and take any subsequence of the harmonic-limit subsequence. The preceding Jacobi compactness argument applies to this restricted sequence and supplies a further subsequence; uniqueness of the already known limit gives $v = -P_*\phi$ there. The profile expansion then applies on that further subsequence, with $P = -v(x', 0)/\phi(0) = P_*$ and $W = v + P\phi = 0$. It gives $R_j(u_j - q_s(\delta_{\Sigma_j})) \rightarrow 0$ in $C^2(K)$. If this convergence failed on the whole harmonic-limit subsequence, choose a subsequence on which the $C^2(K)$ norm stays above a fixed positive constant. The further-subsequence conclusion just obtained contradicts that choice. Since K was arbitrary, this proves (8.7) on the entire harmonic-limit subsequence. $\square$

This transfers curvature across a bounded physical phase displacement. The persistence of a nonzero rescaled curvature on a region of radius proportional to R_j requires an additional argument.

9. A TRACE OPERATOR WITH A FINITE TANGENTIAL SECOND MOMENT

The curvature argument uses an operator on the zero-level trace whose large-scale behavior is of second order. We first construct that operator and the positive extension kernel which connects it to the actual equation. Throughout this section the ambient dimension is three and the tangential variable is $x \in \mathbb{R}^2$. Write

$$V_q(t) = 3q_s(t)^2 - 1, \quad L_0 = |D_t|^p + V_q,$$

$$\Phi = \frac{\phi}{\|\phi\|_2}, \quad \Phi_0 = \Phi(0) > 0, \quad L_k = (D_t^2 + |k|^2)^s + V_q.$$

We use $\widehat{f}(\xi) = (2\pi)^{-d/2} \int e^{-ix \cdot \xi} f(x) dx$ in dimension d . Thus a convolution multiplier m has physical kernel $(2\pi)^{-d/2} \mathcal{F}^{-1}m$. Complex inner products are conjugate-linear in the first argument. For $v \in H^s$ and $f \in H^{-s}$ we use $\langle v, f \rangle = \int \widehat{v} \widehat{f}$ and $\langle f, v \rangle = \overline{\langle v, f \rangle}$; in particular $\langle \delta_0, v \rangle = v(0)$ and $\langle v, \delta_0 \rangle = \overline{v(0)}$. Write $a_z(v, w) = \langle v, T_q(z)w \rangle$ for the corresponding form. The layer tails give $\phi \in L^1 \cap L^2$. The normal cutoff result in Section 8 already supplies smoothness and the distributional identity $|D_t|^p \phi = -V_q \phi$. Since $V_q \phi \in L^2$, Fourier transformation gives $\phi \in H^p \subset H^s$.

Lemma 9.1. *The form kernel of L_0 is $\mathbb{C}\Phi$, and there is $\lambda_0 > 0$ such that*

$$\langle L_0 v, v \rangle \geq \lambda_0 \|v\|_2^2, \quad \|v\|_{H^s}^2 \leq C \langle L_0 v, v \rangle \quad (v \perp \Phi).$$

On the closed subspace $\{v \in H^s(\mathbb{R}) : v(0) = 0\}$, the forms of L_k are uniformly coercive in H^s for bounded k , and $\langle L_k v, v \rangle \geq c_ \|v\|_2^2$ for every k . The full form of $L_q = (-\Delta)^s + V_q(t)$ is coercive in $H^s(\mathbb{R}^3)$ on its zero-trace subspace.*

Proof. The Fourier–Gagliardo identity for the coefficient fixed in Section 2 is, for compact smooth w ,

$$\int |\tau|^p |\widehat{w}(\tau)|^2 d\tau = \frac{c_{1,s}}{2} \iint \frac{|w(t) - w(r)|^2}{|t - r|^{1+p}} dt dr.$$

The equivalence of the normalized Fourier and singular-integral operators, truncated symmetrization and Plancherel give this identity; smoothness makes the near diagonal integrable for $p < 2$, and compact support handles its tail. The ground-state calculation in Section 4, applied separately to the real and imaginary parts, gives for every compact smooth complex w

$$\langle L_0 w, w \rangle = \frac{c_{1,s}}{2} \iint \frac{\phi(t)\phi(r)}{|t - r|^{1+p}} \left| \frac{w(t)}{\phi(t)} - \frac{w(r)}{\phi(r)} \right|^2 dt dr.$$

We write $E_k(v) = \langle L_k v, v \rangle$ and $H_D = \{v \in H^s(\mathbb{R}) : v(0) = 0\}$.

For $w \in H^s$, take a sequence in C_c^∞ converging strongly in H^s , and a subsequence converging almost everywhere. The bounded potential makes E_0 continuous in H^s . Fatou’s lemma bounds the nonnegative quotient integral for w by $E_0(w)$, proving $E_0(w) \geq 0$. If $E_0(w) = 0$, then $w(t)/\phi(t) = w(r)/\phi(r)$ for almost every pair (t, r) , since the kernel weight is positive. Fubini gives a single complex constant, so $w \in \mathbb{C}\Phi$. Conversely the Jacobi equation and $\Phi \in H^p$ from the setting give $E_0(\Phi) = 0$ and $L_0 \Phi = 0$ as a form operator. Nonnegativity also implies that the zero-energy vectors are exactly the form kernel.

For the gap write $V_q = 2 - U_q$, where $U_q = 3(1 - q^2) \geq 0$ is bounded and tends to zero at infinity by the reference-layer result in Section 2. If the L^2 gap failed, choose $v_j \perp \Phi$, $\|v_j\|_2 = 1$, $E_0(v_j) \rightarrow 0$. The kinetic term is bounded because V_q is bounded, so v_j is bounded in H^s . After weak extraction in H^s , Rellich compactness on each bounded interval gives local strong L^2 convergence to v . Cutting off U_q outside a large interval, with error at most $\sup_{|t| > M} U_q (\|v_j\|_2^2 + \|v\|_2^2)$, gives $\int U_q |v_j|^2 \rightarrow \int U_q |v|^2$. The expression $E_0(v) = \| |D_t|^s v \|_2^2 + 2\|v\|_2^2 - \int U_q |v|^2$ and lower semicontinuity imply $0 \leq E_0(v) \leq 0$. Weak convergence preserves $v \perp \Phi$, so the kernel conclusion makes $v = 0$. But then $E_0(v_j) = \| |D_t|^s v_j \|_2^2 + 2 - o(1) \geq 2 - o(1)$, a contradiction. This proves the L^2 gap. Finally $\| |D_t|^s v \|_2^2 \leq E_0(v) + \|V_q\|_\infty \|v\|_2^2$ upgrades it to the H^s estimate on $\Phi^\perp$.

For $v \in H_D$, write $v = a\Phi + w$, with $w \perp \Phi$. Bounded evaluation gives $|a|\Phi_0 = |w(0)| \leq C\|w\|_{H^s}$. The Jacobi equation gives $E_0(v) = E_0(w)$, so $\|v\|_{H^s}^2 \leq CE_0(v)$. The multiplier D_k is

nonnegative; therefore $E_k(v) \geq E_0(v) \geq c_* \|v\|_2^2$ for every k . Writing $K_k(v) = \int (\tau^2 + |k|^2)^s |\widehat{v}|^2$ gives $K_k(v) \leq E_k(v) + \|V_q\|_\infty \|v\|_2^2$. Since $(1 + \tau^2 + |k|^2)^s \leq C[1 + (\tau^2 + |k|^2)^s]$, the asserted all- k weighted coercivity follows.

To pass to dimension three, the tangential Fourier transform of $v \in H^s(\mathbb{R}^3)$ has almost every normal slice in $H^s(\mathbb{R})$ by Fubini and the Fourier Sobolev norm. For smooth v , the tangential transform of its trace is the evaluation at zero of this slice. For general v , approximate strongly in H^s and use both the bounded trace map and the slice estimate, obtained by Fourier inversion and Cauchy–Schwarz,

$$|\widehat{v}_x(k, 0)|^2 \leq C(1 + |k|^2)^{1/2-s} \int_{\mathbb{R}} (1 + \tau^2 + |k|^2)^s |\widehat{v}(k, \tau)|^2 d\tau.$$

After multiplying by $(1 + |k|^2)^{s-1/2}$ and integrating in k , this controls the convergence of the slice traces in $H^{s-1/2}(\mathbb{R}^2)$. The same trace identity follows by density. A zero trace thus puts almost every slice in H_D . Integrating the all- k estimate and applying Plancherel gives the full $H^s(\mathbb{R}^3)$ coercivity. $\square$

For $z \in \mathbb{C} \setminus (-\infty, 0]$, use the principal power and set

$$T_q(z) = (D_t^2 + z)^s + V_q.$$

The inverse from H^{-s} to H^s will be constructed below. When it exists, write

$$G_q(z) = \langle \delta_0, T_q(z)^{-1} \delta_0 \rangle, \quad \theta_q(z) = \frac{\Phi_0^2}{G_q(z)}.$$

The second notation is used after nonvanishing is proved. Point evaluation is bounded on H^s because $s > 1/2$, equivalently $\widehat{\delta}_0 = (2\pi)^{-1/2}$ belongs to the weighted space H^{-s} . For $w = T_q(z)^{-1} \delta_0$, our convention means $G_q(z) = w(0)$. This is a linear evaluation of w and hence is holomorphic when the inverse is. The operator $\Theta_q = \theta_q(-\Delta_x)$ initially denotes the corresponding Fourier multiplier on Schwartz functions.

Lemma 9.2. *The family $T_q(z)$ is invertible on the slit plane. The function G_q is Stieltjes and never zero, and θ_q is a complete Bernstein function. There are $a_*, b_* > 0$ such that, with $\alpha = (p-1)/2$ and $\beta = (p+1)/2$,*

$$\theta_q(\lambda) = a_* \lambda - b_* \lambda^\beta + o(\lambda^\beta) \quad (\lambda \downarrow 0), \quad (9.1)$$

$$\theta_q(\lambda) \sim C_\infty \lambda^\alpha \quad (\lambda \rightarrow \infty). \quad (9.2)$$

The operator $\Theta_q = \theta_q(-\Delta_x)$ has a positive radial kernel j_q satisfying

$$j_q(x) \asymp \begin{cases} |x|^{-2-(p-1)}, & 0 < |x| \leq 1, \\ |x|^{-2-(p+1)}, & |x| \geq 1, \end{cases} \quad (9.3)$$

$$|D^\ell j_q(x)| \leq C_\ell |x|^{-3-p-\ell} \quad (|x| \geq 1), \quad \int |x|^2 j_q(x) dx = 4a_*. \quad (9.4)$$

Proof. For each compact set K in the slit domain, the free multiplier satisfies

$$c_K(1 + \tau^2)^s \leq |(\tau^2 + z)^s + 2| \leq C_K(1 + \tau^2)^s \quad (z \in K).$$

For large $|\tau|$, compare with $|\tau|^{2s}$. On a bounded τ interval, continuity and $|\arg((\tau^2 + z)^s)| < \pi s < \pi$ exclude cancellation with 2; compactness supplies the lower bound. Thus $F(z) = (D_t^2 + z)^s + 2$ is an isomorphism $H^s \rightarrow H^{-s}$, locally uniformly in z . To check operator-norm holomorphy, choose a compact neighborhood K' of K inside the slit plane and set $\delta = \text{dist}(K', (-\infty, 0]) > 0$. For $z \in K'$ and every real τ , $|\tau^2 + z| \geq \delta$. Thus the exact derivatives

$$\partial_z(\tau^2 + z)^s = s(\tau^2 + z)^{s-1}, \quad \partial_z^2(\tau^2 + z)^s = s(s-1)(\tau^2 + z)^{s-2}$$

have modulus at most $s\delta^{s-1}$ and $s(1-s)\delta^{s-2}$, respectively. In particular both are uniformly bounded after division by $(1 + \tau^2)^s$. Choose $|h|$ small enough that $z + uh \in K'$ for $z \in K$,

$0 \leq u \leq 1$. Taylor's integral remainder then gives

$$\sup_{\tau \in \mathbb{R}} \frac{|(\tau^2 + z + h)^s - (\tau^2 + z)^s - hs(\tau^2 + z)^{s-1}|}{(1 + \tau^2)^s} \leq \frac{1}{2}s(1-s)\delta^{s-2}|h|^2.$$

This supremum is exactly the relevant multiplier norm from H^s to H^{-s} , so it proves operator-norm differentiability and holomorphy of F . Its uniformly bounded inverse is also holomorphic: locally $F(z)^{-1}F(z+h) = I + O_K(|h|)$ on H^s , and the inverse Neumann series converges in operator norm.

The multiplier $V_q - 2 = -U_q$ is compact from H^s to H^{-s} . On a fixed interval this follows by Rellich compactness into L^2 , followed by the bounded maps of multiplication and $L^2 \rightarrow H^{-s}$. The discarded part has operator norm at most $C \sup_{|t|>M} U_q(t) \rightarrow 0$. Hence $T_q(z)$ is a Fredholm operator of index zero, being $F(z)$ times identity plus a compact operator. For $\text{Im } z > 0$,

$$\text{Im } a_z(w, w) = \int \text{Im}(\tau^2 + z)^s |\hat{w}(\tau)|^2 d\tau > 0 \quad (w \neq 0).$$

The integrand multiplier is strictly positive at every τ ; the sign is reversed in the lower half-plane. This excludes a kernel off the real axis. On the positive real axis, $L_0 \geq 0$ by Lemma 9.1, while $(\tau^2 + \lambda)^s - |\tau|^{2s} > 0$ for every τ . Their sum has strictly positive energy for every nonzero w , so it also has no kernel. The Fredholm alternative proves invertibility everywhere in the slit plane. Local inversion of bounded operators proves holomorphy of the inverse there.

For $w = T_q(z)^{-1}\delta_0$, the defining equation reads

$$a_z(v, w) = \overline{v(0)} \quad (v \in H^s).$$

In particular

$$G_q(z) = \langle T_q(z)w, w \rangle = \overline{a_z(w, w)}.$$

Thus $\text{Im } G_q(z) < 0$ in the upper half-plane, with the reverse strict sign below, and $G_q(\lambda) > 0$ for $\lambda > 0$. The sign uses the conjugation in this displayed identity; $\langle w, T_q(z)w \rangle$ has the opposite imaginary part. No value of G_q is zero. On the positive axis, form Cauchy–Schwarz applied to Φ and w gives

$$\Phi_0^2 \leq G_q(\lambda)a_\lambda(\Phi, \Phi), \quad 0 < a_\lambda(\Phi, \Phi) \leq \lambda^s.$$

Here $L_0\Phi = 0$, $\|\Phi\|_2 = 1$, and concavity gives $0 \leq (\tau^2 + \lambda)^s - |\tau|^{2s} \leq \lambda^s$. Consequently $G_q(\lambda) \geq \Phi_0^2 \lambda^{-s} \rightarrow \infty$ at zero. The exact hypotheses of [26, Theorem 2.11, preprint version] now hold: holomorphy on the slit plane, a positive extended limit at zero and $\text{Im } z \text{Im } G_q(z) \leq 0$. It follows that G_q is Stieltjes. Its nonzero reciprocal, and hence Φ_0^2/G_q , is complete Bernstein by [26, Theorem 2.10, preprint version].

For the precise low-frequency correction, put $\rho = |k|$, $D_k = L_k - L_0$, and let $P_\perp$ be the L^2 projection onto $\Phi^\perp$. It acts boundedly on H^s and extends to H^{-s} by duality, since $\Phi \in H^s$. Define

$$A_k = P_\perp L_k P_\perp, \quad b_k = P_\perp D_k \Phi, \quad \lambda_k = \langle \Phi, D_k \Phi \rangle, \quad S_k = \lambda_k - \langle b_k, A_k^{-1} b_k \rangle.$$

The domain of A_k is $H^s \cap \Phi^\perp$ and its codomain is the dual subspace annihilating Φ . The gap in Lemma 9.1 and $D_k \geq 0$ make its form uniformly coercive on this space for $\rho \leq 1$; Lax–Milgram therefore supplies the uniformly bounded inverse. Put

$$e_0 = \delta_0 - \Phi_0 \Phi, \quad g_k = \langle e_0, A_k^{-1} e_0 \rangle, \quad c_k = \langle e_0, A_k^{-1} b_k \rangle.$$

All these quantities are real: the operators and data preserve real functions. For $k \neq 0$, $\Phi - A_k^{-1} b_k$ is nonzero, since its Φ coefficient is one, and its L_k energy is $S_k > 0$. To calculate the diagonal inverse, write $T_q(\rho^2)^{-1}\delta_0 = a\Phi + v$, $v \perp \Phi$. The two block equations are

$$a\lambda_k + \langle b_k, v \rangle = \Phi_0, \quad A_k v + ab_k = e_0.$$

They give $a = (\Phi_0 - c_k)/S_k$ and $v = A_k^{-1} e_0 - a A_k^{-1} b_k$. Evaluating at zero yields the exact Schur identity

$$G_q(\rho^2) = g_k + \frac{(\Phi_0 - c_k)^2}{S_k}.$$

Uniform inverse bounds, Sobolev evaluation and the H^{-s} pairing give

$$0 \leq g_k \leq C, \quad |c_k| + \|b_k\|_2 \leq CB_p(\rho), \quad B_p(\rho) = \begin{cases} \rho^{p+1/2}, & p < 3/2, \\ \rho^2 \sqrt{1 + |\log \rho|}, & p = 3/2, \\ \rho^2, & p > 3/2. \end{cases}$$

Indeed, $\widehat{\Phi} \in L^\infty \cap L^2$. On $|\tau| \leq \rho$ the symbol of D_k is at most ρ^p ; on $\rho < |\tau| \leq 1$ it is at most $C\rho^2|\tau|^{p-2}$ by the mean value theorem; on $|\tau| > 1$ it is at most $C\rho^2$, since $p < 2$. Thus

$$\|D_k \Phi\|_2^2 \leq C \left[\rho^{2p+1} + \rho^4 \int_\rho^1 t^{2p-4} dt + \rho^4 \right].$$

Projection does not increase the L^2 norm. Also $|c_k| \leq C\|A_k^{-1}b_k\|_{H^s} \leq C\|b_k\|_2$, and g_k is bounded by $C\|e_0\|_{H^{-s}}^2$.

For λ_k , subtract its quadratic term and set $\tau = \rho z$. Concavity makes

$$F_p(z) = (1 + z^2)^{p/2} - |z|^p - \frac{p}{2}|z|^{p-2}$$

strictly negative for $z \neq 0$. It is integrable at zero because $p > 1$, and at infinity because $F_p(z) = O(|z|^{p-4})$. The exact rescaled remainder is $\rho^{p+1} \int F_p(z) |\widehat{\Phi}(\rho z)|^2 dz$. Since $\Phi \in L^1$, its transform is continuous at zero and bounded. Dominated convergence gives

$$\lambda_k = a_* \rho^2 - b_* \rho^{p+1} + o(\rho^{p+1}), \quad a_* = \frac{p}{2} \int |\tau|^{p-2} |\widehat{\Phi}(\tau)|^2 d\tau, \\ b_* = -|\widehat{\Phi}(0)|^2 \int F_p(z) dz > 0.$$

The coefficient a_* is finite at zero because $p > 1$ and $\widehat{\Phi}$ is bounded, and finite at infinity because $p < 2$ and $\widehat{\Phi} \in L^2$. It is positive since $\Phi \neq 0$. Likewise $\widehat{\Phi}(0) = (2\pi)^{-1/2} \int \Phi > 0$. The inverse bound gives $\langle b_k, A_k^{-1}b_k \rangle = O(B_p(\rho)^2)$. Since $0 < S_k \leq \lambda_k = O(\rho^2)$ and $c_k \rightarrow 0$, the Schur identity gives

$$\theta_q(\rho^2) = \frac{\Phi_0^2 S_k}{(\Phi_0 - c_k)^2 + g_k S_k} = S_k + O(\rho^2 B_p(\rho) + \rho^4).$$

The denominator stays bounded away from zero for small ρ . In each of the three ranges of p , $B_p(\rho)^2 + \rho^2 B_p(\rho) + \rho^4 = o(\rho^{p+1})$. Therefore the Schur terms change neither displayed coefficient, proving (9.1).

For large positive λ , put $H_\lambda = (D_t^2 + \lambda)^s$ and $\epsilon_\lambda = \|V_q\|_\infty \lambda^{-s} < 1$. Because $H_\lambda \geq \lambda^s I$, form order gives $(1 - \epsilon_\lambda)H_\lambda \leq T_q(\lambda) \leq (1 + \epsilon_\lambda)H_\lambda$. For a positive coercive form, its inverse diagonal is the supremum of $|v(0)|^2$ divided by its energy over nonzero $v \in H^s$; this follows by form Cauchy–Schwarz, with equality at the inverse of δ_0 . Hence

$$\frac{G_{\text{free}}(\lambda)}{1 + \epsilon_\lambda} \leq G_q(\lambda) \leq \frac{G_{\text{free}}(\lambda)}{1 - \epsilon_\lambda}, \quad G_{\text{free}}(\lambda) = \frac{1}{2\pi} \int (\tau^2 + \lambda)^{-s} d\tau.$$

The free formula follows by Fourier inversion of $\widehat{\delta}_0 = (2\pi)^{-1/2}$, and its integral is finite because $p > 1$. Scaling gives

$$G_q(\lambda) \sim c_p \lambda^{(1-p)/2}, \quad c_p = \frac{1}{2\pi} \int (1 + \tau^2)^{-p/2} d\tau, \quad C_\infty = \Phi_0^2 / c_p,$$

which proves (9.2).

The complete Bernstein representation in [26, Section 2, formula (2.7) and the definition of completeness] has a nonnegative completely monotone density μ_q . Its killing constant is $\theta_q(0+) = 0$ by (9.1); its linear drift is $\lim_{\lambda \rightarrow \infty} \theta_q(\lambda)/\lambda = 0$ by (9.2), since $0 < \alpha < 1$. Thus

$$\theta_q(\lambda) = \int_0^\infty (1 - e^{-\lambda t}) \mu_q(t) dt, \quad \int_0^\infty t \mu_q(t) dt = a_*.$$

For the second identity divide the first by λ and use monotone convergence as $\lambda \downarrow 0$; the integrand $(1 - e^{-\lambda t})/\lambda$ increases to t , and the left-hand limit is a_* . In particular all first moments below

are finite. Differentiation at a positive λ is legitimate by domination: the Lévy condition controls $t\mu_q(t)$ near zero, and the exponential controls every polynomial factor at infinity.

Put $D(\lambda) = a_*\lambda - \theta_q(\lambda)$. Then $D' \geq 0$ increases and $D''(\lambda) = \int t^2 e^{-\lambda t} \mu_q(t) dt \geq 0$ decreases. For $b > 1$, convexity gives

$$\frac{D(\lambda) - D(\lambda/b)}{(1 - b^{-1})\lambda} \leq D'(\lambda) \leq \frac{D(b\lambda) - D(\lambda)}{(b - 1)\lambda}.$$

Insert $D(\lambda) \sim b_*\lambda^\beta$, let first $\lambda \downarrow 0$, then $b \downarrow 1$, to obtain $D'(\lambda) \sim b_*\beta\lambda^{\beta-1}$. Since the derivative of D' decreases, its secants give

$$\frac{D'(b\lambda) - D'(\lambda)}{(b - 1)\lambda} \leq D''(\lambda) \leq \frac{D'(\lambda) - D'(\lambda/b)}{(1 - b^{-1})\lambda}.$$

The same limits yield $D''(\lambda) \sim b_*\beta(\beta - 1)\lambda^{\beta-2}$. At the other endpoint θ_q is concave, so

$$\frac{\theta_q(b\lambda) - \theta_q(\lambda)}{(b - 1)\lambda} \leq \theta'_q(\lambda) \leq \frac{\theta_q(\lambda) - \theta_q(\lambda/b)}{(1 - b^{-1})\lambda}.$$

First let $\lambda \rightarrow \infty$, then $b \downarrow 1$, to get $\theta'_q(\lambda) \sim C_\infty \alpha \lambda^{\alpha-1}$. These arguments use monotonicity rather than differentiation of an asymptotic remainder.

We recover the density at both endpoints by one positive-measure argument. For large time use

$$\nu_T(du) = T^{\beta+1} u^2 \mu_q(Tu) du, \quad \int e^{-au} \nu_T(du) = T^{\beta-2} D''(a/T) \longrightarrow b_*\beta(\beta - 1)a^{\beta-2} \quad (T \rightarrow \infty).$$

For small time use instead

$$\tilde{\nu}_T(du) = T^{\alpha+1} u \mu_q(Tu) du, \quad \int e^{-au} \tilde{\nu}_T(du) = T^{\alpha-1} \theta'_q(a/T) \longrightarrow C_\infty \alpha a^{\alpha-1} \quad (T \downarrow 0).$$

In either case these limits hold for every $a > 0$. The masses on $[0, M]$ are bounded by e^{aM} times the transform at a . Helly selection on compact intervals therefore provides locally weak subsequential limits of the positive measures. Moreover their exponentially weighted tails satisfy

$$\int_{u>M} e^{-au} \nu_T(du) \leq e^{-aM/2} \int e^{-au/2} \nu_T(du),$$

with the same bound for $\tilde{\nu}_T$. The right side tends uniformly to zero as $M \rightarrow \infty$ along the endpoint limit. Thus the transforms pass to each subsequential limit. Uniqueness of positive-measure Laplace transforms identifies those limits: after multiplying a measure by $e^{-a_0 u}$, they are finite; the substitution $v = e^{-u}$ makes their transform values at $a_0 + n$, $n \geq 0$, the polynomial moments on $[0, 1]$, which uniquely determine the finite measure by polynomial approximation. The gamma integral gives the respective limiting densities

$$\frac{b_*\beta(\beta - 1)}{\Gamma(2 - \beta)} u^{1-\beta}, \quad \frac{C_\infty \alpha}{\Gamma(1 - \alpha)} u^{-\alpha}.$$

Both are locally integrable at zero. Every subsequence has these limits, so the measures themselves converge locally weakly.

To pass from averages to the density, put $I_- = (1 - h, 1)$ and $I_+ = (1, 1 + h)$, $0 < h < 1$. Monotonicity of μ_q gives

$$\frac{\nu_T(I_+)}{\int_{I_+} u^2 du} \leq T^{\beta+1} \mu_q(T) \leq \frac{\nu_T(I_-)}{\int_{I_-} u^2 du}.$$

First let $T \rightarrow \infty$; the interval masses converge because all their endpoints have zero limiting mass. Then let $h \downarrow 0$. Both bounds tend to $b_*\beta(\beta - 1)/\Gamma(2 - \beta)$. For $\tilde{\nu}_T$, replace u^2 by u and β by α , and let $T \downarrow 0$ before $h \downarrow 0$. This proves

$$\mu_q(t) \sim \frac{C_\infty \alpha}{\Gamma(1 - \alpha)} t^{-1-\alpha} \quad (t \downarrow 0), \quad \mu_q(t) \sim \frac{b_*\beta(\beta - 1)}{\Gamma(2 - \beta)} t^{-1-\beta} \quad (t \rightarrow \infty). \quad (9.5)$$

Since the large-time coefficient is positive and μ_q decreases, it is positive at every finite positive time. Because $\beta > \alpha$, these asymptotics and continuity on intervening compact intervals give both global bounds $\mu_q(t) \leq Ct^{-1-\alpha}$ and $\mu_q(t) \leq Ct^{-1-\beta}$.

Define the physical jump kernel by the Gaussian mixture

$$j_q(x) = \int_0^\infty (4\pi t)^{-1} e^{-|x|^2/(4t)} \mu_q(t) dt \quad (x \neq 0).$$

It is positive and radial. With $r = |x|$ and $t = r^2/(4z)$,

$$j_q(r) = \frac{1}{4\pi} \int_0^\infty e^{-z} \mu_q(r^2/(4z)) \frac{dz}{z}.$$

The global bounds just proved give integrable majorants $Ce^{-z}z^\alpha$ and $Ce^{-z}z^\beta$ for the rescaled integrands $r^{2+2\alpha}j_q(r)$ as $r \downarrow 0$ and $r^{2+2\beta}j_q(r)$ as $r \rightarrow \infty$, respectively. Dominated convergence gives strictly positive finite limits. The mixture is continuous away from zero by the same domination; positivity and compactness of intervening radial intervals then prove (9.3).

For every derivative order $\ell \geq 0$, the Gaussian bound

$$|D_x^\ell[(4\pi t)^{-1} e^{-|x|^2/(4t)}]| \leq C_\ell t^{-1-\ell/2} e^{-|x|^2/(8t)}$$

is dominated locally away from $x = 0$ after integration against μ_q . Thus differentiation under the integral is legitimate. The global $t^{-1-\beta}$ bound and a single rescaling give

$$|D^\ell j_q(x)| \leq C_\ell \int_0^\infty t^{-2-\beta-\ell/2} e^{-|x|^2/(8t)} dt = C'_\ell |x|^{-2-2\beta-\ell} = C'_\ell |x|^{-3-p-\ell}.$$

In particular this proves the required tail estimates for $|x| \geq 1$. Tonelli and the exact second moment $4t$ of the mass-one Gaussian in dimension two give

$$\int_{\mathbb{R}^2} |x|^2 j_q(x) dx = 4 \int_0^\infty t \mu_q(t) dt = 4a_*.$$

The near-zero first moment is finite, since polar integration uses $\int_0^1 r^{-(p-1)} dr < \infty$; the mass outside the unit disk is finite by the tail order. Consequently, for $a \in C_b^1$,

$$\Theta_q a(x) = \int_{\mathbb{R}^2} (a(x) - a(x+z)) j_q(z) dz$$

converges absolutely, using $\|Da\|_\infty |z|$ near zero and $2\|a\|_\infty$ outside. Its symbol on Schwartz functions is exactly the one already constructed: the mass-one Gaussian Fourier formula and Tonelli give

$$\int_{\mathbb{R}^2} (1 - \cos(k \cdot z)) j_q(z) dz = \int_0^\infty (1 - e^{-t|k|^2}) \mu_q(t) dt = \theta_q(|k|^2).$$

The sine integral is absolutely convergent and zero by symmetry. For a Schwartz function, translation and the fundamental theorem of calculus give $\|a - a(\cdot + z)\|_1 \leq \min\{|z|\|Da\|_1, 2\|a\|_1\}$. The finite first moment near zero and finite mass far away therefore justify Fourier Fubini and identify the displayed operator with the unitary Fourier multiplier. This fixes its sign and its extension to C_b^1 without Fourier-transforming a bounded function outside the Schwartz class. $\square$

Lemma 9.3. *For a bounded smooth trace a with bounded derivatives satisfying $\Theta_q a = g$ in D_{2L} ,*

$$\text{osc}_{D_\rho} a \leq C \left[(\rho/L)^{2/3} \|a\|_\infty + L^2 \|g\|_{L^\infty(D_{2L})} \right], \quad 1 \leq \rho \leq L/2. \quad (9.6)$$

The constant depends only on the fixed trace symbol.

Proof. The preceding representation has zero killing and drift and $\int_0^\infty t \mu_q(t) dt = a_* < \infty$. Construct its subordinator by summing the positive jumps of a Poisson random measure with intensity $dv \mu_q(t) dt$. The sum on any finite time interval is almost surely finite, since its expected sum is finite. In the compensated Lévy-triplet convention take Gaussian part zero, jump measure $\mu_q(t) dt$ on the positive half-line and drift $\int_0^1 t \mu_q(t) dt$. This cancels the small-jump compensation and gives exactly the uncompensated positive sum. The Poisson construction and Lévy–Itô decomposition are [31, Theorems 7.5 and 9.12]; here the measure is supported on $(0, \infty)$ and

$\int (1 \wedge t) \mu_q(t) dt < \infty$, so this construction has nondecreasing càdlàg paths and no deterministic drift. Its Laplace transform is

$$\mathbb{E} e^{-\lambda S_\tau} = e^{-\tau \theta_q(\lambda)}, \quad \mathbb{E} S_\tau = a_* \tau.$$

Equivalently, the mean follows by monotone convergence as $\lambda \downarrow 0$ in $(1 - \mathbb{E} e^{-\lambda S_\tau})/\lambda$, using $\theta'_q(0+) = a_*$. Since $\theta_q(\lambda) \rightarrow \infty$, $\mathbb{P}(S_\tau = 0) = 0$ for $\tau > 0$.

Let B be independent Brownian motion with generator Δ in $\mathbb{R}^2$, and put $X_\tau = x + B_{S_\tau}$. Conditional Gaussian increments and the independent stationary increments of S show that $X - x$ is a Lévy process with characteristic exponent $\theta_q(|k|^2)$, semigroup P_τ and generator $-\Theta_q$. Its paths are càdlàg. Conditioning also gives

$$\mathbb{E}_x(X_\tau - x) = 0, \quad \mathbb{E}_x |X_\tau - x|^2 = 4a_* \tau.$$

Independence of future increments from the natural filtration then makes $X - x$ a square-integrable martingale; we use the usual completed right-continuous filtration.

The symbol asymptotics and its positive continuous integral representation give uniform constants on the intervening compact frequency ranges. Thus $\theta_q(\lambda) \geq c\lambda$ for $0 < \lambda \leq 1$ and $\theta_q(\lambda) \geq c\lambda^\alpha$ for $\lambda \geq 1$. The Gaussian kernel satisfies $\|\nabla e^{t\Delta}\|_{\infty \rightarrow \infty} \leq Ct^{-1/2}$. Tonelli and the gamma integral yield

$$\mathbb{E} S_\tau^{-1/2} = \frac{1}{\Gamma(1/2)} \int_0^\infty \lambda^{-1/2} e^{-\tau \theta_q(\lambda)} d\lambda \leq C\tau^{-1/2} \quad (\tau \geq 1).$$

Indeed the integral on $(0, 1)$ is at most $\int_0^\infty \lambda^{-1/2} e^{-c\tau\lambda} d\lambda$. On $(1, \infty)$, split the exponential into two equal factors: one is at most $e^{-c\tau\lambda}$ and the other is bounded by $e^{-c\lambda^\alpha/2}$ for $\tau \geq 1$, giving an integrable majorant and an exponentially small bound. These same estimates justify differentiation under the Gaussian mixture. Consequently

$$\|\nabla P_\tau a\|_\infty \leq C\tau^{-1/2} \|a\|_\infty \quad (\tau \geq 1).$$

For a starting point in D_ρ , let τ_D be the first exit from D_{2L} . Before exit the process remains in this disk; an exit requires displacement at least $2L - \rho \geq 3L/2$. Doob's L^2 maximal inequality, applied to the martingale $X - x$, therefore gives

$$\mathbb{P}_x(\tau_D \leq \tau) \leq \frac{4\mathbb{E}_x |X_\tau - x|^2}{(3L/2)^2} \leq C\tau/L^2.$$

We justify Dynkin's formula for the present traces. The preceding jump bounds give $\int_{|z|<1} |z| j_q(z) dz < \infty$ and $\int_{|z|\geq 1} j_q(z) dz < \infty$; the finite second moment also gives $\int_{|z|\geq 1} |z| j_q(z) dz < \infty$. Thus the symmetric process is a pure jump process of finite variation with zero drift. Telescoping its absolutely summable jumps for the bounded Lipschitz function a gives the jump change formula. Writing N for its jump measure, the predictable integrand $H(v, z) = a(X_{v-} + z) - a(X_{v-})$ satisfies, on every finite interval,

$$\int_0^\tau \int \mathbb{E}_x |H(v, z)|^2 j_q(z) dz dv \leq \tau \left[\|Da\|_\infty^2 \int_{|z|<1} |z|^2 j_q(z) dz + 4\|a\|_\infty^2 \int_{|z|\geq 1} j_q(z) dz \right] < \infty.$$

The compensated Poisson measure is the random orthogonal measure in [31, Example 10.14]; its control measure is $dv j_q(z) dz$. Corollary 10.11 there applies to this predictable square-integrable integrand. Approximation by predictable simple integrands and its isometry give a true square-integrable martingale, since the approximating centered integrals are martingales and converge in L^2 at each time. The absolute first-moment bound also permits the uncompensated jump formula. It follows that

$$a(X_\tau) - a(x) + \int_0^\tau \Theta_q a(X_v) dv$$

is a true martingale. Values at X_{v-} or X_v agree for Lebesgue-almost every time along each path. The generator integral is bounded globally by $\|Da\|_\infty \int_{|z|<1} |z| j_q + 2\|a\|_\infty \int_{|z|\geq 1} j_q$, so all these

time integrals are integrable. Bounded optional stopping at $\tau \wedge \tau_D$ is therefore legitimate and, using the local equation, gives

$$a(x) = \mathbb{E}_x a(X_{\tau \wedge \tau_D}) + \mathbb{E}_x \int_0^{\tau \wedge \tau_D} g(X_v) dv.$$

If the local equality is initially almost everywhere, its smooth left side supplies the continuous representative of g without changing its essential bound. Comparing the stopped expectation with $P_\tau a(x)$ costs at most $2\|a\|_\infty \mathbb{P}_x(\tau_D \leq \tau)$. For $x, y \in D_\rho$, the gradient bound and the two stopped formulas now give

$$|a(x) - a(y)| \leq C\|a\|_\infty(\rho\tau^{-1/2} + \tau/L^2) + 2\tau\|g\|_{L^\infty(D_{2L})}.$$

Choose $\tau = (\rho L^2)^{2/3}$. The assumptions imply $L \geq 2$ and $1 \leq \tau \leq L^2$. Each of the first two terms has order $(\rho/L)^{2/3}$, and the forcing term is at most $2L^2\|g\|_{L^\infty(D_{2L})}$, proving the estimate.

The same bound holds for bounded measurable a satisfying the equation distributionally, with essential oscillation. The distribution is well-defined by $\langle \Theta_q a, \eta \rangle = \int a \Theta_q \eta$: for compact smooth η , the first jump moment and finite exterior mass give $\|\Theta_q \eta\|_1 \leq \|D\eta\|_1 \int_{|z|<1} |z| j_q + 2\|\eta\|_1 \int_{|z|\geq 1} j_q < \infty$. Convolve globally with a nonnegative smooth approximate identity supported in D_ε , $0 < \varepsilon < L/4$. Translation invariance and the preceding absolute bounds justify Fubini, yielding $\Theta_q a_\varepsilon = g_\varepsilon$ on $D_{2L-\varepsilon}$, with $\|a_\varepsilon\|_\infty \leq \|a\|_\infty$ and $\|g_\varepsilon\|_{L^\infty(D_{2L-\varepsilon})} \leq \|g\|_{L^\infty(D_{2L})}$. Repeat the smooth argument stopping in $D_{3L/2}$, which is contained in this equation disk and has distance at least L from D_ρ . The same time choice and constants apply. Let $\varepsilon \downarrow 0$ at any two Lebesgue points of a and take essential suprema to obtain the asserted extension. No continuity of this measurable representative is needed; later traces have smooth representatives. $\square$

9.1. An extension kernel with integrable derivative tails. Let $R_D(k)$ denote the inverse of L_k on the form space $\mathcal{H}_D = \{v \in H^s(\mathbb{R}) : v(0) = 0\}$.

Lemma 9.4. *The compatible endpoint extensions satisfy*

$$\|R_D(k)\|_{1 \rightarrow 1} + \|R_D(k)\|_{\infty \rightarrow \infty} \leq C \min\{1, |k|^{-p}\},$$

where the minimum is interpreted as one at $k = 0$. For $|\gamma| \geq 1$, their operator-norm derivatives obey

$$\|\partial_k^\gamma R_D(k)\| \leq \begin{cases} C_\gamma |k|^{p-|\gamma|}, & 0 < |k| \leq 1, \\ C_\gamma |k|^{-p-|\gamma|}, & |k| \geq 1. \end{cases}$$

Proof. Both inverses below are self-adjoint on L^2 : symmetry of their forms, evaluated on two variational solutions, gives $\langle Rf, h \rangle = \langle f, Rh \rangle$; the same argument applies to the restricted free form. Its kinetic part plus $2\|v\|_2^2$ is coercive on H^s . Point evaluation is continuous on $H^s(\mathbb{R})$ since $s > 1/2$. Thus $\mathcal{H}_D$ is closed and is dense in L^2 , because it contains compact smooth functions supported away from zero. The zero-trace coercivity gives a unique variational inverse of L_k on this fixed domain. Let $B_k = (D_t^2 + |k|^2)^s + 2$ and $U_q = 3(1 - q_s^2)$, so $L_k = B_k - U_q$ and $0 \leq U_q \leq 3$. The layer tail gives $U_q \in L^2 \cap L^\infty$. Write $B_{k,D}$ for the form restriction of B_k to $\mathcal{H}_D$, and $\mathcal{B}_k[v, w] = \langle v, B_k w \rangle$ for its form. For real data its unique variational solution is real, by conjugation.

The scalar gamma identity gives the full kinetic form representation, with $\kappa = |k|$,

$$(D_t^2 + \kappa^2)^s = \kappa^p I + \frac{s}{\Gamma(1-s)} \int_0^\infty e^{-\tau\kappa^2} (I - e^{\tau\partial_t^2}) \frac{d\tau}{\tau^{1+s}}.$$

Tonelli proves the quadratic-form identity and polarization gives the mixed form; its absolute convergence follows by Cauchy–Schwarz for the nonnegative heat forms. For disjoint nonnegative v, w , their local pairing is zero, whereas $\langle v, (I - e^{\tau\partial_t^2})w \rangle = -\langle v, e^{\tau\partial_t^2}w \rangle \leq 0$. Positive and negative parts preserve H^s and their zero trace. For every $v \in L^2$, Plancherel for its translated differences, Tonelli, and the normalized scalar identity give

$$\frac{c_{1,s}}{2} \iint \frac{|v(t) - v(z)|^2}{|t - z|^{1+p}} dt dz = \int |\xi|^p |\hat{v}(\xi)|^2 d\xi.$$

The equality permits infinite values on both sides. A finite left side therefore implies H^s membership, since the Fourier Sobolev weight is comparable to $1 + |\xi|^p$. Scalar truncation contracts the difference seminorm and the L^2 norm. Applying this equality to the truncations proves their membership without assuming it; the continuous representatives then agree with the pointwise truncations and preserve zero trace. The kinetic cross term of two nonnegative functions with disjoint supports is therefore nonpositive, and every local potential cross term is zero. If $f \geq 0$, testing $L_k u = f$ with $u_- = \max\{-u, 0\}$ yields $0 \leq \langle f, u_- \rangle \leq -E_k(u_-)$. Zero-trace coercivity forces $u_- = 0$. Hence $R_D(k)$ preserves positivity on L^2 despite the sign-changing potential.

The free inverse B_k^{-1} has a nonnegative convolution kernel b_k with total mass $(2 + \kappa^p)^{-1}$. Indeed use the same Poisson construction for the stable subordinator with jump density $s\tau^{-1-s}/\Gamma(1-s)$, Gaussian part zero and compensated-triplet drift equal to its small-jump first moment, and Laplace transform $\mathbb{E}e^{-\lambda S_v} = e^{-v\lambda^s}$. Small jumps have finite expected sum and large jumps occur finitely often on finite intervals, so the construction applies; $S_v > 0$ almost surely for $v > 0$. Subordination expresses $e^{-v(D_t^2 + \kappa^2)^s}$ as the positive Gaussian mixture $\mathbb{E}[e^{-\kappa^2 S_v} e^{S_v \partial_t^2}]$. Integration against $e^{-2v} dv$ yields an ordinary nonnegative convolution kernel, with mass $\int_0^\infty e^{-2v-v\kappa^p} dv = (2 + \kappa^p)^{-1}$. Fourier transformation identifies it with the free resolvent. For $f \geq 0$ in L^2 , put $u = B_{k,D}^{-1}f$ and $v = B_k^{-1}f$. The same negative-part test gives $u \geq 0$, while $v \geq 0$. The function $w = (u - v)_+$ has zero trace since $u(0) = 0$ and $v(0) \geq 0$. Subtract the free and restricted equations and test with w . Since $u - v = w - (u - v)_-$ and the cross term is nonpositive, $0 = \mathcal{B}_k[u - v, w] \geq \mathcal{B}_k[w, w]$, so $w = 0$. Thus $0 \leq B_{k,D}^{-1}f \leq B_k^{-1}f$.

For each t , evaluation of $B_{k,D}^{-1} : L^2 \rightarrow H^s$ is a bounded functional and has a Riesz row $K_k(t, \cdot) \in L^2$. The H^s embedding makes these rows locally norm-continuous as L^2 -valued functions; they admit a jointly measurable kernel. The comparison, applied to nonnegative L^2 data, gives $0 \leq K_k(t, z) \leq b_k(t - z)$ for almost every z . Self-adjointness, first tested on compactly supported L^2 data, gives $K_k(t, z) = K_k(z, t)$ almost everywhere. Its row and column masses are consequently at most $(2 + \kappa^p)^{-1}$. The kernel therefore defines compatible positive endpoint extensions, adjoint to one another, with

$$\|B_{k,D}^{-1}\|_{\infty \rightarrow \infty} = \|B_{k,D}^{-1}\|_{1 \rightarrow 1} \leq (2 + \kappa^p)^{-1}.$$

For $\kappa \leq 2$, zero-trace coercivity and the variational identity $E_k(u) = \langle f, u \rangle$ give $\|R_D(k)f\|_{H^s} \leq C\|f\|_2$, hence a uniform $L^2 \rightarrow L^\infty$ bound. On L^2 the resolvent identity is

$$R_D(k) = B_{k,D}^{-1} + R_D(k)U_q B_{k,D}^{-1}.$$

Its right side has supremum norm at most $C\|f\|_\infty$ on $L^2 \cap L^\infty$, since $\|U_q B_{k,D}^{-1}f\|_2 \leq \|U_q\|_2 \|B_{k,D}^{-1}f\|_\infty$. For arbitrary bounded f use this same formula, whose second datum lies in L^2 , to define a bounded extension. For $h \in L^1 \cap L^2$, self-adjointness and the bound just proved, tested against bounded compactly supported L^2 functions, give $\|R_D(k)h\|_1 \leq C\|h\|_1$. Density defines its L^1 extension and its adjoint on L^∞ .

These bounded-data constructions agree. For $f_N = f\mathbf{1}_{[-N,N]}$, the positive kernel and its row bound give $B_{k,D}^{-1}f_N \rightarrow B_{k,D}^{-1}f$ pointwise almost everywhere, with a common supremum bound. Dominated convergence after multiplication by $U_q \in L^2$ gives convergence in L^2 , and the $R_D : L^2 \rightarrow L^\infty$ bound gives uniform convergence of the second term. For $h \in L^1 \cap L^2$, dominated convergence and $R_D(k)h \in L^1$ yield

$$\int R_D(k)f \bar{h} = \lim_N \int f_N \overline{R_D(k)h} = \int f \overline{R_D(k)h}.$$

This identifies the bounded formula with the adjoint extension. Both extend the variational inverse on their intersections with L^2 . Positivity passes by positive L^1 approximation and duality.

For $\kappa \geq 2$, the bound $0 \leq U_q \leq 3$ gives $\|B_{k,D}^{-1}U_q\| \leq 3/(2 + \kappa^p) < 1$ on L^2 and both endpoint spaces. The compatible Neumann series gives

$$R_D(k) = (I - B_{k,D}^{-1}U_q)^{-1}B_{k,D}^{-1}, \quad \|R_D(k)\| \leq (\kappa^p - 1)^{-1} \leq C\kappa^{-p}.$$

On L^2 its limit satisfies $u = B_{k,D}^{-1}(f + U_q u)$ and hence the variational equation; uniqueness identifies it with $R_D(k)$. Convergence of the same partial sums on intersecting spaces gives compatibility, and the L^1/L^∞ adjoint relation also follows termwise. The uniform low bound covers $1 \leq \kappa \leq 2$, completing the asserted norm bound.

For $k \in \mathbb{R}^2$, put $\kappa = |k|$ and $D_k = (D_t^2 + \kappa^2)^s - |D_t|^p$. The scalar identity $z^s = \frac{s}{\Gamma(1-s)} \int_0^\infty (1 - e^{-\tau z}) \tau^{-1-s} d\tau$ gives the heat representation

$$D_k = \frac{s}{\Gamma(1-s)} \int_0^\infty (1 - e^{-\tau \kappa^2}) e^{\tau \partial_t^2} \frac{d\tau}{\tau^{1+s}}.$$

This is an operator-norm convergent integral on L^1 , L^2 and L^∞ : the heat operators are contractions, their kernels depend continuously in L^1 on $\tau > 0$, and the scalar weight is integrable at both endpoints. Its total weight is κ^p . It defines compatible positive operators with $\|D_k\| \leq \kappa^p$ and $D_0 = 0$. Fourier transformation on L^2 identifies it with the displayed multiplier, so it is the bounded difference of the two kinetic forms.

For every multi-index γ of positive order, differentiation under this integral is valid on compact sets with $k \neq 0$. At $\tau = 0$ every positive-order derivative of $1 - e^{-\tau |k|^2}$ is $O(\tau)$, uniformly on such sets; at infinity it is a polynomial times an exponential. Hence the differentiated operator integrals converge in norm. Substituting $\tau = \sigma/\kappa^2$ gives

$$\|\partial_k^\gamma D_k\| \leq C_\gamma \kappa^{p-|\gamma|}.$$

More explicitly, with $m = |\gamma|$, the differentiated exponential is a finite sum of terms $C_{\gamma,a} \tau^{m-|a|} k^{\gamma-2a} e^{-\tau |k|^2}$, where $0 \leq a_i \leq \lfloor \gamma_i/2 \rfloor$. Here $m - |a| \geq \lceil m/2 \rceil \geq 1 > s$. After scaling, each term is κ^{p-m} times an integral of $\sigma^{m-|a|-1-s} e^{-\sigma}$ and a bounded monomial in $k/|k|$. This integral is finite uniformly for $k/|k| \in S^1$. The order-zero bound also proves norm continuity at $k = 0$.

On the fixed form domain, subtracting the variational equations gives

$$R_D(k) - R_D(l) = -R_D(k)(D_k - D_l)R_D(l).$$

Both orders of this resolvent identity hold on L^2 . Compatibility and density extend them to L^1 ; taking adjoints, using the self-adjoint compatible heat operators $D_k - D_l$, extends them to L^∞ . Norm continuity of D_k and the endpoint bounds first give norm continuity of $R_D(k)$, including $k = 0$. Difference quotients give $\partial_{k_a} R_D = -R_D(\partial_{k_a} D_k)R_D$; repeated differentiation proves that it is C^∞ away from zero. A derivative of multi-index $\gamma \neq 0$ is a finite linear combination of products

$$R_D(\partial_k^{\gamma_1} D_k) R_D \cdots (\partial_k^{\gamma_m} D_k) R_D, \quad \gamma_i \neq 0, \quad \sum_i \gamma_i = \gamma.$$

For $0 < \kappa \leq 1$ these products have norm at most $C_\gamma \kappa^{mp-|\gamma|} \leq C_\gamma \kappa^{p-|\gamma|}$ because $m \geq 1$. For $\kappa \geq 1$ their norm is at most $C_\gamma \kappa^{-p(m+1)} \kappa^{mp-|\gamma|} = C_\gamma \kappa^{-p-|\gamma|}$. These calculations prove both operator-norm derivative estimates on both endpoint spaces. $\square$

Define

$$\Pi_k(t) = \frac{L_k^{-1} \delta_0(t)}{G_q(|k|^2)} \quad (k \neq 0), \quad \Pi_0(t) = \frac{\phi(t)}{\phi(0)} = \frac{\Phi(t)}{\Phi_0}.$$

Both functions belong to H^s , have trace one, and solve their homogeneous equations against zero-trace tests. The difference identity is

$$\Pi_k - \Pi_0 = -R_D(k) D_k \Pi_0. \tag{9.7}$$

The family is radial in k , continuous into L_t^1 including at zero, and smooth there away from zero. For every multiindex γ , including order zero, it satisfies

$$\|\partial_k^\gamma (\Pi_k - \Pi_0)\|_{L_t^1} \leq C_\gamma |k|^{p-|\gamma|} \quad (0 < |k| \leq 1), \quad \|\partial_k^\gamma \Pi_k\|_{L_t^1} \leq C_\gamma |k|^{-|\gamma|} \quad (|k| \geq 1).$$

Write $\mathcal{E}_k(v, w) = \langle v, L_k w \rangle$ for its sesquilinear form. The diagonal resolvent construction gives $L_k^{-1} \delta_0 \in H^s$ and $G_q(|k|^2) > 0$. Evaluation of the numerator at zero is its denominator, so $\Pi_k(0) = 1$. Its source identity gives the zero-trace form equation on H_D . For $k = 0$, Lemma 9.1

gives $L_0\Phi = 0$, and hence the same zero-trace form equation and trace one for Π_0 . Reality and radial dependence follow from the real coefficients and dependence of L_k only on $|k|^2$.

Subtract the two form equations on H_D . Their difference has trace zero and satisfies

$$L_k(\Pi_k - \Pi_0) = -D_k\Pi_0 \quad \text{against every zero-trace test.}$$

Here $D_k\Pi_0 \in L^2$, since $\Pi_0 \in H^s \cap L^2$ and D_k is bounded on L^2 by the perturbation estimates above. Zero-trace uniqueness from Lemma 9.1 identifies this solution with $-R_D(k)D_k\Pi_0$. The layer mass in the reference-layer estimates of Section 2 gives $\Pi_0 \in L^1$, and compatibility of the L^1 extension with the variational inverse in Lemma 9.4 gives (9.7) also in L^1 .

Lemma 9.4 supplies resolvent positivity. To prove positivity of Π_k , its negative part $\Pi_k^- = \max\{-\Pi_k, 0\}$ belongs to H^s and has trace zero, since the trace of Π_k is one. It is therefore an admissible test in the homogeneous form equation. For real numbers a, b , $(a-b)(a^- - b^-) \leq -|a^- - b^-|^2$. The whole kinetic operator has the jump-plus-killing representation of the perturbation estimates above

$$(D_t^2 + |k|^2)^s = |k|^p I + \frac{s}{\Gamma(1-s)} \int_0^\infty e^{-\tau|k|^2} (I - e^{\tau\partial_t^2}) \frac{d\tau}{\tau^{1+s}}.$$

The form of $I - e^{\tau\partial_t^2}$ is one half the integral of the two differences against the nonnegative symmetric heat kernel. Thus the negative-part inequality applies to every such jump form and to the killing term. The potential pairing equals $-\int V_q(\Pi_k^-)^2$. These observations give

$$0 = \mathcal{E}_k(\Pi_k, \Pi_k^-) \leq -\mathcal{E}_k(\Pi_k^-, \Pi_k^-).$$

The zero-trace coercivity of Lemma 9.1 forces $\Pi_k^- = 0$. The preceding heat formula also shows $D_k\Pi_0 \geq 0$, since $\Pi_0 \geq 0$. Resolvent positivity and the difference identity now give $\Pi_k \leq \Pi_0$. The domination implies a uniform L^2 bound as well as the L^1 bound.

For completeness, use the product rule in the Banach space L_t^1 on $R_D(k)D_k\Pi_0$. The direct endpoint bounds of Lemma 9.4 and the perturbation estimates above justify every derivative. At low frequencies a term with no differentiated resolvent is bounded by $C|k|^{p-|\gamma|}$. A term with a positive-order differentiated resolvent is bounded by $C|k|^{2p-|\gamma|}$, which is no larger for $|k| \leq 1$. At high frequencies every positive-order derivative of that product is bounded by $C|k|^{-p-|\gamma_1|}|k|^{p-|\gamma_2|} = C|k|^{-|\gamma|}$, where $\gamma_1 + \gamma_2 = \gamma$ and the order-zero resolvent bound is included. For order zero use $\Pi_k \leq \Pi_0$. These give the claimed estimates and continuity at zero. Smoothness and continuity away from zero follow from the same endpoint differentiability statements.

Let $\Pi = (2\pi)^{-1}\mathcal{F}_x^{-1}[\Pi_k]$ be the tangential convolution kernel, initially as an L_t^1 -valued tempered distribution. Off $x = 0$ it is a smooth L_t^1 -valued function, with

$$\int |D_x^\ell \Pi(x, t)| dt \leq C_\ell |x|^{-2-p-\ell} \quad (|x| \geq 1, \ell \geq 0). \quad (9.8)$$

In particular, for $b \geq 1$,

$$\int \int_{|x|>b} (|D_x^2 \Pi| + b^{-1}|D_x \Pi| + b^{-2}|\Pi|) dx dt \leq Cb^{-p-2}. \quad (9.9)$$

The uniform L_t^1 bound defines this inverse Fourier transform by Bochner integration against Schwartz test functions. Choose $\psi_0 \in C_c^\infty(\mathbb{R}^2)$ equal to one near zero. Decompose its symbol into

$$\psi_0\Pi_0 + \psi_0(\Pi_k - \Pi_0) + (1 - \psi_0)\Pi_k.$$

The first inverse transform is Schwartz in x with values in L_t^1 . Resolve the second and third terms with a smooth dyadic annular partition, writing $|k| \asymp 2^j$, respectively $j \leq 0$ and $j \geq 0$, after absorbing finitely many transition annuli into either side. Rescale $k = 2^j\xi$; the rescaled annular support is fixed. The derivative estimates of the preceding normalized Poisson family, the product rule for the annular cutoffs, and multiplication by k for each physical derivative show that the rescaled symbol and each of its fixed finite number of derivatives have L_t^1 norm at most

$C_N 2^{j(p+\ell)}$ on low annuli and $C_N 2^{j\ell}$ on high annuli. Integration by parts in ξ (using $(1 - \Delta_\xi)^N$ if desired) gives, for every sufficiently large integer N , inverse-transform bounds

$$C_N 2^{j(2+p+\ell)} (1 + 2^j |x|)^{-N} \quad (j \leq 0), \quad C_N 2^{j(2+\ell)} (1 + 2^j |x|)^{-N} \quad (j \geq 0)$$

in L_t^1 . These are Banach-space integrations over compact supports; their derivatives are integrable, so no scalar pointwise-in- t differentiability is being assumed.

For $r = |x| \geq 1$, split the low sum at $2^j r \asymp 1$. The part below that threshold is a geometric series bounded by $C r^{-2-p-\ell}$. Above it choose $N > 2 + p + \ell$; the resulting geometric sum has the same bound. The high sum is at most $C r^{-N} \sum_{j \geq 0} 2^{j(2+\ell-N)}$, also bounded by $C r^{-2-p-\ell}$. The Schwartz term satisfies the same bound. On every compact set avoiding $x = 0$, these estimates for every derivative show locally uniform convergence of the annular series in L_t^1 , hence smoothness there. The annular expansion also converges to the original symbol in tempered distributions: at low frequencies the remainder tends to zero in L_t^1 uniformly like $|k|^p$, and at high frequencies its pairing with any Schwartz function tends to zero by the bounded symbol and Schwartz decay. Thus the sum identifies the given distribution away from zero. Finally polar integration in dimension two gives

$$\int_{|x|>b} |x|^{-2-p-\ell} dx = C_{p,\ell} b^{-p-\ell}.$$

Use $\ell = 2, 1, 0$ and their respective weights to obtain the integrated estimate.

The physical kernel is positive, and its normal mass has the normalization

$$\Pi_t(\mathbb{R}^2) = \frac{\phi(t)}{\phi(0)} \quad \text{a.e.}, \quad \int_{|t|>d} \Pi_t(\mathbb{R}^2) dt \leq C d^{-p} \quad (d \geq 1). \quad (9.10)$$

Use the full zero-trace coercivity of Lemma 9.1 with the spectral notation above, and write $\mathcal{E}_q(v, w) = \langle v, L_q w \rangle$ for the full form. For $a \in C_c^\infty(\mathbb{R}^2)$ the lift $A(x, t) = a(x) \Pi_0(t)$ is in $H^s(\mathbb{R}^3)$: Plancherel and $(|k|^2 + \tau^2)^s \leq |k|^p + |\tau|^p$ bound its norm by the separate Sobolev norms of a and Π_0 . The bounded Sobolev trace is a . Lax–Milgram on the closed zero-trace subspace, with the bounded full form and this coercivity, gives a unique $U - A$ in that subspace with $\mathcal{E}_q(U, v) = 0$ for every zero-trace v .

Tangential Fourier transformation and localized frequency tests reduce this equation to $\mathcal{E}_k(\widehat{U}(k, \cdot), v) = 0$ for almost every k and all $v \in H_D$. To spell out simultaneity, first use a countable dense subset of H_D and arbitrary compact frequency test functions; the integrated equation gives the slice equation outside one null set, and continuity in v gives every test. Normal slices are in H_t^s almost everywhere and the trace map commutes with this Fourier transformation, as in Lemma 9.1. Thus $\widehat{U}(k, 0) = \widehat{a}(k)$. Subtract $\widehat{a}(k) \Pi_k$ using the preceding normalized Poisson family; zero-trace uniqueness gives $\widehat{U}(k, t) = \widehat{a}(k) \Pi_k(t)$ for almost every $k \neq 0$. This proves $U = \Pi *_x a$ in distributions, with the factor $(2\pi)^{-1}$ in the preceding tangential kernel estimates exactly the convolution normalization in two tangential dimensions.

For real a , uniqueness and conjugation give real U . If $a \geq 0$, the negative part U^- has zero trace and belongs to H^s . The normalized fractional Dirichlet form of the normalization in Section 2 and the pointwise negative-part inequality give

$$0 = \mathcal{E}_q(U, U^-) \leq -\mathcal{E}_q(U^-, U^-).$$

The potential part is exactly $-\int V_q(U^-)^2$, so this does not require $V_q \geq 0$. Full zero-trace coercivity forces $U^- = 0$. Take nonnegative tangential approximate identities $a_\varepsilon \in C_c^\infty$ of integral one. The positive functions $U_\varepsilon = \Pi *_x a_\varepsilon$ converge to Π in distributions, so Π is a positive distribution. Here convolution convergence follows either from the bounded L_t^1 symbol in the preceding normalized Poisson family and dominated convergence against Schwartz tests, or the standard distribution approximation identity. A positive distribution is a Radon measure: for a compact set choose a nonnegative smooth cutoff $\rho \geq 1$ there; positivity gives $|\Pi(f)| \leq \|f\|_\infty \Pi(\rho)$ for real smooth f supported there. Riesz–Markov extends this positive functional locally to continuous compact tests. The resulting locally finite measures agree on overlaps and represent Π .

We establish its mass without presupposing finiteness. For nonnegative $b \in C_c^\infty(\mathbb{R})$, let ν_b be the positive tangential measure obtained by integrating $b(t)$ against Π . Its multiplier is

$$m_b(k) = \int_{\mathbb{R}} b(t) \Pi_k(t) dt.$$

This bounded function is continuous, including at zero, by the preceding normalized Poisson family. In the unitary convention $\widehat{\nu}_b = (2\pi)^{-1} m_b$ as distributions. Positive tempered measures integrate the Gaussian (their polynomially weighted mass is finite). Fourier pairing therefore gives

$$\int e^{-\varepsilon|x|^2} \nu_b(dx) = \int_{\mathbb{R}^2} \frac{e^{-|k|^2/(4\varepsilon)}}{4\pi\varepsilon} m_b(k) dk \rightarrow m_b(0).$$

The frequency Gaussians have integral one and concentrate at zero; boundedness and continuity of m_b justify the limit. On the physical side monotone convergence as $\varepsilon \downarrow 0$ gives

$$\nu_b(\mathbb{R}^2) = \int b(t) \Pi_0(t) dt.$$

Thus the normal marginal, first as a locally finite measure tested against such b and then as a Borel measure, equals $\Pi_0(t) dt$. Increasing compact normal cutoffs and monotone convergence give total mass $\int \Pi_0 = 2/\phi(0)$ by the reference-layer estimates of Section 2. In particular the joint measure is finite.

Apply Radon disintegration to this finite measure and its normal projection: both spaces are Euclidean standard Borel spaces, the normal marginal is the finite Radon measure $\Pi_0(t) dt$, and the projection is inverse-measure-preserving for that marginal. Fremlin, *Measure Theory*, Corollary 452P, supplies Radon probability measures concentrated on the normal fibers. Identify each fiber with $\mathbb{R}^2$ and multiply by $\Pi_0(t)$ to obtain Π_t with respect to Lebesgue dt . Because $\Pi_0(t) > 0$, its null sets coincide with Lebesgue null sets. Essential uniqueness follows by comparing the fiber values on a countable generating family of tangential Borel sets. The normal derivative tail of the reference-layer estimates of Section 2 gives $\int_{|t|>d} \Pi_0(t) dt \leq Cd^{-p}$.

For clarity this disintegration also supplies the needed Fourier fiber identity. Define $m(k, t) = \int e^{-ik \cdot z} \Pi_t(dz)$. It is jointly measurable and $|m(k, t)| \leq \Pi_0(t)$. For every compact smooth normal test b , the Fourier transform of the finite measure ν_b is a continuous function, so its distributional identity gives, for every k ,

$$\int b(t) m(k, t) dt = m_b(k) = \int b(t) \Pi_k(t) dt.$$

Hence $m(k, \cdot) = \Pi_k$ in L_t^1 for each k . Joint measurability and Fubini give equality for almost every (k, t) . The smooth L_t^1 -valued function of the preceding tangential kernel estimates defines a locally integrable joint density on sets avoiding $x = 0$; equality of distributions there identifies it with the restriction of the positive measure. There may be a tangential singular part at $x = 0$, which neither that statement nor the tail estimates discard.

For $g \in L^2(\mathbb{R}^3)$ define

$$Kg = \int_{\mathbb{R}} \Pi_t *_x g(\cdot, t) dt.$$

This is a Bochner integral in L_x^2 , with the precise norm bound and Fourier formula

$$\|Kg\|_{L_x^2} \leq \|\Pi_0\|_{L_t^2} \|g\|_{L_{x,t}^2}, \quad \widehat{Kg}(k) = \int_{\mathbb{R}} \Pi_k(t) \widehat{g}(k, t) dt \quad \text{for almost every } k.$$

Convolution by Π_t has L_x^2 norm at most its mass $\Pi_0(t)$, by translation invariance and Minkowski. Measurability follows first for continuous compact functions, then for general L^2 functions by density and the same bound. Cauchy–Schwarz gives $\int \Pi_0(t) \|g(\cdot, t)\|_2 dt \leq \|\Pi_0\|_2 \|g\|_2$, proving Bochner integrability and its claimed norm bound. Tangential Fourier transformation is bounded on L^2 and commutes with this Bochner integral. The fiber identity just proved gives the stated multiplier formula, initially for simple functions of t with Schwartz tangential factors and then for every $g \in L^2$ by the norm bound. Its right side is also absolutely defined almost everywhere and bounded in L_k^2 by the same Cauchy–Schwarz estimate and $\Pi_k \leq \Pi_0$.

Lemma 9.5. *For smooth w with w, Dw, D^2w bounded and $g = L_q w$,*

$$\Theta_q[w(\cdot, 0)] = \Phi_0^2 \int \Pi_t * g(\cdot, t) dt. \quad (9.11)$$

No decay or global L^2 condition is required.

Proof. The operator and normalization are those of the spectral notation above; the absolutely convergent generator on bounded Lipschitz traces, its first moment near zero, and its finite mass outside the unit disk are supplied by Lemma 9.2. The convolution measures and their Fourier identity are supplied by the positive extension kernel just constructed.

The relation $q'_s = \phi$ is fixed by the spectral notation above, and the preceding normalized Poisson family gives $\Pi_0 = \phi/\phi(0) \in L^1$. Define

$$M = |q_s(0)| + \phi(0) \|\Pi_0\|_1 < \infty.$$

The fundamental theorem of calculus, for every $t \in \mathbb{R}$, gives

$$|q_s(t)| \leq |q_s(0)| + \int_{\min\{0,t\}}^{\max\{0,t\}} |\phi(r)| dr \leq M.$$

The cubic potential $V_q = 3q_s^2 - 1$ in the spectral notation above therefore satisfies $\|V_q\|_\infty \leq 3M^2 + 1$. This supplies all potential bounds used below.

Write $\mathcal{E}_k(v, w) = \langle v, L_k w \rangle$ for the normal form. First take $w \in C_c^\infty(\mathbb{R}^3)$. Then $g = L_q w \in L^2$: the fractional multiplier maps H^p into L^2 and V_q is bounded. Partial Fourier transformation gives $\hat{g}(k, t) = L_k \hat{w}(k, t)$ in L_t^2 for almost every k . The normal source identity of the preceding normalized Poisson family, tested against the smooth compact normal slice $\hat{w}(k, \cdot)$, gives

$$\int \Pi_k(t) \hat{g}(k, t) dt = \mathcal{E}_k(\Pi_k, \hat{w}(k, \cdot)) = \frac{\hat{w}(k, 0)}{G_q(|k|^2)} \quad (k \neq 0).$$

The form is conjugate-linear in its first entry and Π_k is real, so this is exactly this bilinear integral, not its complex conjugate. The trace is $\hat{w}(k, 0)$ because the Fourier transformation here is only tangential. The symbol identity $\theta_q(|k|^2) = \Phi_0^2/G_q(|k|^2)$ in Lemma 9.2 and the L^2 fiber identity of the positive extension kernel just constructed establish the claimed equality in L_x^2 . The left side belongs to L^2 because its compact smooth trace has Schwartz Fourier transform and θ_q has at most polynomial growth, as stated in Lemma 9.2. Equality almost everywhere off $k = 0$ is equality of L^2 functions; it cannot leave a distribution supported at that single frequency.

Both physical sides are continuous. For the left, use the generator integral and the uniform bound $C \min\{|z|, 1\}$ on trace increments against the finite first-moment kernel. For the right, g is bounded continuous; the finite joint measure in the positive extension kernel just constructed permits dominated convergence under its integral as x varies. Thus the compact case holds pointwise everywhere.

We record a cutoff estimate to pass to bounded data. Let v have bounded v, Dv, D^2v , and let $\eta_T(X) = \eta(X/T)$, $T \geq 1$, where η is a fixed smooth bounded cutoff (allowed to depend on only the normal or only the tangential coordinates). The symmetric-increment formula of the normalization in Section 2 gives

$$[(-\Delta)^s, \eta_T]v = v(-\Delta)^s \eta_T - c_{3,s} \int_{\mathbb{R}^3} \frac{(\eta_T(X) - \eta_T(X+H))(v(X) - v(X+H))}{|H|^{3+p}} dH.$$

The product integral is absolutely convergent. The cutoff's own fractional Laplacian has norm at most CT^{-p} by scaling, or by its symmetric-increment bounds $C \min\{|H|^2/T^2, 1\}$. For the product integral, split the radial variable $\rho = |H|$ into $(0, 1)$, $(1, T)$ and (T, ∞) . Use respectively both Lipschitz bounds, boundedness of v with the cutoff Lipschitz bound, and both boundedness bounds. They give

$$C_v \left[T^{-1} \int_0^1 \rho^{1-p} d\rho + T^{-1} \int_1^T \rho^{-p} d\rho + \int_T^\infty \rho^{-1-p} d\rho \right] \leq C_v (T^{-1} + T^{-p}).$$

Here $p < 2$ makes the first integral finite and $p > 1$ bounds the second uniformly. Consequently

$$\|[L_q, \eta_T]v\|_\infty \leq C_v(T^{-1} + T^{-p}),$$

since the potential commutes with multiplication. Every use of L_q on bounded smooth data is its classical symmetric-increment integral, which is bounded by $C(\|v\|_\infty + \|D^2v\|_\infty)$ and continuous by dominated convergence.

Choose $\zeta \in C_c^\infty(\mathbb{R}^2)$ and $\chi \in C_c^\infty(\mathbb{R})$, between zero and one and equal to one near the origin. Set $w_M(x, t) = \zeta(x/M)w(x, t)$ and $w_{M,N}(x, t) = \chi(t/N)w_M(x, t)$. For fixed M the compact identity applies to $w_{M,N}$. Its trace equals that of w_M , while

$$L_q w_{M,N} = \chi(t/N)L_q w_M + [L_q, \chi(t/N)]w_M.$$

The last term tends uniformly to zero as $N \rightarrow \infty$ by the cutoff estimate. The first term converges pointwise to $L_q w_M$ with a uniform bound. Finite joint kernel mass therefore passes the right side to the limit, proving the identity for w_M pointwise. Next

$$L_q w_M = \zeta(x/M)g + [L_q, \zeta(x/M)]w,$$

and the commutator again tends uniformly to zero. The first term converges pointwise to g and stays uniformly bounded; the same finite-measure dominated convergence gives the desired right side. On the left, the traces $w_M(\cdot, 0)$ have uniform boundedness and uniform Lipschitz constants for $M \geq 1$, and converge pointwise to $w(\cdot, 0)$. Their generator increments are bounded by $C \min\{|z|, 1\}$, integrable against j_q by Lemma 9.2. Dominated convergence gives the claimed left side. This completes the identity for the stated bounded class. $\square$

10. FROM THE TRACE HESSIAN TO RETAINED CURVATURE

Fix a zero center and rotate its increasing normal vertically. The chart preparation established in the preceding sections supplies a graph $t = f_\Sigma(x)$ and a signed distance δ_Σ on nested comparison boxes, with

$$\begin{aligned} f_\Sigma(0) = Df_\Sigma(0) = 0, \quad |D^2 f_\Sigma| \leq C_0/R, \quad c_0 R^{2/(p+1)} \leq d \leq c_1 R, \\ u_t(0) \geq \gamma > 0, \quad \|u - q_s(\delta_\Sigma)\|_{C^1} \leq C_0/R. \end{aligned} \quad (10.1)$$

The profile estimate holds on a fixed fraction of the isolation box $|x|, |t| < cd$, with further fixed buffers. On a retained macro-chart these same inputs hold at every recentered zero, with the same R and constants. The comparison boxes include the actual graph domain and its unique normal-footpoint tube; every restriction below reserves a fixed larger box. Nearby zero footpoints retain the same prepared comparison data when recentered and rotated. All constants below may depend on these fixed buffer fractions, p, C_0, c_0, c_1 and γ , and are uniform in R , the center, the branch, and the finite-difference parameters. The preceding entire-solution regularity result gives uniform smooth bounds for u and the reference layer. Only $|u| \leq 1$ is used outside the boxes. Decreasing branches are handled by replacing u by $-u$. In particular $Rd^{-p} \rightarrow 0$.

10.1. Second differences of the actual solution. Use $L_q = (-\Delta)^s + 3q_s(t)^2 - 1$ and $\Phi_0 = q'_s(0)/\|q'_s\|_2$ with the normalization fixed in the preceding section. Set $v = u - q_s(t)$ and $N_q(v) = -3q_s v^2 - v^3$. The equation gives exactly $L_q v = N_q(v)$. Choose $1 \ll \ell \ll \min(d, \sqrt{R})$. Fix a sufficiently small buffer fraction $\beta > 0$ once and for all. The source and trace bounds below will hold on $D_{8\beta\ell} \times (-2\beta d, 2\beta d)$ and $D_{8\beta\ell}$, respectively, after making the preceding comparison boxes larger by fixed factors. Evaluation points lie in $D_{\beta\ell}$; none of these fractions depends on h or R . The normal footpoint equations, with footpoint y , are

$$x = y - \frac{\delta_\Sigma Df_\Sigma(y)}{\sqrt{1 + |Df_\Sigma(y)|^2}}, \quad t = f_\Sigma(y) + \frac{\delta_\Sigma}{\sqrt{1 + |Df_\Sigma(y)|^2}}.$$

Shrinking the fixed comparison fraction so that $cc_1 C_0$ is small gives $|\delta_\Sigma| \leq Ccd$. The footpoint equations and $|Df_\Sigma(y)| \leq C|y|/R$ imply

$$|x - y| \leq C|\delta_\Sigma||y|/R, \quad |\delta_\Sigma - t| \leq C|y|^2/R + C|\delta_\Sigma||y|^2/R^2.$$

Absorb the first inequality to get $|y| \leq 2|x|$, and then $|\delta_\Sigma - t| \leq C|x|^2/R$. The signed-distance identity $D_x \delta_\Sigma = -Df_\Sigma(y)/\sqrt{1+|Df_\Sigma(y)|^2}$ also gives $|D_x \delta_\Sigma| \leq C|x|/R$. Consequently, on $|x| \leq 16\beta\ell$, $|t| \leq 4\beta d$,

$$|v| \leq C\ell^2/R, \quad |D_x v| \leq C\ell/R, \quad |D_x^2 v| \leq C(\ell/R + \ell^{-p-1}) \quad (10.2)$$

on the smaller cylinder $D_{8\beta\ell} \times (-2\beta d, 2\beta d)$. To obtain the last estimate, localize $w = \partial_a u$ on a ball of radius comparable to ℓ . Its Jacobi equation $[(-\Delta)^s + 3u^2 - 1]w = 0$ is the already established differentiation identity; no new differentiation argument is needed here. Choose a smooth cutoff supported in this ball and equal to one on its inner half; the balls used here lie inside the preceding cylinder. The localized function has global supremum $C\ell/R$. At interior evaluation points, the exterior forcing has the signed representation

$$\int \frac{(1 - \zeta(Y))\partial_a u(Y)}{|X - Y|^{3+p}} dY = - \int u(Y) \partial_{Y_a} \left[\frac{1 - \zeta(Y)}{|X - Y|^{3+p}} \right] dY.$$

Nested buffers separate its support by $c\ell$. Both the kernel derivative and cutoff derivative cost $C\ell^{-p-1}$, with the same or better bounds for fixed-order evaluation derivatives. To justify this integration by parts, first insert a cutoff at radius M . Its boundary-annulus term is bounded by CM^{-p-1} and tends to zero, while the differentiated kernel is integrable. This proves the formula without a decay assumption on u . The dimensional normalization factor $c_{3,s}$ is retained in the localized forcing. On fixed unit balls in the inner region, apply the $C^{1,\alpha}$ localization estimate from (3.2), with $0 < \alpha < p - 1$, to this bounded localized equation and its separated forcing. On each such B_4 , the weighted norm of the localized function is at most $C\ell/R$, because its global supremum has this bound and the weight is integrable. Its forcing is at most $C\ell^{-p-1}$ there, and its coefficient $3u^2 - 1$ is uniformly bounded. This controls one derivative of w by $C(\ell/R + \ell^{-p-1})$ and proves (10.2); it does not require a second derivative bound on the pinned error.

For $e \in S^1$ and $0 < h \leq 1$, write

$$\delta_{h,e}^2 F = \frac{F(\cdot + he) + F(\cdot - he) - 2F}{h^2}, \quad w_{h,e} = R\delta_{h,e}^2 u.$$

Tangential differences commute with L_q , giving

$$L_q w_{h,e} = g_{h,e} = R\delta_{h,e}^2 N_q(v), \quad |g_{h,e}| \leq C(\ell^3/R + \ell^{1-p}) \quad (10.3)$$

on the smaller cylinder, uniformly in h . Indeed $R|D_x^2 N_q(v)| \leq CR(|v||D_x^2 v| + |D_x v|^2)$, which is bounded by $C(\ell^3/R + \ell^{1-p} + \ell^2/R)$. Here the cubic derivative terms are absorbed since $|v| \leq 2$ globally; ℓ^2/R is absorbed by ℓ^3/R for $\ell \geq 1$. The integral formula for a second difference uses only points at distance at most $h \leq 1$, so the reserved buffers give the same bound uniformly in h . The entire-solution derivative bounds give $|g_{h,e}| \leq CR$ globally.

We need a uniform bound on the trace, rather than on the whole normal cylinder. At a zero footpoint P let $q_P(X) = q_s(n(P) \cdot (X - P))$ and $V_P = R(u - q_P)$. Recenter and rotate at P . For the nearby centers used below, the remaining buffers contain a comparison box enclosing a ball of radius comparable to d , with the same constants. In these coordinates the tangent plane is $t = 0$ and the column estimate below applies. The pinned profile and tangent geometry imply

$$\|V_P\|_{L^\infty(B_4(P))} + \int \frac{|V_P(P + Y)|}{(1 + |Y|)^{3+p}} dY \leq C(1 + Rd^{-p}) \leq C. \quad (10.4)$$

For the graph contribution inside the comparison box use

$$\int_{\text{comparison interval}} R|q_s(\delta_\Sigma(x, t)) - q_s(t)| dt \leq C|x|^2. \quad (10.5)$$

Monotonicity bounds the difference by $q_s(t + b) - q_s(t - b)$ with $b = C|x|^2/R$, whose full integral is $4b$, by Tonelli and $\int q'_s = 2$. Indeed $q_s(t + b) - q_s(t - b) = \int_{t-b}^{t+b} q'_s(z) dz$; this is the same normal-column calculation used in the physical normalized-tail estimate of Theorem 8.3. The comparison interval is a subset of that full normal line, so positivity permits this upper bound without extending the actual profile comparison outside its box. The weighted tangential radial integral is $C \int_0^\infty \rho^3(1 + \rho)^{-3-p} d\rho < \infty$; no assertion about entire normal columns through other

layers is used. The pinned part has magnitude at most C after multiplication by R and therefore has uniformly bounded weighted integral. Outside the comparison box its distance from the center is at least a fixed multiple of d , and $|V_P| \leq 2R$, giving the cost CRd^{-p} . The same decomposition on $B_4(P)$ bounds its local supremum, since $d \rightarrow \infty$ and the tangent displacement there is $O(R^{-1})$. The planar-profile conclusion for the reference layer gives that q_P solves the normalized equation. Thus both u and q_P solve the equation, hence

$$[(-\Delta)^s + c_P(X)]V_P = 0, \quad c_P = u^2 + uq_P + q_P^2 - 1.$$

The coefficient has uniform smooth bounds by the entire-solution derivative estimates. Applying (3.3) with (10.4) gives $\|V_P\|_{C^j(B_1(P))} \leq C_j$ for each fixed j . Returning to the original coordinates, for $|x| \leq 8\beta\ell$ the point $(x, 0)$ is within $o(1)$ of its zero footpoint; indeed its distance is at most $|f_\Sigma(x)| \leq C\ell^2/R = o(1)$. The horizontal normal component there is $O(\ell/R)$, so $R|D_x^2 q_P| \leq C\ell^2/R$ because q_s'' is bounded. Thus $R|D_x^2 u(x, 0)| \leq C(1 + \ell^2/R) \leq C$, including neighboring points in the second difference. We have proved

$$|w_{h,e}(x, 0)| \leq C \quad (|x| \leq 8\beta\ell). \quad (10.6)$$

The integral identity $\delta_{h,e}^2 u(X) = \int_{-1}^1 (1 - |v|) \partial_{ee} u(X + hve) dv$ shows that $w_{h,e}$ and its first two derivatives have global bounds CR , independent of h . Lemma 9.5 applies to $a_{h,e} = w_{h,e}(\cdot, 0)$ and gives the exact normalized identity

$$\Theta_q a_{h,e}(x) = \Phi_0^2 \int_{\mathbb{R}} \int_{\mathbb{R}^2} g_{h,e}(x - z, t) \Pi_t(dz) dt.$$

This convolution measure integral includes any tangential singular part at $z = 0$. The fixed factor Φ_0^2 is absorbed only in subsequent bounds. The normal range $|t| > \beta d$ costs CRd^{-p} by (9.10). In the complementary range $|t| \leq \beta d$, choose $0 \leq \chi_\ell \leq 1$, equal to one on $D_{2\beta\ell}$, supported in $D_{4\beta\ell}$, with $|D^j \chi_\ell| \leq C_j \ell^{-j}$. These disks lie in the source region $D_{8\beta\ell}$; their separations from $D_{\beta\ell}$ are fixed multiples of ℓ . The near contribution is bounded using that source estimate and the finite joint mass of Π . For $|x| \leq \beta\ell$, the far kernel

$$K_{x,t}(y) = (1 - \chi_\ell(y)) \Pi(x - y, t)$$

is separated from the tangential origin of Π . It has an ordinary density there, smooth in the tangential variables with L_t^1 bounds on its derivatives. Equation (9.9), including the derivatives of χ_ℓ , gives

$$\int_{|t| \leq \beta d} \int |D_y^2 K_{x,t}(y)| dy dt \leq C\ell^{-p-2}.$$

Transfer the second difference from the bounded function $N_q(v)$ to this kernel. The identity

$$\int K_\ell(y) \delta_{h,e}^2 F(y) dy = \int F(y) \delta_{h,e}^2 K_\ell(y) dy, \quad \delta_{h,e}^2 K_\ell(y) = \int_{-1}^1 (1 - |v|) \partial_{ee} K_\ell(y + hve) dv$$

holds by translation and absolute integrability. The same derivative bound holds after the shifts hve , since $h \leq 1 \ll \ell$ and the buffers are larger than $2h$. The far contribution is $CR\ell^{-p-2}$, including derivatives of the cutoff. Therefore

$$\|\Theta_q a_{h,e}\|_{L^\infty(D_{\beta\ell})} \leq CE_R, \quad E_R = \ell^3/R + \ell^{1-p} + R\ell^{-p-2} + Rd^{-p}. \quad (10.7)$$

Choose $0 \leq \zeta_\ell \leq 1$ equal to one on $D_{2\beta\ell}$, supported in $D_{4\beta\ell}$, and satisfying $|D^j \zeta_\ell| \leq C_j \ell^{-j}$. Its support is within the trace region of (10.6). Then $\zeta_\ell a_{h,e}$ is globally bounded by C . For $x \in D_{\beta\ell}$, where $\zeta_\ell(x) = 1$, the generator formula gives the exact identity

$$\Theta_q(\zeta_\ell a_{h,e})(x) - \Theta_q a_{h,e}(x) = \int (1 - \zeta_\ell(y)) a_{h,e}(y) j_q(x - y) dy.$$

Transferring its second difference as above rewrites it as

$$R \int u(y, 0) \delta_{h,e,y}^2 [(1 - \zeta_\ell(y)) j_q(x - y)] dy.$$

The L_y^1 norm of the second derivatives of this separated cutoff kernel is $C\ell^{-p-3}$ by (9.4). It bounds the display by $CR\ell^{-p-3}$, uniformly in h , which is smaller than the retained far Poisson

error. The transfer is legitimate for the trace kernel for the same reason as for $K_{x,t}$: the separated cutoff kernel and its first two derivatives belong to L^1 , u is bounded, and translation preserves the integral. All difference paths remain separated for $h \leq 1$ and large ℓ . Hence

$$\|\zeta_\ell a_{h,e}\|_\infty \leq C, \quad \|\Theta_q(\zeta_\ell a_{h,e})\|_{L^\infty(D_{\beta\ell})} \leq C(E_R + R\ell^{-p-3}) \leq CE_R.$$

No separate factor h^{-2} enters either estimate.

Choose

$$\ell = R^{(p+5)/(6(p+2))}, \quad \kappa = \frac{p-1}{2(p+2)}, \quad L = R^{\kappa/8}, \quad \rho_R = R^{\kappa/16}. \quad (10.8)$$

If $a = (p+5)/(6(p+2))$, then $1/(p+2) < a < 1/3$, $1-3a = \kappa$, $a(p+2) - 1 = (p-1)/6 \geq \kappa$, and $a(p-1) \geq \kappa$. Also $Rd^{-p} \leq CR^{-(p-1)/(p+1)} \leq CR^{-\kappa}$. Thus $E_R \leq CR^{-\kappa}$ and $L \ll \ell \ll \sqrt{R} \ll d$. In particular D_{2L} lies inside the common interior forcing region, and the trace cutoff is one on it for large R . Apply (9.6) with radius $4\rho_R$ to the bounded cut trace. For large R this radius lies between one and $L/2$, and

$$\text{osc}_{D_{4\rho_R}} a_{h,e} \leq C(R^{-\kappa/24} + R^{-3\kappa/4}) =: \omega_R.$$

The factor $(4\rho_R/L)^{2/3}$ has order $R^{-\kappa/24}$, and $L^2 E_R \leq CR^{-3\kappa/4}$. Since the trace cutoff is one on $D_{4\rho_R}$, this is the oscillation of the uncut trace as well. Let $h \downarrow 0$ in the coordinate directions and $(e_1 + e_2)/\sqrt{2}$. The bounds are uniform in h and the traces are smooth, so the limits are their directional second derivatives. Polarization yields, with $a_R(x) = Ru(x, 0)$,

$$\text{osc}_{D_{4\rho_R}} D^2 a_R \leq C\omega_R, \quad \|D^2 a_R(0)\| \leq C. \quad (10.9)$$

At the origin $a_R(0) = 0$ and $Da_R(0) = 0$, because the origin is an actual zero and its gradient is vertical. Integrating the Hessian along the segment from zero to x consequently gives

$$\left| a_R(x) - \frac{1}{2} D^2 a_R(0)[x, x] \right| \leq C\omega_R |x|^2 \quad (|x| \leq 4\rho_R).$$

The source was multiplied by L^2 , rather than ℓ^2 ; this is where the separation of these two tangential scales is used.

10.2. A quadratic test for the mean-curvature coefficient. Put $Z_R = R(u - q_s(t))$ and $P_R(x) = \frac{1}{2} D^2 a_R(0)[x, x]$. At the tangent center, $a_R(0) = Da_R(0) = 0$. The integral Taylor formula and (10.9) therefore give $|a_R - P_R| \leq C\omega_R |x|^2$ on $D_{4\rho_R}$. The exact rescaled equation is

$$L_q Z_R = -3q_s Z_R^2/R - Z_R^3/R^2. \quad (10.10)$$

For $|x| \leq 4\rho_R$ and $|t| \leq cd$, put $e_R = \delta_\Sigma - (t - f_\Sigma(x))$. The prepared footpoint equations and the Hessian bound give

$$|e_R| \leq C(|t| + |f_\Sigma(x)|)|x|^2/R^2, \quad |f_\Sigma(x)| \leq C\rho_R^2/R = o(1).$$

Here the fixed comparison fraction is chosen as in the preceding subsection. In particular $|\delta_\Sigma|/R$ is small, and the horizontal footpoint equation gives $|y| \leq 2|x|$ and $|y - x| \leq C|\delta_\Sigma||x|/R$. Write

$$\delta_\Sigma = (t - f_\Sigma(y))\sqrt{1 + |Df_\Sigma(y)|^2}.$$

The bounds $|Df_\Sigma(y)| \leq C|x|/R$ and Taylor's formula for $f_\Sigma(y) - f_\Sigma(x)$ bound the error in this equality by $C|\delta_\Sigma||x|^2/R^2$. Absorption also gives $|\delta_\Sigma| \leq C(|t| + |f_\Sigma(x)|)$, proving the displayed estimate for e_R . Since $\rho_R^2/R^2 \rightarrow 0$ and $\rho_R^2/R \rightarrow 0$, uniformly for $0 \leq \vartheta \leq 1$, $1 + |t - f_\Sigma(x) + \vartheta e_R| \asymp 1 + |t|$. The reference-layer bound $\sup(1 + |t|)\phi(t) < \infty$ and the mean value theorem therefore give

$$R|q_s(t - f_\Sigma + e_R) - q_s(t - f_\Sigma)| \leq C\rho_R^2/R.$$

A separate Taylor expansion of $q_s(t - f_\Sigma)$, using the earlier uniform bound for q_s'' , has remainder $CRf_\Sigma^2 \leq C\rho_R^4/R$. Thus

$$R[q_s(\delta_\Sigma) - q_s(t)] = -Rf_\Sigma(x)\phi(t) + O(\rho_R^2/R + \rho_R^4/R).$$

Its value at $t = 0$, together with the pinned error, gives $a_R(x) = -Rf_\Sigma(x)\phi(0) + O(1 + \rho_R^4/R)$. Substitution back into the preceding expansion proves, precisely on $D_{4\rho_R} \times (-cd, cd)$,

$$Z_R = \frac{a_R(x)}{\phi(0)}\phi(t) + W_R, \quad |W_R| \leq C(1 + \rho_R^4/R). \quad (10.11)$$

Choose a nonnegative $\eta_0 \in C_c^\infty(D_1)$ with positive integral, and set $\eta_\rho(x) = \eta_0(x/\rho)$ and $\psi_\rho(x, t) = \eta_\rho(x)\phi(t)$, with $\rho \geq 1$. The exact horizontal kernel reduction from the normalized constants, and the cancellation $L_0\phi = 0$, give

$$L_q\psi_\rho = \frac{c_{3,s}}{2} \int \frac{\phi(v)[2\eta_\rho(x) - \eta_\rho(x+z) - \eta_\rho(x-z)]}{(|z|^2 + (t-v)^2)^{(3+p)/2}} dz dv. \quad (10.12)$$

At a fixed x , the second increment is bounded by $C_\rho \min(|z|^2, 1)$. Boundedness of ϕ and integration in v therefore prove absolute convergence at each point, since $p < 2$ at zero and $p > 0$ at infinity. The test is smooth and belongs to the weighted operator class used in Lemma 8.1, so the previously established principal-value and compact-test consistency give the same identity in distributions. The second increment has L_x^1 norm at most $C \min(|z|^2, \rho^2)$, by its integral second-derivative formula and its L^1 bound. Normal integration of the kernel leaves $C|z|^{-2-p}$. Splitting the tangential radial integral at ρ therefore gives

$$\|L_q\psi_\rho\|_1 \leq C\rho^{2-p}, \quad \int_{\{|x|>cd\} \cup \{|t|>cd\}} |L_q\psi_\rho| \leq C\rho^2 d^{-p} \quad (cd \geq 4\rho). \quad (10.13)$$

Indeed, for $|x| > 2\rho$ the trace cutoff vanishes and

$$|L_q\psi_\rho(x, t)| \leq C\rho^2 |x|^{-3-p}, \quad \int_{\mathbb{R}} |L_q\psi_\rho(x, t)| dt \leq C\rho^2 |x|^{-2-p}.$$

The second inequality, integrated over $|x| > cd$, proves the horizontal tail. For the normal tail split the source variable at $|v| = cd/2$. The far part has ϕ -mass at most Cd^{-p} and costs $C\rho^{2-p}d^{-p}$. In the near part $|t-v| \geq cd/2$; use the increment bound $C\rho^2$, integrate the kernel in z , and then integrate $|t-v|^{-1-p}$ over $|t| > cd$. This costs $C\rho^2 d^{-p}$. Since $\rho \geq 1$, these bounds give the stated normal tail.

Lemma 8.1 supplies the exact identity

$$L_q(P_R\phi) = -(\Delta P_R)\Xi_p(t), \quad \Xi_p(t) = \frac{c_{1,s}}{2(p-1)} \int |t-v|^{1-p}\phi(v) dv, \quad \sigma_p = \int \phi\Xi_p \in (0, \infty).$$

We use the same moment σ_p throughout the surface argument; its finite ϕ pairing suffices even when $\Xi_p \notin L^2$. For the pairing with a quadratic define

$$k_\phi(z) = c_{3,s} \iint \frac{\phi(t)\phi(v)}{(|z|^2 + (t-v)^2)^{(3+p)/2}} dt dv.$$

For $0 < |z| \leq 1$, the convolution bound $\int \phi(t)\phi(t-a) dt \leq \|\phi\|_2^2$ and integration in a give $k_\phi(z) \leq C|z|^{-2-p}$. For $|z| \geq 1$, the finite normal masses give $k_\phi(z) \leq C|z|^{-3-p}$. The kernel is radial and nonnegative. Tonelli and the definition of Ξ_p give

$$\int z_i z_j k_\phi(z) dz = 2\sigma_p \delta_{ij}.$$

The diagonal moments are finite at zero because $p < 2$ and at infinity because $p > 1$; reflection makes the mixed moment zero. For fixed ρ and any quadratic P , the integral in x of $|P(x+z) - P(x)| |\eta_\rho(x+z) - \eta_\rho(x)|$ is at most $C_{P,\rho}|z|^2$. Below $|z| = 1$ use two first differences on a fixed compact neighborhood of the trace support. Above one, one endpoint lies in that support, and quadratic growth gives the same bound. Thus the bilinear integral is absolutely convergent.

Normal projection in (10.12) gives the operator

$$T_\phi\eta(x) = \frac{1}{2} \int [2\eta(x) - \eta(x+z) - \eta(x-z)] k_\phi(z) dz, \quad \int \phi(t) L_q\psi_\rho(x, t) dt = T_\phi\eta_\rho(x).$$

The normal projection permits Fubini even at a fixed x , because the absolute second increment against k_ϕ is bounded by $C_\rho \int \min(|z|^2, 1) k_\phi(z) dz < \infty$. Symmetric truncation, the preceding

bilinear integrability, and the finite second moment now allow the second increment to be transferred from η_ρ to P_R . Since $2P_R(x) - P_R(x+z) - P_R(x-z) = -D^2a_R(0)[z, z]$, this proves

$$\int \frac{P_R(x)\phi(t)}{\phi(0)} L_q \psi_\rho dx dt = -\frac{\sigma_p}{\phi(0)} \Delta a_R(0) \rho^2 \int \eta_0. \quad (10.14)$$

The ordinary pairing is also absolutely convergent: on a compact horizontal disk use $\phi \in L^\infty$ and $L_q \psi_\rho \in L^1$. Outside the trace support the integral in (10.12) has one sign. After weighting by $\phi(t)$, the far bound for k_ϕ bounds it by $C\rho^2|x|^{-3-p}$. Quadratic growth then leaves the integrable radial tail $\int_1^\infty r^{-p} dr$.

The finite second moment also gives

$$\|T_\phi \eta_\rho\|_1 \leq \frac{1}{2} \|D^2 \eta_\rho\|_1 \int |z|^2 k_\phi(z) dz \leq C.$$

Here the tangential dimension is two, so $\|D^2 \eta_\rho\|_1$ is independent of ρ . These noncompact normal tests are admissible against any bounded distributional solution $L_q Z = F$ with bounded F . Indeed, take the normal cutoffs $\phi_N = \phi\chi(t/N)$ from the normal-cutoff result in Section 8. Its conclusion is $\|L_0 \phi_N\|_1 \rightarrow 0$, and $\phi_N \rightarrow \phi$ in L^1 . The same normalized separation used in (10.12) gives

$$L_q(\eta_\rho \phi_N) = \eta_\rho L_0 \phi_N + \frac{c_{3,s}}{2} \int \frac{\phi_N(v)[2\eta_\rho(x) - \eta_\rho(x+z) - \eta_\rho(x-z)]}{(|z|^2 + (t-v)^2)^{(3+p)/2}} dz dv.$$

The increment estimate above therefore proves

$$\|L_q(\eta_\rho \phi_N) - L_q \psi_\rho\|_1 \leq \|\eta_\rho\|_1 \|L_0 \phi_N\|_1 + C\rho^{2-p} \|\phi_N - \phi\|_1 \rightarrow 0.$$

Each $\eta_\rho \phi_N$ is compact smooth. Testing with it and passing to the limit using boundedness of Z, F gives $\int Z L_q \psi_\rho = \int F \psi_\rho$. This argument uses no decay hypothesis on Z and does not test an unbounded quadratic by a compact-cutoff limit.

Use $\rho = \rho_R$, and choose a cutoff ξ_ρ , between zero and one, supported in $D_{4\rho}$ and equal to one on $D_{2\rho}$. Put $E_* = (a_R - P_R)\xi_\rho$. Then $\|E_*\|_\infty \leq C\omega_R \rho^2$ and, on its support, $|E_*(x)| \leq C\omega_R |x|^2$. The signed normal projection just proved gives

$$\left| \int \frac{E_*(x)\phi(t)}{\phi(0)} L_q \psi_\rho dx dt \right| \leq C\omega_R \rho^2.$$

Let $b = cd$, and write the comparison box as $\mathcal{B} = D_b \times (-b, b)$; for large R , $b \geq 4\rho$. On $D_{2\rho} \times (-b, b)$, the residual $Z_R - (P_R + E_*)\phi/\phi(0)$ is W_R . Its pairing is bounded by $C(1 + \rho^4/R)\rho^{2-p}$. In the rest of $\mathcal{B}$, decompose $Z_R = R(u - q_s(\delta_\Sigma)) + R(q_s(\delta_\Sigma) - q_s(t))$. The first term is uniformly bounded and costs $C\rho^{2-p}$. For the second, use (10.5) and the pointwise horizontal bound for $L_q \psi_\rho$; its contribution is at most

$$C\rho^2 \int_{2\rho}^b r^2 r^{-3-p} r dr \leq C\rho^2 \rho^{1-p}.$$

The subtracted quadratic and E_* have the same bound, since their normal L^1 norms are at most $C|x|^2$ on this region.

Outside $\mathcal{B}$, the bound $|Z_R| \leq 2R$ and (10.13) cost $CR\rho^2 d^{-p}$. For $|x| > b$, $E_* = 0$ and the far horizontal quadratic contribution is at most $C\rho^2 b^{1-p}$, using the normal mass of ϕ and the pointwise horizontal test bound. For $|x| \leq b$ and $|t| > b$, use $|P_R| + |E_*| \leq Cb^2$ and $\phi(t) \leq Cb^{-1-p}$, followed by the normal test tail, to obtain $C\rho^2 b^{1-2p}$. Both terms are at most $C\rho^2 \rho^{1-p}$ because $b \geq 4\rho \geq 4$ and $p > 1$.

On the support of η_ρ within the normal comparison interval, the trace Hessian bound and (10.11) give $|Z_R| \leq C\rho^2$. Pairing the right side of (10.10) against ψ_ρ and dividing by ρ^2 costs $C(\rho^4/R + \rho^6/R^2) \leq C\rho^4/R$, since $\rho^2/R \rightarrow 0$. For $|t| > b$, that right side is bounded by CR globally, while the normal ϕ -mass is Cd^{-p} ; the normalized contribution is CRd^{-p} . The admissibility argument applies for each fixed R , since both Z_R and its exact nonlinear source are bounded

globally. Subtracting the quadratic pairing and using $\sigma_p > 0$, $\phi(0) > 0$ and $\int \eta_0 > 0$ gives

$$|\Delta a_R(0)| \leq C[\omega_R + \rho_R^{1-p} + \rho_R^{-p} + \rho_R^4/R + Rd^{-p}] \leq CR^{-\eta_p}, \quad \eta_p = \frac{p-1}{32(p+2)} \min\{2/3, p-1\} > 0. \quad (10.15)$$

To check the last inequality, the two possibly slowest exponents are $\kappa/24$ from ω_R and $\kappa(p-1)/16$ from ρ_R^{1-p} ; their minimum is exactly η_p . The other exponents are $3\kappa/4$, $\kappa p/16$, $1 - \kappa/4$ and $(p-1)/(p+1)$, each at least η_p .

Differentiate $u(x, f_\Sigma(x)) = 0$ twice at the tangent center. Since $Df_\Sigma(0) = 0$ and $u_t(0) \geq \gamma$, this gives $u_{ab}(0) + u_t(0)f_{\Sigma,ab}(0) = 0$. The increasing graph convention has $H(0) = \Delta f_\Sigma(0)$, so

$$RH(0) = R\Delta f_\Sigma(0) = -\frac{\Delta a_R(0)}{u_t(0)}, \quad |H(0)| \leq CR^{-1-\eta_p}. \quad (10.16)$$

Every estimate used the uniform prepared inputs and fixed cutoffs, so the same constants and threshold apply at all buffered centers.

10.3. Vanishing Dini data imply convergence of the Hessian. Let $F_j(y) = f_{\Sigma_j}(R_j y)/R_j$ on a fixed buffered rescaled domain, with uniformly bounded slopes and Hessians, and suppose $F_j \rightarrow F_\infty$ locally in C^1 . For nested compact domains $K_{\text{int}} \Subset K_{\text{buf}}$, put $h_j(y) = R_j H_j(R_j y)$ and $\delta_j = \|h_j\|_{L^\infty(K_{\text{buf}})} \leq CR_j^{-\eta_p}$. Physical C^3 bounds on u_j , slope control, and the nondegenerate vertical derivative give $|DH_j| \leq C$. In fact the fixed chart orientation has $(u_j)_t = |\nabla u_j|/\sqrt{1 + |Df_{\Sigma_j}|^2}$ uniformly bounded below on its zero graph: the recentered nondegeneracy gives $|\nabla u_j| \geq \gamma$, and the graph slope is uniformly bounded. Differentiating the implicit equation three times expresses each third graph derivative in terms of derivatives of u_j of order at most three, bounded slopes and second graph derivatives, divided by this positive vertical derivative. This bounds $D^3 f_{\Sigma_j}$ and hence DH_j uniformly. Rescaling gives $|Dh_j| \leq CR_j^2$. The modulus on K_{int} consequently satisfies

$$\omega_{h_j}(\rho) \leq \min(2\delta_j, CR_j^2\rho), \quad \|h_j\|_\infty + \int_0^1 \frac{\omega_{h_j}(\rho)}{\rho} d\rho \leq C\delta_j \left[1 + \log \left(2 + \frac{CR_j^2}{\delta_j} \right) \right] \rightarrow 0. \quad (10.17)$$

For $\delta_j = 0$ this expression is zero. For positive δ_j , the power upper bound suffices: the function $x[1 + \log(2 + CR_j^2/x)]$ is increasing for $x > 0$, so the right side is at most $CR_j^{-\eta_p}(1 + \log R_j) \rightarrow 0$. No positive lower bound on δ_j is used. Moduli can be computed on a fixed interior ball cover; the buffers ensure the gradient estimate along the short segments used above.

The graph equation is

$$A(DF_j) : D^2 F_j = h_j, \quad A^{ab}(z) = \frac{\delta_{ab}}{\sqrt{1 + |z|^2}} - \frac{z_a z_b}{(1 + |z|^2)^{3/2}}.$$

Write $M(z) = z/\sqrt{1 + |z|^2}$, so $DM = A$ and the equation is $\text{div } M(DF_j) = h_j$. Convergence in C^1 and $h_j \rightarrow 0$ permit passage against compact smooth tests, giving $\text{div } M(DF_\infty) = 0$. The uniform Hessian bound also gives $F_\infty \in C_{\text{loc}}^{1,1} \subset W_{\text{loc}}^{2,2}$. The Sobolev chain rule converts the weak equation into $A(DF_\infty) : D^2 F_\infty = 0$ almost everywhere. On a fixed buffer with $|DF_\infty| \leq S$, this coefficient matrix is Lipschitz and has eigenvalues in $[(1 + S^2)^{-3/2}, 1]$. The interior Hölder version of [23, Theorem 9.19, pp. 243–244] therefore applies directly to this strong $W_{\text{loc}}^{2,2}$ solution, with $k = 1$, any fixed $0 < \alpha < 1$, zero lower-order coefficients and zero forcing. It gives $F_\infty \in C_{\text{loc}}^{2,\alpha}$. Now $A(DF_\infty) \in C_{\text{loc}}^{1,\alpha}$, so the same theorem with $k = 2$ gives $C_{\text{loc}}^{3,\alpha}$; iteration on nested interior balls proves smoothness. Put $w_j = F_j - F_\infty$, $A_j = A(DF_j)$ and $b_j = h_j + [A(DF_\infty) - A_j] : D^2 F_\infty$. The matrices are uniformly elliptic and Lipschitz. Indeed bounded slopes fix the ellipticity constants, while the uniform Hessian bound controls their Lipschitz constants. If $c_j = [A(DF_\infty) - A_j] : D^2 F_\infty$, then on each fixed smaller buffered domain $\epsilon_j = \|c_j\|_\infty \rightarrow 0$ and $\text{Lip}(c_j) \leq C$, since F_∞ is smooth there. Thus $\omega_{c_j}(t) \leq \min(2\epsilon_j, Ct)$, and its Dini norm is at most $C\epsilon_j[1 + \log(2 + C/\epsilon_j)] \rightarrow 0$,

with the value zero if $\epsilon_j = 0$. Adding (10.17) proves the required vanishing Dini norm of b_j . Also $\|D^2w_j\|_\infty \leq C$ and $w_j \rightarrow 0$ in C^1 .

At y_0 in a compact $K \Subset K' \Subset K_{\text{int}}$, freeze the matrix and localize in $B_{2\rho}(y_0) \Subset K'$:

$$A_j(y_0) : D^2w_j = b_j + [A_j(y_0) - A_j(y)] : D^2w_j.$$

Put $B = A_j(y_0)$ and

$$\Gamma_B(z) = \frac{1}{2\pi\sqrt{\det B}} \log |B^{-1/2}z|, \quad B : D^2\Gamma_B = \delta_0.$$

The fundamental-solution identity follows from the linear change of variables $z = B^{1/2}\zeta$ and $\Delta_\zeta((2\pi)^{-1} \log |\zeta|) = \delta_0$; the determinant factor accounts for the transformed point mass. The off-origin Hessian $K_B = D^2\Gamma_B$ is homogeneous of degree -2 , satisfies $|K_B(z)| \leq C|z|^{-2}$, and has zero integral on each centered annulus. The last assertion follows from the divergence theorem: for each a, b , the integral $\int_{\partial B_r} \partial_a \Gamma_B \nu_b$ is independent of r by homogeneity. Distributionally,

$$D^2\Gamma_B = \text{PV } K_B + C_B \delta_0, \quad C_B = \int_{S^1} D\Gamma_B(\vartheta) \otimes \vartheta \, d\vartheta, \quad |C_B| \leq C.$$

To see the atom explicitly, integrate by parts outside B_ε : the inner boundary term is $\int_{S^1} D\Gamma_B(\vartheta) \otimes \vartheta \psi(\varepsilon\vartheta) \, d\vartheta$, which tends to $C_B \psi(0)$. The remaining integral converges in radial principal value by annular cancellation. The first derivative of Γ_B has no atom, since its preceding integration-by-parts boundary term is $O(\varepsilon |\log \varepsilon|)$. All constants are uniform for the matrices B at issue. Take a radial cutoff χ , equal to one on $B_\rho(y_0)$ and supported in $B_{2\rho}(y_0)$, with its usual derivative bounds. The compactly supported function χw_j has frozen forcing

$$\chi b_j + \chi(B - A_j) : D^2w_j + 2(BD\chi) \cdot Dw_j + (B : D^2\chi)w_j.$$

Extend $z_j = \chi w_j$ by zero to $\mathbb{R}^2$ and denote the displayed forcing by $g_j = B : D^2z_j$. The localized function is smooth and compactly supported. Convolution of distributions is consequently well defined, and constant-coefficient differentiation commutes with it. Thus

$$\Gamma_B * g_j = \Gamma_B * (B : D^2z_j) = (B : D^2\Gamma_B) * z_j = z_j, \quad D^2z_j = (\text{PV } K_B + C_B \delta_0) * g_j.$$

This is an exact identity, with no undetermined harmonic summand and no L^2 requirement. At y_0 , $D^2z_j = D^2w_j$ and $g_j(y_0) = b_j(y_0)$. Radiality of χ and annular cancellation permit subtraction of $b_j(y_0)$ in the first principal-value integral; the local term is $C_B b_j(y_0)$. The resulting integral is bounded by $C \int_0^{2\rho} \omega_{b_j}(t) \, dt/t$. The coefficient residual vanishes at y_0 and is at most $C|y - y_0| \|D^2w_j\|_\infty$. Its atom is therefore zero, and its kernel integral is absolutely convergent and bounded by $C\rho \|D^2w_j\|_\infty$. The last two terms also have zero atom at y_0 , are supported in the annulus, and cost $C(\rho^{-1} \|Dw_j\|_\infty + \rho^{-2} \|w_j\|_\infty)$. Consequently

$$|D^2w_j(y_0)| \leq C\rho \|w_j\|_{C^1(K')} + C \left[\|b_j\|_\infty + \int_0^{2\rho} \frac{\omega_{b_j}(t)}{t} \, dt \right] + C\rho \|D^2w_j\|_{L^\infty(K')}. \quad (10.18)$$

The last term uses the radial integral $\int_0^{2\rho} dt$. For fixed ρ the first two terms vanish as $j \rightarrow \infty$, while the uniform Hessian bound bounds the last term by $C\rho$. Thus $\limsup_j \sup_K |D^2w_j| \leq C\rho$ for every sufficiently small fixed ρ ; only then let $\rho \downarrow 0$. The estimates are uniform in $y_0 \in K$, so they apply to the given C^1 -convergent subsequence without another extraction.

Theorem 10.1. *Under the uniform prepared inputs (10.1), $|H| \leq CR^{-1-\eta_p}$ at every buffered zero center, where η_p is defined in (10.15). If such inputs hold on charts of radius cR_j , $R_j \rightarrow \infty$, and the rescaled graphs have uniformly bounded slopes and Hessians, every locally C^1 convergent subsequence converges in C_{loc}^2 to a smooth minimal graph along that same subsequence. A nonzero normalized second fundamental form at a marked center is preserved.*

Proof. The curvature rate is (10.16). Equations (10.17)–(10.18) prove the convergence. The second fundamental form depends continuously on the slope and Hessian. More precisely, if the marked horizontal coordinates $y_j \rightarrow y_\infty$ remain in a compact interior set, graph rescaling gives

$$R_j |A_{\Sigma_j}|(R_j y_j, f_{\Sigma_j}(R_j y_j)) = |A_{F_j}|(y_j) \longrightarrow |A_{F_\infty}|(y_\infty).$$

Indeed, in graph coordinates the metric is $g_{ab} = \delta_{ab} + F_a F_b$ and the second fundamental form is $\Pi_{ab} = F_{ab}/\sqrt{1+|DF|^2}$, up to the common orientation sign. Its squared norm is $g^{ac}g^{bd}\Pi_{ab}\Pi_{cd}$, a continuous function of the slope and Hessian. The identities $DF_j(y) = Df_{\Sigma_j}(R_j y)$ and $D^2F_j(y) = R_j D^2f_{\Sigma_j}(R_j y)$ show that the metric coefficients are unchanged and the second fundamental form is multiplied by R_j . Thus a normalization tending to a positive value, in particular $R_j|A_{\Sigma_j}|(P_j) \rightarrow 1$, is retained with that value. Mere nonvanishing for each j , without a positive limiting normalization, is not needed or asserted. The constants may depend on s and the fixed prepared-chart constants; no uniform assertion as $s \downarrow 1/2$ is made. $\square$

11. TURNING AMBIENT STABILITY INTO A SURFACE INEQUALITY

Let u be a bounded stable solution in $\mathbb{R}^3$. Lemma 2.1 gives $|u| \leq 1$, and Section 3 gives every fixed global derivative bound. On a buffered regular portion Σ of $\{u = 0\}$ include every local branch and orient it by $n = \nabla u/|\nabla u|$. We use the following inputs at a positive scale r , which is the restriction of a positive ambient 1-Lipschitz function, or a constant:

$$\begin{aligned} \text{complete small-slope charts of radius } c_0 r, \quad |A| &\leq C_0/r, \quad \text{distinct branch gaps} \geq c_1 r^{2/(p+1)}, \\ |\nabla u(P + tn(P)) - \phi(t)n(P)| &\leq C_N/r(P) \quad (|t| \leq N), \\ 1 - |u(X)| &\leq C(1 + \text{dist}(X, \{u = 0\}))^{-p}, \quad |\nabla u(X)| \leq C(1 + \text{dist}(X, \{u = 0\}))^{-p-1}. \end{aligned} \tag{11.1}$$

The profile bound is uniform for every fixed N . Local branches which reconnect outside their charts remain distinct branches in the gap condition. The preceding chart, separation, pinned-profile and clearing results provide these inputs at a selected curvature scale, before a distance-to-bad-set curvature bound has been obtained.

11.1. Injective tubes and the full vector identity. Choose the intrinsic-horizon constant $c > 0$ below the fixed chart buffer constants. Next fix a geometric $\tau > 0$ sufficiently small relative to the branch-gap constant c_1 , and set $T(P) = \tau r(P)^{2/(p+1)}$. For a large lower scale threshold,

$$T/r = o(1), \quad |\nabla_\Sigma T| \leq CT/r \quad \text{a.e.}$$

All tube assertions in this subsection are on $r(P) > r_\theta$; this contains the supports of κ and ξ with a fixed margin. The normal map $X(P, t) = P + tn(P)$ is injective there for $|t| < 4T(P)$ and has Jacobian $J(P, t) = \det(I - tA) = 1 + O(T/r)$. Indeed, a collision implies $|P - Q| \leq 4(T(P) + T(Q))$. Once $T \leq \epsilon r$ with $4\epsilon < 1/3$, ambient Lipschitz continuity gives

$$r_+ - r_- \leq 4\epsilon(r_+ + r_-), \quad r_+ \leq \frac{1 + 4\epsilon}{1 - 4\epsilon} r_- \leq 2r_-.$$

Both bases lie in one complete buffered chart because their distance is $O(\tau r_-^{2/(p+1)}) = o(r_-)$. In this chart a vertical gap h between two graphs of slope at most σ gives Euclidean separation at least $h/\sqrt{1 + \sigma^2}$: for points with base separation a , their vertical separation is at least $(h - \sigma a)_+$, and minimizing $a^2 + (h - \sigma a)_+^2$ over $a \geq 0$ gives this bound. Indeed, for $0 \leq a \leq h/\sigma$ the minimum is $h^2/(1 + \sigma^2)$, while for $a \geq h/\sigma$ the horizontal term alone is at least this value; the case $\sigma = 0$ is immediate. On the retained common base disk, the branch gap is at least a fixed multiple of $c_1 r_-^{2/(p+1)}$. Choose τ so that the collision bound is smaller than this separation. Thus different local graphs are excluded, including graphs connected outside the chart. On one graph, $|n(P) \cdot (Q - P)| + |n(Q) \cdot (Q - P)| \leq Cr(P)^{-1}|P - Q|^2$. Here the normal variation on one graph has a consistent sign, because $n = \nabla u/g$ is continuous on that regular graph. The two Taylor remainders therefore have the stated bounds regardless of its vertical orientation. Dotting $Q - P = tn(P) - vn(Q)$ with $Q - P$ gives $|P - Q|^2 \leq C(|t| + |v|)r_-^{-1}|P - Q|^2$. The coefficient is less than one at the large lower threshold, so $P = Q$, and then $t = v$. The differential in the tangent directions is $I - tA$, where $A = -D_\Sigma n$, and in the normal direction it is n . Its inverse has a uniform bound when $4T|A| < 1/2$; thus the injective normal map has smooth local inverses and positive Jacobian. The same graph estimate gives

$$|X(P, t) - Q|^2 \geq t^2 + (1 - C|t|/r(P))|P - Q|^2.$$

If a zero Q were closer to $X(P, t)$ than P , then $|P - Q| < 2|t| \leq 8T(P)$. Hence it lies in the complete chart; a zero outside that chart is farther away by the fixed buffer. The same branch separation excludes the other graphs, while the displayed same-graph bound makes $|X(P, t) - Q| \geq |t|$, with equality only at $Q = P$. Thus the distance is to the entire zero set, not just the selected surface portion. Consequently

$$\text{dist}(X(P, t), \{u = 0\}) = |t| \quad (|t| < 4T(P)), \quad g(X(P, t)) \leq Cw(t), \quad (11.2)$$

where $g = |\nabla u|$ and $w(t) = (1 + |t|)^{-p-1}$.

We will need a uniform Fermi distance comparison on the intrinsic horizon. A path of length below $2cr(P)$ remains in the buffered cylinder and on its starting graph: each complete local graph is relatively open and closed in that cylinder's zero set. Thus a remote reconnection cannot shorten a selected path by switching local branches. For $\rho = d_\Sigma(P, Q) < cr(P)$, the same graph's straight base-segment lift stays in the common buffered disk and gives $|P - Q| \leq \rho \leq C|P - Q|$. Along these paths $r \asymp r(P)$, so $|n(Q) - n(P)| \leq C\rho/r(P)$. Graph Taylor expansion gives $|(Q - P) \cdot n(P)| \leq C\rho^2/r(P)$, while its tangential component is comparable to ρ . In

$$X(P, t) - X(Q, v) = P - Q + (t - v)n(P) + v[n(P) - n(Q)]$$

the error from the orthogonal tangent/normal leading vector is at most $C[c + T(Q)/r(P)]\rho$. Choose c small and the threshold large so this is absorbed by that leading length. Consequently, on selected pairs with $|t| < 3T(P)$ and $|v| < 3T(Q)$,

$$|X(P, t) - X(Q, v)| \asymp [\rho^2 + (t - v)^2]^{1/2}.$$

For a nonnegative compact Lipschitz ξ supported in $r > 4r_\theta$, fix once and for all a smooth $0 \leq b \leq 1$, equal to one on $[-1, 1]$ and supported in $(-3/2, 3/2)$, and define $\eta(X(P, t)) = b(t/T(P))\xi(P)$, with zero extension. This is one compact Lipschitz ambient function. On the compact base support, r and T have positive minima and finite maxima. The finite buffered chart cover has bounded inverse-coordinate derivatives, and the normal support $|t| < 3T/2$ has a fixed margin inside $|t| < 4T$. The base support also has a margin from $r = r_\theta$ and the surface chart boundary. The weak chain rule gives, in the tube, $|D\eta| \leq C(|\nabla_\Sigma \xi| + |\xi|(T^{-1} + r^{-1}))$. Consequently its zero extension is compact and globally Lipschitz. Bounded Hessians make $g = |\nabla u|$ bounded Lipschitz as well, so $g\eta$ is a compact Lipschitz ambient function. Constants may depend on the fixed cutoff profiles. Take a smooth even cutoff $0 \leq \beta \leq 1$ equal to one on $[-2, 2]$, supported in $(-3, 3)$, nonincreasing on the positive half-line. Let ζ be a nonincreasing intrinsic-horizon cutoff equal to one near zero and supported below one, and let κ vanish for $r \leq 2r_\theta$, equal one for $r \geq 3r_\theta$, with $|\nabla \kappa| \leq C/r_\theta$. Both ζ and κ take values in $[0, 1]$; ζ is smooth and κ may be Lipschitz when r is Lipschitz. The symmetric selector is

$$\chi(X(P, t), X(Q, v)) = \kappa(P)\kappa(Q)\beta(t/T(P))\beta(v/T(Q))\zeta\left(\frac{d_\Sigma(P, Q)}{c \min(r(P), r(Q))}\right), \quad (11.3)$$

with zero extension; intrinsic distance is infinite between components. The intrinsic-horizon argument above keeps selected targets on the source's single local graph. At an active source $r(P) > 4r_\theta$, selected targets satisfy $|r(Q) - r(P)| \leq |P - Q| \leq \rho < cr(P)$ and hence $r(Q) > (1 - c)r(P) > 3r_\theta$ for $c < 1/4$, so both base factors are one. The same applies on the support of the horizon derivative, whose argument is supported strictly below the end of the horizon. The inverse normal map and the source margin $|t| < 3T/2$ show that, for a fixed $c_* > 0$,

$$\chi(x, y) = 1 \quad \text{if } \eta(x) \neq 0 \text{ and } |x - y| < c_*T(P). \quad (11.4)$$

For a point y in this ball, choose a nearest actual zero Q ; it exists because the entire zero set is nonempty and closed. Since $|x - P| < 3T(P)/2$, the inequalities $|y - Q| \leq |y - P| \leq (3/2 + c_*)T(P)$ and $|P - Q| \leq |P - y| + |y - Q| \leq (3 + 2c_*)T(P)$ put Q in the same complete buffered cylinder. The fixed small τ makes the latter distance less than the different-branch separation just proved, so Q lies on the source's local graph. Its nearest-point displacement is normal, so $y = X(Q, v)$ and tube injectivity makes these coordinates unique. Ambient Lipschitz continuity gives $r(Q) = r(P) + O(T(P))$ and thus $T(Q) = T(P)[1 + O(T(P)/r(P))]$. Choose

$c_* < 1/8$, and then increase the threshold until $T(P)/T(Q) < 16/15$. The preceding normal-displacement bound gives $|v| < 2T(Q)$. Join the horizontal projections of P and Q by a straight segment in the common buffered base disk and lift it to the same graph. Its length is at most $\sqrt{1+\sigma^2}|P-Q| \leq CT(P) < cr(P)$ at the large threshold. Thus the short-intrinsic-distance condition is established before using the chord/intrinsic comparison, and the base displacement is at most $C|P-Q| \leq CT(P) \ll cr(P)$. The horizon argument is in the plateau of ζ . The two κ factors and two β factors are therefore one, proving the claim.

We also record selector derivatives needed for the signed row. At an active source, the base cutoff $\kappa(Q)$ is identically one wherever the horizon factor or its derivatives are nonzero. Intrinsic distance is locally 1-Lipschitz in its base point, and r is 1-Lipschitz. Writing $\rho = d_\Sigma(P, Q)$, the weak target derivative of the horizon factor is therefore bounded by $C/r(P)$: with $m = \min(r(P), r(Q)) \asymp r(P)$, the weak derivative of $\rho/(cm)$ has bound $1/(cm) + \rho/(cm^2) \leq C/r(P)$. The support of ζ' puts ρ between fixed positive multiples of $cr(P)$. Pullback through the bounded inverse normal differential retains this bound. The normal cutoff derivative has the decomposition

$$D_y[\beta(v/T(Q))] = T(Q)^{-1}\beta'(v/T(Q))n(Q) + E(y), \quad |E(y)| \leq Cr(Q)^{-1}\mathbf{1}_{\{2T(Q) < |v| < 3T(Q)\}}.$$

Indeed $D_y v = n(Q)$ and the tangential inverse normal differential is uniformly bounded, while $|D_y T(Q)| \leq CT(Q)/r(Q)$. The chain-rule remainder is $-\beta'(v/T(Q))vT(Q)^{-2}D_y T(Q)$; it has magnitude at most $C/r(Q)$ on the stated normal annulus. Since T/r is small, it is absorbed by the normal-shell bound $CT(Q)^{-1}$. Thus target rows of χ are Lipschitz, with derivatives bounded by the normal-shell term $CT(Q)^{-1}$ and the horizon-shell term $Cr(P)^{-1}$. These assertions are weak derivative statements when r is only Lipschitz; the zero extension causes no boundary term because the horizon lies in a compact interior graph patch, its bases satisfy $r(Q) > 3r_\theta$, and the normal cutoff vanishes before $|v| = 4T(Q)$.

Write $K(x, y) = |x - y|^{-3-p}$ and $\nu = \nabla u/g$ where $g > 0$. Set

$$\begin{aligned} P_\chi &= \frac{c_{3,s}}{2} \iint \chi g(x)g(y)(\eta(x) - \eta(y))^2 K(x, y) \, dx \, dy, \\ D_\chi &= \frac{c_{3,s}}{2} \iint \chi g(x)g(y)\eta(x)^2 |\nu(x) - \nu(y)|^2 K(x, y) \, dx \, dy, \\ V_\chi(x) &= \int (1 - \chi(x, y)) \nabla u(y) K(x, y) \, dy \quad (\eta(x) \neq 0). \end{aligned}$$

The row is only asserted at active sources, where (11.4) removes its diagonal singularity. Terms weighted by η^2 are zero elsewhere. At zeros of g interpret the defect through $g(x)g(y)|\nu(x) - \nu(y)|^2 = 2[g(x)g(y) - \nabla u(x) \cdot \nabla u(y)]$ and interpret $g\nu \cdot V_\chi$ as $\nabla u \cdot V_\chi$.

Apply the already proved vector identity of Lemma 5.1 to this compact Lipschitz η : $Q_u(g\eta) = P_{\text{all}} - D_{\text{all}}$. Its universal identity uses no two-layer hypotheses. Splitting by χ gives

$$\begin{aligned} Q_u(g\eta) &= P_\chi - D_\chi + c_{3,s} \int g\eta^2 \nu \cdot V_\chi \, dx \\ &\quad - c_{3,s} \iint (1 - \chi)g(x)g(y)\eta(x)\eta(y)K(x, y) \, dx \, dy. \end{aligned} \tag{11.5}$$

To see the new selector algebra explicitly, put $\mathcal{D}(x, y) = g(x)g(y) - \nabla u(x) \cdot \nabla u(y)$. Symmetry of χ gives

$$\begin{aligned} P_{1-\chi} &= c_{3,s} \iint (1 - \chi)g(x)g(y)[\eta(x)^2 - \eta(x)\eta(y)]K \, dx \, dy, \\ D_{1-\chi} &= c_{3,s} \iint (1 - \chi)\mathcal{D}(x, y)\eta(x)^2 K \, dx \, dy. \end{aligned}$$

Their difference is precisely the two exterior terms in (11.5). Every exterior target is retained in V_χ , including targets outside all normal coordinates. The last term in (11.5) is nonpositive, including all remote and returning branches, because the actual amplitude η is nonnegative. To check the diagonal, use the Lipschitz bound for η and the pointwise inequality

$$0 \leq 2[g(x)g(y) - \nabla u(x) \cdot \nabla u(y)] \leq |\nabla u(x) - \nabla u(y)|^2 \leq C|x - y|^2.$$

Its product with K is locally integrable in three dimensions because $p < 2$. Compact support and bounded gradients control the tails, while (11.4) removes the diagonal in every signed row at an active source. The compact-test extension of stability in Section 3 makes $g\eta$ admissible even at zeros of g . In particular $Q_u(g\eta) \geq 0$.

11.2. Signed rows retain every exterior target.

Lemma 11.1. *For fixed $\theta > 0$, increasing r_θ if necessary gives*

$$Q_u(g\eta) \leq P_\chi - (1 - \theta)D_\chi + C_\theta \int_\Sigma [T^{-1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}] \xi^2 dA. \quad (11.6)$$

Proof. Fix an active source $x = X(P, t)$ and write $r = r(P)$, $T = T(P)$, $a = 1 + |t|$. All comparison constants below are fixed by the prepared charts, the cutoffs, p and τ . Increase the lower threshold so that $T \geq 4$, T/r is as small as required, and every chart used below lies in the supplied buffers. At active sources $|t| < 3T/2$, hence $a \leq CT$. The source factors in χ equal one, and the target base factor κ equals one on its entire intrinsic horizon. Thus only the target normal cutoff and intrinsic-horizon factor are differentiated below. Every selector-derivative term vanishes on $B_{c_*T}(x)$, where all these factors are in their constant plateaux.

The selector-ball plateau gives absolute convergence of $V_\chi(x)$, since the global gradient is bounded and $1 - \chi(x, y) = 0$ near x . Componentwise integration by parts on the whole ambient space gives

$$V_{\chi,i}(x) = - \int u(y) \partial_{y_i} [(1 - \chi(x, y)) K(x, y)] dy.$$

For precision, first insert an ambient cutoff at radius M centered at x . The term from its derivative is $O(M^{-1-p})$, because $|u| \leq 1$; it tends to zero, and the differentiated kernel is integrable off the selector ball. The target selector has compact support on the selected local branch. Its inverse tube coordinates have uniform Lipschitz bounds, so its weak derivatives and their zero extensions are legitimate. Lipschitz widths and intrinsic distances can be approximated in these coordinates while retaining the interior plateaux; alternatively the same calculation uses weak derivatives. All estimates below depend on first derivatives only.

The kernel derivative outside $B_{c_*T}(x)$ costs CT^{-p-1} . The target normal-cutoff derivative is supported where $2T(Q) \leq |v| \leq 3T(Q)$. There $r(Q) \asymp r$, $T(Q) \asymp T$, and its distance from x is bounded below in source tangent coordinates by $c(|z|^2 + T^2)^{1/2}$. For $|z| \leq 4T$ the graph height and width variation are $O(T^2/r)$, so the margin between $2T(Q)$ and $|t| < 3T/2$ gives separation cT . For $|z| > 4T$, the tangential projection of $Q + vn(Q) - x$ has length at least $(1 - CT/r)|z|$. Integrating first in z and then over a normal interval of length CT bounds the integral of K there by CT^{-p} . The normal derivative C/T therefore costs CT^{-p-1} ; the base part caused by the variable width is C/r and costs CT^{-p}/r . The intrinsic-horizon derivative is C/r on a base annulus of area Cr^2 and normal length CT , at distance at least cr ; it costs CTr^{-p-2} . Since $T \leq cr$, all these contributions give

$$|V_\chi(x)| \leq CT^{-p-1}. \quad (11.7)$$

The core $\mathcal{C}_x = \{X(Q, v) : a < d_\Sigma(P, Q) < 2a, |v| < 1\}$ is in all selector plateaux. A complete small-slope chart identifies its intrinsic annulus with an annulus of comparable area a^2 ; for example a fixed narrower Euclidean annulus lies inside it. For the annulus comparison use only $|z| \leq d_\Sigma(P, Q) \leq \sqrt{1 + \sigma^2}|z|$ on the short chart paths, where the fixed chart slope is at most σ . Choose a Euclidean subannulus between a and $2a/\sqrt{1 + \sigma^2}$ for the lower area bound, and the disk $|z| < 2a$ for the upper bound; the slopes are small enough that the subannulus has positive fixed width. No cut-locus or geodesic-polar assertion is needed. The normal Jacobian is uniformly comparable to one. Its ambient distance from x is between ca and Ca : the tangential base separation is comparable to a , the target normal offset is at most one, and $|t| \leq a$. On this core $r(Q) \asymp r$ and the fixed- N profile, with $N = 1$, gives

$$\nabla u(X(Q, v)) = \phi(v)n(Q) + O(r^{-1}), \quad |n(Q) - n(P)| \leq Ca/r.$$

Define $m_x = \int_{\mathcal{C}_x} gK$, $A_x = \int_{\mathcal{C}_x} \nabla u K$ and $I_x = \int_{\mathcal{C}_x} \phi(v)K$. Since $\min_{[-1,1]} \phi > 0$, these geometric estimates give $I_x \asymp a^{-p-1}$ and $\int_{\mathcal{C}_x} K \asymp a^{-p-1}$. Moreover

$$|m_x - I_x| \leq Cr^{-1} \int_{\mathcal{C}_x} K, \quad |A_x - n(P)I_x| \leq C(a/r) \int_{\mathcal{C}_x} K.$$

For large thresholds it follows that

$$m_x \asymp a^{-p-1}, \quad cm_x \leq |A_x| \leq m_x, \quad e_x = A_x/|A_x|, \quad |e_x - n(P)| \leq Ca/r.$$

The upper bound $|A_x| \leq m_x$ is the triangle inequality. The moving core is measurable in the source coordinates; its kernel is separated from the source, so integration and normalization give a measurable direction even when $g(x) = 0$. It is never differentiated. For $g(x) > 0$, retain this core in the nonnegative defect. The exact identity

$$\int_{\mathcal{C}_x} g(y)|\nu(x) - \nu(y)|^2 K dy = 2m_x - 2\nu(x) \cdot A_x \geq |A_x||\nu(x) - e_x|^2$$

therefore gives

$$\eta(x)^2 d_\chi(x) \geq cg(x)\eta(x)^2 a^{-p-1} |\nu(x) - e_x|^2. \quad (11.8)$$

At $g(x) = 0$, $\nabla u(x) = 0$ and the vector expression for the defect makes $d_\chi(x) = 0$. Thus the weighted conclusion has that same interpretation without defining $\nu(x)$ there.

We next replace the intrinsic horizon in this one branch by a physical cylinder. Rotate so $n(P)$ is vertical, and choose a small fixed $c_h > 0$ whose disk $D_P \subset T_P \Sigma$ of radius $c_h r$ lies well inside the horizon plateau and chart buffers. The auxiliary target weight is the indicator of the straight cylinder $\pi_P(y - P) \in D_P$, multiplied by the unchanged normal factor $\beta(v/T(Q))$ on this branch, and is zero elsewhere. If $y = Q + vn(Q)$ belongs to it and $|v| < 3T(Q)$, then with $z = \pi_P(Q - P)$,

$$|z| \leq c_h r + C(T/r)|z|, \quad |z| \leq 2c_h r.$$

Here $|\pi_P n(Q)| \leq C|z|/r$, and $T(Q) \asymp T$ on the buffered chart. The projected tube therefore stays on the same branch and in the old horizon plateau. Conversely the old and new weights both equal the same normal factor near x . Their difference is supported at ambient distance comparable to r , over base area Cr^2 and normal length CT .

Integrate this difference against $\nabla u K$ by parts. Its volume kernel term costs $C(Tr^2)r^{-4-p} = CT r^{-p-2}$; old horizon and width-base derivatives have the same bound. Its cylindrical wall has area CTr , so its variation costs $C(Tr)r^{-3-p} = CT r^{-p-2}$. Its normal-cutoff derivative costs $CT^{-1}(Tr^2)r^{-3-p} = Cr^{-p-1}$. The product rule for the Lipschitz normal weight and the cylinder indicator is a BV product rule; the wall term just listed is its surface variation. Smooth approximations give the same bounds and converge because the difference is separated from x . Thus the replacement changes the row by at most Cr^{-p-1} . This comparison subtracts two full complement weights $1 - \chi$ and $1 - \tilde{\chi}$; it removes no other branch or exterior target.

Write $d\omega(\lambda) = -\beta'(\lambda) d\lambda$ on $[2, 3]$. It is a positive measure of mass one and

$$\beta(v/T(Q)) = \int_2^3 \mathbf{1}_{|v| < \lambda T(Q)} d\omega(\lambda).$$

Consequently the auxiliary complement row is the average of complements of the actual variable-width tubes Ω_λ on this single branch, clipped by the physical cylinder.

We verify that each clipped tube is exactly a region between two caps. Write its zero graph as $Q(z) = P + z + f_\Sigma(z)n(P)$ on a disk containing $D_{2c_h r}$. It has $f_\Sigma(0) = Df_\Sigma(0) = 0$, $|Df_\Sigma(z)| \leq C|z|/r$ and $|D^2 f_\Sigma| \leq C/r$. For a scaled normal parameter $v \in [-3, 3]$, its physical projection is

$$F_v(z) = z + E_v(z), \quad E_v(z) = vT(Q(z))\pi_P n(Q(z)).$$

The local Lipschitz bounds for T and n give $\text{Lip}(E_v) \leq CT/r < 1/4$, and $E_v(0) = 0$. Thus F_v is injective. For each physical base $z_{\text{phys}} \in D_P$, the map $z \mapsto z_{\text{phys}} - E_v(z)$ is a contraction of the closed buffer disk $\overline{D}_{2c_h r}$ into itself, since $|E_v(z)| \leq (CT/r)|z|$. It has a unique fixed point z_v . This proves coverage and a uniformly Lipschitz inverse, not just local invertibility. The inverses are Lipschitz jointly in their physical base and scaled normal parameter.

At a fixed physical base let $H(v) = f_\Sigma(z_v) + vT(Q(z_v))n(Q(z_v)) \cdot n(P)$. Subtracting the projection equations at $v+h$ and v gives $|z_{v+h} - z_v| \leq C|h|T|Df_\Sigma(z_v)|$. The smoothness of f_Σ, n and the first Lipschitz bound CT/r for T then give, for $h > 0$,

$$H(v+h) - H(v) = hT(Q(z_v))[1 + O(|Df_\Sigma|^2 + T/r)].$$

The constants are uniform; this follows directly by estimating the change of f_Σ, T and n separately in the difference. Choose c_h small and then the threshold large so the bracket lies between $1/2$ and 2 . The height is continuous and strictly increasing, so every height between the two cap values is filled and no interior piece is missing. This increment argument remains valid for Lipschitz widths and implies the corresponding derivative statement almost everywhere, without a multidimensional chain rule on exceptional paths.

The cap heights at physical base z satisfy

$$h_\pm(z) = \pm\lambda T + O(|z|^2 + T|z| + T^2/r). \quad (11.9)$$

Indeed $|z_v - z| \leq C(T/r)|z|$, $|f_\Sigma(z_v)| \leq C|z|^2/r$, $|T(Q(z_v)) - T(P)| \leq CT|z|/r$ and $|n(Q(z_v)) \cdot n(P) - 1| \leq C|z|^2/r^2$. Their Lipschitz constants are at most $L_H = Cc_h + CT/r$. The same height bound, with the displayed $T|z|/r$ and T^2/r terms, holds after projecting along e_x : its angle from $n(P)$ is at most $Ca/r \leq CT/r$, and the cap Lipschitz bounds make this small change of projection injective and onto a smaller buffered disk.

Consider $y = x + w + \ell e_x$, $w \perp e_x$. Membership in the physical cylinder is an interval in ℓ , possibly the whole line or empty. Inside it, the cap difference has the increment bound

$$[y_3(\ell_2) - h_\pm(z(\ell_2))] - [y_3(\ell_1) - h_\pm(z(\ell_1))] \geq [e_x \cdot n(P) - L_H|\pi_P e_x|](\ell_2 - \ell_1) \geq \frac{1}{2}(\ell_2 - \ell_1)$$

for $\ell_2 > \ell_1$. Hence the line meets the tube in one interval or not at all. This assertion holds for every line, using differences of Lipschitz functions rather than differentiating on that line. The clipped tube is bounded, so a nonempty intersection is a bounded interval. Its complement in the whole ambient line is precisely the two full complementary rays; when the intersection is empty it is the whole line. Points beyond the cylinder wall remain in this complement. For $\rho = |w| < c_0\sqrt{rT}$, the relation $w \perp e_x$ gives $|w \cdot n(P)| \leq C(T/r)\rho$. On $|\ell| \leq MT$, for a fixed large M , the physical base has size at most $\rho + CMT^2/r = o(r)$. It therefore stays strictly inside the cylinder wall. Substitution in (11.9), with $e_x \cdot n(P) = 1 + O(T^2/r^2)$, gives unique cap parameters

$$b_\pm(w) = \pm\lambda T - t + O(\rho^2/r + T\rho/r + T^2/r).$$

Choose c_0 small first and then increase the threshold. Since $|t| < 3T/2$ and $\lambda \geq 2$, the leading endpoint margins are $T/2$ and the error is smaller than $T/4$. The two crossings precede the wall and obey

$$-CT \leq b_-(w) \leq -cT < cT \leq b_+(w) \leq CT.$$

They are the normal endpoints $Q_\pm \pm \lambda T(Q_\pm)n(Q_\pm)$, not artificial wall points. Their distance from the full zero set is $\lambda T(Q_\pm) \asymp T$ by (11.2). The derivative of u along $n(Q_\pm)$ at the zero is positive. On the remaining positive or negative normal segment there is no zero, by the same actual-distance identity; continuity propagates the initial sign to the cap. Clearing now gives

$$1 - u(x + w + b_+e_x) \leq CT^{-p}, \quad 1 + u(x + w + b_-e_x) \leq CT^{-p}.$$

Apply the signed-ray calculation from the proof of Lemma 5.2. Set $U(\ell) = u(x + w + \ell e_x)$ and $k_\rho(\ell) = (\rho^2 + \ell^2)^{-(3+p)/2}$. With the actual endpoints just verified its two identities are

$$\begin{aligned} \int_{b_+}^{\infty} U'k_\rho &= (1 - U(b_+))k_\rho(b_+) + \int_{b_+}^{\infty} (1 - U)k'_\rho, \\ \int_{-\infty}^{b_-} U'k_\rho &= (1 + U(b_-))k_\rho(b_-) - \int_{-\infty}^{b_-} (1 + U)k'_\rho. \end{aligned}$$

Boundedness of U makes the terms at infinity vanish. On the upper ray $k'_\rho \leq 0$, and on the lower ray $k'_\rho \geq 0$, while $1 \pm U \geq 0$ everywhere. Thus both last terms are nonpositive even if the solution later crosses any number of other branches. The sum is at most $CT^{-p}(\rho^2 + cT^2)^{-(3+p)/2}$; integrating over the transverse plane gives CT^{-1-2p} .

For $\rho \geq c_0\sqrt{rT}$, the complement of the interval consists of at most two rays, or it is the whole line. Integration by parts and $|U| \leq 1$ bound its signed integral in absolute value by $C\rho^{-3-p}$; there are at most two finite endpoint terms, each at most $k_\rho(0)$, and $\int |k'_\rho| = 2k_\rho(0)$. The transverse integral is $C(rT)^{-(p+1)/2}$. The line integrals and endpoints are measurable; absolute row integrability permits orthogonal-coordinate Fubini. Averaging in λ also uses absolute convergence and its finite probability measure. Adding the horizon replacement error proves

$$e_x \cdot V_\chi(x) \leq C[T^{-1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}]. \quad (11.10)$$

For $g(x) > 0$, split $\nu \cdot V_\chi = e_x \cdot V_\chi + (\nu - e_x) \cdot V_\chi$. The absolute row and variance bounds, with Young's inequality, give

$$c_{3,s}g\eta^2|(\nu - e_x) \cdot V_\chi| \leq \theta\eta^2d_\chi + C_\theta g\eta^2a^{p+1}T^{-2p-2}.$$

For example apply Young to $\sqrt{g}a^{-(p+1)/2}|\nu - e_x|\eta$ and $\sqrt{g}a^{(p+1)/2}T^{-p-1}\eta$, absorbing the first square through (11.8). At $g = 0$, both the gradient source and defect vanish and the same weighted inequality is zero. Using the uniformly bounded normal Jacobian and $|b| \leq 1$, the supported normal integration is

$$\int_{|t| < 3T/2} g(X(P, t))(1 + |t|)^{p+1} dt \leq CT, \quad \int_{|t| < 3T/2} g(X(P, t)) dt \leq C.$$

Thus the Young remainder is $C_\theta T^{-1-2p}\xi(P)^2$. The directional term is controlled by (11.10) and the second normal bound. Integrate over P , insert both conclusions in (11.5), and discard its nonpositive last term. This proves (11.6) for every fixed $\theta > 0$, with threshold and constants independent of ξ . $\square$

11.3. The exact coefficient of the positive cutoff energy. The normal moment from Lemma 8.1 can also be written as

$$\sigma_p = \frac{c_{3,s}}{2} \int \frac{z_1^2 \phi(t) \phi(v)}{(|z|^2 + (t - v)^2)^{(3+p)/2}} dz dt dv. \quad (11.11)$$

This is the already proved normal moment, paired with ϕ ; no new computation of its normalization is needed. For the comparison weight $w(t) = (1 + |t|)^{-p-1}$, its convolution is bounded and integrable. Consequently the corresponding integrated kernel is bounded by $k_0(\rho) := C \min(\rho^{-2-p}, \rho^{-3-p})$. Its second moment is finite: near zero the radial integrand is $O(\rho^{1-p})$, and at infinity it is $O(\rho^{-p})$. These are integrable precisely in the present range $1 < p < 2$.

Lemma 11.2. *For fixed $\theta > 0$ and sufficiently large r_θ ,*

$$P_\chi \leq (1 + \theta)\sigma_p \int_\Sigma |\nabla_\Sigma \xi|^2 dA + C_\theta \int_\Sigma [T^{-1-2p} + r^{-2}T^{1-2p}]\xi^2 dA. \quad (11.12)$$

Proof. Write $b_P(t) = b(t/T(P))$. Split the lifted difference as

$$|b_P(t)\xi(P) - b_Q(v)\xi(Q)|^2 \leq (1 + \epsilon)b_P(t)^2|\xi(P) - \xi(Q)|^2 + C_\epsilon \xi(Q)^2|b_P(t) - b_Q(v)|^2.$$

On selected pairs, physical distance is comparable to $[d_\Sigma(P, Q)^2 + (t - v)^2]^{1/2}$ and the normal weights are at most $Cw(t)w(v)$. Their integrated tangential kernel is bounded by $k_0(\rho) = C \min(\rho^{-2-p}, \rho^{-3-p})$. It has finite second moment. The second-moment portions with $\rho > L$ or $|t| > N$ or $|v| > N$ tend to zero uniformly in the supplied scale. For example the normal-tail kernel is dominated by k_0 , and its second moment is a multiple of

$$\iint_{\{|t| > N \text{ or } |v| > N\}} w(t)w(v)|t - v|^{1-p} dt dv \rightarrow 0.$$

Choose L, N first. On the fixed core, the profile comparison is relative because $\min_{[-N, N]} \phi > 0$. For $\rho \leq L$ and $|t|, |v| \leq N$, the curvature bound along the joining geodesic gives

$$Q - P = \rho\omega + O(\rho^2/r), \quad n(Q) - n(P) = O(\rho/r).$$

Consequently

$$X(P, t) - X(Q, v) = -\rho\omega + (t - v)n(P) + E, \quad |E| \leq C(L + N)\rho/r.$$

The leading tangent and normal vectors are orthogonal, so it has length $[\rho^2 + (t - v)^2]^{1/2} \geq \rho$. Thus the physical kernel has a relative $O((L + N)/r)$ error even arbitrarily close to the diagonal. The normal Jacobians have relative $O_N(r^{-1})$ errors as well.

Use geodesic polar coordinates $Q = \exp_P(\rho\omega)$ on the source's single buffered graph. Such coordinates exist without a completeness assumption on Σ : decrease c so every path of length below $2cr(P)$ lies in a compact interior chart patch. Reparametrize a minimizing sequence on $[0, 1]$ with constant speed. Its lengths are bounded below the buffered horizon, so its images stay in a fixed compact interior patch. In smooth coordinates the sequence has uniformly bounded H^1 norms, hence a weak H^1 and uniformly convergent subsequence. Uniform convergence of the metric coefficients and weak lower semicontinuity of its positive quadratic energy give an energy minimizer. Cauchy–Schwarz relates this energy to squared length, with equality for constant-speed paths, so it also minimizes length. The interior Euler–Lagrange equation for the smooth positive metric makes it a smooth geodesic. The geodesic ODE preserves unit speed. Up to length $2cr(P)$ its ambient displacement is at most this length, so its position and unit tangent stay in a compact subset of the smooth buffered chart's unit tangent bundle. The ODE therefore extends throughout that interval. Along them $r \asymp r(P)$ and $|K_\Sigma| \leq C/r(P)^2$. The transverse scalar Jacobi field satisfies

$$j'' + K_\Sigma j = 0, \quad j(0) = 0, \quad j'(0) = 1, \quad j(\rho) = \rho - \int_0^\rho (\rho - a)K_\Sigma(\gamma(a))j(a) da.$$

On this interval $|K_\Sigma| \leq C/r(P)^2$. Taking the supremum of $|j(a)|/a$ in the integral equation gives $\sup |j(a)|/a \leq 1 + Cc^2 \sup |j(a)|/a$. For small c this is at most two, and substitution back yields $j(\rho) = \rho + O(\rho^3/r(P)^2) > 0$ for $\rho \leq 2cr(P)$. Parallel transport along a geodesic, viewed in the ambient space, changes a unit tangent vector by at most $C\rho/r(P)$. Together with the radial and transverse Jacobi fields, this proves $D(\pi_P \exp_P) = I + O(|z|/r(P))$ on the convex tangent disk $|z| < 2cr(P)$. Integrating this derivative along straight segments in that disk shows that $\pi_P \exp_P$ is bi-Lipschitz for small c . The minimizing geodesics already constructed must therefore be unique on the smaller selected disk. There is no cut locus there, and its polar area factor is bounded by a fixed multiple of ρ on the whole horizon. For fixed S , writing $Q = \exp_P z$ with $|z| \leq S$ gives the area formula $dA_Q = J_P(z) dz$, where $J_P(z) = j(|z|)/|z| = 1 + O_S(r(P)^{-2})$ and $J_P(0) = 1$. For first variation, vary the initial point smoothly while fixing the final endpoint. The derivative of the geodesic length is minus the inner product of its initial unit tangent with the initial velocity: the interior term vanishes by the geodesic equation, leaving this boundary term. The exponential map's smooth inverse supplies these variations, giving $\nabla_P d_\Sigma(P, Q) = -z/|z|$ for $Q \neq P$. Only curvature, not its derivatives, enters these estimates. We will also use a fixed-core consequence: for $|z| \leq S$ and $|v| \leq N$, the profile bound and normal variation give

$$g(X(Q, v)) \geq c_N > 0, \quad |\nu(X(Q, v)) - n(P)| \leq C_{N,S}/r(P)$$

at bases $r(P) > 2r_\theta$ and a sufficiently large threshold $r_\theta > 2S$. Indeed, set $m_N = \min_{[-N, N]} \phi > 0$ and increase the threshold until $C_N/r(Q) < m_N/2$. Then $g \geq m_N/2$, and normalizing the gradient costs at most $2C_N/(m_N r(Q))$. Add the normal variation $|n(Q) - n(P)| \leq CS/r(P)$, using $r(Q) \asymp r(P)$. Then $r(Q) > r_\theta$, so the core tube coordinates are valid. For bases $r(P) > 4r_\theta$, the fixed-core selector is also one: both base factors are one, $N/T \rightarrow 0$ and $S/(c \min(r(P), r(Q))) \rightarrow 0$ put the other cutoffs in their plateaux.

Along its geodesic, $|\xi(P) - \xi(Q)|^2 \leq \rho \int_0^\rho |d\xi_{\gamma(a)}(\dot{\gamma}(a))|^2 da$. For completeness, the geodesic flow preserves $dA d\omega$ on the unit tangent bundle even when it is defined only locally. In cotangent coordinates (x, π) its Hamiltonian is $\mathcal{H} = \frac{1}{2}g^{ij}(x)\pi_i\pi_j$. The vector field $(\partial_\pi \mathcal{H}, -\partial_x \mathcal{H})$ has zero divergence and preserves $\mathcal{H}$, hence preserves $dx d\pi$ on every domain where its fixed-time map exists. Cotangent polar coordinates in dimension two give $dx d\pi = dA d\mathcal{H} d\omega$; restriction to $\mathcal{H} = 1/2$ gives the asserted invariant measure. For the set $\mathcal{A}_\rho$ of starting vectors whose selected

length- ρ segment stays in the buffer, the nonnegative flow change of variables therefore yields

$$\int_{\mathcal{A}_\rho} \rho \int_0^\rho |\mathrm{d}\xi_{\gamma(a)}(\dot{\gamma}(a))|^2 \mathrm{d}a \mathrm{d}A \mathrm{d}\omega \leq \pi \rho^2 \int_\Sigma |\nabla \xi|^2 \mathrm{d}A.$$

Here $\int_{S^1} |v \cdot \omega|^2 \mathrm{d}\omega = \pi |v|^2$. For the flat radial kernel

$$k_\phi(\rho) = \iint \phi(t)\phi(v)[\rho^2 + (t-v)^2]^{-(3+p)/2} \mathrm{d}t \mathrm{d}v,$$

the full coefficient is exactly $(c_{3,s}/2)\pi \int_0^\infty \rho^3 k_\phi(\rho) \mathrm{d}\rho = \sigma_p$. Since $b_P^2 \leq 1$, the fixed flat core contributes at most this coefficient. Its geometric error tends to zero as the threshold increases. On the remaining tails, the finite second moment $\int \rho^3 k_0(\rho) \mathrm{d}\rho$ and the chosen L, N give an arbitrarily small additional Dirichlet coefficient. The segment restriction is dropped only after the positive flow change of variables. The image need not equal the original selected set: it is enlarged to the whole unit tangent bundle only after changing variables in the nonnegative integral.

For the normal difference insert $b(t/T(Q))$. At fixed comparable width T , tangential integration reduces the cost to

$$\iint w(t)w(v) \frac{|b(t/T) - b(v/T)|^2}{|t-v|^{1+p}} \mathrm{d}t \mathrm{d}v \leq CT^{-1-2p}. \quad (11.13)$$

If one modulus is at most $T/2$, a nonzero difference forces the other above T and their separation is comparable to its modulus. The integral is bounded by $C \int_T^\infty v^{-2p-2} \mathrm{d}v$. If both moduli exceed $T/2$, at least one is below a fixed multiple of T . Scaling $t = T\tau$, $v = Tv$ gives CT^{-1-3p} times

$$\iint_{\substack{|\tau|, |v| \geq 1/2 \\ \min(|\tau|, |v|) \leq 2}} \frac{|b(\tau) - b(v)|^2 |\tau v|^{-p-1}}{|\tau - v|^{1+p}} \mathrm{d}\tau \mathrm{d}v.$$

This is finite by the Lipschitz numerator near the diagonal and the weights at infinity. Since $T \geq 1$ this is bounded by the stated power. Finally, with $\rho = d_\Sigma(P, Q)$, the ambient Lipschitz bound gives $|r(P) - r(Q)| \leq \rho < c \min(r(P), r(Q))$. The widths are comparable, and $|\log(T(P)/T(Q))| \leq C\rho/r(Q)$. Apply the fundamental theorem of calculus in the logarithm of the width to $b(t/T)$; its derivative is $-(t/T)b'(t/T)$, supported in a fixed annulus of comparable widths. Therefore

$$|b(t/T(P)) - b(t/T(Q))| \leq C\rho/r(Q) \mathbf{1}_{\{cT(Q) < |t| < CT(Q)\}}.$$

For this contribution fix Q and integrate P in the geodesic polar coordinates centered at Q . The bounded area factor and Fermi distance comparison reduce the factor ρ^2 to the flat second moment; the selected scales are comparable. The resulting base weight is exactly $r(Q)^{-2}T(Q)^{1-2p}\xi(Q)^2$, because the moment identity bounds the normal factor by

$$Cr^{-2} \int_{cT < |t| < CT} w(t) \int w(v) |t-v|^{1-p} \mathrm{d}v \mathrm{d}t \leq Cr^{-2}T^{1-2p}.$$

Indeed, for $|t| \asymp T \geq 1$, split the v -integral into $|v| \leq |t|/2$, $|v-t| \leq |t|/2$, and their complement. Their bounds are respectively CT^{1-p} , CT^{1-2p} and CT^{1-2p} . For the middle region use $w(v) \leq CT^{-p-1}$ and $\int_{|h| \leq CT} |h|^{1-p} \mathrm{d}h \leq CT^{2-p}$. Thus the inner integral is at most CT^{1-p} ; multiplying by $w(t) \leq CT^{-p-1}$ and the length CT proves the stated power. Choose ϵ so the splitting factor is less than $1 + \theta/4$, then L, N so the remaining second moment is less than the required fixed multiple of $\theta\sigma_p$. Finally increase the threshold so the fixed-core profile, physical kernel and area errors together supply the remaining margin in $1 + \theta$. The normal and variable-width errors have the stated powers, with their constants absorbed into C_θ . This proves (11.12) first for smooth ξ . For a nonnegative compact Lipschitz ξ , use a finite smooth chart partition and nonnegative mollification to obtain nonnegative smooth ξ_j with a common compact support in $r > 4r_\theta$, $\xi_j \rightarrow \xi$ uniformly and in $H^1(\Sigma)$. Their lifted amplitudes converge pointwise. Fatou's lemma for the nonnegative pair integrand gives $P_\chi(\xi) \leq \liminf_j P_\chi(\xi_j)$; the Dirichlet and weighted L^2 terms converge, proving the same inequality. This also avoids any pointwise differentiability claim for a Lipschitz amplitude along an exceptional geodesic. $\square$

11.4. A positive average of the normal and a curvature conversion. Fix nontrivial smooth cutoffs $0 \leq b_N, a_0 \leq 1$, with b_N supported in $(-N, N)$ and the radial cutoff a_0 supported in (ϵ_0, S) , $\epsilon_0 > 0$. Set

$$\mathcal{W}(z, t, v) = b_N(t)\phi(t)b_N(v)\phi(v)a_0(|z|)(|z|^2 + (t - v)^2)^{-(3+p)/2}, \quad C_f = \int z_1^2 \mathcal{W} dz dt dv,$$

$$h_f(\rho, v) = b_N(v)\phi(v) \int b_N(t)\phi(t) \int_\rho^\infty \lambda a_0(\lambda)(\lambda^2 + (t - v)^2)^{-(3+p)/2} d\lambda dt.$$

The positive smooth profile and the excision $\lambda \geq \epsilon_0$ make $h_f(|z|, v)$ smooth and nonnegative. It is constant for $|z| < \epsilon_0$, zero near and beyond S , and compactly supported in v . Differentiating its lower radial limit gives

$$-\partial_{z_i} h_f = z_i \int \mathcal{W} dt.$$

Compactly supported integration by parts then gives, for either i ,

$$\int h_f dz dv = \int -z_i \partial_{z_i} h_f dz dv = \int z_i^2 \mathcal{W} dz dt dv = C_f > 0.$$

Radial symmetry makes the two diagonal moments equal and the mixed moment zero. Monotone exhaustion of the normal line and punctured tangent plane lets us fix N, S, ϵ_0 so that $c_{3,s} C_f / 2 \geq (1 - \epsilon) \sigma_p$.

For a bounded vector-valued field $\mathbf{v}$, the flat average $m_f(x) = C_f^{-1} \int h_f(|z|, v) \mathbf{v}(x + z, v) dz dv$ has derivative

$$\partial_i m_f = C_f^{-1} \int z_i \mathcal{W} [\mathbf{v}(x + z, v) - \mathbf{v}(x, t)] dz dt dv.$$

Here derivatives fall on the compact smooth kernel after the change of variables $y = x + z$. The source term can be subtracted because $\int z_i \mathcal{W} dz = 0$ at every fixed t, v . The coordinate functions $z_i / \sqrt{C_f}$ are orthonormal in the weighted L^2 space, so componentwise Bessel gives $C_f |Dm_f|^2 \leq \int \mathcal{W} |\mathbf{v}(x + z, v) - \mathbf{v}(x, t)|^2$, with no extra factor two.

Fix these filter parameters before increasing r_θ . Construct the following maps on the buffered exterior neighborhood $U_\theta = \{P \in \Sigma : r(P) > 3r_\theta\}$ of the test support. Choose the threshold so that every radius- $2S$ disk based there lies in a complete chart, $N < T$ throughout its normal core, and all fixed disks based on the test support lie in the plateau of the selector. Use $\mathbf{v}(Q, v) = \nu(X(Q, v))$ only on the fixed core $r(Q) > 2r_\theta$, $|v| < N$, extending it by zero elsewhere. Where h_f vanishes, define the integrands as zero. For $P \in U_\theta$ and $d_\Sigma(P, Q) \leq S$, the ambient Lipschitz bound gives $r(Q) \geq r(P) - S > 2r_\theta$ once $r_\theta > S$; thus every nonzero kernel contribution uses only this core, where the tube map and the positive normal gradient are available. Define, for $P \in U_\theta$,

$$C_f(P) = \int_\Sigma \int h_f(d_\Sigma(P, Q), v) dv dA(Q),$$

$$m_f(P) = C_f(P)^{-1} \int_\Sigma \int h_f(d_\Sigma(P, Q), v) \mathbf{v}(Q, v) dv dA(Q).$$

The geodesic-disk construction in Lemma 11.2 puts each fixed disk inside one buffered embedded exponential chart, with no cut locus there. Set the kernel to zero on different intrinsic components, equivalently using distance infinity between them. A returning branch outside the retained chart cannot enter this fixed disk through a short intrinsic path. On the fixed normal core, the profile and $\min_{[-N, N]} \phi > 0$ give $g \geq c_N > 0$, so $\mathbf{v}$ is smooth on every part of the integral that is used. The kernel is constant near the diagonal and zero near its outer edge. These facts make the averages smooth even at the diagonal distance. Differentiation holds Q fixed and differentiates the compact kernel, rather than the profile error or a coordinate Jacobian. Its support on a neighborhood of any center lies in one fixed compact buffered patch. In geodesic coordinates, the curvature bound gives

$$dA(Q) = J_P(z) dz, \quad J_P(z) = 1 + O_S(r^{-2}), \quad C_f(P) = C_f + O(r^{-2}),$$

$$G_{ij} := \int z_i z_j \mathcal{W} J_P dz dt dv = C_f \delta_{ij} + O(r^{-2}).$$

Write $d\mu_P = \mathcal{W}(z, t, v) J_P(z) dz dt dv$. The preceding area comparison gives $C_f(P) \geq C_f/2$ after a threshold increase. The exact first variation $\nabla_P d_\Sigma(P, Q) = -z/|z|$, together with the radial kernel derivative, yields

$$\nabla_i C_f(P) = \int z_i d\mu_P, \quad \nabla_i m_f(P) = C_f(P)^{-1} \int z_i [\mathbf{v}(Q, v) - m_f(P)] d\mu_P.$$

These formulas were differentiated before changing to coordinates centered at P ; writing them in those coordinates introduces no P derivative of J_P . Subtracting a source value gives

$$\nabla_i m_f(P) = C_f(P)^{-1} \int z_i \mathcal{W}[\mathbf{v}(Q, v) - \mathbf{v}(P, t)] dA(Q) dt dv + e_i(P), \quad |e| \leq Cr^{-2}.$$

More explicitly,

$$e_i(P) = C_f(P)^{-1} \int z_i [\mathbf{v}(P, t) - m_f(P)] d\mu_P.$$

For each fixed t, v , the part with $J_P = 1$ is zero by oddness in z_i . Since $|\mathbf{v}|, |m_f| \leq 1$, replacing J_P by $J_P - 1$ bounds this expression by Cr^{-2} . Neither J_P nor $\mathbf{v}(P, t)$ is differentiated. If L_i is the integral numerator in the formula for $\nabla_i m_f - e_i$, apply the Gram-matrix form of Bessel's inequality separately to its three ambient components:

$$\int |\mathbf{v}(Q, v) - \mathbf{v}(P, t)|^2 d\mu_P \geq \sum_{\alpha=1}^3 (L^\alpha)^T G^{-1} L^\alpha \geq (1 - Cr^{-2}) C_f^{-1} |L|^2.$$

Since $L = C_f(P)(\nabla m_f - e)$, Young's inequality proves

$$\int \mathcal{W} |\mathbf{v}(Q, v) - \mathbf{v}(P, t)|^2 dA(Q) dt dv \geq (1 - \epsilon) C_f |\nabla m_f|^2 - C_\epsilon r^{-4}.$$

On this fixed disk, the profile bound and $|Dn| \leq C/r$ give $|\mathbf{v}(Q, v) - n(P)| \leq C/r$. Since both vectors are unit,

$$\mathbf{v}(Q, v) \cdot n(P) = 1 - \frac{1}{2} |\mathbf{v}(Q, v) - n(P)|^2 \geq 1 - C/r^2.$$

Positivity of the normalized average therefore gives $|m_f - n| \leq C/r$ and $1 - C/r^2 \leq m_f \cdot n \leq |m_f| \leq 1$. Thus $M = m_f/|m_f|$ is one smooth map $U_\theta \rightarrow S^2$ satisfying

$$|M - n| \leq C/r, \quad |\nabla M|^2 \leq (1 + C/r^2) |\nabla m_f|^2, \quad |\nabla M| + |\nabla n| \leq C/r.$$

For precision, normalization has the exact derivative $dM = |m_f|^{-1} (I - M \otimes M) dm_f$. For the last derivative estimate, subtract the fixed vector $n(P)$ inside the already differentiated normalized average: $|\mathbf{v}(Q, v) - n(P)| + |m_f(P) - n(P)| \leq C/r$, and kernel derivatives have uniformly finite mass. This estimates the derivative at P ; it does not differentiate $n(P)$ or a profile error. The estimate for ∇n is the supplied curvature bound. Restrict D_χ to the filter support for sources in $\text{supp } \xi$. There $\chi = 1$ and $b(t/T(P)) = 1$, so the source amplitude is $\xi(P)$. Because $0 \leq b_N, a_0 \leq 1$, inserting the filter can only decrease this nonnegative energy. The normal Jacobians, physical kernel and fixed-core profile have relative $O_{N, S, \epsilon_0}(r^{-1})$ errors; the excision $|z| \geq \epsilon_0$ avoids its diagonal. First choose the moment and Bessel losses small enough in terms of θ , and then enlarge r_θ so that their product with the geometric and normalization factors is at least $1 - \theta$. We obtain

$$D_\chi \geq (1 - \theta) \sigma_p \int_\Sigma |\nabla M|^2 \xi^2 dA - C_\theta \int_\Sigma r^{-4} \xi^2 dA. \quad (11.14)$$

Let ω_{S^2} be the sphere area form and use on U_θ the globally defined normalized homotopy $H_M(\tau, P) = ((1 - \tau)n(P) + \tau M(P))/|(1 - \tau)n(P) + \tau M(P)|$. Writing $a_\tau = (1 - \tau)n + \tau M$, its denominator is at least $1 - |M - n| \geq 1/2$ at the enlarged threshold. If $\Pi_H = I - H_M \otimes H_M$, its derivatives are

$$\partial_\tau H_M = |a_\tau|^{-1} \Pi_H(M - n), \quad d_P H_M = |a_\tau|^{-1} \Pi_H[(1 - \tau)dn + \tau dM].$$

The one-form

$$\alpha_M(v) = \int_0^1 \omega_{S^2, H_M}(\partial_\tau H_M, d_P H_M[v]) d\tau$$

satisfies $M^*\omega_{S^2} - n^*\omega_{S^2} = d\alpha_M$ and $|\alpha_M| \leq C|M - n|(|\nabla M| + |\nabla n|) \leq C_\theta r^{-2}$. The sign can be checked in local coordinates: writing H_i for $\partial_i H_M$ and H_τ for $\partial_\tau H_M$, closedness of the area form and commuting mixed derivatives give

$$\partial_\tau[\omega_{S^2, H_M}(H_i, H_j)] = \partial_i[\omega_{S^2, H_M}(H_\tau, H_j)] - \partial_j[\omega_{S^2, H_M}(H_\tau, H_i)].$$

Integration in τ gives the stated transgression identity. The displayed derivative formulas and the unit comass of the sphere area form give the one-form bound. With the orientation induced by n , the Gauss map satisfies $n^*\omega_{S^2} = K_\Sigma dA$: in two dimensions $\det(Dn) = \det(-A) = K_\Sigma$. Using also $-2 \operatorname{Jac}(M) \leq |\nabla M|^2$, integration by parts against ξ^2 gives

$$-2 \int K_\Sigma \xi^2 dA \leq \int |\nabla M|^2 \xi^2 dA + \theta \int |\nabla \xi|^2 dA + C_\theta \int r^{-4} \xi^2 dA. \quad (11.15)$$

Indeed $\int \xi^2 d\alpha_M = -\int d(\xi^2) \wedge \alpha_M$; the resulting error is bounded by $4 \int |\xi| |\nabla \xi| |\alpha_M|$ and Young's inequality. For compact Lipschitz tests, approximate in H^1 by smooth tests with support in one compact subset of the homotopy domain. Their squares and differentiated squares converge in L^1 , by Cauchy-Schwarz. The smooth geometric fields are bounded there, so the identity and estimate pass to the limit. The homotopy is defined on a neighborhood of the compact support, so no boundary contribution from ∂U_θ occurs. Its global definition avoids local connection choices and introduces neither a topology condition nor a mean-curvature remainder.

Theorem 11.3. *Under (11.1), every sufficiently small fixed $\theta > 0$ has constants r_θ, C_θ such that every compact Lipschitz test in the buffered region $r > 4r_\theta$ satisfies*

$$\int_\Sigma [|\nabla \Sigma \xi|^2 + a_\theta K_\Sigma \xi^2 + W_\theta \xi^2] dA \geq 0, \quad (11.16)$$

where

$$a_\theta = \frac{2}{(1+\theta)/(1-\theta)^2 + \theta} \rightarrow 2, \quad W_\theta = C_\theta[r^{-4} + T^{-1-2p} + r^{-2}T^{1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}] \leq C_\theta r^{-2-(p-1)/2}. \quad (11.17)$$

Constants are independent of branch count, connectivity, bad-cluster diameter and test support. In particular $a_\theta > 1/2$ for small θ .

Proof. First let $\xi \geq 0$. Stability and (11.6) give $(1-\theta)D_\chi \leq P_\chi + C_\theta \int [T^{-1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}] \xi^2$. Insert (11.12) and then (11.14). Since $\sigma_p > 0$, division gives

$$\int |\nabla M|^2 \xi^2 \leq \frac{1+\theta}{(1-\theta)^2} \int |\nabla \xi|^2 + C_\theta \int [r^{-4} + T^{-1-2p} + r^{-2}T^{1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}] \xi^2.$$

Equation (11.15) adds exactly $\theta \int |\nabla \xi|^2$ to this Dirichlet coefficient. Move the Gaussian term to the left and divide by $(1+\theta)/(1-\theta)^2 + \theta$, yielding precisely (11.16) and its stated a_θ . For signed tests apply the result to $|\xi|$, which has the same Dirichlet energy. Substituting $T = \tau r^{2/(p+1)}$, the five decay exponents beyond two are 2, $2p/(p+1)$, $2(2p-1)/(p+1)$, $(p-1)/2$ and $p-1$; their minimum is $(p-1)/2$. Every estimate is a one-source estimate on its unique selected graph, or an integral over the full ambient complement. Thus no summation over branches enters a constant; remote reconnections and the size of any excluded region do not affect the constants either. $\square$

12. COMPLETE MARKED LIMITS AND DISTANCE-SCALE CURVATURE

Consider stable solutions u_j with retained zero centers P_j and scales $R_j, L_j \rightarrow \infty$. Assume their regular zero surfaces in $B_{L_j R_j}(P_j)$ have (11.1) with the constant scale $r = R_j$, uniformly on each fixed buffered subball, and

$$R_j |A_{\Sigma_j}(P_j)| \rightarrow 1. \quad (12.1)$$

Assume also the prepared inputs (10.1) at every zero center of those buffered subballs, with the same R_j and uniform constants. The earlier point selection and all-center preparation provide this quantifier. Recentring consumes only fixed fractions of the complete-chart, flat-band, gap and pinned-profile buffers; the isolation scale remains at least $cR_j^{2/(p+1)}$.

Lemma 12.1. *After subsequence extraction, the rescaled components marked at P_j have a C^2 limit in graph coordinates. Their limit is an immersion $\iota_\infty : \Sigma_\infty \rightarrow \mathbb{R}^3$ of a complete connected smooth surface; it is minimal, carries a global normal, and satisfies $|A_\infty(o_\infty)| = 1$. Each $\xi \in C_c^\infty(\Sigma_\infty)$ can be transferred to a single smooth compact intrinsic test ξ_j on the approximating surface. These tests stay inside a fixed intrinsic-radius buffer and have physical support area $O_\xi(R_j^2)$. With $\widehat{g}_j = R_j^{-2}g_{\Sigma_j}$ and $\widehat{K}_j = R_j^2K_{\Sigma_j}$,*

$$\int_{\widehat{\Sigma}_j} |\nabla \xi_j|^2 dA_{\widehat{g}_j} \rightarrow \int_{\Sigma_\infty} |\nabla \xi|^2 dA_\infty, \quad (12.2)$$

$$\int_{\widehat{\Sigma}_j} \widehat{K}_j \xi_j^2 dA_{\widehat{g}_j} \rightarrow \int_{\Sigma_\infty} K_\infty \xi^2 dA_\infty. \quad (12.3)$$

No ambient embedding or properness of the limiting immersion is required.

Proof. Write $\widehat{\iota}_j(X) = R_j^{-1}(X - P_j)$, regard the component with metric $R_j^{-2}g_{\Sigma_j}$ as the abstract rescaled surface $\widehat{\Sigma}_j$, and use its intrinsic distance d_j . Choose the exhaustion slowly: for each integer m , the radius- m data and the finitely many larger buffers used at all preceding stages hold for all sufficiently large j . Choose increasing thresholds J_m also making $L_j > 4m$, and retain $\Lambda_j = m$ for $J_m \leq j < J_{m+1}$, with any fixed buffer restriction required. Discard the finite initial segment. Then $\Lambda_j \rightarrow \infty$, $\Lambda_j < L_j/2$, and the whole retained region is regular; no regularity in an unprepared outer portion is assumed. Restrict to the component through the marked zero inside this rescaled ambient ball. Its artificial boundary is at intrinsic distance at least Λ_j , because ambient displacement does not exceed path length. Put $Z_j = \{u_j = 0\}$. The rescaled mark is o_j , corresponding to P_j , with $\widehat{\iota}_j(o_j) = 0$, and the increasing unit normal is $n_j = \nabla u_j / |\nabla u_j|$ on this regular component; rescaling leaves its values unchanged.

First fix j and $L < \Lambda_j$. A sequence in the closed intrinsic ball has ambient positions in the compact Euclidean ball of radius L . An ambient convergent subsequence has limit Q in the interior regular zero set, since Z_j is closed and $L < \Lambda_j$. The implicit-function theorem at this regular zero identifies the *whole* nearby zero set with one graph. The sequence eventually lies on that graph; its points connect to Q by graph paths of length tending to zero. Consequently Q is in the marked component, $d_j(o_j, Q) \leq L$, and the sequence converges intrinsically to Q . This proves compactness and does not merge remote returning portions of the component. The same proof applies to $\overline{B}_s(P)$ whenever $|\widehat{\iota}_j(P)| + s < \Lambda_j$: every ambient cluster lies strictly inside the regular region, and local graph paths from the sequence to its limit show $d_j(P, Q) \leq s$. This argument needs only the center's ambient boundary margin, not a bound on $d_j(o_j, P)$, which can be large for returning portions.

Fix now an integer radius stage and its slightly larger fixed buffer. Let C_A bound the rescaled shape $\widehat{A}_j = R_j A_j$ on that larger buffer. All estimates below use only this bound, the chart fractions and preparation thresholds, with j sufficiently large; they can be chosen afresh at each radius stage. On the union of all regular zeros in that buffer, apply the small exponential disks of the local geodesic-disk construction in the proof of Lemma 11.2 in (11.1) with constant physical $r = R_j$. Rescaling gives injective exponential disks of a fixed positive radius and unique minimizing geodesics from their centers on a smaller fixed disk. Choose $16\rho_0$ below that latter radius, below all regular buffer fractions, and below $1/4$. Applying the result to the union rather than just its marked component keeps its complete-chart hypotheses intact; the geodesics starting in the marked component remain in that component. This reuses the previously proved local geodesic existence and injectivity.

For later longer paths from P to Q , choose $\epsilon > 0$ with $|\widehat{\iota}_j(P)| + d_j(P, Q) + \epsilon < \Lambda_j$. Every sufficiently near-minimizing path of length at most $d_j(P, Q) + \epsilon$ stays in the compact recentered ball about P of that radius, by its length bound. This also handles arbitrary ambient-buffer centers without a uniform intrinsic distance from the mark. Reparametrize near-minimizing paths with constant speed. Their images lie in one compact intrinsic ball and are uniformly Lipschitz in its metric. A uniformly convergent subsequence exists; length, defined as the supremum of finite distance sums, is lower semicontinuous, so it minimizes length. For distinct endpoints, reparametrize

the minimizing curve by arclength, removing any constant subintervals. It has constant metric speed; in a compact smooth coordinate neighborhood it is Lipschitz and hence belongs to $W^{1,\infty}$. Every subinterval minimizes length, since a shorter replacement would shorten the full path. For a short interval $[a, b]$ in one chart, Cauchy–Schwarz gives $E(v) \geq \text{Length}(v)^2/(b-a)$ for every fixed-endpoint coordinate variation, with equality for this constant-speed curve. It therefore minimizes the energy $E = \int_a^b g_{kl}(\gamma) \dot{\gamma}^k \dot{\gamma}^l$. Its first variation against compactly supported smooth coordinate variations gives the distributional equation

$$\frac{d}{dt}(g_{km}(\gamma) \dot{\gamma}^m) = \frac{1}{2} \partial_k g_{mn}(\gamma) \dot{\gamma}^m \dot{\gamma}^n.$$

The right side is bounded, so each momentum $g_{km}(\gamma) \dot{\gamma}^m$ has a Lipschitz representative. Multiplying by the smooth inverse metric along the Lipschitz curve makes $\dot{\gamma}$ continuous and Lipschitz. The right side is now continuous, yielding $\gamma \in C^2$; the resulting smooth geodesic ODE bootstraps to C^∞ . The identical-endpoint case is the constant curve. Thus every constructed length minimizer has a smooth constant-speed geodesic parametrization, without using uniqueness to infer regularity. Similarly a unit-speed geodesic from P extends to every required fixed length s for which $|\widehat{\ell}_j(P)| + s < \Lambda_j$: its position stays in the compact recentered ball $\overline{B}_s(P)$ and its tangent in that ball's compact unit tangent bundle, so continuation of the smooth geodesic ODE applies. At an ambient-buffer center $|\widehat{\ell}_j(P)| \leq M$, any fixed required s satisfies this margin for all sufficiently large j . No geodesic is extended through the artificial boundary.

Here is the additional convexity estimate. In an injective exponential disk the polar metric is $d\rho^2 + j(\rho, \omega)^2 d\omega^2$. The transverse Jacobi equation and its initial conditions are

$$j'' = -\widehat{K}_j j, \quad j(0) = 0, \quad j'(0) = 1, \quad j(\rho) = \rho - \int_0^\rho (\rho - t) \widehat{K}_j(\gamma(t)) j(t) dt.$$

With $|\widehat{K}_j| \leq \kappa^2 = C_A^2/2$, the integral equation on a small interval gives $|j| \leq 2\rho$, and then $j = \rho + O(\kappa^2 \rho^3)$ and $j' = 1 + O(\kappa^2 \rho^2)$ by its once-integrated equation. The constants are uniform on the radius stage. Thus

$$\text{Hess}(\rho^2/2) = d\rho \otimes d\rho + \frac{\rho j'}{j} (\widehat{g}_j - d\rho \otimes d\rho) \geq \frac{1}{2} \widehat{g}_j$$

after decreasing ρ_0 . At the center the squared-distance function is smooth with Hessian exactly the metric. A minimizing geodesic between two points in $B_{4\rho_0}(P)$ has length less than $8\rho_0$, remains in $B_{12\rho_0}(P)$, and exists by the strict compact-ball argument. The displayed Hessian makes its squared distance from P convex, hence no greater than the maximum of the endpoint values. It therefore stays in $B_{4\rho_0}(P)$. Injectivity at its initial endpoint gives uniqueness, because its length is below $16\rho_0$. This proves strong convexity without any derivative of curvature.

For clarity the uniform graph domains can be obtained in the same disks. If v has length ρ , decompose a tangent vector into its radial and transverse components. The differential of $\exp_P$ is intrinsic parallel transport of the radial component and of $j(\rho, \omega)/\rho$ times the transverse component. Viewed in $\mathbb{R}^3$, parallel transport differs from the initial fixed ambient vector by at most $C_A \rho$ times its norm, by the Gauss formula. Therefore

$$\|D(\pi_P \widehat{\ell}_j \exp_P)(v) - I\| \leq C(C_A \rho + \kappa^2 \rho^2) < 1/4 \quad (|v| \leq 16\rho_0)$$

for small ρ_0 . Its difference from the identity is Lipschitz with constant $1/4$ on this convex tangent disk. Its image contains $D_{12\rho_0}$: for a target w there, the map $v \mapsto w - [\pi_P \widehat{\ell}_j \exp_P(v) - v]$ maps the closed $16\rho_0$ disk into itself and is a contraction. The inverse function theorem then makes its graph smooth. The normal variation bound $C_A \rho$ gives a uniformly small graph slope, denoted by σ . For such a graph the identity between its second fundamental form and its Hessian bounds the Hessian by CC_A . Choose $b = b(M) < \rho_0/100$, so all D_{8b} charts and their fixed larger graph buffers are available. Small intrinsic balls are inside this graph because their unique radial minimizing geodesics are inside the injective disk. Ambient chords bound intrinsic distance below, and straight base-segment lifts bound it above on the graph. In particular the graph image of a base disk of radius $\delta/\sqrt{1+\sigma^2}$ lies in $B_\delta(P)$; its area is at least the Euclidean disk area. This proves the local area lower bound.

For the area upper bound at radius L , put $\kappa = C_A/\sqrt{2}$ and

$$S_\kappa(t) = \frac{\sinh(\kappa t)}{\kappa}, \quad S_0(t) = t.$$

All radial geodesics of length L exist in the compact strict buffer. Along each of them the absolute Jacobi equation gives

$$|j(t)| \leq t + \kappa^2 \int_0^t (t-a)|j(a)| da \quad (0 \leq t \leq L).$$

Iterating this positive integral inequality yields $|j(t)| \leq S_\kappa(t)$: its successive terms are $\kappa^{2n}t^{2n+1}/(2n+1)!$, and the remainder is at most $\sup_{[0,L]} |j| \kappa^{2N}L^{2N}/(2N)! \rightarrow 0$. Every point of $B_L(o_j)$ is reached by a minimizing radial geodesic, so the exponential map of the tangent disk covers that ball. The area formula for this smooth two-dimensional map, counting multiplicity, bounds image area by its integral absolute Jacobian, even when the map is not injective. Polar integration consequently gives

$$\text{Area}_{\widehat{g}_j}(B_L^{d_j}(o_j)) \leq 2\pi \int_0^L S_\kappa(t) dt =: V_{C_A}(L).$$

This avoids a cut-locus measure assertion. Apply the lower area bound to disjoint $\delta/2$ balls centered in B_L ; they lie in $B_{L+\delta}$, so their number is at most $CV_{C_A}(L+\delta)/\delta^2$. A greedy disjoint selection stops after that many steps, and maximality forces its concentric δ balls to cover B_L . All constants and all buffer requirements are chosen at the fixed radius stage before letting j grow.

Use the compact balls, finite nets, strongly convex disks and buffered graphs just proved. All choices in the following construction are made at fixed integer radius stages. Enlarge the stage curvature constants monotonically and decrease their positive local radii if necessary; no single curvature or chart-radius bound at infinity is used.

For every integer N and every dyadic resolution tending to zero below the stage- N local radius, choose a finite net in $B_N(o_j)$ using the preceding packing construction. Include o_j . Retain the finitely many earlier centers in a refinement, then add a greedy separated family; the sum of the old cardinality bound and the new packing bound is still uniform in j . The integer cardinalities have only finitely many possibilities at each fixed stage and resolution. Extract diagonally so they stabilize; label the centers consistently, or pad unused entries by the mark. This gives a countable collection of representatives $p_{j,a}$, each at bounded intrinsic distance from the mark. At each label choose an oriented orthonormal tangent frame whose third vector is $n_j(p_{j,a})$ and a buffered graph from the preceding graph construction. Choose its positive size b_a from the radius stage containing that center. Retaining centers from smaller stages does not shrink their charts; on a fixed finite-radius cover these sizes have a positive lower bound. The graphs have bounded slopes and Hessians on D_{8b_a} and a still larger reserved graph disk. Denote these graph maps by $\Gamma_{j,a} : D_{8b_a} \rightarrow \widehat{\Sigma}_j$ and write their ambient parametrizations as

$$\widehat{\iota}_j \Gamma_{j,a}(z) = \widehat{\iota}_j(p_{j,a}) + E_{a,j}z + n_j(p_{j,a})F_{j,a}(z), \quad F_{j,a}(0) = DF_{j,a}(0) = 0,$$

where $E_{a,j}$ consists of the two oriented unit tangent vectors. Their centers and frames have convergent subsequences. Arzelà–Ascoli gives C^1 graph convergence on compact disks, after a countable diagonal extraction.

Apply Theorem 10.1 with the prepared inputs (10.1) at every such center and its fixed larger physical chart. Rotating the increasing normal preserves the normalized equation; the graphs occupy fixed positive radii in curvature-scale coordinates, have bounded slopes and Hessians, and retain the prescribed interior margins. The all-center hypothesis supplies the same R_j and comparison data throughout these fixed chart buffers. Thus the already chosen C^1 subsequence converges in C^2 on compact chart domains to a smooth minimal graph, and $R_j H_j \rightarrow 0$ there. This applies independently at every fixed radius stage and uses its constants only; the power decay in R_j still tends to zero. Keep this one diagonal subsequence for all charts.

Add to the countable label set every rational-coordinate point in every graph disk. Its representatives have bounded intrinsic distance from the mark, since a straight graph lift joins it to the center with uniformly bounded length. Pairwise intrinsic distances of all labels are bounded by the sum of their distances to the mark. Extract once more so every pair distance has a limit. Ambient positions converge at the same time (or already do by graph convergence) and have the same distance bounds. Write $d_\infty(a, b) = \lim_j d_j(p_{j,a}, p_{j,b})$. Its triangle inequalities pass to the limit. Quotient this pseudometric by zero distance and complete it to a metric space X with marked point o_∞ . The nets make its dense labeled quotient countable. Since ambient chord never exceeds intrinsic distance, the limiting ambient positions are 1-Lipschitz and extend uniquely to a map $\iota_\infty : X \rightarrow \mathbb{R}^3$.

This completion is proper. Fix a ball radius L and choose an integer stage $N > L + 1$. A fixed label at limiting distance at most $L + 1/2$ has its finite- j representative in B_N for large j . At each fixed small resolution its distance from one of the finitely many labeled stage- N net centers is at most that resolution. Passing to a subsequence of these finitely many choices gives the same assertion in X . It holds also for limits of labels, since the minimum distance to a finite family is continuous. Thus every closed bounded ball has finite nets at resolutions tending to zero. It is totally bounded and complete, hence compact.

The metric is geodesic. For two labels choose M bounding both of their distances from o_j for all large j . Their mutual distance is at most $2M$. Every near-minimizing path with length below $2M + 1$ lies in $\overline{B}_{3M+1}(o_j)$, strictly below Λ_j for large j . This is one fixed compact ball, so constant-speed parametrization, uniform convergence and lower semicontinuity of length attain the infimum. The minimizing-geodesic regularity conclusion of the preceding compact-ball and minimizing-geodesic construction applies, since $|\widehat{\iota}_j(p_{j,a})| + d_j(p_{j,a}, p_{j,b}) \leq 3M < \Lambda_j$. Their midpoints remain in such a buffer. Approximate these midpoints by fixed-stage nets at resolutions tending to zero. For each resolution there are finitely many choices; diagonal extraction of the choices gives labels whose limiting mutual distances are at most the sum of those resolutions. They converge to a point of X with distances exactly half the original limiting distance to the two endpoints. This pairwise extraction does not alter any previously fixed distance limits. Approximating arbitrary endpoints by labels and using compactness of the bounded midpoint ball extends the midpoint property to all of X . Recursively choose midpoints at dyadic subdivisions. Triangle inequalities give distance $(t - s)d(x, y)$ between any two dyadic points, and completeness extends this to an isometric interval. Consequently X is a connected proper geodesic space. Positive limiting intrinsic distance, rather than ambient position, distinguishes its points.

We next identify the completion with a surface. In each smaller graph disk, uniform graph bounds give

$$|z - z'| \leq d_j(\Gamma_{j,a}(z), \Gamma_{j,a}(z')) \leq C_a |z - z'|.$$

The lower bound follows by projecting the ambient chord onto its tangent base; the upper bound is the length of the straight base-segment lift, wholly inside the reserved graph. These bounds define, first on rational coordinates and then by completion, an injective bi-Lipschitz map $\Gamma_{\infty,a}$ from the disk into X . They also imply for *any* fixed coordinates in any two charts that their finite- j distance converges to the corresponding distance in X : approximate both coordinates by rationals and use these uniform Lipschitz errors before letting the rational errors tend to zero. Variable coordinates converging in compact disk interiors obey the same statement.

Here is the uniform first-exit argument used below. Fix one buffered graph chart and write its reserved base disk as D_B , for example $B = 8b_a$. The buffered graph image $\Gamma_{j,a}(D_B)$ is open in the finite- j surface: its graph parametrization is an injective local diffeomorphism between two-dimensional smooth manifolds. For compact concentric base subdisks $K_0 \subset \text{int } K \subset K \subset D_B$, the image $\Gamma_{j,a}(K)$ is compact, hence closed in that Hausdorff surface, while $\Gamma_{j,a}(\text{int } K)$ is open. Its relative surface boundary is exactly $\Gamma_{j,a}(\partial K)$, because the graph chart is a homeomorphism and K lies strictly inside its domain. A continuous path starting at $\Gamma_{j,a}(z)$ with $z \in K_0$ cannot leave this inner graph image without first hitting that boundary. At such a first exit its ambient orthogonal projection onto the fixed graph plane changes by at least $m_a = \text{dist}(K_0, \partial K) > 0$. Its intrinsic length is therefore at least m_a . This lower bound is uniform in j : the base domains and

their margins are fixed and the projection frame is orthonormal. Paths of length less than m_a stay in this actual chart branch even if other branches or distant returns have arbitrarily close ambient images.

Here is the reverse inclusion needed for an atlas. Near an interior coordinate z choose an intrinsic radius much smaller than its reserved graph and base-boundary margins. A label whose limiting distance from $\Gamma_{\infty,a}(z)$ is below that radius has representatives at a slightly larger short distance for large j . A minimizing path from $\Gamma_{j,a}(z)$ exists by the recentered boundary-margin condition just checked and has length below m_a . The first-exit argument therefore keeps it in this reserved graph disk. The label therefore has coordinates in this same finite- j graph, in a compact subdisk about z . Extract these coordinates; the preceding uniform local Lipschitz distance bound identifies the label with the resulting limiting chart point. For any point of the metric completion, approximate it by labels and use compactness of that coordinate subdisk to obtain the same inclusion. Thus every chart interior maps to an open subset of X and contains a smaller intrinsic ball. At a fixed radius stage choose a net resolution below these uniform smaller chart radii. Its finitely many centers cover the corresponding limiting bounded ball by the same reverse-inclusion argument. The retained graph interiors therefore cover X , with finite covers on compact balls.

Their metric is the induced graph length metric. Fix an interior coordinate and choose a small radius well below its graph-boundary margins and the stage injectivity radius. For endpoints sufficiently near that coordinate, a minimizing path has length at most their short straight-lift distance; its ambient displacement from the initial endpoint is bounded by this length. The reverse-inclusion argument just proved keeps this path on the same reserved graph, and its base projection remains inside a compact subdisk of the coordinate domain. The endpoint approximations have the same fixed margins for large j . In fixed coordinates C^1 graph convergence gives

$$(1 - \varepsilon_j)g_\infty \leq \Gamma_{j,a}^* \hat{g}_j \leq (1 + \varepsilon_j)g_\infty, \quad \varepsilon_j \rightarrow 0$$

on the compact path buffer. Upper bounds for distances follow by comparing any smooth limiting graph path and its finite- j coordinate path. For the lower bound, compare finite- j minimizing paths with their limiting-metric lengths; their endpoints converge, and small coordinate segments correct the endpoints at vanishing cost. Taking the path infima proves that the local X distance is precisely the limiting graph distance. Globally every smooth path has the same metric and Riemannian length. Conversely a metric geodesic interval is covered by finitely many such neighborhoods; subdivision and nearly minimizing local graph paths give smooth piecewise paths of arbitrarily close total length. Hence the global induced path metric is also d_∞ .

Now let two interior coordinates represent the same point of X . Their representative distances tend to zero. Choose smaller neighborhoods with a positive common graph margin. For all large j , the two representatives and every point in a sufficiently small neighborhood of either are joined by short minimizing paths. Their lengths are below the compact base first-exit margins, so the preceding relative-boundary argument keeps these paths within the other's actual graph image. Thus the actual finite- j transition exists there. In ambient graph frames it is just projection onto the target base plane:

$$T_{ba,j}(z) = E_{b,j}^\top [\hat{\iota}_j \Gamma_{j,a}(z) - \hat{\iota}_j(p_{j,b})],$$

where $E_{b,j}$ is the matrix of its two oriented tangent vectors. The existence just proved ensures that these projected coordinates belong to the target graph and represent the identical *intrinsic* point, rather than a different nearby sheet. The C^2 graph and frame convergence gives C^2 transition convergence. The inverse transitions satisfy the same formula in the other direction, and their exact inverse and cocycle identities pass to the limit. Compactness permits a finite subcover for every compact portion of an overlap, giving the claimed uniform eventual neighborhoods. Conversely, if compact-interior coordinate sequences have d_j distance tending to zero, subsequential coordinate convergence and the all-pair distance convergence identify the two limiting points. This includes every finite- j intersection of compact chart supports. Equal limiting ambient images at positive intrinsic distance give no overlap and no identification.

Use these transition maps to give X its surface atlas, denoted Σ_∞ . Its topology is the metric topology just established, so it is Hausdorff, second countable and without boundary (the atlas domains are open disks). The limiting ambient graph maps agree on actual overlaps and are smooth minimal graphs. Their projection transitions are therefore smooth, and ι_∞ is a smooth immersion inducing g_∞ . Its local mean curvature is zero. The finite- j increasing normals converge in C^1 on compact graph interiors by the C^2 graph convergence. At representatives with vanishing intrinsic distance they have the same limiting value, since along the connecting short paths their variation is at most the fixed-buffer curvature constant times the path length. Thus these normals glue to a global normal, smooth in the minimal graph atlas. Properness and metric completeness of X , and the induced-path identity, give completeness of the immersed surface. The marked chart and Theorem 10.1 preserve the prescribed normalized curvature in the marked rescaling and all-center hypotheses above, giving $|A_\infty(o_\infty)| = 1$.

Finally a fixed finite union of compact chart images has centers at bounded distance from the marks and bounded within-chart path lengths, so it lies in one fixed intrinsic-radius buffer. The uniformly bounded graph slopes bound each compact chart's rescaled area by a fixed multiple of its Euclidean base area. Their finite union has bounded rescaled area, and rescaling back multiplies it by R_j^2 . Pulled-back metrics are C^1 convergent, normals are C^1 convergent, and Gaussian curvature and area density are uniformly convergent functions of the graph derivatives through order two. These conclusions hold intrinsically and do not require the limiting immersion to be embedded or proper.

For test transfer, cover a compact neighborhood of $\text{supp } \xi$ by finitely many marked charts $\Gamma_{\infty,a} : D_a \rightarrow \Sigma_\infty$, with their larger domains retained as buffers. Choose a nonnegative smooth partition ζ_a supported compactly in these charts, with $\sum_a \zeta_a = 1$ near $\text{supp } \xi$, and set $\xi_a = \zeta_a \xi$. Define

$$\xi_{j,a}(\Gamma_{j,a}(z)) = \xi_a(\Gamma_{\infty,a}(z)), \quad \xi_j = \sum_a \xi_{j,a},$$

using smooth zero extension outside each chart. Compact support of ξ_a inside the chart leaves a fixed margin on which the function and all its derivatives vanish. These are smooth intrinsic functions on the actual approximating surface, and their finite sum is the one global test. The construction uses the already selected marked subsequence, with no further extraction depending on ξ . For the finitely many indices before these fixed charts are available, set $\xi_j = 0$; this affects neither the limits nor the support bounds.

We check its convergence using genuine intrinsic overlaps. If a finite- j point lies both in a compact support of a second chart and in a fixed compact interior of the first, its two representatives have zero intrinsic distance. Compactness of the coordinate supports and the marked distance convergence force their limits to represent the same intrinsic point. The reserved overlap buffers then give actual transition maps there for large j , converging in C^2 . Thus each nonzero summand, pulled back through the first chart, converges in C^1 to the corresponding limiting summand. A disappearing overlap is handled by the fixed zero-extension margin. Uniformity follows by contradiction from compactness of the finite coordinate supports. No summand can arrive from a distinct marked point merely because its immersed image coincides with the first image. Consequently the whole sum converges in C^1 on compact chart interiors to $\sum_a \zeta_a \xi = \xi$. The finite number of pieces also gives $\|\xi_j\|_\infty \leq C_\xi$, and the finite chart union lies in a fixed intrinsic-radius buffer and has uniformly bounded rescaled area.

For the integral passage, choose $0 \leq \beta_a \leq 1$ smooth, supported compactly in the chart used for ξ_a , and equal to one on a neighborhood of $\text{supp } \xi_a$. Transfer them through the same charts and write $B_j = \sum_a \beta_{j,a}$. This is a smooth intrinsic function and $B_j \geq 1$ on an open neighborhood of the union of the transferred piece supports. To make the weights globally defined, fix a nonnegative smooth function F on $[0, \infty)$, equal to $1/t$ for $t \geq 1/2$ and zero near zero, and put

$$\omega_{j,a} = \beta_{j,a} F(B_j).$$

On that neighborhood $\omega_{j,a} = \beta_{j,a}/B_j$ and $\sum_a \omega_{j,a} = 1$. Outside it both integrands under consideration vanish, so no division at $B_j = 0$ is involved. The limiting weights are defined by the same

formula. The overlap argument just given applies to the transferred β_a as well, and therefore their pullbacks and these weights converge in C^1 on the fixed chart interiors. This is a partition for intrinsic integration; coincident ambient images do not identify different marked points.

Compute each weighted integral in its own fixed compact chart domain. The graph metrics, inverse metrics and area densities converge uniformly; their Gaussian curvatures converge uniformly by marked C^2 convergence. The pulled-back test functions and their first derivatives are uniformly bounded and converge. All weighted integrands are supported in a fixed compact coordinate set, so the area formula and dominated convergence give (12.2)–(12.3). The bounded rescaled support area becomes physical area $O_\xi(R_j^2)$. The support remains inside a fixed intrinsic-radius buffer. We use this single intrinsic function in the exterior inequality below. $\square$

Theorem 12.2. *A sequence with the uniform geometric and prepared inputs of this section and (12.1) cannot exist.*

Proof. Lemma 12.1 supplies a complete connected two-sided smooth minimal immersion and, on the same subsequence, one global test ξ_j for each $\xi \in C_c^\infty(\Sigma_\infty)$. Fix such a ξ . Its transferred support has bounded intrinsic radius and hence lies in a fixed rescaled ambient ball. Choose a larger fixed buffered subball containing that support, every selected-target neighborhood and the larger complete-chart buffers. The intrinsic selector horizon is a fixed multiple of R_j ; the normal thickness is $o(R_j)$, and every fixed filter radius also fits in the reserved margin for large j . All inputs on this fixed buffer have constants independent of j ; they need not be uniform over buffers of arbitrarily large radius.

For fixed sufficiently small $\theta > 0$, first fix the filter parameters and the corresponding threshold and constants in Theorem 11.3. Apply it with $r = R_j$ to the whole buffered union of regular zero branches just chosen. Use the transferred function on the marked component and zero on other components. Its pieces were summed before this application, and their fixed interior support margins give a single compact Lipschitz test on the union for each j . The exterior theorem already permits signed compact Lipschitz tests. For large j this buffer is inside the supplied growing regular region and $R_j > 4r_\theta$. All test integrals lie on the marked component, while the signed row retains every ambient target.

Using the uniform test bound and physical support area, its error is

$$\int W_\theta(R_j)\xi_j^2 dA \leq C_{\theta,\xi}R_j^2R_j^{-2-(p-1)/2} \rightarrow 0.$$

The constants here may depend on the fixed buffered geometric class chosen for ξ . The Dirichlet energy and Gaussian-curvature integral are invariant under surface rescaling:

$$|\nabla \xi_j|_{g_j}^2 = R_j^2 |\nabla \xi_j|_{g_{\Sigma_j}}^2, \quad dA_{\widehat{g_j}} = R_j^{-2} dA_{g_{\Sigma_j}}, \quad \widehat{K_j} = R_j^2 K_{\Sigma_j}.$$

Thus (12.2) and (12.3), first with θ and all filter parameters fixed, give

$$\int_{\Sigma_\infty} (|\nabla \xi|^2 + a_\theta K_\infty \xi^2) dA \geq 0.$$

The geometric subsequence and its atlas were selected independently of θ and ξ . Let $\theta \downarrow 0$ on this same surface. Since $a_\theta \rightarrow 2$ and minimality gives $2K_\infty = -|A_\infty|^2$,

$$\int_{\Sigma_\infty} (|\nabla \xi|^2 - |A_\infty|^2 \xi^2) dA \geq 0 \quad (\xi \in C_c^\infty(\Sigma_\infty)).$$

This is stability for every compact intrinsic normal variation.

The limit has no boundary, since its atlas consists of open graph disks. Its induced metric is complete, it is connected and minimal, and its global unit normal gives the orientation and two-sidedness required for the classification theorem. It is noncompact: otherwise the coordinate functions of its minimal immersion into $\mathbb{R}^3$ would be harmonic on a compact connected surface without boundary, hence constant, contradicting immersion. Fischer-Colbrie–Schoen [21, Corollary 4, p. 207] now makes its image a plane. Their Section 3 setup, pp. 205–206, is for complete oriented minimally immersed surfaces with a global unit normal, and equation (10) is the compact

intrinsic stability inequality just established. It requires no embedding, ambient properness or additional area-growth hypothesis. Planarity contradicts $|A_\infty(o_\infty)| = 1$. $\square$

Fix $a \in (1/\sqrt{3}, 1)$ and retain the earlier comparison radius $L_* > D_a + 2$, where D_a is the clearing distance for this phase band. For the bad zeros and good region write

$$B_\epsilon(u) = \left\{ z \in \{u = 0\} : \inf_{e \in S^2} \|u(z + \cdot) - q_s(e \cdot \cdot)\|_{C^2(B_{L_*})} \geq \epsilon \right\}, \quad \Omega_\epsilon = \{\text{dist}(\cdot, B_\epsilon) > D_a + 1\}.$$

The earlier wide-band preparation supplies D_a , nondegeneracy and $O(\epsilon)$ curvature on $|u| \leq a$ in this region.

The preparation also has a uniform fixed-buffer form. Fix the lower gradient bound $\gamma > 0$. There are a fixed $B > 2$ and a lower threshold such that, whenever an actual zero Q has

$$|\nabla u| \geq \gamma, \quad |A| \leq 2/R \quad \text{on } B_{BR}(Q) \cap \{|u| \leq a\},$$

all the complete-chart, separation and profile data used below are available at Q , with constants depending only on this fixed class. No marked lower curvature bound is required for this preparation. Indeed Theorem 4.3, with $K = 2/R$ and a prescribed very small slope, supplies complete phase families with fixed buffers proportional to R . Choose B once to contain its comparison ball and every fixed enlargement needed later. The families have the fixed ratios, signs, nondegeneracy, bounded phase offsets and Hessian bound used in Theorems 4.4 and 7.2. The qualitative normal comparison is uniform in this class: a sequence violating a compact-core modulus would retain precisely those fixed chart data and contradict Theorem 4.4.

Reserve larger and smaller fixed interiors before applying Theorem 7.2. On the larger one neighboring gaps are at least $c_{\text{pow}}R^\beta$, $\beta = 2/(p+1)$. Keep the zero graph through Q in the smaller interior, and put

$$d_0 = \bar{c}R^\beta, \quad r_1 = \sqrt{Rd_0}, \quad 0 < \bar{c} < c_{\text{pow}}/4.$$

Then $r_1/R, d_0/r_1 \rightarrow 0$, and $d_0/(R \log R)^{1/p} \rightarrow \infty$ because $\beta > 1/p$. The graph on D_{8r_1} remains a distance proportional to R from its original domain and completeness boundaries. Every other listed zero at the same base, if inside the larger power-gap interior, is separated by at least d_0 using ordering and neighboring gaps. A listed zero outside that interior instead has vertical distance at least cR : the retained bases are deep inside its base disk and the selected graph has height $O(r_1) = o(R)$. It too has separation at least $d_0/2$ for large R . This includes listed returning branches; with only one listed graph the separation condition is vacuous.

The single-sheet local comparison established in the proof of Theorem 7.2 therefore applies. It gives an actual selected graph on $D_{8\ell}$, zero completeness in $B_{4\ell}$, Hessian bound C/R , reach at least cR , unique actual normal footpoints and a $C^{1,\alpha}$ profile error C_α/R on a fixed smaller ball, where $\ell = c_{\text{iso}}d_0$. Choose the effective isolation length d to be a fixed smaller multiple of ℓ so all required nested boxes are retained. This supplies (10.1), including its actual-distance and recentring requirements. For precision, reduce the fixed box fractions so the largest retained box lies in $B_{\ell/32}(Q)$. For any X in that box, a nearest actual zero Q' to X exists because the zero set is closed and nonempty. Using Q as a competing zero gives

$$|X - Q'| \leq |X - Q|, \quad |Q' - Q| \leq 2|X - Q| \leq \ell/16.$$

Thus every minimizing zero is in the zero-complete $B_{4\ell}(Q)$, belongs to the selected graph, and has old base coordinate in D_{4r_1} because $\ell/r_1 \rightarrow 0$. A remote zero cannot minimize. The preceding distance construction, with $\ell/R \rightarrow 0$, provides actual/patched agreement and a unique actual normal footpoint there. The single-sheet local comparison is uniform at every such base coordinate. It therefore supplies all recentered boxes, profile data and complete margins at that footpoint with the same R, d and constants; a fixed further reduction retains all nested boxes and their finite larger buffers. On any fixed normal ray $|t| \leq N$, the actual signed distance is t and its gradient is $n(Q)$, once that ray is inside the retained reach and comparison box. Differentiating the profile comparison gives $\nabla u(Q + tn(Q)) = \phi(t)n(Q) + O_N(R^{-1})$. Clearing and all global solution derivative bounds are the earlier uniform estimates. Thus these are also exactly (11.1) with

constant scale R . All thresholds are uniform over centers whose fixed larger buffer is retained, and no number of branches enters a constant.

Corollary 12.3. *There are $\epsilon_* > 0$ and $C < \infty$, depending only on s, a, L_* and the uniform preparation constants, such that every bounded stable entire solution satisfies*

$$|A_{\{u=u(x)\}}(x)| \leq \frac{C}{\text{dist}(x, \mathbb{R}^3 \setminus \Omega_{\epsilon_*})}, \quad x \in \Omega_{\epsilon_*}, \quad |u(x)| \leq a. \quad (12.4)$$

An infinite denominator means zero curvature.

Proof. Negate the existence of one pair (ϵ_*, C) uniform over all solutions and band points, and choose prescribed $\epsilon_j \downarrow 0$ below the good-band threshold and positive $H_j \rightarrow \infty$. For a nonempty bad set, $B_\epsilon \subset \Omega_\epsilon^c$, so

$$0 < \text{dist}(X, \Omega_\epsilon^c) \leq \text{dist}(X, B_\epsilon) \quad (X \in \Omega_\epsilon).$$

The complement is empty exactly when the bad set is empty. Consequently the stronger uniform assertion using distance to the bad set, stated before (4.5), would imply the present corollary, including its infinite-distance case. Its uniform negation follows. The earlier whole-band selection therefore supplies solutions u_j , phase centers Y_j and scales R_j with

$$|A|(Y_j) = R_j^{-1} > 0, \quad B_{H_j R_j}(Y_j) \subset \Omega_{\epsilon_j}, \quad |A| \leq 2/R_j, \quad |\nabla u_j| \geq \gamma_a$$

on the whole phase band there. By (4.3), $R_j^{-1} \leq C_a \epsilon_j$, hence $R_j \rightarrow \infty$. This invokes the already proved selection in both the finite- and empty-complement cases; it neither forms $0 \cdot \infty$ nor assumes an attained maximum on an unbounded band.

Apply Theorem 4.3 to the phase family through Y_j , using its retained buffer. Continue the phase $u_j(Y_j)$ to zero at the same old horizontal coordinate and call that zero P_j . The uniform nondegeneracy and phase interval $[-a, a]$ give $|P_j - Y_j| \leq C = O(1) = o(R_j)$. For each fixed M , every zero $Q \in B_{(M+1)R_j}(Y_j)$ has its whole required BR_j comparison ball inside the selected ball for large j , uniformly over Q . The fixed-buffer preparation just checked supplies the same power isolation, profile constants and nested boxes there. In particular it covers every zero in $B_{MR_j}(P_j)$, including all local branches of returning portions of the marked component.

At P_j rotate its increasing normal vertically and choose $\ell_j = cR_j^{2/(p+1)}$ below all preparation fractions. The selected graph is defined on $D_{8\ell_j}$, contains every zero in $B_{4\ell_j}$, and has $f_j(0) = Df_j(0) = 0$, $|D^2 f_j| \leq C/R_j$. On the fixed smaller comparison ball, the cutoff graph used in Theorem 8.2 has the same signed distance as the actual graph. Indeed distance to either graph is at most $|X|$ using its zero at the origin, so a nearest point has norm at most $2|X|$; points outside the unchanged disk $D_{4\ell_j}$ are farther away. The prepared profile comparison thus gives precisely its pinned supremum hypothesis. Also

$$\ell_j/R_j \rightarrow 0, \quad R_j \ell_j^{-p} = CR_j^{(1-p)/(p+1)} \rightarrow 0.$$

The harmonic-limit theorem requires no nonflat normalization at this zero. Apply it and then Corollary 8.3 on its subsequence, uniformly for $\tau_j = u_j(Y_j) \in [-a, a]$.

In the new frame Y_j has bounded physical coordinates (y'_j, t_j) ; after extraction y'_j converges in a fixed compact disk. The inverse phases $t_\tau = q_s^{-1}(\tau)$ are uniformly bounded for $\tau \in [-a, a]$. Choose T_0 larger by two than their bound and the bound on $|t_j|$. Positivity and continuity of the reference derivative give $m_0 = \min_{|t| \leq T_0} q'_s(t) > 0$. The local C^1 convergence in Theorem 8.2 gives $\partial_t u_j \geq m_0/2$ on $D_{C+1} \times [-T_0, T_0]$ for all large j , where C bounds the physical coordinates of Y_j . The phase-transfer expansion places its phase graphs inside that interval uniformly for these horizontal coordinates and $\tau \in [-a, a]$. Since Y_j is also there and has value τ_j , strict monotonicity identifies it with the unique selected phase root at y'_j . The phase transfer compares its shape operator at y'_j with that of the zero graph at the same coordinate. By Theorem 8.2, the latter's rescaled graph limit is a polynomial of degree at most two, hence has constant Hessian. Its local C^k convergence compares that value to the zero graph's shape operator at its origin. Both induced metrics tend to the same Euclidean metric on these fixed cores. Therefore

$$R_j |A_{\{u_j=0\}}|(P_j) - R_j |A|(Y_j) \rightarrow 0, \quad R_j |A_{\{u_j=0\}}|(P_j) \rightarrow 1.$$

The constant Hessian makes the bounded coordinate displacement harmless for this curvature comparison.

To supply one growing sequence of prepared balls, for each integer m choose increasing indices j_m so that, for every $j \geq j_m$, the original selected ball contains all fixed enlarged buffers about $B_{(m+1)R_j}(P_j)$, and the fixed-core thresholds for $N \leq m$ and all fixed preparation restrictions hold uniformly at its zero centers. This is possible by $H_j, R_j \rightarrow \infty$ and the uniform fixed-buffer preparation. Set $L_j = m$ for $j_m \leq j < j_{m+1}$, with a fixed further buffer reduction if needed. Then $L_j \rightarrow \infty$. On each fixed buffered rescaled subball the whole zero set is regular, and every zero center has (11.1) with constant $r = R_j$ and (10.1) at that same R_j , with constants independent of the center and branch. Clearing remains global. Theorem 10.1 applies at all these recentered zeros, providing the curvature-scale compactness, while the bounded physical phase transfer supplied the marked zero and its positive normalization. These are all the hypotheses of Theorem 12.2; its contradiction proves the corollary. $\square$

Decrease ϵ_* , if needed, to the earlier wide-band regularity threshold. The smaller threshold enlarges the bad set and shrinks the good region. Its complement distance is no larger, so the old bound implies (12.4) there with the same constant. If the old complement was empty, its zero-curvature conclusion also implies the bound on the smaller region.

We record the uniform conversion to a distance scale for later use. Let P be a zero and $D = \text{dist}(P, \mathbb{R}^3 \setminus \Omega_{\epsilon_*})$ be large and finite. On $B_{D/2}(P)$ that distance is at least $D/2$, and the whole wide band has curvature at most $2C/D$ and the same lower gradient bound γ_a . Enlarge C_{buf} once to cover every fixed larger buffer in the preparation above. Choose one fixed

$$M \geq \max(2C, 2C_{\text{buf}}, 1), \quad R = D/M.$$

Then $1/R$ is an admissible curvature upper bound and $B_{C_{\text{buf}}R}(P) \subset B_{D/2}(P)$. The fixed-buffer preparation applies, giving the complete-chart, isolation and profile data, and thus (11.1) with $R \asymp D$. This application uses the general fixed complete-chart class checked above.

More precisely, choose $c < 1/4$ below all chart and target-buffer fractions and put $d_\Omega(X) = \text{dist}(X, \mathbb{R}^3 \setminus \Omega_{\epsilon_*})$. Every zero $Q \in B_{cD}(P)$ satisfies $(1 - c)D \leq d_\Omega(Q) \leq (1 + c)D$. Recenter preparation at Q with $R_Q = d_\Omega(Q)/M$. The same M gives

$$B_{C_{\text{buf}}R_Q}(Q) \subset \{X : d_\Omega(X) \geq d_\Omega(Q)/2\} \subset \Omega_{\epsilon_*}.$$

Throughout this ball the whole wide band has curvature at most $2C/d_\Omega(Q) \leq 1/R_Q$ and gradient at least γ_a . Thus the same preparation constants, nested buffers and thresholds work at every center and on every entering or returning local branch. Replacing R_Q by $d_\Omega(Q) = MR_Q$ multiplies curvature and profile constants by fixed factors, gap constants by $M^{-2/(p+1)}$, and divides chart fractions by M . Clearing stays global. No threshold depends on the number of centers or branches.

Here the complement is nonempty because D is finite. Distance to this closed nonempty set is finite and ambient 1-Lipschitz. For the positive scale in Theorem 11.3, use $r = \max(d_\Omega, r_0)$, where $r_0 > 0$ is fixed below the retained threshold. This scale is positive and ambient 1-Lipschitz. A selected-target horizon of length at most cD has $d_\Omega(Q) \geq (1 - c)D$, and its normal thickness is $O(D^{2/(p+1)}) = o(D)$. The complete-chart buffers use fixed smaller distance fractions. After increasing the retained threshold, all source and selected-target neighborhoods therefore have distance above r_0 , and $r = d_\Omega$ throughout them.

If the good region is all of $\mathbb{R}^3$, the corollary makes the whole regular wide band flat. Every fixed comparison ball is then available, so the same preparation and exterior inequality use arbitrarily large positive constant scales. If that band is empty, there is no surface application. An alternative positive ambient 1-Lipschitz scale may be used once its fixed two-sided comparability with d_Ω is checked on every required complete-chart, source and selected-target buffer of the selected region. Replacing the scale in the estimates changes only those fixed comparison constants and permits a fixed reduction of horizon and chart fractions. A fixed larger Lipschitz constant is normalized by dividing the scale by that constant. The distance-scale use of the exterior inequality thus follows the curvature-scale contradiction.

13. REMOVING THE EXCEPTIONAL ZEROS

Let u be a bounded stable entire solution in $\mathbb{R}^3$. Throughout this section,

$$p = 2s \in (1, 2), \quad P = 3 - \frac{2}{p+1}, \quad g = |\nabla u|, \quad Z = \{u = 0\}.$$

Fix the planar-profile constants ε_*, L_* used in Corollary 12.3, with $\varepsilon_* \leq \varepsilon_0$ from the earlier good-band construction. Such a decrease preserves the curvature bound: it enlarges the exceptional set, shrinks the good region, and decreases the complement distance. The old zero-curvature assertion also handles an originally empty complement. The exceptional set is

$$B = \left\{ z \in Z : \inf_{e \in S^2} \|u(z + \cdot) - q_s(e \cdot \cdot)\|_{C^2(B_{L_*})} \geq \varepsilon_* \right\}.$$

In particular, this definition is independent of all the large scales introduced below. The variation and clearing estimates of Propositions 3.1 and 4.1 give

$$\int_{B_\rho(x)} g \leq M\rho^P \quad (\rho \geq 1), \quad 1 - |u| \leq C(1 + \text{dist}(\cdot, Z))^{-p}, \quad g \leq C(1 + \text{dist}(\cdot, Z))^{-p-1}. \quad (13.1)$$

The constants are uniform in the center and in the solution. Uniform derivative bounds enlarge the range of the variation estimate to $1 \leq \rho < 2$ if necessary.

13.1. A fixed amount of defect at every exceptional zero. Write

$$\mathcal{D}(x, y) = g(x)g(y) - \nabla u(x) \cdot \nabla u(y), \quad D_u(x) = c_{3,s} \int_{\mathbb{R}^3} \mathcal{D}(x, y) |x - y|^{-3-p} dy.$$

The vector identity established in Lemma 5.1 is

$$Q_u(g\eta) = \frac{c_{3,s}}{2} \iint g(x)g(y)(\eta(x) - \eta(y))^2 |x - y|^{-3-p} dx dy - \int D_u \eta^2 \quad (13.2)$$

for every compactly supported Lipschitz amplitude η . The uniform derivative bounds give

$$0 \leq \mathcal{D}(x, y) \leq \frac{1}{2} |\nabla u(x) - \nabla u(y)|^2 \leq C \min\{|x - y|^2, 1\}.$$

The radial bounds are ρ^{1-p} near zero and ρ^{-1-p} at infinity, so D_u is nonnegative and uniformly bounded. The universal vector identity and the compact-test extension in Section 3 apply to every real compact Lipschitz amplitude, including where $g = 0$.

Two consequences of the uniform variation bound will control interactions with arbitrary targets:

$$\sup_x \int g(y) |x - y|^{-1-p} dy \leq C, \quad \sup_x \int_{|x-y| \geq R} g(y) |x - y|^{-3-p} dy \leq CR^{P-3-p} \quad (R \geq 1). \quad (13.3)$$

For the first inequality, the gradient bound handles the unit ball; the shells $2^j \leq |x - y| < 2^{j+1}$ cost at most $C2^{j(P-p-1)}$. Their sum converges because

$$p + 1 - P = \frac{p(p-1)}{p+1} > 0.$$

The shells $2^j R \leq |x - y| < 2^{j+1} R$ similarly prove the second inequality. Both estimates retain uniformity in the source point.

Choose a cutoff equal to one on $B_R(x_0)$, supported in $B_{2R}(x_0)$, and bounded between zero and one, with Lipschitz constant C/R . A pair at which its values differ has an endpoint in $B_{2R}(x_0)$. Symmetry and the first estimate in (13.3) bound the positive term in (13.2) by $CR^{-2} \int_{B_{2R}(x_0)} g$. Stability therefore yields

$$\int_{B_R(x_0)} D_u \leq CR^{-2} \int_{B_{2R}(x_0)} g \leq CR^{P-2} = CR^{(p-1)/(p+1)} = o(R). \quad (13.4)$$

Lemma 13.1. *There exist $b \geq 1$ and $b_* > 0$, depending only on the fixed profile thresholds and the uniform estimates, such that*

$$c_{3,s} \iint_{B_b(z)^2} \mathcal{D}(x, y) |x - y|^{-3-p} dx dy \geq b_* \quad (z \in B). \quad (13.5)$$

Proof. Suppose the assertion fails. For each integer j , choose a bounded stable solution and an exceptional zero whose pair defect on the radius- j ball is less than $1/j$. Translate the zero to the origin; change of variables in the normalization of Section 2 preserves equation and stability. By the smooth compactness statement of Section 3 a subsequence converges in every fixed local derivative norm to a bounded stable entire solution u_∞ , with $u_\infty(0) = 0$.

For each fixed L , the nonnegative limit pair integral is zero by Fatou: the radius- L charges are bounded by the radius- j charges once $j > L$. Gradient convergence makes the limiting defect continuous. Positivity of its kernel off the diagonal implies $\mathcal{D}_\infty(x, y) = 0$ there; on the diagonal it is zero by definition. This holds for all pairs.

A constant limit would be zero; this contradicts Lemma 2.2, which makes every nonnegative stable entire solution the positive well. Choose a nonzero gradient and let e be its oriented unit direction. Equality in Cauchy–Schwarz with that anchor gives $\nabla u_\infty(x) = h(x)e$ with $h(x) \geq 0$, including at critical points. Integrating the vanishing transverse derivatives along straight segments gives $u_\infty(x) = Q(e \cdot x)$ with $Q' \geq 0$ and $Q \in C_b^2(\mathbb{R})$.

Apply the cylindrical equation and stability reduction of Lemma 3.2 with $m = 1, k = 2$ and its orthogonal invariance: both the equation and stability descend to Q . The classification Lemma 4.2 gives the increasing layer, its decreasing reflection or a constant well. Nonconstancy and $Q' \geq 0$ leave $Q = q_s(\cdot - t_b)$; its zero at zero and the unique zero of the normalized reference layer give $t_b = 0$. Convergence in C^2 on a larger ball than B_{L_*} makes the comparison in this single direction less than ϵ_* , a contradiction. No variation or cluster premise is used. $\square$

Lemma 13.2. *For every fixed $A > 0$, every connected component of $\bigcup_{z \in B} B_A(z)$ is bounded. If $B \neq \emptyset$, each such component determines a nonempty compact cluster K_{cl} with*

$$\text{dist}(K_{\text{cl}}, B \setminus K_{\text{cl}}) \geq 2A. \quad (13.6)$$

*There is a finite maximal disjoint family $B_b(z_i)$, $z_i \in K_{\text{cl}}$, $1 \leq i \leq N$, such that the doubled balls cover K_{cl} and their retained pair defects sum to at least b_*N .*

Proof. Apply Lemma 13.1 and the sublinear estimate (13.4). The earlier good-band construction implies that B is closed, since the fixed threshold is at most ϵ_0 .

Fix $A > 0$ and suppose a component U of the indicated open union is unbounded. It contains an exceptional center z_0 : any one of its points belongs to an A -ball, and that connected ball belongs wholly to the same component. An open connected Euclidean set is path connected. For every large L choose a path in U from z_0 to a point farther than L away. Fix $D > 4A + 4b$ and use its crossings of the spheres of radii $jD \leq L$. At a crossing choose an exceptional center whose A -ball contains it; that center is in U and its distance from z_0 is between $jD - A$ and $jD + A$. Distinct selected centers have separation at least $D - 2A > 2b$. There are at least c_AL of their disjoint detector balls for all large L , and all lie in $B_{L+A+b}(z_0)$.

The retained charge of each is at least b_* . Disjointness in the source variable and nonnegativity of all target defects therefore give

$$b_*c_AL \leq c_{3,s} \sum_i \iint_{B_b(z_i)^2} \mathcal{D}(x, y) |x - y|^{-3-p} dx dy \leq \int_{B_{L+A+b}(z_0)} D_u \leq C(L + A + b)^{P-2}.$$

Because $P - 2 < 1$, this is impossible as $L \rightarrow \infty$ with A, b fixed. Thus each U is bounded.

For a bounded U , $K_{\text{cl}} = B \cap U$ is nonempty. If $z_j \in K_{\text{cl}}$ converges to z , closedness gives $z \in B$, and for large j one has $|z - z_j| < A$. Hence $z \in B_A(z_j) \subset U$. The cluster is closed and bounded, thus compact. If $z \in K_{\text{cl}}$ and $w \in B \setminus K_{\text{cl}}$ had separation less than $2A$, their open A -balls would meet and belong to the same component, a contradiction. This proves isolation.

A greedy disjoint selection of B_b centers in the compact cluster terminates: all such balls lie in one bounded Euclidean ball and each has the same positive volume. It is therefore finite and maximal. Any uncovered cluster point at distance at least $2b$ from all selected centers could supply another disjoint open b -ball, so the doubled balls cover. Each selected center satisfies Lemma 13.1; summing yields b_*N . These arguments hold for any fixed prescribed A and give no uniform diameter bound. $\square$

13.2. Geometry and summable errors around an isolated cluster. Assume temporarily that $B \neq \emptyset$ and take a cluster from Lemma 13.2. Put $r(x) = \text{dist}(x, K_{\text{cl}})$. For $r(x) < A$, isolation gives

$$\text{dist}(x, B \setminus K_{\text{cl}}) \geq 2A - r(x) > r(x), \quad \text{dist}(x, B) = r(x).$$

The distance to an empty complementary cluster is understood to be infinite. Choose $d_0 \geq D_a + 1$, where D_a is the excluded-neighborhood threshold of Corollary 12.3, and define $E_{d_0} = \{\text{dist}(\cdot, B) \leq d_0\}$. Outside this set,

$$\text{dist}(x, E_{d_0}) = \text{dist}(x, B) - d_0.$$

To see the upper bound, move toward a nearest exceptional point by the indicated distance. The reverse bound is the Lipschitz property of distance. Enlarging the original excluded neighborhood weakens the curvature bound, so it remains valid with this denominator. In particular,

$$\frac{1}{2}r \leq r - d_0 \leq r \quad (2d_0 \leq r < A). \quad (13.7)$$

Denote by A_Z the shape of the regular zero set. Put $h = D_a + 1$, $\Omega = \{\text{dist}(\cdot, B) > h\}$ and $E_h = \{\text{dist}(\cdot, B) \leq h\}$. The original good region of the preceding good-band construction has complement E_h . Since B is nonempty and closed, its complement distance at a point with bad distance $d > h$ is $d - h$. By Corollary 12.3, at every zero in the isolated regular region,

$$|A_Z| \leq C_d/(r - h) \leq C_d/(r - d_0) \leq 2C_d/r \quad (r > 2d_0).$$

The gradient lower bound there is γ_a from the good-band estimate (4.3). Both apply on larger actual-zero buffers, not merely on the truncated set Σ .

Fix $r_{\text{reg}} > 16d_0$ above the fixed preparation thresholds and retain

$$\Sigma = Z \cap \{r_{\text{reg}} < r < A/2\}, \quad n = \nabla u / |\nabla u|.$$

The preceding gradient bound makes this an actual smooth regular oriented zero surface. Write $a_T = 2/(p + 1)$ and define

$$T(z) = \tau r(z)^{2/(p+1)}, \quad X(z, t) = z + tn(z). \quad (13.8)$$

The shape A_Σ is the restriction of A_Z . For positive-scale applications put $r_{\text{scale}} = \max\{r, r_{\text{reg}}\}$. Here are the preparation and boundary checks. At a retained base z put $r_z = r(z)$ and $D = d_\Omega(z) = r_z - h$. It is comparable to r_z . Apply the distance-scale preparation following Corollary 12.3 and restrict its fixed chart fractions once. Choose a fixed small $c_{\text{buf}} < 1/20$ and denote the three retained ambient windows by

$$\mathcal{W}_z = B_{c_{\text{buf}}r_z}(z), \quad \mathcal{W}_z^+ = B_{2c_{\text{buf}}r_z}(z), \quad \mathcal{W}_z^- = B_{c_{\text{buf}}r_z/2}(z).$$

Here an actual-zero extension means the whole set $Z \cap \mathcal{W}_z^+$, before intersecting with the artificial annulus defining Σ . The fixed complete-chart, normal-source and selected-target constructions are retained in these nested windows by first shrinking their fractions and then increasing the lower threshold so their sublinear normal widths fit. All data are supplied at every zero center of these windows; no profile comparison to one fixed plane on an entire macroscopic ball is asserted. Points w in the largest windows satisfy

$$9r_z/10 \leq r(w) \leq 11r_z/10, \quad r(w) < 11A/20 < A, \quad r(w) > r_{\text{reg}}/4.$$

Hence bad distance is still $r(w)$, $d_\Omega(w) = r(w) - h$ is between fixed positive multiples of r_z , and $r_{\text{scale}}(w) = \max(r(w), r_{\text{reg}})$ is comparable to that same distance, with fixed comparison constants. The positive scale is ambient 1-Lipschitz. These checks meet exactly the alternative-scale clause of the distance-scale preparation following Corollary 12.3 on every complete-chart, source and selected-target window. It supplies the surface inputs (11.1) with uniform curvature, gap and profile constants. In particular the branch gaps are at least cr^{a_T} and, with $\phi = q'_s$,

$$|\nabla u(z + tn(z)) - \phi(t)n(z)| \leq C_H/r(z) \quad (|t| \leq H)$$

for each fixed finite H . In particular all actual zero branches, including entering and returning branches, are retained in these complete windows. At an artificial edge of Σ the window extends to actual regular zeros outside Σ ; the crop is imposed only after constructing the geometry.

Choose the geometric τ for this fixed prepared class as in the surface inputs (11.1) and the tube construction of Section 11. Since $a_T < 1$, increasing the fixed r_{reg} makes $4T/r$ smaller than every chosen chart and normal-window fraction throughout the needed buffers. The normal-map conclusions of the actual-tube construction of Section 11 then apply to the collectively retained bases, with these actual-zero extensions. To check that this is global rather than chartwise injection, a possible collision has base displacement at most $4(T(z)+T(w))$. The ambient Lipschitz scale and T/r small make the scales comparable and put both bases in one prepared window. Likewise a zero closer to $X(z, t)$ than z has distance from z less than $2|t|$. It is in that same complete actual-zero window. Thus the hypotheses for the previous global collision and whole-zero-distance conclusions are retained even at the artificial crop edges. The resulting normal map is globally injective on $|t| < 4T(z)$, and

$$J(z, t) = \det(I - tA_\Sigma) > 0, \quad J(z, t) \asymp 1, \quad \text{dist}(X(z, t), Z) = |t|.$$

The entire-zero gradient estimate in Proposition 4.1 gives $g(X(z, t)) \leq C(1 + |t|)^{-p-1}$. Because $r_{\text{scale}} = r$ at bases in Σ , its tube width is exactly the stated T . The tangential weak chain rule gives $|\nabla_\Sigma T| \leq a_T T/r$ almost everywhere.

For the area estimate restrict this globally injective tube to one fixed short normal interval. Let $H_0 \geq 1$ bound the global ambient Hessian by the smooth compactness statement of Section 3. Choose one

$$0 < t_* < \min\{1, \gamma_a/(2H_0), \tau r_{\text{reg}}^{a_T}, r_{\text{reg}}/(8C_A)\},$$

where $|A_\Sigma| \leq C_A/r$ above; enlarge C_A to a positive constant if necessary. The short tube is a restriction of the injective variable tube. Its gradient is at least $\gamma_a - H_0 t_* \geq \gamma_a/2$, and its Jacobian is at least $1/2$ by the two eigenvalue factors of $I - tA_\Sigma$. Its image above $\Sigma \cap B_\rho(x)$ lies in $B_{\rho+t_*}(x)$. Change of variables, first on compact chart pieces and then by monotone exhaustion, gives

$$\frac{\gamma_a t_*}{2} \text{Area}(\Sigma \cap B_\rho(x)) \leq \int_{X((\Sigma \cap B_\rho(x)) \times (-t_*, t_*))} g \leq \int_{B_{\rho+t_*}(x)} g.$$

Use the growth bound (13.1) and $t_* \leq 1$ to obtain the stated local estimate for $\rho \geq 1$. Global injectivity counts each returning or distinct branch only once.

$$\text{Area}(\Sigma \cap B_\rho(x)) \leq C \int_{B_{\rho+t_*}(x)} g \leq C\rho^P \quad (\rho \geq 1). \quad (13.9)$$

Finally the clip scale need not equal r in a preparation window crossing $r = r_{\text{reg}}$; the preceding comparison, rather than equality, is what gives its prepared inputs there. In later active neighborhoods with base radii at least $1.8R$ and R above a fixed multiple of r_{reg} , the chosen horizons and normal offsets change r by at most a fixed small fraction. They are entirely above r_{reg} , so the clip equals r throughout those used windows. All choices were made from the fixed distance/preparation and analytic constants, before A, N or the cluster diameter were known.

The cover $K_{\text{cl}} \subset \bigcup_i B_{2b}(z_i)$ implies $\{r < \rho\} \subset \bigcup_i B_{\rho+2b}(z_i)$. Applying this single cover to volume, variation, and (13.9) gives

$$\int_{\{r < \rho\}} g \leq CN\rho^P, \quad |\{r < \rho\}| \leq CN\rho^3, \quad \text{Area}(\Sigma \cap \{r < \rho\}) \leq CN\rho^P \quad (\rho \geq b). \quad (13.10)$$

Hence, for $e > P$, $R \geq b$, and $A' < A$,

$$\int_{\Sigma \cap \{R < r < A'\}} r^{-e} dA \leq C_e N R^{P-e}. \quad (13.11)$$

For the weighted area bound partition $R < r < A'$ into the available shells with lower radii $2^j R$. A truncated last shell is a subset of its full shell. Its contribution is at most

$$(2^j R)^{-e} \text{Area}(\Sigma \cap \{r < 2^{j+1} R\}) \leq CN(2^j R)^{P-e},$$

by the buffered geometry and area estimates just proved, even if the numerical outer radius exceeds the controlled annulus, because the set still intersects only the original Σ . Summing the numerical series to infinity gives $C_e N R^{P-e}$ for $e > P$ and does not add any surface point.

Fix one sufficiently small filter tolerance and set $\lambda = a_\theta > 1/2$ in Theorem 11.3. Increase the fixed r_{reg} if needed for this tolerance. Its exterior inequality is

$$\int_{\Sigma} (|\nabla v|^2 + \lambda K_{\Sigma} v^2 + W v^2) dA \geq 0 \quad (13.12)$$

with

$$\begin{aligned} 0 \leq W &\leq C[r^{-4} + T^{-1-2p} + r^{-2}T^{1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}], \\ E(r, T) &= T^{-1-2p} + (rT)^{-(p+1)/2} + r^{-p-1}, \\ E_0(r, T) &= T^{-1-2p} + r^{-2}T^{1-2p}. \end{aligned} \quad (13.13)$$

Apply Theorem 11.3 in the surface inputs (11.1) and the tube construction of Section 11 to compact tests on this buffered regular region. If necessary extend their base tests by zero to the larger actual-zero prepared neighborhood before applying the result; compact support inside the retained open region gives zero-extension margins and no interface derivative. The complete windows supplied above remain available across the artificial crop. Keep the fixed choice $\lambda = a_\theta > 1/2$ and increase the lower threshold as needed. At every tested base the clip scale is r , so the resulting nonnegative potential has precisely the displayed five-term bound with $T = \tau r^{a_T}$. Enlarging a nonnegative potential preserves the inequality.

The signed-row estimate uses E , and the selected positive form uses E_0 . Their decay exponents exceed P by the following amounts:

term	decay exponent minus P
r^{-4}	$(p+3)/(p+1)$
T^{-1-2p}	1
$r^{-2}T^{1-2p}$	$(3p-1)/(p+1)$
$(rT)^{-(p+1)/2}$	$(p-1)^2/(2(p+1))$
r^{-p-1}	$p(p-1)/(p+1)$.

Set $\alpha = (p-1)^2/(2(p+1)) > 0$, the smallest of these margins. Insert $T = \tau r^{2/(p+1)}$. The five decay exponents are 4, $2(1+2p)/(p+1)$, $6p/(p+1)$, $(p+3)/2$, and $p+1$. Subtracting $P = 3 - 2/(p+1)$ gives exactly the stated five margins. The fourth is $\alpha > 0$; it is smaller than one, the first and third margins exceed one, and the last margin divided by α is $2p/(p-1) > 1$. Thus every term in W, E, E_0 has exponent at least $P + \alpha$. Their fixed τ coefficients are absorbed into constants. Apply the weighted bound to each term and add them. For $R \geq 1$, every larger margin gives $R^{-\text{margin}} \leq R^{-\alpha}$, proving the final estimate including its partial last shell and its single factor N .

$$\int_{\Sigma \cap \{R < r < A/2\}} (W + E + E_0) dA \leq CNR^{-\alpha}. \quad (13.14)$$

The numerical series is extended to infinity only as an upper bound; the surface and every used geometric buffer remain in the controlled isolated annulus.

13.3. A complete surface with controlled attachment cost. Take $R > 10r_{\text{reg}}$ larger than the detector and geometric thresholds, choose a final cutoff radius $S > 0$, and suppose $A > 1000(S + R)$, with the fixed factor increased when chart buffers require it. Convolve the ambient distance at scale $R/20$ to obtain a smooth function f_{Σ} satisfying

$$|f_{\Sigma} - r| \leq R/20, \quad |\nabla f_{\Sigma}| \leq 1, \quad |D^2 f_{\Sigma}| \leq C/R.$$

The probability kernel preserves the first-derivative bound of the 1-Lipschitz distance, its support gives the displayed error, and one derivative of the convolution of that bounded weak derivative gives C/R . The slab $5R \leq f_{\Sigma} \leq 6R$ lies in $4.95R \leq r \leq 6.05R$. Compactness of K_{cl} makes this ambient slab bounded, and its positive distance from both artificial edges makes its zero portion closed and regular, hence compact. On this portion, $|\text{Hess}_{\Sigma} f_{\Sigma}| \leq C/R$: the ambient Hessian

contributes C/R , and its normal-gradient term is bounded by $|\nabla f_\Sigma| |A_\Sigma| \leq C/R$. For regular levels $C_t = \Sigma \cap \{f_\Sigma = t\}$, coarea gives

$$\int_{5R}^{6R} \text{Length}(C_t) dt \leq C N R^P, \quad \int_{5R}^{6R} \int_{C_t} |k_g| ds dt \leq C N R^{P-1}.$$

For the second estimate, the unit tangent T_C of a regular level satisfies

$$|k_g| |\nabla_\Sigma f_\Sigma| = |\text{Hess}_\Sigma f_\Sigma(T_C, T_C)| \leq C/R;$$

the coarea factor cancels the denominator. Each set of levels at which one of the two quantities exceeds four times its average has measure at most $R/4$. Their union leaves a set of measure at least $R/2$. Removing $(5R, 5.2R]$ discards only $R/5$ of this set. Sard's theorem therefore allows a regular $t_0 \in (5.2R, 6R)$ satisfying both estimates. This choice also leaves the plateau margin used below. The compact regular one-manifold $C = \Sigma \cap \{f_\Sigma = t_0\}$ consists of finitely many circles. Fenchel's theorem [19] and $|\kappa_{\mathbb{R}^3}| \leq |k_g| + |A_\Sigma|$ give

$$\mathcal{L} := \text{Length}(C) \leq C N R^{P-1}, \quad \mathcal{G} := \int_C |k_g| \leq C N R^{P-2}, \quad \#\pi_0(C) \leq C N R^{P-2}, \quad (13.15)$$

since $2\pi\#\pi_0(C) \leq \mathcal{G} + (C/R)\mathcal{L}$. For completeness, the total-curvature fact used here has a short geometric proof. If the unit tangent loop of a closed regular curve had spherical length $\ell_T < 2\pi$, choose two tangent values a, b splitting this loop into arcs of equal length. Every tangent value v then satisfies $d_{S^2}(v, a) + d_{S^2}(v, b) \leq \ell_T/2 < \pi$. It follows that $v \cdot (a + b) > 0$, since the sum of the two cosines is positive. This contradicts the vanishing integral of the unit tangent over a closed curve. A constant tangent is excluded by the same integral. Thus each component contributes at least 2π .

If C is nonempty, choose a regular $t_1 \in (A/3, A/3 + R)$ and let Y be the union of those components of $\Sigma \cap \{t_0 \leq f_\Sigma \leq t_1\}$ which meet C . The inequalities $t_0 - R/20 > r_{\text{reg}}$ and $t_1 + R/20 < A/2$ place the entire closed slab and its fixed buffers in the regular region. Regularity of both levels makes the slab a compact smooth surface with boundary, with finitely many components. Thus Y is also such a surface, and every point of C lies on the inner boundary of a retained component. No annular topology is assumed.

Lemma 13.3. *One can attach a connected surface along the inner boundary circles C , matching the metric smoothly, with a point o such that $d(o, z) \leq D_0 = 5$ for every $z \in C$. The attachment has a complete noncompact end and satisfies*

$$\text{Area}(\text{attachment}) + \int_{\text{attachment}} |K| dA \leq C(1 + \mathcal{L} + \mathcal{G} + \#\pi_0(C)). \quad (13.16)$$

Proof. Enumerate the finitely many circles as $C_1, \dots, C_m$. At a boundary circle of length ℓ , take the Fermi coordinate $t \geq 0$ to point away from Y into the new attachment. The metric is $dt^2 + h(t, z)^2 dz^2$, with $z \in \mathbb{R}/\ell\mathbb{Z}$, $h(0, z) = 1$, and $h_t(0, z) = b(z)$, where $|b| = |k_g|$. The original regular surface continues smoothly across this transverse circle, providing a positive extension h_e matching every boundary jet in this normal coordinate. The metric on Y is left unchanged. The normal exponential map exists for a uniform small time by compactness and has invertible differential along the circle. Collisions at widths tending to zero would force their base points to converge to the same boundary point, contradicting local invertibility there. Thus each collar has a positive embedded width; the finitely many widths may also be reduced to make the collars disjoint. By compactness, shorten each collar further to $0 \leq t \leq 2d$, $2d \leq 1/2$, so that

$$3/4 \leq h_e \leq 5/4, \quad |(h_e)_t - b| \leq 1, \quad \int_0^{2d} |(h_e)_{tt}| dt \leq 1$$

uniformly in z . The width may depend on the boundary jets. Choose a smooth cutoff $0 \leq \theta \leq 1$, one near zero and zero near two, and put $h_{\text{flat}} = 1 + \theta(t/d)(h_e - 1)$. This retains the metric near the boundary and is the exact product $dt^2 + dz^2$ near the other end. Positivity is preserved.

Thus the same construction provides a finite metric-matched product collar for any of the given boundary continuations. The bounds

$$|h_e - 1| \leq t(1 + |b|), \quad |(h_e)_t| \leq 1 + |b|$$

show that the three terms $\theta(h_e)_{tt}$, $2d^{-1}\theta'(h_e)_t$, and $d^{-2}\theta''(h_e - 1)$ in $(h_{\text{flat}})_{tt}$ have total integral at most $C(1 + |b(z)|)$. In particular, the inverse-width factors do not enlarge the cost, because

$$\int_0^{2d} d^{-1}|\theta'(t/d)| dt \leq C, \quad \int_0^{2d} t d^{-2}|\theta''(t/d)| dt \leq C.$$

For this metric, including its z -dependence,

$$K = -h_{tt}/h, \quad \int |K| dA = \int |h_{tt}| dt dz.$$

Indeed the orthonormal coframe $dt, h dz$ has connection form $-h_t dz$, whose exterior derivative is $-h_{tt} dt \wedge dz$. The z -derivative contributes a factor $dz \wedge dz = 0$. Thus arbitrarily large tangential boundary jets do not add curvature terms to this cost. Its area and absolute-curvature cost are therefore $C(\ell + \int |k_g|)$, independent of the small collar width.

Choose positive radii r_i with $2\pi r_i \leq \ell_i$ and $2r_i \leq 2^{-i-5}$. On a length-one collar, shrink the i th circumference ℓ_i to $2\pi r_i$. With arclength period ℓ_i , take

$$h(t) = 1 - (1 - 2\pi r_i/\ell_i)\zeta_0(t),$$

where ζ_0 is a fixed smooth step on $[0, 1]$. Then $h > 0$, $\ell_i \int h \leq \ell_i$, and $\ell_i \int |h''| \leq C\ell_i$. Every angular coordinate traverses this collar in length one. Join the resulting product end to a Euclidean hole of radius $2r_i$ by

$$dt^2 + r_i^2 h_{\text{fl}}(t/r_i)^2 d\vartheta^2, \quad 0 \leq t \leq 2r_i.$$

Here $h_{\text{fl}} = 1$ near zero, $h_{\text{fl}}(\tau) = \tau$ near two, and h'_{fl} increases from zero to one. For an explicit choice, take a smooth nondecreasing step ζ on $[0, 2]$, constant zero and one near its endpoints, with $\zeta(\tau) + \zeta(2 - \tau) = 1$, and set $h_{\text{fl}}(\tau) = 1 + \int_0^\tau \zeta(v) dv$. Then $\int_0^2 \zeta = 1$, so the endpoint conditions hold. The area of this joining piece is $2\pi r_i^2 \int_0^2 h_{\text{fl}}$, and its absolute-curvature integral is $2\pi \int_0^2 |h''_{\text{fl}}| = 2\pi$.

Place the holes at $(2^{-i}, 0)$ in the unit disk. The dyadic radius bounds put their closures strictly inside the disk and make their horizontal projections disjoint. A segment from a hole boundary point to $o = (0, 1/2)$ has length less than two and crosses each convex hole in at most one interval. Replace each intercepted interval, including one in the starting hole, by the shorter boundary arc. The total increase is bounded by $4\pi \sum_i r_i \leq \pi/16$. Keep the angular coordinate fixed while traversing the Fermi collar, the compression, and the flare. The resulting path from every point of C to o has length at most

$$\frac{1}{2} + 1 + \frac{1}{64} + 2 + \frac{\pi}{16} < 5.$$

Thus $d(o, z) \leq D_0 = 5$ for every $z \in C$, without a trip around an original long boundary circle.

Complete the attachment at the outer circle of the unit disk with the rotational metric $dt^2 + H(t)^2 d\vartheta^2$. Prescribe $H = 1 + t$ close to $t = 0$ and $H = e^{2-t}$ on $t \geq 2$, joining the two profiles smoothly and positively on the intervening compact interval. For example, $H(t) = (1 - \theta_e(t))(1 + t) + \theta_e(t)e^{2-t}$ with a fixed smooth step θ_e equal to zero near zero and one for t near and beyond two has all the required matching jets. The finitely many collars, compression pieces, flares, punctured closed disk and cusp segment $0 \leq t \leq 2$ form a compact smooth core with retained boundary C . Its only end is the cusp $t \geq 2$. A path reaching cusp height t from the core has length at least $t - 2$, because its metric contains dt^2 . Every intrinsic Cauchy sequence has bounded distance from the core and hence bounded cusp height. Each bounded-height truncation, with C included, is compact, so the sequence converges there. The attachment is therefore complete with its attaching boundary included and has no additional incomplete escape. Moreover, $\int_0^\infty (H + |H''|) dt < \infty$ controls both its area and its absolute-curvature integral. Use orientation-reversing identifications at the boundary circles so that the orientations of all components of Y extend over the attachment. Summing the preceding costs proves (13.16). $\square$

In this nonempty- C case, apply the metric-matched finite collar construction just proved to the smooth continuation at each outer boundary circle of Y , and extend its exact product end by an infinite product cylinder. The original metric on all of Y remains unchanged, including every boundary jet. No uniform cost is needed for these outer ends. Every component of Y meets C and is connected to the common inner attachment. Orientation-reversing boundary identifications extend its orientation, and every attaching circle is filled, so the resulting M is connected, oriented and without boundary. Its inner compact core, Y and the finitely many outer collars form a compact union. Outside it there are exactly the inner cusp and finitely many product cylinders. Each has a height coordinate with metric containing dt^2 , so reaching height t from this union costs at least the height increment. A Cauchy sequence has bounded distance from a fixed point of the union, remains in a compact bounded-height truncation, and converges. Thus M is complete, and its cusp makes it noncompact. A component of Y without an outer boundary causes no exception.

13.4. Protected distance balls and logarithmic capacity. We extend (13.12) only to compact Lipschitz tests whose supports avoid the outer attachments and their attaching circles. On Y , choose a nondecreasing smooth angle depending on f_Σ/R , zero for $f_\Sigma \leq 7R$ and equal to $\pi/2$ for $f_\Sigma \geq 8R$. Let χ_c and ψ_c be its sine and cosine, so that $\chi_c^2 + \psi_c^2 = 1$ and both gradients are bounded by C/R . Extend $\chi_c = 0, \psi_c = 1$ over the inner attachment. Since χ_c vanishes near the inner circles, $\chi_c v$ extends by zero to a compact test on the original surface. Apply (13.12) to that product, and use

$$\int (|\nabla(\psi_c v)|^2 + \lambda K \psi_c^2 v^2 + \lambda K_- \psi_c^2 v^2) dA \geq 0, \quad K_- = \max\{-K, 0\},$$

for the remaining product. Pointwise almost everywhere,

$$|\nabla(\chi_c v)|^2 + |\nabla(\psi_c v)|^2 = |\nabla v|^2 + (|\nabla \chi_c|^2 + |\nabla \psi_c|^2) v^2,$$

because the cross terms contain $\chi_c \nabla \chi_c + \psi_c \nabla \psi_c = 0$. Adding the two inequalities gives

$$\int_M (|\nabla v|^2 + \lambda K v^2 + W_c v^2) dA \geq 0, \quad W_c = \chi_c^2 W + |\nabla \chi_c|^2 + |\nabla \psi_c|^2 + \lambda K_- \psi_c^2. \quad (13.17)$$

Here W is zero off Y . We can extend the partition constantly over the outer ends, where v vanishes, and set $W_c = 0$ there. This asserts no spectral positivity for an arbitrary outer end. On $Y \cap \{f_\Sigma \leq 8R\}$, $r \asymp R$, $|K| \leq C/R^2$, and the area is at most CNR^P . Thus the original collar curvature and IMS terms cost CNR^{P-2} . The W term is controlled by (13.14); the inner attachment costs $C(1 + \mathcal{L} + \mathcal{G} + \#\pi_0(C)) \leq CNR^{P-1}$. Consequently, on the region relevant to these tests,

$$\int_M W_c dA \leq CNR^{P-1}. \quad (13.18)$$

Extend f_Σ by the constant t_0 on the inner attachment and t_1 on every outer end. These values agree at the attaching circles. In a chart across a circle the two smooth pieces have matching traces and bounded first derivatives, so the extension is locally Lipschitz. On Y its metric gradient has norm at most one, and it is constant on every new piece. For a piecewise smooth path γ , $f_\Sigma \circ \gamma$ is absolutely continuous. Off the attaching circles its derivative is bounded by the metric speed of γ ; on their preimages it equals one of the finitely many constant level values t_0, t_1 , and has derivative zero almost everywhere there. Integrating and taking the infimum of path lengths proves that f_Σ is globally 1-Lipschitz on M . Since $f_\Sigma(o) = t_0$,

$$d_M(o, \text{outer attaching circles}) \geq t_1 - t_0 > A/3 - 6R.$$

Every closed ball $\overline{B}_\rho(o)$ of radius $\rho < A/3 - 7R$ therefore has a positive buffer from the uncontrolled ends, even when a path visits the inner connector repeatedly. The assumption on A protects all radii up to S .

For these balls, the integrated Fiala inequality is

$$\int_{B_\rho(o)} K(1 - d_M(o, \cdot)/\rho)^2 dA \leq 2\pi - \frac{2}{\rho^2} \text{Area}_M(B_\rho(o)). \quad (13.19)$$

Put $\zeta_\rho(t) = (1 - t/\rho)_+$. This is the triangular-profile case of [3, Lemma 2.3, Eq. (2.12), p. 1246]: the profile is nonnegative, decreasing, and convex on $[0, \rho]$, vanishes at ρ , and equals one at zero. The geometric inequality requires no global spectral positivity. Its surface hypotheses hold for the complete, connected, oriented, noncompact boundaryless surface M . More explicitly, the profile on $[0, \rho]$ is C^1 and piecewise C^2 , with derivative $-1/\rho$ and second derivative zero, and $(\zeta_\rho^2)'' = 2/\rho^2$. Substitution into the cited geometric formula gives exactly (13.19). Its one-sided length comparison already includes arbitrary topology and the cut locus; a global polar chart and spectral positivity outside the protected region are not required.

Completeness makes the closed distance balls compact, so the radial triangular test is admissible in (13.17). Writing $\mathcal{A}(\rho) = \text{Area}_M(B_\rho(o))$, its gradient energy is at most $\mathcal{A}(\rho)/\rho^2$. Equations (13.18) and (13.19) imply

$$0 \leq 2\pi\lambda - (2\lambda - 1)\frac{\mathcal{A}(\rho)}{\rho^2} + CNR^{P-1}.$$

Because $\lambda > 1/2$, we obtain

$$\mathcal{A}(\rho) \leq \frac{2\pi\lambda + CNR^{P-1}}{2\lambda - 1}\rho^2 \leq CNR^{P-1}\rho^2. \quad (13.20)$$

Fix $S_0 > D_0 + 2$ and $S > S_0e$. Take the radial cutoff equal to one up to S_0 , equal to $\log(S/t)/\log(S/S_0)$ for $S_0 < t < S$, and zero beyond S . Its Dirichlet energy is at most

$$\frac{1}{\log^2(S/S_0)} \int_{S_0}^S t^{-2} d\mathcal{A}(t).$$

Stieltjes integration by parts and (13.20) give

$$\begin{aligned} \int_{S_0}^S t^{-2} d\mathcal{A}(t) &= \frac{\mathcal{A}(S)}{S^2} - \frac{\mathcal{A}(S_0)}{S_0^2} + 2 \int_{S_0}^S \frac{\mathcal{A}(t)}{t^3} dt \\ &\leq CNR^{P-1}(1 + 2\log(S/S_0)). \end{aligned}$$

Restrict this cutoff to Y , and denote the resulting surface field by ξ . Extend it on Σ by one where $f_\Sigma < t_0$, by zero on slab components not meeting C , and by zero where $f_\Sigma > t_1$. Since $d_M(o, z) \leq D_0 < S_0 - 2$ on C , the field is one in a fixed intrinsic neighborhood of every inner circle. It vanishes near every outer circle by the protected-distance bound. Compactness and regularity prevent a separate slab component from accumulating at an attaching circle without belonging to its local component. The metric on Y is exactly the original metric. The radial cutoff on M is Lipschitz with constant $1/[S_0 \log(S/S_0)]$, and its restriction has that same local Lipschitz bound on Y . It is constant on neighborhoods of all the gluing interfaces just described. The constant extensions are therefore globally Lipschitz in the original intrinsic path metric, with the same bound: integrate the local bound along any original surface path. They create no interface energy. Components wholly inside carry the constant one, and $0 \leq \xi \leq 1$ everywhere. All cutoff differences along selected geodesics use those original surface paths and their original metric; shortcuts through the new attachment are not substituted for them. We have

$$\xi = 1 \quad (r < 4.5R), \quad \text{supp } \xi \subset \{r < 7R + S\}, \quad \int_\Sigma |\nabla_\Sigma \xi|^2 dA \leq \frac{CNR^{P-1}}{\log(S/S_0)}. \quad (13.21)$$

Indeed $r < 4.5R$ implies $f_\Sigma < 4.55R < t_0$. Our choice $t_0 > 5.2R$ gives the stronger plateau $\xi = 1$ at every regular base with $r \leq 5R$, including components not meeting C . On the closure of the nonconstant support, the Lipschitz extension of f_Σ gives $f_\Sigma \leq t_0 + S$, so $r \leq 6.05R + S < 7R + S$. The gradient vanishes on every original piece outside Y , while on Y the metric and area form coincide with those in M . Hence the original gradient integral is at most the complete-surface cutoff energy computed above. The surface field may approach the artificial inner edge; the base cutoff below removes that edge before an ambient test is formed.

If C is empty, no completion is needed. Continuity of f_Σ on each connected component prevents that component from crossing the level t_0 . Set $\xi = 1$ on components with $f_\Sigma < t_0$ and zero on those with $f_\Sigma > t_0$. The same plateau, including $r \leq 5R$, and support statements hold. This field

is globally Lipschitz in the intrinsic path metric (constant on each component), and its gradient energy is zero.

13.5. Joining the cluster to its regular exterior. Fix $c_{\text{sel}} < 1/20$ smaller than the chart constant, with an intrinsic selector plateau on half its horizon. Enlarge R so that for every $r \geq R$,

$$T(r)/r \leq \min\{1/100, c_{\text{sel}}/100\}, \quad T(r) \geq 1. \quad (13.22)$$

This is possible because $T(r)/r = \tau r^{-(p-1)/(p+1)}$ decreases. Increase the same fixed threshold further so that $T(r) \geq 4$ for every $r \geq R$, as required by the pointwise core-row estimate. All one-variable cutoff profiles below are smooth with values in $[0, 1]$; their compositions with the Lipschitz distance r need only be Lipschitz. Choose a base cutoff j , zero for $r \leq R$, one for $r \geq 3R/2$, with $|j'| \leq C/R$. On the disjoint actual tubes define

$$\eta_0(X(z, t)) = j(r(z))b(t/T(z))\xi(z),$$

where $b = 1$ on $[-1, 1]$ and zero outside $(-3/2, 3/2)$; extend the amplitude by zero elsewhere. Let θ be one on $[0, 2]$ and zero on $[3, \infty)$, with values in $[0, 1]$, and set

$$\eta(x) = \theta(r(x)/R) + (1 - \theta(r(x)/R))\eta_0(x). \quad (13.23)$$

By the actual-tube construction preceding (11.2), smallness of $|tA_\Sigma|$ gives $D_x t = n(z)$ and $D_x z = (I - tA_\Sigma)^{-1}\pi_{T_z\Sigma}$, with uniform differential control of the normal base-point map. The buffered intrinsic and chord distances are comparable. Both T and the chosen surface field ξ are Lipschitz, the latter in the original intrinsic metric. The capacity construction gives $\xi = 1$ at every base with $r \leq 5R$ and $\text{supp}_\Sigma \xi \subset \{r < 7R + S\}$; it need not have compact intrinsic support at the artificial inner edge. The three boundaries have separate zero buffers: the normal cutoff is already zero when the tube ends, j is zero on a neighborhood of the inner artificial boundary, and ξ is zero near the outer one. Thus extension by zero preserves local Lipschitz regularity. The base points at which the amplitude can be nonzero belong to the regular compact set $Z \cap \{R \leq r \leq 7R + S\}$. Covering this set by finitely many buffered charts supplies a common local Lipschitz bound and a positive Lebesgue number there; boundedness controls more distant pairs. Thus η_0 and η are globally Lipschitz with compact ambient support. Tube disjointness makes them single functions on returning branches. Both amplitudes satisfy $0 \leq \eta_0, \eta \leq 1$. The amplitude η is one throughout $r \leq 2R$, so no regular surface inside the cluster is required.

For a fixed $c_* > 0$ chosen after τ , its matching properties are

$$\begin{aligned} \eta(x) &= 1 \quad \text{if } r(x) < 4R, \quad \text{dist}(x, Z) \leq c_* R^{2/(p+1)}, \\ |\eta(x) - \eta(y)| &\leq CR^{-2/(p+1)}|x - y| \quad \text{if } r(x) \leq 3R, \quad |x - y| < R. \end{aligned} \quad (13.24)$$

Beyond $r = 2R$, a nearest zero exists because Z is nonempty and closed. Its base radius is between $1.9R$ and $4.1R$ after the fixed lower threshold for R is enlarged. It is a regular zero, so minimizing squared distance makes the displacement normal to its tangent plane. Choose $c_* > 0$ sufficiently small that $c_* < \tau 1.9^{2/(p+1)}$, retaining strict chart buffers. Then the offset is below $T(z)$, and $j = \xi = b = 1$ there. Injectivity of the actual tubes identifies this normal representation with the assigned tube coordinates. This nearest point is chosen from the whole closed set Z , not from a selected branch. These margins follow from the Lipschitz property of r and (13.22). For the second assertion, the segment from x to y stays in $r < 4R$. Every contributing tube in $2R < r < 4R$ has the same base margins, so $j = \xi = 1$ and $T \asymp R^{2/(p+1)}$. Normal differentiation costs C/T , width variation C/r , and the mixing cutoff C/R , all bounded by $CR^{-2/(p+1)}$. These are uniform local Lipschitz bounds, including across the buffered zero extension, rather than only almost-everywhere gradient bounds. Cover the segment by finitely many such neighborhoods and sum their increment bounds, passing to an endpoint by continuity if needed. This gives the displayed estimate for every stated pair without an exceptional-segment chain-rule issue.

Put $I = \{r \leq 3R\}$ and $O = \mathbb{R}^3 \setminus I$. Denote by $\mathcal{J}_R$ the positive term of (13.2) on pairs having at least one endpoint in I , retaining its factor $c_{3,s}/2$. For the pairs with $|x - y| \geq R$, symmetry,

the tail moment, and (13.10) give

$$\mathcal{J}_R^{\text{far}} \leq CR^{P-3-p} \int_I g \leq CNR^{2P-3-p} = CNR^{-(p-1)^2/(p+1)}.$$

For a contributing pair with $|x - y| < R$, both endpoints have $r < 4R$. If their amplitudes differ, one endpoint belongs to

$$\mathcal{W}_R = \{r < 4R : \text{dist}(x, Z) > c_* R^{2/(p+1)}\}.$$

Charge the pair to that endpoint, with multiplicity at most two if both endpoints qualify. Extend the resulting positive quadratic majorant to all targets of the charged point; the local increment claim itself is used only for the original near pair. The unordered pair still has an endpoint in I , even when the charged endpoint lies outside I . Thus the symmetric Lipschitz bound in (13.24) remains available after this choice. The second moment in (13.3), clearing $g \leq CR^{-2}$ on $\mathcal{W}_R$, and its volume bound yield

$$\mathcal{J}_R^{\text{near}} \leq CR^{-4/(p+1)} \int_{\mathcal{W}_R} g \leq CNR^3 R^{-2(p+3)/(p+1)} = CNR^{-(3-p)/(p+1)}.$$

Consequently,

$$\mathcal{J}_R \leq CN[R^{-(p-1)^2/(p+1)} + R^{-(3-p)/(p+1)}]. \quad (13.25)$$

The far estimate includes every ambient target, including remote clusters. Its center-uniform moment uses only one factor N .

13.6. Keeping all targets in the exterior source algebra. The detector balls lie in $\{r < b < R\} \subset I$. On these balls $\eta = 1$ because $r \leq 2R$. Their defect gives

$$\int_I D_u \eta^2 \geq c_{3,s} \sum_{i=1}^N \iint_{B_b(z_i)^2} \mathcal{D}(x, y) |x - y|^{-3-p} dx dy \geq b_* N.$$

Let P_{O^2} denote the positive pair form restricted to O^2 . The exact pair partition in (13.2) implies

$$Q_u(g\eta) \leq -b_* N + \mathcal{J}_R + P_{O^2}(\eta) - \int_O D_u \eta^2. \quad (13.26)$$

Choose the symmetric selector from Section 11 in the following form, zero off the actual tubes:

$$\chi(X(z, t), X(w, v)) = \kappa(z)\kappa(w)\beta(t/T(z))\beta(v/T(w))\zeta\left(\frac{d_\Sigma(z, w)}{c_{\text{sel}} \min\{r(z), r(w)\}}\right). \quad (13.27)$$

Here β is even and nonincreasing on $[0, \infty)$, equals one on $[-2, 2]$, and vanishes outside $(-3, 3)$; ζ is nonincreasing, equals one on $[0, 1/2]$, and vanishes on $[1, \infty)$. These monotonicity choices are the ones required for the positive layer-cake selector in Section 11. The intrinsic distance between components is infinite. Put $B_0 = 7R + S$. Take $\kappa = \kappa_{\text{in}}\kappa_{\text{out}}$, with the inner factor zero for $r \leq 2R$ and one for $r \geq 5R/2$, and the outer factor one for $r \leq 2B_0$ and zero for $r \geq 3B_0$. Their derivative bounds are C/R and C/B_0 . The scale choice puts every selected tube and its buffers inside the regular annulus.

An active exterior source has $x = X(z, t)$, $|t| < 3T(z)/2$, and $2.9R < r(z) < B_0$. The lower bound follows from $r(x) > 3R$ and $T/r \leq 1/100$; the upper bound follows from (13.21). A selected target base satisfies

$$(1 - c_{\text{sel}})r(z) < r(w) < (1 + c_{\text{sel}})r(z).$$

Thus its base lies above $2.5R$ and below $2B_0$, making all base factors one in each selected neighborhood of an active source. The normal plateau has a fixed buffer beyond $|t| < 3T/2$, and (13.22) puts the core annuli of the signed-row estimate inside the intrinsic plateau. Here is the diagonal plateau needed for the source algebra. Starting at $x = X(z, t)$, lift a short straight ambient segment by the local inverse of the actual normal map. The inverse-coordinate bounds above give, with one fixed C_g ,

$$|v - t| \leq |y - x|, \quad d_\Sigma(z, w) \leq C_g |y - x|, \quad |r(w) - r(z)| \leq C_g |y - x|,$$

where $y = X(w, v)$; the intrinsic bound is the length of this lifted base path on the original surface. Choose a fixed $c_0 > 0$ small in terms of these chart constants. If $|y - x| < c_0 T(z)$, the scale margin and $T(w) = \tau r(w)^{2/(p+1)}$ give

$$2.5R < r(w) < 2B_0, \quad |T(w)/T(z) - 1| < 1/20, \quad |v| < 2T(w), \quad d_\Sigma(z, w) < \frac{1}{2}c_{\text{sel}} \min\{r(z), r(w)\}.$$

These estimates hold along a partial inverse lift and keep it strictly inside its compact complete graph buffer and normal strip. At a putative stopping time its limiting coordinates are still interior, so the inverse-function theorem continues the lift; injectivity makes the continuations agree. Thus every point of the stated physical ball has these coordinates. Both base factors and both normal factors are one, and the intrinsic factor lies in its half-horizon plateau. Consequently $\chi(x, y) = 1$ for $|y - x| < c_0 T(z)$. The uniform lower bound $T(z) \geq \tau(2.9R)^{2/(p+1)} > 0$ removes a common positive neighborhood of the diagonal.

Write $k(x, y) = |x - y|^{-3-p}$ and define

$$D_{\chi, O}(\eta) = c_{3,s} \int_O \eta(x)^2 \int_{\mathbb{R}^3} \chi(x, y) \mathcal{D}(x, y) k(x, y) dy dx,$$

$$\mathcal{R}_O(\eta) = c_{3,s} \int_O g(x) \eta(x)^2 \nu(x) \cdot \int_{\mathbb{R}^3} (1 - \chi(x, y)) \nabla u(y) k(x, y) dy dx, \quad \nu = \nabla u / g.$$

The source integrand is zero when $g\eta = 0$; an inactive source requires no definition of its unselected row. At active sources, the plateau $\chi = 1$ removes a fixed positive neighborhood of the diagonal. Boundedness of g and the integrable kernel tail make both the scalar row with target $g(y)$ and the vector row with target $\nabla u(y)$ absolutely integrable. In particular the unselected scalar quantity required below satisfies

$$c_{3,s} \int_O g(x) \eta(x)^2 \int_{\mathbb{R}^3} (1 - \chi) g(y) k(x, y) dy dx \leq C[\tau(2.9R)^{2/(p+1)}]^{-p} \int_O g\eta^2 < \infty.$$

The active source support is compact, and the plateau radius has a positive lower bound there. The selected defect is integrable by the quadratic diagonal bound preceding (13.3); selected positive forms are integrable because the amplitudes are Lipschitz. Thus every term in the following rearrangement is finite, and Fubini is justified. If P_{χ, O^2} is the selected positive form, symmetry gives

$$P_{1-\chi, O^2}(\eta) = c_{3,s} \int_O g(x) \eta(x)^2 \int_O (1 - \chi) g(y) k dy dx$$

$$- c_{3,s} \iint_{O^2} (1 - \chi) g(x) g(y) \eta(x) \eta(y) k dx dy.$$

On the other hand, $\int_O D_u \eta^2 - D_{\chi, O}$ has the scalar source term over all targets, minus $\mathcal{R}_O$. Subtracting these identities leaves

$$P_{O^2}(\eta) - \int_O D_u \eta^2 = P_{\chi, O^2}(\eta) - D_{\chi, O}(\eta) + \mathcal{R}_O(\eta)$$

$$- c_{3,s} \iint_{O \times I} (1 - \chi) g(x) g(y) \eta(x)^2 k dx dy \quad (13.28)$$

$$- c_{3,s} \iint_{O^2} (1 - \chi) g(x) g(y) \eta(x) \eta(y) k dx dy.$$

Both remaining pair terms are nonpositive because $0 \leq \chi \leq 1$ and $\eta \geq 0$. In particular, the $O \times I$ term preserves the inner targets in the signed row.

The adapted base cutoff is not literally the fixed-threshold cutoff in Section 11. To transfer its pointwise estimates, use the positive ambient scale $\max\{r, r_{\text{reg}}\}$ and the larger regular complete buffers already supplied by the cluster exterior. Fix the earlier lower threshold r_θ independently of the cluster and increase R so that $R > 10r_\theta$ and all fixed pointwise-row thresholds hold. At each active source choose a legitimate compact smooth nonnegative base bump ξ_{std} , supported in $r > 4r_\theta$ and positive at z . Use the same $c_{\text{sel}}, T, b, \beta, \zeta$ with the standard base cutoff, which equals one for $r > 3r_\theta$. On every nonzero row horizon, the two bases lie above $2.5R$ and below $2B_0$, so both the adapted and standard base factors equal one. Off this horizon or the selected normal

strips, both selector values vanish. If the auxiliary surface is restricted to a larger retained annulus, every original path shorter than the horizon stays in that annulus by the ambient Lipschitz bound for r . Its short intrinsic distance is unchanged; beyond the original horizon the restricted distance can only increase. Thus the two selectors agree for every ambient target, including those for which both are zero.

Define $V_\chi(x) = \int_{\mathbb{R}^3} (1 - \chi) \nabla u(y) k(x, y) dy$ and $d_\chi(x) = c_{3,s} \int_{\mathbb{R}^3} \chi \mathcal{D}(x, y) k(x, y) dy$. Put $a = 1 + |t|$ at this source. The absolute core row and coherent direction of the pointwise argument in Lemma 11.1, followed by its directional signed-row bound, therefore apply to this identical row. The legitimate source amplitude $b(t/T)\xi_{\text{std}}(z)$ is positive whenever the actual amplitude is positive. Divide its squared defect bound by this positive square and multiply by $\eta(x)^2$. With the resulting coherent direction e_x this yields

$$d_\chi(x)\eta(x)^2 \geq cg(x)\eta(x)^2 a^{-p-1} |\nu(x) - e_x|^2, \quad |V_\chi(x)| \leq CT^{-p-1}, \quad e_x \cdot V_\chi(x) \leq CE(r, T).$$

Young's inequality gives

$$c_{3,s}g\eta^2\nu \cdot V_\chi \leq \varepsilon d_\chi\eta^2 + C_\varepsilon g\eta^2 [E(r(z), T(z)) + a^{p+1}T(z)^{-2p-2}].$$

This controls the signed vector row and retains its vector targets. When $g(x) = 0$, one has $\nabla u(x) = 0$ and $d_\chi(x) = 0$, so the source expression is zero without choosing $\nu(x)$. Multiplication by $\mathbf{1}_O$ differentiates nothing. On the supported portion of a normal fiber, $\eta = b(t/T)\xi(z)$ and $g \leq C(1 + |t|)^{-p-1}$, so

$$\int_{\substack{|t| < 3T/2 \\ X(z,t) \in O}} g dt \leq C, \quad \int_{\substack{|t| < 3T/2 \\ X(z,t) \in O}} ga^{p+1}T^{-2p-2} dt \leq CT^{-1-2p}.$$

The Jacobian is bounded, and only this subset of the normal interval is integrated. Thus

$$\mathcal{R}_O(\eta) \leq \varepsilon D_{\chi,O}(\eta) + C_\varepsilon \int_{\Sigma \cap \{r > 2.8R\}} E(r, T)\xi^2 dA. \quad (13.29)$$

All targets outside the isolated annulus remain in V_χ . There is neither a discontinuous equation test nor a boundary term on ∂O . Since $\eta = \eta_0$ on O and the selected positive integrand is nonnegative, $P_{\chi,O^2}(\eta) \leq P_\chi(\eta_0)$. This restricts a pair domain; it does not use monotonicity under a general amplitude replacement. Fixing $\varepsilon < 1$, the source identities and the error sum give

$$Q_u(g\eta) \leq -b_*N + \mathcal{J}_R + P_\chi(\eta_0) + CNR^{-\alpha}. \quad (13.30)$$

The selected form has no cost from the artificial base cutoff j . Indeed, a nonzero base factor κ_{in} requires $r > 2R$, where $j = 1$. A joining geodesic of length $\rho < c_{\text{sel}} \min\{r(z), r(w)\}$ stays in $r \geq (1 - c_{\text{sel}})r(z) > 1.8R$, where $j = 1$. Hence its tangential difference is estimated directly by

$$|\xi(z) - \xi(w)|^2 \leq \rho \int_0^\rho |d\xi_{\gamma(a)}(\dot{\gamma}(a))|^2 da.$$

These geodesics belong to the original surface, not to the completion with its added shortcuts. After normal integration, the tangential kernel has a majorant

$$k_0(\rho) = C \min\{\rho^{-2-p}, \rho^{-3-p}\}, \quad \int_0^\infty \rho^3 k_0(\rho) d\rho < \infty.$$

Indeed $\rho^3 k_0(\rho) \leq C\rho^{1-p}$ for $\rho \leq 1$ and $\rho^3 k_0(\rho) \leq C\rho^{-p}$ for $\rho \geq 1$, so both integrals converge precisely in the range $1 < p < 2$. The local polar Jacobian bound and the geodesic-flow change of variables in Section 11 give the required gradient cost. Explicitly, let $\mathcal{A}_\rho$ denote the admissible starting unit vectors, let Φ_a be geodesic flow, put $h_\xi(z, \omega) = |d\xi_z(\omega)|^2$, and use $d\mu = dA d\omega$. Keeping the admissible-segment restriction until the intermediate point is the integration variable gives

$$\int_{\mathcal{A}_\rho} \int_0^\rho h_\xi(\Phi_a(z, \omega)) da d\mu \leq \rho \int_{UT(\{r > 1.8R\})} h_\xi d\mu = \pi\rho \int_{\{r > 1.8R\}} |\nabla \xi|^2 dA.$$

The polar factor and the difference estimate supply $\rho^2 k_0(\rho) d\rho$, leaving precisely the finite third moment above. This argument requires no completeness of Σ . For Lipschitz ξ , use the measure-preservation clause of the geodesic-flow argument in Lemma 11.2, rather than its compact-smooth-test corollary. Define h_ξ by the differential almost everywhere and set it zero on the area-null nondifferentiability set. The inverse image of that set under each admissible flow map has measure zero. Fubini implies that almost every admissible geodesic meets it only for a null set of times. The composition of ξ with a unit-speed path is Lipschitz and absolutely continuous, so its chain rule and the fundamental theorem of calculus give the displayed difference estimate for almost every starting vector. This uses the integrable differential of the single original-surface field and introduces no auxiliary inner cutoff. For the normal term, split $b(t/T(z)) - b(v/T(w))$ through $b(t/T(w))$. The equal-width cutoff moment of the proof of Lemma 11.2 contributes $CT(w)^{-1-2p}\xi(w)^2$. The variable-width difference is bounded by $C\rho/r(w)$ and supported where $a_1T(w) < |t| < a_2T(w)$, with fixed $a_1, a_2 > 0$. Its polar integral uses

$$\int_0^\infty \rho^3 [\rho^2 + (t-v)^2]^{-(3+p)/2} d\rho = C_p |t-v|^{1-p} \quad (t \neq v),$$

and the earlier annular normal moment gives $Cr(w)^{-2}T(w)^{1-2p}\xi(w)^2$. Tonelli justifies these nonnegative majorants; only their scalar normal and radial integrals are extended, not the actual surface or its controlled tubes. Both are charged to selected bases with $r > 2R$. Therefore

$$P_\chi(\eta_0) \leq C \int_{\Sigma \cap \{r > 1.8R\}} |\nabla \xi|^2 dA + C \int_{\Sigma \cap \{r > 2R\}} E_0(r, T) \xi^2 dA. \quad (13.31)$$

For any nonzero selected amplitude difference, one endpoint has base radius below B_0 and the other has radius below $(1 + c_{\text{sel}})B_0 < 2B_0$. Thus the actual outer base cutoff is one throughout every contributing pair and contributes no cost.

Theorem 13.4. *The fixed exceptional set B is empty.*

Proof. Combine (13.30), (13.25), (13.31), (13.14), and (13.21). Stability yields

$$0 \leq Q_u(g\eta) \leq -b_*N + CN[R^{-(p-1)^2/(p+1)} + R^{-(3-p)/(p+1)} + R^{-\alpha}] + \frac{CNR^{P-1}}{\log(S/S_0)}.$$

Since

$$\frac{3-p}{p+1} - \alpha = \frac{5-p^2}{2(p+1)} > 0,$$

we can use one fixed constant to write

$$0 \leq Q_u(g\eta) \leq N \left[-b_* + CR^{-\alpha} + \frac{CR^{P-1}}{\log(S/S_0)} \right]. \quad (13.32)$$

The constants are independent of the cluster, its diameter, its packing number, and its isolation width. First fix the profile, variation, clearing, and geometric constants; this fixes b, b_* . Then fix τ , the filter tolerance with $\lambda > 1/2$, and their lower thresholds. Choose R above these thresholds and a fixed multiple of $r_{\text{reg}} + d_0 + b$, satisfying $CR^{-\alpha} < b_*/4$ and (13.22) for every $r \geq R$. Fix $S_0 > 7$, for example $S_0 = 10$. Next choose $S > S_0e$ with $CR^{P-1}/\log(S/S_0) < b_*/4$, and finally choose $A > 1000(S+R)$, enlarged by the fixed chart-buffer factor. Only now select the cluster and its packing number N . The transverse levels and tiny collar widths may depend on the cluster; their cost bounds and D_0 do not. All selected original-surface geodesics and normal tubes stay in the controlled annulus. The attached completion is used only to construct the scalar field ξ and its cost; no distance or geodesic from that completion is substituted in a selector or positive-energy estimate. The signed rows and far pairs already account for all targets beyond the annulus. The right side of (13.32) is at most $-b_*N/2 < 0$, a contradiction. For empty C , the tangential cost is zero and the same contradiction holds. $\square$

14. FROM PLANAR PHASE LEVELS TO STABLE RIGIDITY

Continue with a bounded stable entire solution u of the normalized cubic equation in $\mathbb{R}^3$, and write $p = 2s \in (1, 2)$ and $Z = \{u = 0\}$. Lemma 2.1 gives $|u| \leq 1$, and Section 3 supplies smoothness and uniform global derivative bounds. Choose exactly the fixed wide-band level $a_0 = a \in (1/\sqrt{3}, 1)$ and the threshold ϵ_* used in the preceding section; this threshold was decreased to the good-band threshold when chosen. Theorem 13.4 gives $B = B_{\epsilon_*}(u) = \emptyset$. Distance to this empty set is $+\infty$, so $\Omega_{\epsilon_*} = \mathbb{R}^3$. Its complement is also empty and the denominator in the distance-curvature estimate is therefore $+\infty$ at every point. The good-band estimate (4.3) supplies nondegeneracy, while Corollary 12.3, with the prescribed interpretation $C/\infty = 0$, supplies zero curvature. Thus

$$|\nabla u| \geq \gamma > 0, \quad |A_{\{u=u(x)\}}(x)| = 0 \quad (|u(x)| \leq a_0). \quad (14.1)$$

Lemma 14.1. *At any phase $|\tau| \leq a_0$, every existing connected component of $\{u = \tau\}$ is a whole affine plane. Planes belonging to any two of these phase components have the same plane direction.*

Proof. Let S be a nonempty connected component of $\{u = \tau\}$. A component is closed in its level set because its closure there is connected and hence, by maximality, remains in the component. The level set is closed by continuity, so S is closed in $\mathbb{R}^3$. Nondegeneracy at every phase $|\tau| \leq a_0$, including the endpoints, gives a smooth graph chart over an open disk at each point. This chart is connected and belongs to S , so S is an embedded smooth surface without boundary. It has the single increasing normal $n = \nabla u / |\nabla u|$. For every tangent vector v , differentiation of $|n|^2 = 1$ makes $D_v n$ tangent; the shape operator is $-D_v n$. Its vanishing in (14.1) therefore gives $D_v n = 0$. A connected smooth manifold is connected by piecewise smooth paths; integrating along such paths shows that n is a constant vector e_S on S . The same argument applied to the tangential derivative of $e_S \cdot x$ puts S in a plane $\Pi_S = \{e_S \cdot x = c_S\}$. At each point its tangent space is $e_S^\perp = T\Pi_S$. The inclusion $S \hookrightarrow \Pi_S$ therefore has invertible differential, so the inverse function theorem makes its image open in Π_S . It is also closed there because S is closed in $\mathbb{R}^3$. Since S is nonempty and Π_S is connected, $S = \Pi_S$. Distinct components, including components from different phase levels, are disjoint. Nonparallel affine planes in $\mathbb{R}^3$ intersect, so all these planes have the same unoriented normal line. Their increasing normals may have opposite signs; their tangent directions nevertheless agree. $\square$

Lemma 14.2. *If $z \in Z$ and $e = \nabla u(z) / |\nabla u(z)|$, then $\partial_v u \equiv 0$ for every $v \perp e$.*

Proof. Nondegeneracy in (14.1) makes e well defined. The case $v = 0$ is immediate; otherwise, $\partial_v u = |v| \partial_{v/|v|} u$, so it suffices to take $|v| = 1$ and $v \perp e$. The zero plane through z has normal line $\mathbb{R}e$. By Lemma 14.1, every point of $\{|u| \leq a_0\}$ lies on a phase plane with this same unoriented normal line. Its nonzero gradient is therefore parallel to e , regardless of the sign of its increasing normal, and $w = \partial_v u$ vanishes on the whole closed band.

The bounded smooth Jacobi derivative identity (4.2), applied to the bounded entire solution u in dimension three and the constant unit vector v , gives bounded smooth w with bounded derivatives. Its dimension-independent difference-quotient formulation is useful below: for $h \neq 0$, put $u_h(x) = u(x + hv)$ and $w_h = (u_h - u)/h$. Translation invariance and subtraction of the cubic equations give

$$(-\Delta)^s w_h + (u_h^2 + u_h u + u^2 - 1)w_h = 0.$$

The formula $w_h(x) = \int_0^1 \partial_v u(x + \theta hv) d\theta$ and the earlier global derivative bounds give $w_h \rightarrow w$ in C_{loc}^2 , with common global bounds through two derivatives. The symmetric-increment limit argument of Section 2 then passes to the limit: its integrands are dominated by $C \min\{|y|^2, 1\}|y|^{-n-p}$, which is integrable for $0 < p < 2$. This quotient argument and (4.2) work in every fixed dimension, including dimension four. Here they give

$$(-\Delta)^s w + (3u^2 - 1)w = 0, \quad w = 0 \text{ on } \{|u| \leq a_0\}. \quad (14.2)$$

The set $D = \{|u| > a_0\}$ is open, possibly unbounded. Both w and $-w$ belong to $C_b(\mathbb{R}^3) \cap C_{\text{loc}}^2(D)$ and vanish on D^c . Their equation has potential $c(X) = 3u(X)^2 - 1 \geq 3a_0^2 - 1 > 0$ on D . Thus

the positive-potential comparison proved at the start of Section 4 applies to each of them, with the same normalized operator and $s \in (1/2, 1)$. It gives $w \leq 0$ and $-w \leq 0$ everywhere, hence $w = 0$. Linearity gives the conclusion for every $v \perp e$. $\square$

Proof of Theorem 1.2. If Z is empty, continuity on the connected space $\mathbb{R}^3$ gives one fixed strict sign: opposite signs at two points would give a zero on the intervening segment by the intermediate value theorem. Lemma 2.2 therefore gives $u \equiv 1$ or $u \equiv -1$. Otherwise choose $z \in Z$. The nondegeneracy in (14.1) applies at every zero, so $e = \nabla u(z)/|\nabla u(z)| \in S^2$ is well defined. Put $Q(t) = u(z + te)$. Every $x \in \mathbb{R}^3$ is uniquely $x = z + te + y$, where $t = e \cdot (x - z)$ and $y \perp e$. Lemma 14.2 gives

$$u(x) - Q(t) = \int_0^1 \nabla u(z + te + \theta y) \cdot y \, d\theta = 0.$$

The global derivative bounds of Section 3 give $Q \in C_b^2(\mathbb{R})$, with bounded derivatives of every order, and $Q(0) = 0$, $Q'(0) = |\nabla u(z)| > 0$. Choose an orthogonal matrix O with first column e , so that $U(t, y') = u(z + O(t, y')) = Q(t)$ on $\mathbb{R} \times \mathbb{R}^2$. Translations preserve the normalized operator and stability by change of variables: the kernel depends on the difference of the points, and Lebesgue measure and the compact smooth test class are translation invariant. The orthogonal invariance is included in Lemma 3.2. Thus U solves the same normalized equation and is stable. That lemma with the exact integers $m = 1$, $k = 2$, using $Q \in C_b^2$, gives $(-\partial_t^2)^s Q = Q - Q^3$ with the normalized one-dimensional coefficient and transfers stability to Q . Lemma 4.2 lists all such stable bounded profiles. The value $Q(0) = 0$ excludes the constant wells, and $Q'(0) > 0$, together with $q'_s > 0$, excludes every decreasing layer. Hence $Q(t) = q_s(t - t_0)$ for some $t_0 \in \mathbb{R}$. Since q_s is strictly increasing and its unique zero is at zero, $Q(0) = 0$ forces $t_0 = 0$. Therefore $u(x) = q_s(e \cdot x - a)$ with $a = e \cdot z$, as asserted. The constant alternatives are stable in $\mathbb{R}^3$ as well: Lemma 4.2 includes their stability in dimension one, and the two-way stability in Lemma 3.2 preserves it under constant cylindrical extension with $m = 1$, $k = 2$. $\square$

15. DIRECTIONAL LIMITS AND GLOBAL MINIMIZATION

Let u now be a bounded entire solution in $\mathbb{R}^4$, written $x = (x', t) \in \mathbb{R}^3 \times \mathbb{R}$, with $\partial_t u > 0$. Throughout this section $p = 2s \in (1, 2)$. Lemma 2.1 gives $-1 < u < 1$, since attainment of either well would make u constant, contrary to $\partial_t u > 0$. The all-dimensional regularity and Jacobi statement established in Section 4 supplies smoothness and global bounds for every derivative; it requires only the equation and boundedness.

Lemma 15.1. *The solution u is stable in $\mathbb{R}^4$.*

Proof. The bounded smooth positive function $w = \partial_t u$ satisfies $(-\Delta)^s w + (3u^2 - 1)w = 0$ and its compact weak pairing by the all-dimensional Jacobi statement of Section 4, applied with the unit vector e_4 . That statement needs no stability hypothesis. For real $\varphi \in C_c^\infty(\mathbb{R}^4)$, both φ/w and φ^2/w belong to $C_c^\infty(\mathbb{R}^4)$: w is smooth and has a positive minimum on a compact neighborhood of $\text{supp } \varphi$. Thus the latter quotient is a legitimate test in the Jacobi equation. Writing subscripts for evaluations, the exact pair identity is

$$(\varphi_x - \varphi_y)^2 - (w_x - w_y) \left(\frac{\varphi_x^2}{w_x} - \frac{\varphi_y^2}{w_y} \right) = w_x w_y \left(\frac{\varphi_x}{w_x} - \frac{\varphi_y}{w_y} \right)^2.$$

Testing the Jacobi equation and substituting this identity yields

$$Q_u^{(4)}(\varphi) = \frac{c_{4,s}}{2} \iint \frac{w(x)w(y)}{|x-y|^{4+p}} \left(\frac{\varphi(x)}{w(x)} - \frac{\varphi(y)}{w(y)} \right)^2 dx dy \geq 0. \quad (15.1)$$

For completeness, the zero test gives zero on both sides. For a nonzero test put $K = \text{supp } \varphi$ and fix $\delta > 0$. On the compact neighborhood $K_{2\delta} = \{x : \text{dist}(x, K) \leq 2\delta\}$, positivity gives $m = \min w > 0$. The quotients $h = \varphi/w$ and $\zeta = \varphi^2/w$ are supported in K and have bounded derivatives on this neighborhood. If $|x - y| < \delta$ and a quotient increment is nonzero, one endpoint lies in K and the entire connecting segment lies in $K_{2\delta}$. Hence each quotient increment and the

increment of w are bounded by $C|x - y|$. Boundedness of w and the finite measure of K control both the weighted square on the right of (15.1) and the absolute mixed Jacobi pairing near the diagonal by $C \int_0^\delta \rho^{1-p} d\rho$. For $|x - y| \geq \delta$, a nonzero quotient increment still has an endpoint in K ; boundedness of w, h, ζ gives the finite tail $C \int_\delta^\infty \rho^{-1-p} d\rho$. The potential pairing is bounded and compactly supported, and the ordinary kinetic integral of φ has the same near and far estimates. Thus every integral in the subtraction is finite. In particular the weak test gives $\mathcal{B}_4(w, \varphi^2/w) = -\int (3u^2 - 1)\varphi^2$, so the pair identity indeed has the sign and coefficient in (15.1). These estimates require no lower bound for w at infinity. $\square$

Lemma 15.2. *The directional limits $v^\pm(x') = \lim_{t \rightarrow \pm\infty} u(x', t)$ exist and are bounded smooth stable solutions of the same Fourier-normalized equation in $\mathbb{R}^3$. Moreover, $v^- < v^+$ pointwise, and $v^- < u < v^+$ on $\mathbb{R}^4$.*

Proof. For each fixed x' , monotonicity and boundedness give the two limits as the infimum and supremum of $t \mapsto u(x', t)$. For every fixed finite t , and every real sequence T_j tending to either end, $t + T_j$ tends to the same end. Thus the pointwise limit of $u(x', t + T_j)$ is $V^\pm(x', t) = v^\pm(x')$, independent of t .

Lemma 15.1 gives stability of u . Apply the all-dimensional translation compactness result of Section 2 in dimension four, with the common bound one. Translates are stable solutions, and every such sequence has a further subsequence converging locally in C^2 to a stable entire classical solution. Its pointwise limit must be $V^\pm$. Using first $T_j = j$ and $T_j = -j$ shows that these functions are C^2 stable solutions of the dimension-four equation. Their first two derivative bounds pass pointwise through this local convergence, so $v^\pm = V^\pm(\cdot, 0)$ belong to $C_b^2(\mathbb{R}^3)$.

The convergence holds for the full real-parameter translation family. Otherwise some compact ball, some $\varepsilon > 0$, and a sequence T_j tending to the relevant end would have $\|u(\cdot, \cdot + T_j) - V^\pm\|_{C^2}$ on that ball at least ε . The same compactness gives a further subsequence converging locally in C^2 , whose pointwise limit is again $V^\pm$, a contradiction. Every compact set lies in a ball. Equation and stability pass to these limits by that earlier compactness result.

Lemma 3.2, with $m = 3$ and $k = 1$, now transfers the exact equation from $V^\pm$ to $v^\pm$, with coefficient $c_{3,s}$, and its two-way stability statement transfers their stability. The all-dimensional regularity of Section 4, applied in dimension three, makes the bounded endpoints smooth with global derivative bounds. Finally,

$$v^-(x') \leq u(x', t - 1) < u(x', t) < u(x', t + 1) \leq v^+(x')$$

holds because each intermediate difference is an integral of the continuous positive $\partial_t u$ over an interval of length one. It proves all the strict inequalities. Theorem 1.2 classifies each endpoint separately as a well or an increasing planar layer $q_s(e_\pm \cdot x' - b_\pm)$, without prescribing either endpoint or a relation between any endpoint directions. $\square$

For a bounded open set $\Omega \subset \mathbb{R}^4$, the energy (2.5) takes the form

$$\begin{aligned} \mathcal{Q}_\Omega &= \mathbb{R}^8 \setminus (\Omega^c \times \Omega^c), \\ E_\Omega(v) &= \frac{c_{4,s}}{4} \iint_{\mathcal{Q}_\Omega} \frac{|v(x) - v(y)|^2}{|x - y|^{4+p}} dx dy + \int_\Omega F(v) dx. \end{aligned}$$

For $\kappa > 0$, write E_Ω^κ for this same functional with $c_{4,s}$ replaced by κ , and put $\beta = 2 - p > 0$. The integrals are nonnegative extended integrals. A finite-energy competitor with exterior datum U is a measurable real function equal to U almost everywhere on Ω^c , with finite energy. This definition does not assume the existence of a minimizer. The energy of any smooth bounded function is finite on Ω : its gradient is bounded on a fixed bounded neighborhood of Ω , so small squared increments are bounded by $C|x - y|^2$; boundedness controls the larger increments. For a source in Ω , the radial integrals are bounded by $C((2 - p)^{-1} + p^{-1})$, and symmetry controls the whole pair domain. In particular this applies to u .

We first match the normalization of the monotone-to-minimizer comparison in Cabré–Cinti–Serra [6, Proposition 7.9 and the sentence following its statement]. Their kernel coefficient is $2 - p$.

Put

$$I_p f(x) = \text{PV} \int \frac{f(x) - f(y)}{|x - y|^{4+p}} dy, \quad \lambda_0 = \left(\frac{c_{4,s}}{2-p} \right)^{1/p}, \quad \tilde{u}(x) = u(\lambda_0 x). \quad (15.2)$$

Here I_p acts on bounded measurable real functions locally C^2 at the evaluation point. Equations (2.1) and (2.3) give $c_{4,s} I_p u = f(u)$. Changing variables in the truncated integral and then taking its principal value gives $I_p[v(\lambda_0 \cdot)](x) = \lambda_0^p (I_p v)(\lambda_0 x)$. Hence $\tilde{u}$ solves $\beta I_p \tilde{u} = \tilde{u} - \tilde{u}^3$. The corresponding comparison functional is

$$E_B^{\text{CCS}}(v) = E_B^\beta(v) = \frac{\beta}{4} \iint_{Q_B} \frac{|v(x) - v(y)|^2}{|x - y|^{4+p}} dx dy + \int_B F(v) dx.$$

The exact localized-energy relation and its change-of-variables calculation are recorded in (15.5) below. Dilation bijects the exterior-data finite-energy classes, without a global L^2 assumption. It preserves strict increase in the fourth coordinate directly: $t_1 < t_2$ implies $\lambda_0 t_1 < \lambda_0 t_2$, so the order of the two values is preserved even for a measurable strictly increasing function. Since $\lambda_0 t \rightarrow \pm\infty$, the endpoints become $v^\pm(\lambda_0 x')$. Although the cited proposition is stated for orders in $(0, 1)$, the authors explicitly observe immediately afterward that its comparison argument works for orders in $(0, 2)$. We give the variational, endpoint and contact details for the present $p \in (1, 2)$ in this normalization.

For finite-energy competitors with exterior datum in $[-1, 1]$, clipping by $T(r) = \max\{-1, \min\{r, 1\}\}$ preserves that datum and decreases the energy: T is 1-Lipschitz and $F(T(r)) \leq F(r)$. If a, b are bounded finite-energy functions equal almost everywhere outside a bounded open set B , then

$$\iint \frac{|(a-b)(x) - (a-b)(y)|^2}{|x - y|^{4+p}} dx dy \leq 2 \iint_{Q_B} \frac{|a(x) - a(y)|^2 + |b(x) - b(y)|^2}{|x - y|^{4+p}} dx dy. \quad (15.3)$$

Indeed the increment is zero off Q_B , and on that domain the square of the difference of two increments is at most twice the sum of their squares. The difference is bounded and supported in the finite-measure set B , so it is also in L^2 and therefore belongs to $H^s(\mathbb{R}^4)$. Put

$$X_B = \{h \in H^s(\mathbb{R}^4) : h = 0 \text{ almost everywhere on } B^c\}.$$

This is a closed subspace of H^s , since Sobolev convergence implies L^2 convergence.

Let U be any smooth bounded entire solution of $\beta I_p U = f(U)$, with $|U| \leq 1$, and let B be a smooth ball. The preceding finite-energy estimate applies to U . Take a clipped minimizing sequence z_j with exterior value U and $E_B^{\text{CCS}}(z_j) \leq E_B^{\text{CCS}}(U) + 1$. For $h_j = z_j - U$, the bound $|h_j| \leq 2$ and the zero exterior values give a uniform L^2 bound; (15.3) gives the uniform Sobolev seminorm bound. Weak compactness in H^s gives a subsequence $h_j \rightharpoonup h \in X_B$. Rellich compactness on a larger ball containing $\bar{B}$ gives strong L^2 convergence there and, since all differences and the limit vanish outside B , on all of $\mathbb{R}^4$. After another extraction the convergence is almost everywhere. Consequently $z_j \rightarrow z = U + h$ almost everywhere globally and $|z| \leq 1$ almost everywhere. Fatou for the nonnegative pair term and dominated convergence for the bounded potentials on B prove that z is a minimizer. Clipping all competitors makes it minimizing against every finite-energy competitor, including unbounded ones. Only the differences, rather than U or z , have been placed in the global Sobolev space.

We justify the weak equation on all of X_B , since unbounded Sobolev variations need not have finite quartic potential energy. For real $h \in X_B$, put $h_K = \max\{-K, \min\{h, K\}\}$. Lipschitz contraction gives $h_K \in X_B$. For each fixed K and ε , the function $z + \varepsilon h_K$ has finite pair energy and bounded potential on B , and is therefore admissible. Differentiation at $\varepsilon = 0$, using Cauchy–Schwarz for the mixed pair term, gives

$$\frac{\beta}{2} \iint_{Q_B} \frac{(z(x) - z(y))(h_K(x) - h_K(y))}{|x - y|^{4+p}} dx dy + \int_B (z^3 - z) h_K dx = 0.$$

Moreover $h_K \rightarrow h$ in H^s . The remainder $r_K(t) = t - \max\{-K, \min\{t, K\}\}$ is 1-Lipschitz, so the increments of $r_K(h)$ are dominated by those of h , and $|r_K(h)| \leq |h|$. Both remainders tend pointwise to zero; dominated convergence applies to their L^2 norm and Gagliardo seminorm. The

pair term is continuous in this norm by Cauchy–Schwarz and the finite localized pair energy of z . Since $z^3 - z \in L^2(B)$, passing to the limit proves the same weak equation for every $h \in X_B$.

The datum U satisfies this identity on the full test class directly. Write $I_{p,\delta}U(x) = \int_{|x-y|>\delta} (U(x) - U(y))|x-y|^{-4-p} dy$. For each $\delta > 0$, boundedness of U and $h \in L^1(B)$ justify Fubini and symmetry, giving

$$\frac{1}{2} \iint_{|x-y|>\delta} \frac{(U(x) - U(y))(h(x) - h(y))}{|x-y|^{4+p}} dx dy = \int_B h I_{p,\delta}U dx.$$

The mixed integrand is absolutely integrable by Cauchy–Schwarz, using the finite localized pair energy of U and the Sobolev seminorm of h . On B , the Hessian bound on a fixed larger compact set and the bounded tail control the symmetric increment and give a uniform bound for $I_{p,\delta}U$, $0 < \delta < 1$. Dominated convergence on both sides, multiplication by β and the equation for U give the same full weak identity. No density assertion at the boundary is used.

Subtracting the identities shows that $d = z - U \in X_B$ solves

$$(-\Delta)^s d = \frac{c_{4,s}}{\beta} [f(z) - f(U)] \quad \text{weakly in } B.$$

The coefficient here is exact: translation Plancherel and Tonelli, with the multiplier normalization in (2.1), identify the $c_{4,s}/2$ Gagliardo pairing with $\int (-\Delta)^{s/2} d (-\Delta)^{s/2} h$ for every $d, h \in H^s$. Thus this is the full zero-exterior weak formulation of [29, Definition 2.1], with bounded forcing. The ball is a bounded Lipschitz domain with an exterior ball condition, so [29, Proposition 1.1]¹ gives $d \in C^s(\mathbb{R}^4)$. Its continuous representative is zero on the open exterior because it is zero there almost everywhere, and on ∂B by continuity. Hence $z = U + d$ is globally continuous, equals U on B^c including the boundary, and is clamped everywhere.

For interior regularity write

$$(-\Delta)^s d = G(x, d), \quad G(x, r) = \frac{c_{4,s}}{\beta} [f(r + U(x)) - f(U(x))].$$

Here $G \in C_{\text{loc}}^{0,1}(B \times \mathbb{R})$, d is bounded and the ball is bounded and $C^{1,1}$. Corollary 1.6(a) of the same version gives $d \in C_{\text{loc}}^\alpha(B)$ for every $s \leq \alpha < 1 + p$. The noninteger $\alpha_{\text{reg}} = 2 + (p - 1)/2$ lies in this range because $p > 1$, so z is locally C^2 . Its symmetric integral is continuous on compact subballs by the local Hessian bound and global boundedness. Truncated symmetry against compact smooth tests identifies that integral with the weak operator, making $\beta I_p z = f(z)$ pointwise. This construction applies in particular with $U = \tilde{u}$, and with every other smooth bounded entire comparison solution having values in $[-1, 1]$.

We will repeatedly use one contact observation. Suppose $U, V \in C_b(\mathbb{R}^4) \cap C_{\text{loc}}^2(B)$ solve the same βI_p equation in the open set B , $D = U - V \geq 0$ globally, and $D(x_0) = 0$ at an interior point. Then $\nabla D(x_0) = 0$ and $D(y) = O(|y - x_0|^2)$ near x_0 . Boundedness controls the tail, so

$$\beta I_p D(x_0) = -\beta \int_{\mathbb{R}^4} \frac{D(y)}{|x_0 - y|^{4+p}} dy = f(U(x_0)) - f(V(x_0)) = 0$$

is absolutely convergent. The positive kernel and $D \geq 0$ force $D = 0$ almost everywhere, hence everywhere by continuity. In particular, positivity on any exterior ball forbids interior contact. This single observation applies to the wells, endpoint slides and the parent-solution slides below.

Every endpoint model minimizes E_B^{CCS} on every smooth ball B . A well has zero energy. We record the ordered-family comparison used for a planar endpoint and for the monotone solution.

Lemma 15.3 (Ordered-family comparison). *Write $\beta = 2 - p$, and let $B \subset \mathbb{R}^4$ be nonempty, bounded and open. Suppose $z \in C_b(\mathbb{R}^4) \cap C_{\text{loc}}^2(B)$ solves $\beta I_p z = f(z)$ in B , and let $\{U_\tau : \tau \in \mathbb{R}\} \subset C_b(\mathbb{R}^4) \cap C_{\text{loc}}^2(B)$ solve the same equation there. Assume $U_\sigma(x) < U_\tau(x)$ for every $x \in \mathbb{R}^4$ whenever $\sigma < \tau$, the family is continuous in the parameter in $C_{\text{loc}}^0(\mathbb{R}^4)$, and has limits $L^\pm \in C(\overline{B})$ uniformly on $\overline{B}$ as $\tau \rightarrow \pm\infty$. If $z = U_0$ on B^c , including ∂B , and $L^- < z < L^+$ on $\overline{B}$, then $z \equiv U_0$ on $\mathbb{R}^4$.*

¹Here and for Corollary 1.6(a), the numbering refers to [arXiv:1207.5985v1](https://arxiv.org/abs/1207.5985v1): Proposition 1.1 is on printed p. 2, and Definition 2.1 and Corollary 1.6(a) are on printed p. 6.

Proof. The two continuous endpoint gaps have positive minima on $\overline{B}$. Uniform convergence to L^+ therefore makes a sufficiently large positive translate dominate z on this compact set. Every positive parameter already gives $U_\tau > U_0 = z$ on B^c , by strict ordering. Thus

$$A_+ = \{\tau \geq 0 : U_\tau \geq z \text{ on } \mathbb{R}^4\}$$

is nonempty and upward closed. Put $\tau_+ = \inf A_+$. For a sequence of parameters in A_+ tending to the infimum, local parameter continuity passes the inequality at every fixed point to the limit. Hence $U_{\tau_+} \geq z$ globally and the threshold is attained.

If $\tau_+ > 0$, the difference is strictly positive on B^c , including the boundary. If it has no zero in B , it has a positive minimum on $\overline{B}$. Continuity in the parameter uniformly on this compact set permits a small downward change, still positive, preserving domination there. On the entire exterior ordering relative to U_0 still gives domination, so the smaller parameter belongs to A_+ , a contradiction. Consequently there is an interior zero. The preceding contact observation applies to $U_{\tau_+} - z \geq 0$: both functions are globally bounded and continuous, locally C^2 and solve the same equation in B . Strict ordering and continuity give a strictly positive difference on a small ball outside $\overline{B}$, so that contact is impossible. Therefore $\tau_+ = 0$ and $U_0 \geq z$.

For the lower comparison,

$$A_- = \{\tau \leq 0 : U_\tau \leq z \text{ on } \mathbb{R}^4\}$$

is nonempty by uniform convergence to L^- , and is downward closed. Its supremum $\tau_- \leq 0$ is attained by pointwise parameter continuity. A negative threshold gives $z - U_{\tau_-} > 0$ on B^c . If the difference has no interior zero, its positive compact minimum allows a small upward change still below zero, contradicting the supremum. An interior zero contradicts the same contact observation. Thus $\tau_- = 0$, giving $U_0 \leq z$. The two comparisons prove equality. The parameter changes use only compact gaps; exterior ordering is pointwise and requires no uniform gap at infinity. $\square$

For a planar endpoint

$$U_e(x', t) = q_s(\lambda_0 a \cdot x' - b), \quad a \in S^2,$$

the normalized layer equation in $\mathbb{R}^3$, the exact cylindrical identity with $m = 3$, $k = 1$, and (15.2) give $\beta I_p U_e = f(U_e)$ in $\mathbb{R}^4$. The global three-dimensional regularity gives the C_b^2 hypothesis of the cylindrical identity; the model is smooth and strictly between the wells. Its localized energy is finite by boundedness and the local gradient bound, using the radial integrals $\int_0^1 r^{1-p} dr$ and $\int_1^\infty r^{-1-p} dr$. Take the continuous clamped minimizer z_e supplied by the preceding Dirichlet construction with exterior datum U_e . If it attained 1 inside B , apply contact to $1 - z_e \geq 0$; the exterior layer gives a strictly positive exterior ball and a contradiction. Contact for $z_e + 1$ similarly excludes -1 . On the boundary $z_e = U_e$ is strictly between the wells. Thus $-1 < z_e < 1$ on $\overline{B}$, with positive compact margins.

The family $U_{e,\tau} = q_s(\lambda_0 a \cdot x' - b + \tau)$ consists of solutions of the same equation, is strictly ordered by $q'_s > 0$, and is continuous in C_{loc}^0 with respect to τ . Its limits are ± 1 uniformly on $\overline{B}$, since the affine argument without τ ranges over a bounded interval there. Lemma 15.3 gives $z_e = U_e$. Its unrestricted finite-energy minimality proves endpoint minimality.

The ordering of a general minimizer uses the exact lattice identity. For two functions a, b , put $M = \max(a, b)$, $m = \min(a, b)$, $h = a - b$, and $h_+ = \max\{h, 0\}$, $h_- = \max\{-h, 0\}$. Since $M = b + h_+$, $m = b - h_-$ and $h_+ h_- = 0$ pointwise, expanding the increments gives

$$\begin{aligned} (M_x - M_y)^2 + (m_x - m_y)^2 - (a_x - a_y)^2 - (b_x - b_y)^2 \\ = -2[h_+(x)h_-(y) + h_-(x)h_+(y)] \leq 0. \end{aligned} \tag{15.4}$$

The potential values are exchanged exactly: $F(M) + F(m) = F(a) + F(b)$. When the two original localized energies are finite, each new pair integral and potential integral is finite, because its nonnegative integrand is bounded by the sum of the two original nonnegative integrands. The cross-product integral in (15.4) is finite for the same reason. Thus the integrated identities concern finite quantities.

Let $U = \tilde{u}$, and take its continuous clamped minimizer z on a ball B . The actual endpoints $\tilde{v}^\pm(x', t) = v^\pm(\lambda_0 x')$ are separately wells or planar models by Lemma 15.2. The endpoint comparison above therefore makes each a minimizer for its own exterior datum. Their energies are finite. The original localized-energy bound for u above and the independent algebraic dilation (15.5) give

$$E_B^{\text{CCS}}(U) = \lambda_0^{-4} E_{\lambda_0 B}(u) < \infty.$$

The minimizer z has no greater energy than this admissible datum. Thus its localized energy is finite as well. Outside B , including its boundary,

$$\tilde{v}^- < z = U < \tilde{v}^+.$$

For the upper endpoint use $a = z$, $b = \tilde{v}^+$. Then $m = U$ and $M = \tilde{v}^+$ on B^c . Both are admissible finite-energy competitors, so

$$E_B^{\text{CCS}}(z) \leq E_B^{\text{CCS}}(m), \quad E_B^{\text{CCS}}(\tilde{v}^+) \leq E_B^{\text{CCS}}(M).$$

If $z > \tilde{v}^+$ somewhere inside, continuity gives a ball compactly contained in B where h_+ has a positive lower bound. The strict exterior sandwich gives a ball outside $\bar{B}$ where h_- has a positive lower bound. The two balls have positive separation, and their products in both orders belong to $\mathcal{Q}_B$. The kernel is strictly positive there. Consequently, with the quarter ordered interaction in the energy,

$$\begin{aligned} & E_B^{\text{CCS}}(M) + E_B^{\text{CCS}}(m) - E_B^{\text{CCS}}(z) - E_B^{\text{CCS}}(\tilde{v}^+) \\ &= -\frac{2-p}{2} \iint_{\mathcal{Q}_B} \frac{h_+(x)h_-(y) + h_-(x)h_+(y)}{|x-y|^{4+p}} dx dy < 0. \end{aligned}$$

This contradicts the sum of the two minimality inequalities.

For the lower endpoint take $a = z$, $b = \tilde{v}^-$. Now $M = U$ and $m = \tilde{v}^-$ on the exterior. An interior violation produces a ball where $h_- > 0$ and the exterior sandwich a ball where $h_+ > 0$. The identical negative decrement contradicts the inequalities $E_B^{\text{CCS}}(z) \leq E_B^{\text{CCS}}(M)$ and $E_B^{\text{CCS}}(\tilde{v}^-) \leq E_B^{\text{CCS}}(m)$. Thus $\tilde{v}^- \leq z \leq \tilde{v}^+$ globally. Contact excludes an interior equality with either endpoint, since the corresponding nonnegative difference is strictly positive on an exterior ball. The boundary trace already gives strict inequalities. Compactness and continuity therefore give

$$\delta_B = \min_{\bar{B}} \min\{z - \tilde{v}^-, \tilde{v}^+ - z\} > 0.$$

Apply the same single comparison lemma to

$$U_\tau(x', t) = U(x', t + \tau) = u(\lambda_0 x', \lambda_0(t + \tau)).$$

Every member solves the same βI_p equation by translation invariance. The family is strictly ordered and continuous locally uniformly in its real parameter, by the smoothness and strict monotonicity of u . The full real-parameter C_{loc}^2 translation convergence of Lemma 15.2, with parameter $\lambda_0 \tau$, supplies uniform convergence on $\bar{B}$ to $\tilde{v}^\pm$. The compact margins just proved and the boundary condition $z = U_0$ verify the remaining hypotheses. Lemma 15.3 therefore yields $z = U$. It follows that $U = \tilde{u}$ is minimizing on every smooth ball.

Proposition 15.4. *The strictly monotone solution u is a global minimizer of E_Ω against every finite-energy competitor equal to u almost everywhere outside a bounded open set Ω .*

Proof. The ball comparisons imply comparisons on every bounded open set in the same βI_p normalization. Choose a ball B with $\bar{\Omega} \subset B$. If $v = U$ almost everywhere on Ω^c and has finite $E_\Omega^{\text{CCS}}(v)$, every pair in $\mathcal{Q}_B \setminus \mathcal{Q}_\Omega$ has both endpoints in Ω^c , where its increments agree with those of U . The added potential on $B \setminus \Omega$ also agrees. Their common nonnegative sum $A_{B,\Omega}$ is finite because it is bounded by the finite $E_B^{\text{CCS}}(U)$. Thus

$$E_B^{\text{CCS}}(v) = E_\Omega^{\text{CCS}}(v) + A_{B,\Omega} < \infty, \quad E_B^{\text{CCS}}(U) = E_\Omega^{\text{CCS}}(U) + A_{B,\Omega} < \infty.$$

Subtracting these finite identities and using the ball comparison gives $E_\Omega^{\text{CCS}}(U) \leq E_\Omega^{\text{CCS}}(v)$. An empty Ω has zero energy and is included.

Return to the original normalization. For any possibly unbounded finite-energy competitor $v_0 = u$ almost everywhere on Ω^c , put $v = T(v_0)$ with $T(r) = \max\{-1, \min\{r, 1\}\}$. The preceding clipping estimate preserves the exterior datum and gives $E_\Omega(v) \leq E_\Omega(v_0)$. Set $v_\lambda(x) = v(\lambda_0 x)$ and $\Omega_\lambda = \lambda_0^{-1}\Omega$. The change of variables in both points sends $\mathcal{Q}_{\Omega_\lambda}$ exactly onto $\mathcal{Q}_\Omega$. The bare interaction scales by λ_0^{p-4} and the potential by λ_0^{-4} . Since $(2-p)\lambda_0^p = c_{4,s}$,

$$E_{\Omega_\lambda}^{\text{CCS}}(v_\lambda) = \lambda_0^{-4} E_\Omega(v). \quad (15.5)$$

The same identity holds for $U = \tilde{u}$ and u . Apply the bounded-set comparison to the scaled clipped competitor and multiply by λ_0^4 . This gives

$$E_\Omega(u) \leq E_\Omega(v) \leq E_\Omega(v_0).$$

The dilation uses only nonnegative localized integrals, and the domain extension adds only finite common terms. Global Sobolev norms in the Dirichlet construction concern differences from exterior data; no global H^s or L^2 condition on the entire solution or on the original competitor is needed. $\square$

16. THE FOUR-DIMENSIONAL CONCLUSION

Continue with the strictly monotone solution in $\mathbb{R}^4$ from Section 15, with the same $s \in (1/2, 1)$ and $p = 2s$. Divide the energy by $c_{4,s} > 0$ and write

$$J_\Omega(v) = c_{4,s}^{-1} E_\Omega(v) = \frac{1}{4} \iint_{\mathcal{Q}_\Omega} \frac{|v(x) - v(y)|^2}{|x - y|^{4+2s}} dx dy + \int_\Omega \tilde{F}(v) dx, \quad \tilde{F}(r) = \frac{(1 - r^2)^2}{4c_{4,s}}. \quad (16.1)$$

Positive multiplication preserves every exterior-data energy comparison. The finite localized energy established in Section 15 and Proposition 15.4 therefore make u a global minimizer of J : for every bounded open Ω and every finite-energy $v = u$ almost everywhere on Ω^c , $J_\Omega(u) \leq J_\Omega(v)$. Strict monotonicity makes u nonconstant, so Lemma 2.1 gives $-1 < u < 1$. Since both energy terms are nonnegative and $c_{4,s} > 0$, the identity gives $J_\Omega(v) < \infty$ if and only if $E_\Omega(v) < \infty$, so the finite exterior-data class is unchanged. It also agrees with the comparison-energy class of the preceding section. Write E_Ω^β for the functional with kinetic coefficient $\beta = 2 - p > 0$ and the same potential F . Nonnegativity of both terms gives

$$E_\Omega^\beta(v) \leq \max\{\beta/c_{4,s}, 1\} E_\Omega(v), \quad E_\Omega(v) \leq \max\{c_{4,s}/\beta, 1\} E_\Omega^\beta(v).$$

In particular these comparisons include every competitor with values in $[-1, 1]$ used by Savin; infinite-energy competitors satisfy the comparison trivially.

The potential $\tilde{F}$ is a polynomial, with

$$\tilde{F}'(r) = \frac{r^3 - r}{c_{4,s}}, \quad \tilde{F}''(r) = \frac{3r^2 - 1}{c_{4,s}}.$$

It is therefore C^2 on $[-1, 1]$, vanishes with its first derivative at ± 1 , is positive in $(-1, 1)$, and has $\tilde{F}''(\pm 1) = 2/c_{4,s} > 0$. Thus (16.1) is exactly Savin's functional [33, pp. 1881–1882], with $W = \tilde{F}$. With I_p defined in (15.2), his operator is $\Delta_s = -I_p$, and hence has the opposite sign to our positive fractional Laplacian:

$$\Delta_s u(x) = \text{PV} \int_{\mathbb{R}^4} \frac{u(y) - u(x)}{|x - y|^{4+2s}} dy = -c_{4,s}^{-1} (-\Delta)^s u(x) = \frac{u(x)^3 - u(x)}{c_{4,s}} = \tilde{F}'(u(x)).$$

The exponent is still $2s$, and the ambient dimension remains four. The applicable hypotheses in [33, pp. 1881–1882, Theorem 1.2] are the range $|u| \leq 1$, locally finite energy with every bounded exterior-data comparison in this range, the C^2 double-well conditions just checked, $1/2 < s < 1$, and $n \leq 7$. The comparison class, all potential conditions, and the unchanged order have been verified above, and $n = 4 \leq 7$.

Savin's global-minimizer classification [33, Theorem 1.2, p. 1882] therefore yields $u(x) = H(e \cdot x)$ for some $e \in S^3$, with any additive offset absorbed into the profile H . This theorem does not require prescribed directional well limits.

Proof of Theorem 1.1. The preceding Savin application, using the global minimality of Proposition 15.4 in the exact functional (16.1), gives

$$u(x) = H(e \cdot x) \quad (x \in \mathbb{R}^4)$$

for some $e \in S^3$ and a function H . The profile is recovered pointwise as $H(\tau) = u(\tau e)$. The earlier global regularity therefore makes H smooth with bounded derivatives of every order, in particular $H \in C_b^2(\mathbb{R})$. Differentiating the representation gives

$$0 < \partial_4 u(x) = e_4 H'(e \cdot x).$$

It follows that $e_4 \neq 0$, and evaluating at $x = \tau e$ gives $e_4 H'(\tau) > 0$ for every real τ . If $e_4 < 0$, replace (e, H) by $(-e, \hat{H})$, where $\hat{H}(\tau) = H(-\tau)$. The represented function u is unchanged. After relabeling, $e_4 > 0$ and $H' > 0$ everywhere. Lemma 15.1 makes the original u stable. Choose an orthogonal matrix O with first column e , so that $u(O(\tau, y')) = H(\tau)$ on $\mathbb{R} \times \mathbb{R}^3$. Lemma 3.2, with $m = 1$, $k = 3$, orthogonal invariance and the established C_b^2 regularity, gives both

$$(-\partial_\tau^2)^s H = H - H^3 \quad \text{on } \mathbb{R} \quad \text{and stability of } H,$$

with the one-dimensional Fourier normalization $c_{1,s}$. Lemma 4.2 applies. Since $H' > 0$, it excludes the constant wells and every decreasing layer; therefore $H(\tau) = q_s(\tau - a)$ for some $a \in \mathbb{R}$. This proves $u(x) = q_s(e \cdot x - a)$ with $e \in S^3$ and $e_4 > 0$.

Finally write $e = (e', e_4)$. For every fixed $x' \in \mathbb{R}^3$, $e' \cdot x' + e_4 t - a$ tends to the corresponding end of the real line as $t \rightarrow \pm\infty$. The defining limits of q_s give

$$\lim_{t \rightarrow -\infty} u(x', t) = -1, \quad \lim_{t \rightarrow +\infty} u(x', t) = 1.$$

□

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